

Do not open this Test Booklet until you are asked to do so.

This Booklet contains 32 pages, including Rough Page.

Important Instructions:

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**.
3. Wherever the symbols/constants are not mentioned, they are to be considered as per their standard meaning/value.
4. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
8. **The CODE for this Booklet is “45”. Make sure to enter this code in the OMR answer sheet.**
9. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
10. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
11. Each candidate must show on-demand his/her Admit Card to the Invigilator.
12. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
13. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
14. Use of Electronic/Manual Calculator is prohibited.
15. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination along with Public Examinations (Prevention of unfair means act 2024).
16. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
17. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
18. If a candidate marks more than one answers for a question in the **OMR Sheet**, it will be treated as incorrect and negative marking will be applicable.

Name of the Candidate (in Capitals) :

Roll Number : in figures

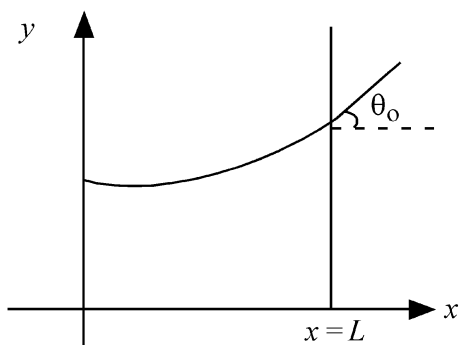
: in words

Centre of Examination (in Capitals) :

Candidate's Signature : Invigilator's Signature :

Fascimile signature stamp of Centre Superintendent :

- 1 Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z direction. When filled with a liquid of surface tension S and density ρ , the liquid surface makes angle θ_0 ($\theta_0 \ll 1$) with the x -axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is :



(take $\theta(x) = \sin \theta(x) = \tan \theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity)

- 2 A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is
- (1) 100 (2) 125
(3) 150 (4) 250

- 3 An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) moving with speed $c/100$ (c = speed of light) is injected into a magnetic field $\vec{B}$ of magnitude 9×10^{-4} T perpendicular to its direction of motion. We wish to apply a uniform electric field $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then (speed of light $c = 3 \times 10^8$ ms $^{-1}$)

- (1) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^4 V m $^{-1}$
(2) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^2 V m $^{-1}$
(3) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^2 V m $^{-1}$
(4) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^4 V m $^{-1}$

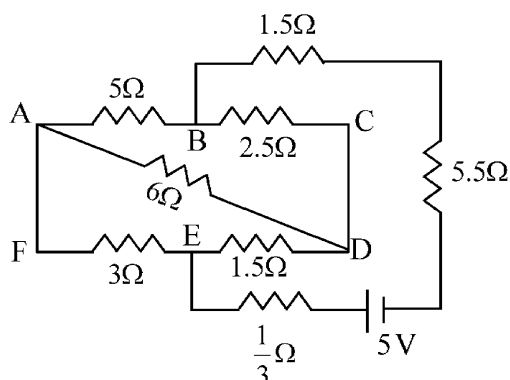
- 4 There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The coefficient of kinetic friction (μ_k) between the object and the rough surface is close to

- (1) 0.25 (2) 0.40
(3) 0.5 (4) 0.75

- 5 The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying breaks, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the breaks on cars A and B, respectively, then the ratio F_A/F_B is

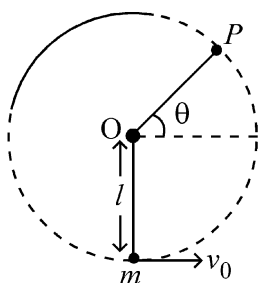
- (1) $\frac{3}{2}$ (2) $\frac{2}{3}$
(3) $\frac{1}{3}$ (4) $\frac{1}{2}$

- 6 The current passing through the battery in the given circuit, is:



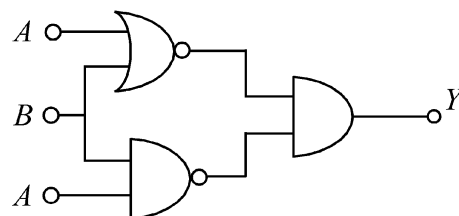
- (1) 2.0 A (2) 0.5 A
(3) 2.5 A (4) 1.5 A

- 7 A bob of heavy mass m is suspended by a light string of length l . The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:



- (1) $(\sin \theta)^{1/2}$
(2) $\left(\frac{1}{2 + 3 \sin \theta} \right)^{1/2}$
(3) $\left(\frac{\cos \theta}{2 + 3 \sin \theta} \right)^{1/2}$
(4) $\left(\frac{\sin \theta}{2 + 3 \sin \theta} \right)^{1/2}$

- 8 The output (Y) of the given logic implementation is similar to the output of an/a _____ gate.



- (1) AND (2) NAND
(3) OR (4) NOR

- 9 The electric field in a plane electromagnetic wave is given by

$$E_z = 60 \cos (5x + 1.5 \times 10^9 t) \text{ V/m}.$$

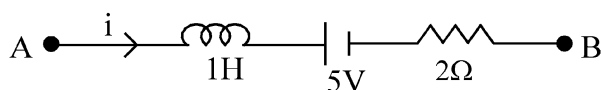
Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field) :

- (1) $B_y = 2 \times 10^{-7} \cos (5x + 1.5 \times 10^9 t) T$
(2) $B_x = 2 \times 10^{-7} \cos (5x + 1.5 \times 10^9 t) T$
(3) $B_z = 60 \cos (5x + 1.5 \times 10^9 t) T$
(4) $B_y = 60 \sin (5x + 1.5 \times 10^9 t) T$

- 10 A ball of mass 0.5 kg is dropped from a height of 40 m . The ball hits the ground and rises to a height of 10 m . The impulse imparted to the ball during its collision with the ground is (Take $g = 9.8\text{ m/s}^2$)

- (1) 21 NS (2) 7 NS
(3) 0 (4) 84 NS

- 11 AB is a part of an electrical circuit (see figure). The potential difference " $V_A - V_B$ ", at the instant when current $i = 2$ A and is increasing at a rate of 1 amp / second is:



- (1) 5 volt
(2) 6 volt
(3) 9 volt
(4) 10 volt
- 12 A 2 amp current is flowing through two different small circular copper coils having radii ratio 1:2. The ratio of their respective magnetic moments will be
- (1) 1:4 (2) 1:2
(3) 2:1 (4) 4:1
- 13 In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively –
- (1) $4p$ and $4m$
(2) p^4 and $4m$
(3) $4p$ and m^4
(4) p^4 and m^4

- 14 An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperature 27°C . The mass of the oxygen withdrawn from the cylinder is nearly equal to :

[Given, $R = \frac{100}{12} \text{ J mol}^{-1}\text{K}^{-1}$, and

molecular mass of $\text{O}_2 = 32$,

1 atm pressure = $1.01 \times 10^5 \text{ N/m}^2$]

- (1) 0.125 kg (2) 0.144 kg
(3) 0.116 kg (4) 0.156 kg
- 15 In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is
- (1) $-\frac{2}{(x+2)^3}$ (2) $-\frac{2}{(2x+1)^3}$
(3) $+\frac{2}{(x+1)^3}$ (4) $+\frac{2}{2x+1}$
- 16 To an ac power supply of 220 V at 50 Hz, a resistor of 20Ω , a capacitor of reactance 25Ω and an inductor of reactance 45Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively -
- (1) 7.8 A and 30°
(2) 7.8 A and 45°
(3) 15.6 A and 30°
(4) 15.6 A and 45°

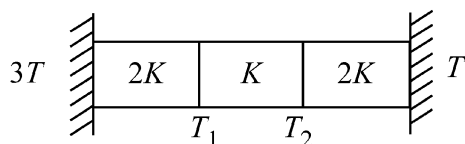
- 17 The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

- (1) 100 days (2) 105 days
(3) 115 days (4) 108 days

- 18 A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

- (1) $\frac{he}{\pi m}$ (2) $\frac{he}{2\pi m}$
(3) $\frac{heB}{\pi m}$ (4) $\frac{heB}{2\pi m}$

- 19 Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K . The left end of the combination is maintained at temperature $3T$ and the right end at T . The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2 . The ratio T_1/T_2 is



- (1) $\frac{3}{2}$ (2) $\frac{4}{3}$
(3) $\frac{5}{3}$ (4) $\frac{5}{4}$

- 20 The plates of a parallel plate capacitor are separated by d . Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates.

If $K_1 = 1.25 K_2$, the value of K_1 is :

- (1) 2.66 (2) 2.33
(3) 1.60 (4) 1.33

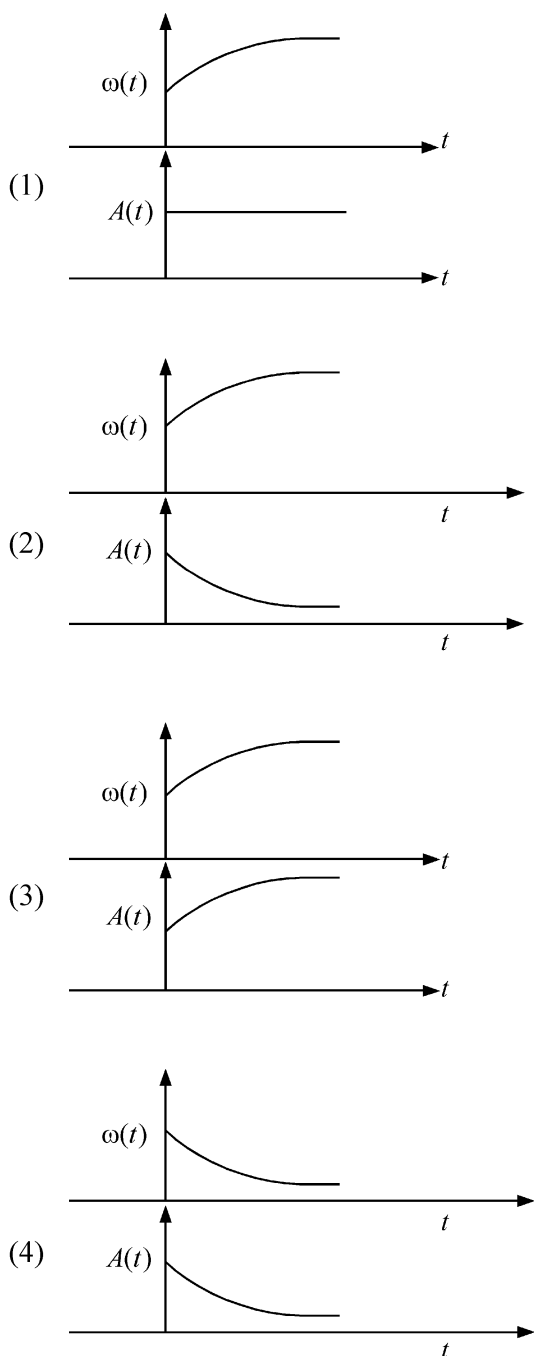
- 21 Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooter with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

- (1) 9 min, 40 km/h
(2) 25 min, 100 km/h
(3) 10 min, 90 km/h
(4) 15 min, 120 km/h

- 22 A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (take $g = 10 \text{ m/s}^2$)

- (1) 100 N
(2) $100\sqrt{3}$ N
(3) 200 N
(4) $200\sqrt{3}$ N

- 23 In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t . Which one of the following options schematically depicts these changes correctly?



- 24 A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A . It is filled with a gas of density ρ and takes a spherical shape of radius R . When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T . If the speed $v(r)$ of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

- (1) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$
- (2) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$
- (3) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$
- (4) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

- 25 Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed.

If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is

- (1) 5.18 cm (2) 5.08 cm
- (3) 4.98 cm (4) 5.00 cm

- 26 A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is :

- (1) zero at all places
- (2) constant between the plates and zero outside the plates
- (3) non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates
- (4) zero between the plates and non-zero outside

- 27 An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then-
- (1) reflected light is completely polarized and the angle of reflection is close to 60°
 - (2) reflected light is partially polarized and the angle of reflection is close to 30°
 - (3) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30° , respectively.
 - (4) transmitted light is completely polarized with angle of refraction close to 30°
- 28 Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F . A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as :
- (1) $\frac{3F}{5}$
 - (2) $\frac{2F}{3}$
 - (3) $\frac{F}{2}$
 - (4) $\frac{3F}{8}$
- 29 A container has two chambers of volumes $V_1 = 2$ litres and $V_2 = 3$ litres separated by a partition made of a thermal insulator. The chambers contains $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures $p_1 = 1$ atm and $p_2 = 2$ atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of :
- (1) 1.3 atm
 - (2) 1.6 atm
 - (3) 1.4 atm
 - (4) 1.8 atm
- 30 A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius r of the n^{th} orbit and the particle's speed v in the orbit depend on n as
- (1) $r \propto n^{1/3}$; $v \propto n^{1/3}$
 - (2) $r \propto n^{1/3}$; $v \propto n^{2/3}$
 - (3) $r \propto n^{2/3}$; $v \propto n^{1/3}$
 - (4) $r \propto n^{4/3}$; $v \propto n^{-1/3}$
- 31 The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of 1 year on Mercury ?
- (1) 88 earth days
 - (2) 225 earth days
 - (3) 172 earth days
 - (4) 124 earth days
- 32 A body weighs 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is :
- (1) 16 N
 - (2) 27 N
 - (3) 32 N
 - (4) 36 N
- 33 A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is :
- (1) $\frac{R}{64}$
 - (2) $\frac{R}{32}$
 - (3) $\frac{R}{16}$
 - (4) $\frac{R}{8}$

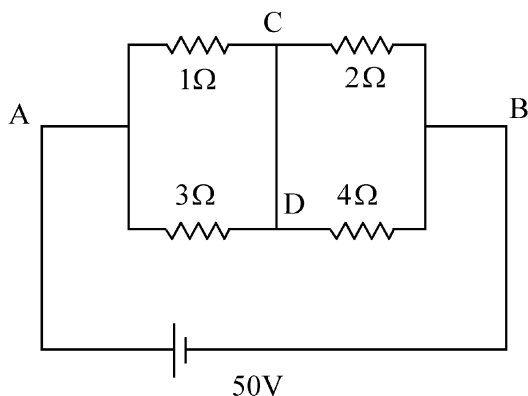
34 De-Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to (Given Bohr radius = 0.052 nm)

- (1) 0.067 nm (2) 0.67 nm
(3) 1.67 nm (4) 2.67 nm

35 An electric dipole with dipole moment $5 \times 10^{-6} \text{ Cm}$ is aligned with the direction of a uniform electric field of magnitude $4 \times 10^5 \text{ N/C}$. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is :

- (1) 0.8 J (2) 1.0 J
(3) 1.2 J (4) 1.5 J

36 A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is :

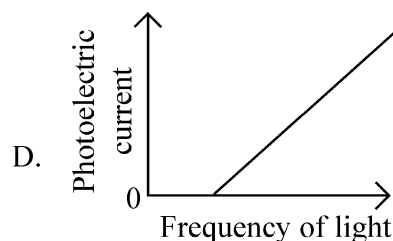
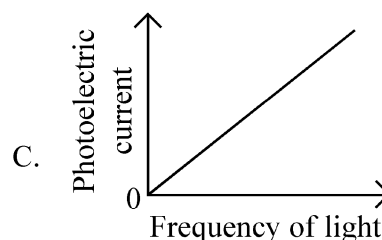
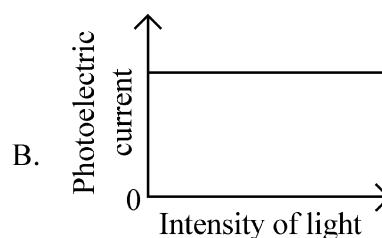
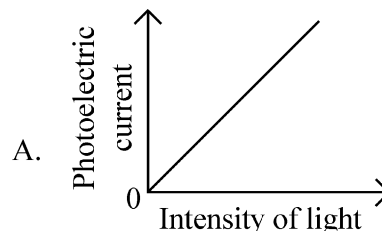


- (1) 1.5 A (2) 2.0 A
(3) 2.5 A (4) 3.0 A

37 A photon and an electron (mass m) have the same energy E . The ratio $(\lambda_{\text{photon}}/\lambda_{\text{electron}})$ of their de Broglie wavelengths is: (c is the speed of light)

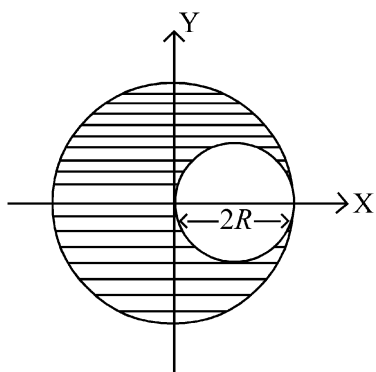
- (1) $\sqrt{E/2m}$ (2) $c\sqrt{2mE}$
(3) $c\sqrt{2m/E}$ (4) $\frac{1}{c}\sqrt{E/2m}$

38 Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?



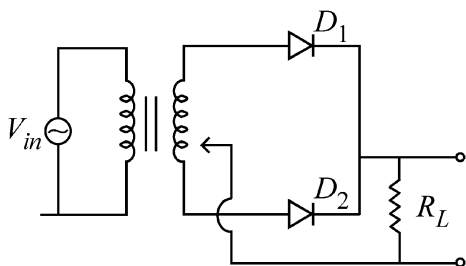
- (1) A only
(2) A and C
(3) A and D
(4) B and D

- 39 A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is :



- (1) $\frac{7}{8}$ (2) $\frac{7}{40}$
 (3) $\frac{7}{57}$ (4) $\frac{7}{64}$

- 40 A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{in} = 220\sin(100\pi t)$ volt, then at $t = 15 \text{ msec}$



- (1) D_1 is forward biased, D_2 is reverse biased
 (2) D_1 is reverse biased, D_2 is forward biased
 (3) D_1 and D_2 both are forward biased
 (4) D_1 and D_2 both are reverse biased

- 41 Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B , respectively.

On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm , respectively. If the change in their internal energy is the same, then the ratio

r_A/r_B is equal to

- (1) $\frac{4}{3}$ (2) $\frac{3}{4}$
 (3) $\frac{2}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2}$

- 42 A physical quantity P is related to four observations a , b , c and d as follows:

$$P = a^3 b^2 / c \sqrt{d}$$

The percentage errors of measurement in a , b , c and d are 1%, 3%, 2%, and 4% respectively.

The percentage error in the quantity P is

- (1) 10% (2) 2%
 (3) 13% (4) 15%

- 43 The intensity of transmitted light when a polaroid sheet, placed between two crossed polaroids at 22.5° from the polarization axis of one of the polaroid, is (I_0 is the intensity of polarised light after passing through the first polaroid):

- (1) $\frac{I_0}{2}$ (2) $\frac{I_0}{4}$
(3) $\frac{I_0}{8}$ (4) $\frac{I_0}{16}$

- 44 Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2 , respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q/A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is:

- (1) $\frac{k_2}{k_1}$ (2) $\frac{k_1}{k_2}$
(3) $\sqrt{\frac{k_2}{k_1}}$ (4) $\sqrt{\frac{k_1}{k_2}}$

- 45 A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to :

- (1) $\frac{f}{2}$ (2) f
(3) $\frac{3f}{2}$ (4) $2f$

- 46 The ratio of the wavelengths of the light absorbed by a Hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions, respectively, is

- (1) $\frac{1}{36}$ (2) $\frac{1}{16}$
(3) $\frac{1}{9}$ (4) $\frac{1}{4}$

- 47 Which of the following statements are true?

- A. Unlike Ga that has a very high melting point, Cs has a very low melting point.
B. On Pauling scale, the electronegativity values of N and Cl are not the same.
C. Ar, K^+ , Cl^- , Ca^{2+} , and S^{2-} are all isoelectronic species.
D. The correct order of the first ionization enthalpies of Na, Mg, Al, and Si is $Si > Al > Mg > Na$.
E. The atomic radius of Cs is greater than that of Li and Rb.

Choose the **correct** answer from the options given below :

- (1) A, B, and E only
(2) C and E only
(3) C and D only
(4) A, C, and E only

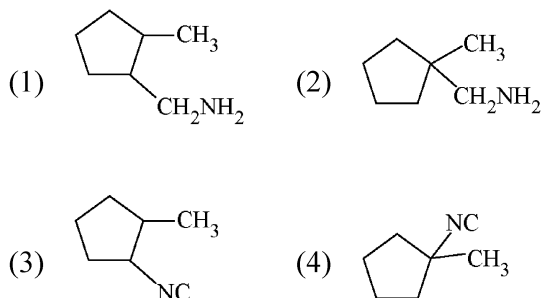
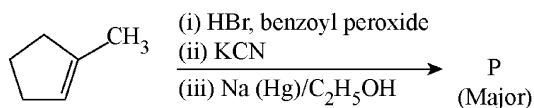
- 48 Match List I with List II

List I (Ion)	List II (Group Number in Cation Analysis)
A. Co^{2+}	I. Group-I
B. Mg^{2+}	II. Group-III
C. Pb^{2+}	III. Group-IV
D. Al^{3+}	IV. Group-VI

Choose the **correct** answer from the options given below :

- (1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-III, B-II, C-IV, D-I
(4) A-III, B-II, C-I, D-IV

- 49 Predict the major product 'P' in the following sequence of reactions -



- 50 Energy and radius of first Bohr orbit of He^+ and Li^{2+} are

[Given $R_H = 2.18 \times 10^{-18} \text{ J}$, $a_0 = 52.9 \text{ pm}$]

- (1) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-18} \text{ J}$;
 $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -8.72 \times 10^{-18} \text{ J}$;
 $r_n(\text{He}^+) = 26.4 \text{ pm}$
 (2) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-18} \text{ J}$;
 $r_n(\text{Li}^{2+}) = 26.4 \text{ pm}$
 $E_n(\text{He}^+) = -19.62 \times 10^{-18} \text{ J}$;
 $r_n(\text{He}^+) = 17.6 \text{ pm}$
 (3) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-16} \text{ J}$;
 $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -8.72 \times 10^{-16} \text{ J}$;
 $r_n(\text{He}^+) = 26.4 \text{ pm}$
 (4) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-16} \text{ J}$;
 $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -19.62 \times 10^{-16} \text{ J}$;
 $r_n(\text{He}^+) = 17.6 \text{ pm}$

- 51 Which of the following are paramagnetic?

- A. $[\text{NiCl}_4]^{2-}$ B. $\text{Ni}(\text{CO})_4$
 C. $[\text{Ni}(\text{CN})_4]^{2-}$ D. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 E. $\text{Ni}(\text{PPh}_3)_4$

Choose the **correct** answer from the options given below:

- (1) A and C only (2) B and E only
 (3) A and D only (4) A, D and E only

- 52 Given below are two statements :

Statement I : Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II : Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both Statement I and Statement II are correct
 (2) Both Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Statement I is incorrect but Statement II is correct

- 53 Which among the following electronic configurations belong to main group elements?

- A. $[\text{Ne}]3s^1$ B. $[\text{Ar}]3d^3 4s^2$
 C. $[\text{Kr}]4d^{10} 5s^2 5p^5$ D. $[\text{Ar}]3d^{10} 4s^1$
 E. $[\text{Rn}]5f^0 6d^2 7s^2$

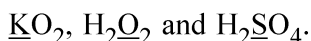
Choose the **correct** answer from the option given below :

- (1) B and E only (2) A and C only
 (3) D and E only (4) A, C and D only

54 Dalton's Atomic theory could not explain which of the following?

- (1) Law of conservation of mass
- (2) Law of constant proportion
- (3) Law of multiple proportion
- (4) Law of gaseous volume

55 Consider the following compounds :



The oxidation states of the underlined elements in them are, respectively,

- (1) +1, -1, and +6
- (2) +2, -2, and +6
- (3) +1, -2, and +4
- (4) +4, -4, and +6

56 If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closest to :

- (1) 2 minutes (2) 4 minutes
- (3) 5 minutes (4) 10 minutes

57 The correct order of the wavelength of light absorbed by the following complexes is,

- A. $[Co(NH_3)_6]^{3+}$ B. $[Co(CN)_6]^{3-}$
- C. $[Cu(H_2O)_4]^{2+}$ D. $[Ti(H_2O)_6]^{3+}$

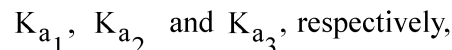
Choose the **correct** answer from the options given below:

- (1) $B < D < A < C$ (2) $B < A < D < C$
- (3) $C < D < A < B$ (4) $C < A < D < B$

58 Which one of the following compounds can exist as cis-trans isomers?

- (1) Pent-1-ene
- (2) 2-Methylhex-2-ene
- (3) 1,1-Dimethylcyclopropane
- (4) 1,2-Dimethylcyclohexane

59 Phosphoric acid ionizes in three steps with their ionization constant values



while K is the overall ionization constant.

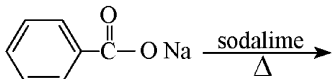
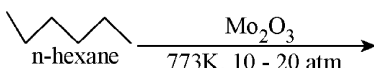
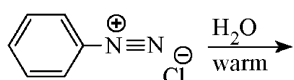
Which of the following statements are true?

- A. $\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$
- B. H_3PO_4 is a stronger acid than $H_2PO_4^-$ and HPO_4^{2-} .
- C. $K_{a_1} > K_{a_2} > K_{a_3}$
- D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$

Choose the **correct** answer from the options given below:

- (1) A and B only
- (2) A and C only
- (3) B, C and D only
- (4) A, B and C only

60 Which one of the following reactions does **NOT** give benzene as the product ?

- (1) 
- (2) 
- (3) $H-C \equiv C-H \xrightarrow[\text{at } 873 \text{ K}]{\text{red hot Iron Tube}}$
- (4) 

61 If the molar conductivity (Λ_m) of a 0.050 mol L⁻¹ solution of a monobasic weak acid is 90 S cm² mol⁻¹, its extent (degree) of dissociation will be

[Assume $\Lambda_+^\circ = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and

$\Lambda_-^\circ = 50.4 \text{ S cm}^2 \text{ mol}^{-1}$.]

- (1) 0.115 (2) 0.125
- (3) 0.225 (4) 0.215

62 Given below are two statements :

Statement I : A hypothetical diatomic molecule with bond order zero is quite stable.

Statement II : As bond order increases, the bond length increases.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

63 Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?

- (1) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- (2) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
- (3) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (4) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}$

64 Match List - I with List - II

List-I	List-II
A. XeO_3	I. sp^3d ; linear
B. XeF_2	II. sp^3 ; pyramidal
C. XeOF_4	III. sp^3d^3 ; distorted octahedral
D. XeF_6	IV. sp^3d^2 ; square pyramidal

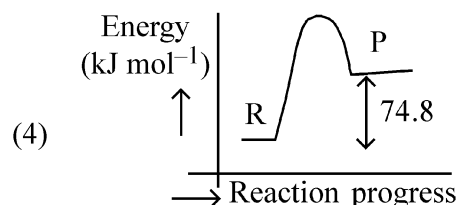
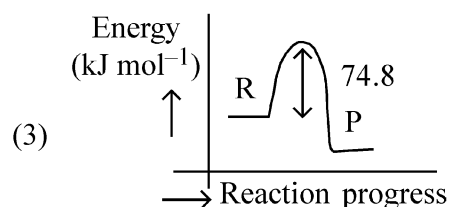
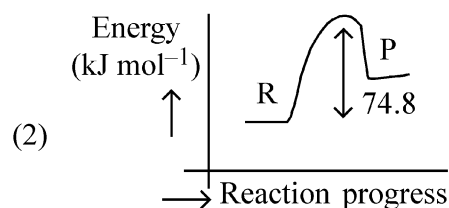
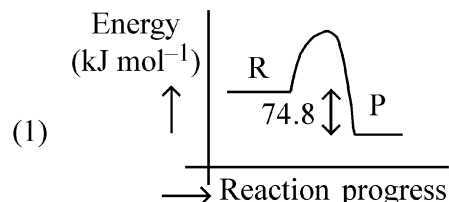
Choose the **correct** answer from the options given below :

- (1) A-II, B-I, C-IV, D-III
- (2) A-II, B-I, C-III, D-IV
- (3) A-IV, B-II, C-III, D-I
- (4) A-IV, B-II, C-I, D-III

65 $\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g}); \Delta H = -74.8 \text{ kJ mol}^{-1}$

Which of the following diagrams gives an accurate representation of the above reaction?

[R $\rightarrow$ reactants; P $\rightarrow$ products]



66 Match List - I with List - II

List-I (Example)	List-II (Type of Solution)
A. Humidity	I. Solid in solid
B. Alloys	II. Liquid in gas
C. Amalgams	III. Solid in gas
D. Smoke	IV. Liquid in solid

Choose the **correct** answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B-II, C-I, D-IV

67 The correct order of decreasing basic strength of the given amines is :

- (1) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine
- (2) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline
- (3) N-ethylethanamine > ethanamine > N-methylaniline > benzenamine
- (4) benzenamine > ethanamine > N-methylaniline > N-ethylethanamine

68 Among the following, choose the ones with equal number of atoms.

- A. 212 g of Na_2CO_3 (s) [molar mass = 106 g]
- B. 248 g of Na_2O (s) [molar mass = 62 g]
- C. 240 g of NaOH (s) [molar mass = 40 g]
- D. 12 g of H_2 (g) [molar mass = 2 g]
- E. 220 g of CO_2 (g) [molar mass = 44 g]

Choose the **correct** answer from the options given below :

- (1) A, B, and C only
- (2) A, B, and D only
- (3) B, C, and D only
- (4) B, D, and E only

69 Match List I with List II.

List I (Name of Vitamin)	List II (Deficiency disease)
A. Vitamin B_{12}	I. Cheilosis
B. Vitamin D	II. Convulsions
C. Vitamin B_2	III. Rickets
D. Vitamin B_6	IV. Pernicious anaemia

Choose the **correct** answer from the options given below :

- (1) A-I, B-III, C-II, D-IV
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-III, C-I, D-IV
- (4) A-IV, B-III, C-II, D-I

70 The correct order of decreasing acidity of the following aliphatic acids is :

- (1) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
- (2) $\text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH} > \text{HCOOH}$
- (3) $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$
- (4) $\text{HCOOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH}$

71 Given below are two statements :

Statement I : Ferromagnetism is considered as an extreme form of paramagnetism.

Statement II : The number of unpaired electrons in a Cr^{2+} ion ($Z = 24$) is the same as that of a Nd^{3+} ion ($Z = 60$).

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

72 Match List I with List II

List I (Mixture)	List II (Method of Separation)
A. $\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	I. Distillation under reduced pressure
B. Crude oil in petroleum industry	II. Steam distillation
C. Glycerol from spent-lye	III. Fractional distillation
D. Aniline - water	IV. Simple distillation

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-I, D-II
 (2) A-IV, B-III, C-II, D-I
 (3) A-III, B-IV, C-I, D-II
 (4) A-III, B-IV, C-II, D-I

73 For the reaction $\text{A(g)} \rightleftharpoons 2\text{B(g)}$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K.

[Given : $R = 0.0831 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

K_p for the reaction at 1000 K is

- (1) 83.1 (2) 2.077×10^5
 (3) 0.033 (4) 0.021

74 Given below are two statements :

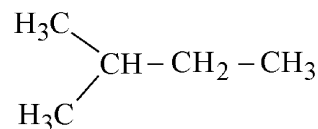
Statement I : Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at 273 - 278 K. It decomposes easily in the dry state.

Statement II : Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

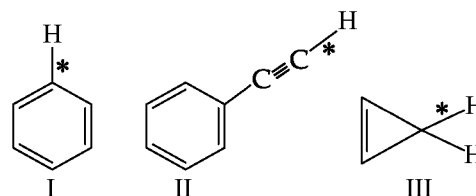
- (1) Both Statement I and Statement II are correct
 (2) Both Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Statement I is incorrect but Statement II is correct

75 How many products (including stereoisomers) are expected from monochlorination of the following compound?



- (1) 2 (2) 3
 (3) 5 (4) 6

76 Among the given compounds I-III, the correct order of bond dissociation energy of C-H bond marked with * is :

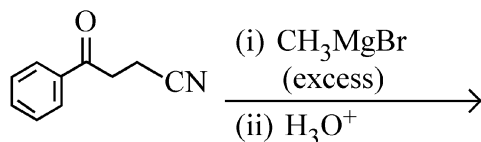


- (1) $\text{II} > \text{I} > \text{III}$ (2) $\text{I} > \text{II} > \text{III}$
 (3) $\text{III} > \text{II} > \text{I}$ (4) $\text{II} > \text{III} > \text{I}$

77 Which one of the following compounds **does not** decolourize bromine water?

- (1) (2)
 (3) (4)

- 78 The major product of the following reaction is :



- (1)
- (2)
- (3)
- (4)

- 79 Which of the following aqueous solution will exhibit highest boiling point?

- (1) 0.01M Urea
 (2) 0.01M KNO_3
 (3) 0.01M Na_2SO_4
 (4) 0.015M $\text{C}_6\text{H}_{12}\text{O}_6$

- 80 Match List - I with List - II

List-I	List-II
A. Haber process	I. Fe catalyst
B. Wacker oxidation	II. PdCl_2
C. Wilkinson catalyst	III. $[(\text{PPh}_3)_3\text{RhCl}]$
D. Ziegler catalyst	IV. TiCl_4 with $\text{Al}(\text{CH}_3)_3$

Choose the **correct** answer from the options given below :

- (1) A-I, B-II, C-IV, D-III
 (2) A-II, B-III, C-I, D-IV
 (3) A-I, B-II, C-III, D-IV
 (4) A-I, B-IV, C-III, D-II

- 81 5 moles of liquid X and 10 moles of liquid Y make a solution having a vapour pressure of 70 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Which of the following is true regarding the described solution?

- (1) The solution shows positive deviation.
 (2) The solution shows negative deviation.
 (3) The solution is ideal.
 (4) The solution has volume greater than the sum of individual volumes.

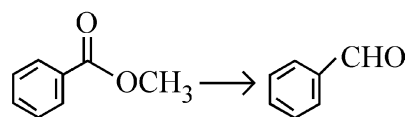
- 82 Sugar 'X'

- A. is found in honey.
 B. is a keto sugar.
 C. exists in α and β - anomeric forms.
 D. is laevorotatory.

'X' is :

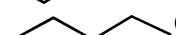
- (1) D-Glucose (2) D-Fructose
 (3) Maltose (4) Sucrose

- 83 Identify the suitable reagent for the following conversion.



- (1) (i) LiAlH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$
 (2) (i) $\text{AlH}(\text{iBu})_2$, (ii) H_2O
 (3) (i) NaBH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$
 (4) $\text{H}_2 / \text{Pd-BaSO}_4$

- 84 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) :  I undergoes S_N2 reaction faster than  Cl.

Reason (R) : Iodine is a better leaving group because of its large size.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

- 85 The standard heat of formation, in kcal/mol of Ba^{2+} is :

[Given : standard heat of formation of SO_4^{2-} ion (aq) = -216 kcal/mol, standard heat of crystallisation of

$BaSO_4(s)$ = -4.5 kcal/mol, standard heat of formation of $BaSO_4(s)$ = -349 kcal/mol]

- (1) -128.5 (2) -133.0
- (3) $+133.0$ (4) $+220.5$

- 86 Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula C_4H_8O is :

- (1) 6 (2) 8
- (3) 10 (4) 11

- 87 Identify the correct orders against the property mentioned

- A. $H_2O > NH_3 > CHCl_3$ – dipole moment
- B. $XeF_4 > XeO_3 > XeF_2$ – number of lone pairs on central atom
- C. $O-H > C-H > N-O$ – bond length
- D. $N_2 > O_2 > H_2$ – bond enthalpy

Choose the **correct** answer from the options given below :

- (1) A, D only (2) B, D only
- (3) A, C only (4) B, C only

- 88 Higher yield of NO in

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ can be obtained at

[ΔH of the reaction = $+180.7$ kJ mol $^{-1}$]

- A. higher temperature
- B. lower temperature
- C. higher concentration of N_2
- D. higher concentration of O_2

Choose the **correct** answer from the options given below:

- (1) A, D only (2) B, C only
- (3) B, C, D only (4) A, C, D only

- 89 If the rate constant of a reaction is 0.03 s $^{-1}$, how much time does it take for 7.2 mol L $^{-1}$ concentration of the reactant to get reduced to 0.9 mol L $^{-1}$?

(Given: $\log 2 = 0.301$)

- (1) 69.3 s (2) 23.1 s
- (3) 210 s (4) 21.0 s

- 90 Which one of the following reactions does **NOT** belong to “Lassaigne’s test”?

- (1) $Na + C + N \xrightarrow{\Delta} NaCN$
- (2) $2Na + S \xrightarrow{\Delta} Na_2S$
- (3) $Na + X \xrightarrow{\Delta} + NaX$
- (4) $2CuO + C \xrightarrow{\Delta} 2Cu + CO_2$

- 91 The complex II of mitochondrial electron transport chain is also known as
 (1) Cytochrome bc_1
 (2) Succinate dehydrogenase
 (3) Cytochrome c oxidase
 (4) NADH dehydrogenase
- 92 Polymerase chain reaction (PCR) amplifies DNA following the equation.
 (1) N^2 (2) 2^n
 (3) $2n + 1$ (4) $2N^2$
- 93 What are the potential drawbacks in adoption of the IVF method?
 A. High fatality risk to mother
 B. Expensive instruments and reagents
 C. Husband/wife necessary for being donors
 D. Less adoption of orphans
 E. Not available in India
 F. Possibility that the early embryo does not survive
 Choose the correct answer from the options given below :
 (1) B, D, F only
 (2) A, C, D, F only
 (3) A, B, C, D only
 (4) A, B, C, E, F only
- 94 What is the name of the blood vessel that carries deoxygenated blood from the body to the heart in a frog ?
 (1) Aorta
 (2) Pulmonary artery
 (3) Pulmonary vein
 (4) Vena cava
- 95 Which one of the following statements refers to Reductionist Biology?
 (1) Physico-chemical approach to study and understand living organisms.
 (2) Physiological approach to study and understand living organisms.
 (3) Chemical approach to study and understand living organisms.
 (4) Behavioural approach to study and understand living organisms.

- 96 Given below are two statements :
Statement I : In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.
Statement II : DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.
 In the light of the above statements, choose the **most appropriate** answer from the options given below :
 (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct
- 97 Epiphytes that are growing on a mango branch is an example of which of the following?
 (1) Commensalism
 (2) Mutualism
 (3) Predation
 (4) Amensalism
- 98 From the statements given below choose the **correct** option :
 A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
 B. Each ribosome has two sub-units.
 C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S.
 D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.
 E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S.
 (1) A, B, C are true
 (2) A, B, D are true
 (3) A, B, E are true
 (4) B, D, E are true

99 Which one of the following is an example of ex-situ conservation?

- (1) National Park
- (2) Wildlife Sanctuary
- (3) Zoos and botanical gardens
- (4) Protected areas

100 Given below are two statements :

Statement I : The primary source of energy in an ecosystem is solar energy.

Statement II : The rate of production of organic matter during photosynthesis in an ecosystem is called net primary productivity (NPP).

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

101 Match **List - I** with **List - II**.

List - I	List - II
A. Emphysema	I. Rapid spasms in muscle due to low Ca^{++} in body fluid
B. Angina Pectoris	II. Damaged alveolar walls and decreased respiratory surface
C. Glomerulo-nephritis	III. Acute chest pain when not enough oxygen is reaching to heart muscle
D. Tetany	IV. Inflammation of glomeruli of kidney

Choose the **correct** answer from the options given below :

- (1) A-III, B-I, C-IV, D-II
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-II, B-III, C-IV, D-I

102 Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Both wind and water pollinated flowers are not very colourful and do not produce nectar.

Reason (R) : The flowers produce enormous amount of pollen grains in wind and water pollinated flowers.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

103 Which of the following is an example of non-distilled alcoholic beverage produced by yeast?

- (1) Whisky
- (2) Brandy
- (3) Beer
- (4) Rum

104 Given below are two statements :

Statement I : In a floral formula $\oplus$ stands for zygomorphic nature of the flower, and $\underline{\text{G}}$ stands for inferior ovary.

Statement II : In a floral formula $\oplus$ stands for actinomorphic nature of the flower and $\underline{\text{G}}$ stands for superior ovary.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

105 Streptokinase produced by *bacterium Streptococcus* is used for

- (1) Curd production
- (2) Ethanol production
- (3) Liver disease treatment
- (4) Removing clots from blood vessels

106 Which chromosome in the human genome has the highest number of genes?

- (1) Chromosome X
- (2) Chromosome Y
- (3) Chromosome 1
- (4) Chromosome 10

107 Which of the following statement is **correct** about location of the male frog copulatory pad ?

- (1) First and Second digit of fore limb
- (2) First digit of hind limb
- (3) Second digit of fore limb
- (4) First digit of the fore limb

108 Which one of the following phytohormones promotes nutrient mobilization which helps in the delay of leaf senescence in plants?

- (1) Ethylene (2) Absciscic acid
- (3) Gibberellin (4) Cytokinin

109 While trying to find out the characteristic of a newly found animal, a researcher did the histology of adult animal and observed a cavity with presence of mesodermal tissue towards the body wall but no mesodermal tissue was observed towards the alimentary canal. What could be the possible coelome of that animal ?

- (1) Acoelomate
- (2) Pseudocoelomate
- (3) Schizocoelomate
- (4) Spongocoelomate

110 Match **List - I** with **List - II**.

List - I

List - II

- | | |
|-----------------|----------------------|
| A. Head | I. Enzymes |
| B. Middle piece | II. Sperm motility |
| C. Acrosome | III. Energy |
| D. Tail | IV. Genetic material |

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-I, D-IV

111 Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.

- A. Prothallus stage
- B. Meiosis in spore mother cells
- C. Fertilisation
- D. Formation of archegonia and antheridia in gametophyte.
- E. Transfer of antherozoids to the archegonia in presence of water.

Choose the **correct** answer from the options given below :

- (1) B, A, D, E, C
- (2) B, A, E, C, D
- (3) D, E, C, A, B
- (4) E, D, C, B, A

112 Cardiac activities of the heart are regulated by :

- A. Nodal tissue
- B. A special neural centre in the medulla oblongata
- C. Adrenal medullary hormones
- D. Adrenal cortical hormones

Choose the **correct** answer from the options given below :

- (1) A, B and C Only
- (2) A, B, C and D
- (3) A, C and D Only
- (4) A, B and D Only

113 Which of following organisms *cannot* fix nitrogen?

- A. *Azotobacter* B. *Oscillatoria*
C. *Anabaena* D. *Volvox*
E. *Nostoc*

Choose the **correct** answer from the options given below:

- (1) A only (2) D only
(3) B only (4) E only

114 Given below are two statements :

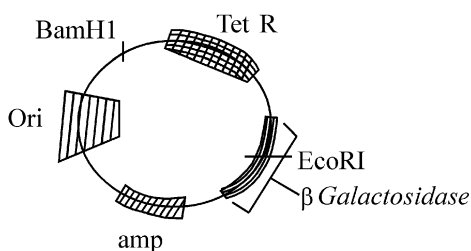
Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

115



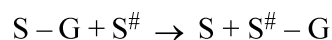
In the above represented plasmid an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

- (1) Using ampicillin & tetracyclin containing medium plate.
(2) Blue color colonies will be selected.
(3) White color colonies will be selected.
(4) Blue color colonies grown on ampicillin plates can be selected.

116 Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

- (1) Bacterium (2) Yeast
(3) Virus (4) Phage

117 Name the class of enzyme that usually catalyze the following reaction :



Where, G $\rightarrow$ a group other than hydrogen

S $\rightarrow$ a substrate

S[#] $\rightarrow$ another substrate

- (1) Hydrolase (2) Lyase
(3) Transferase (4) Ligase

118 Find the statement that is **NOT** correct with regard to the structure of monocot stem.

- (1) Hypodermis is parenchymatous.
(2) Vascular bundles are scattered.
(3) Vascular bundles are conjoint and closed.
(4) Phloem parenchyma is absent.

119 The correct sequence of events in the life cycle of bryophytes is

- A. Fusion of antherozoid with egg.
B. Attachment of gametophyte to substratum.
C. Reduction division to produce haploid spores.
D. Formation of sporophyte.
E. Release of antherozoids into water.

Choose the **correct** answer from the options given below :

- (1) D, E, A, C, B
(2) B, E, A, C, D
(3) B, E, A, D, C
(4) D, E, A, B, C

120 Which are correct:

- A. Computed tomography and magnetic resonance imaging detect cancers of internal organs.
- B. Chemotherapeutics drugs are used to kill non-cancerous cells.
- C. α -interferon activate the cancer patients' immune system and helps in destroying the tumour.
- D. Chemotherapeutic drugs are biological response modifiers.
- E. In the case of leukaemia blood cell counts are decreased.

Choose the **correct** answer from the options given below:

- (1) B and D only (2) D and E only
(3) C and D only (4) A and C only

121 Match List - I with List - II.

List - I	List - II
A. Centromere	I. Mitochondrion
B. Cilium	II. Cell division
C. Cristae	III. Cell movement
D. Cell membrane	IV. Phospholipid Bilayer

Choose the **correct** answer from the options given below :

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-II, C-III, D-I
(4) A-II, B-III, C-I, D-IV

122 Match List I with List II :

List I	List II
A. Chlorophyll a	I. Yellow-green
B. Chlorophyll b	II. Yellow
C. Xanthophylls	III. Blue-green
D. Carotenoids	IV. Yellow to Yellow-orange

Choose the option with all **correct** matches.

- (1) A-III, B-IV, C-II, D-I
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-II, C-IV, D-III
(4) A-I, B-IV, C-III, D-II

123 Find the correct statements :

- A. In human pregnancy, the major organ systems are formed at the end of 12 weeks.
- B. In human pregnancy the major organ systems are formed at the end of 8 weeks.
- C. In human pregnancy heart is formed after one month of gestation.
- D. In human pregnancy, limbs and digits develop by the end of second month.
- E. In human pregnancy the appearance of hair is usually observed in the fifth month.

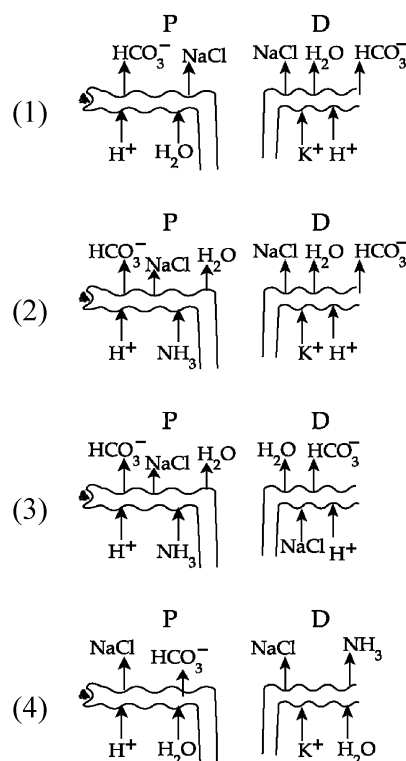
Choose the **correct** answer from the options given below :

- (1) A and E Only
(2) B and C Only
(3) B, C, D and E Only
(4) A, C, D and E Only

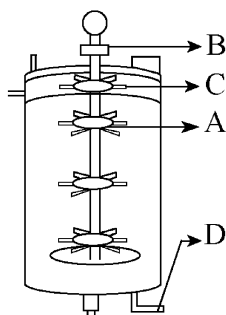
124 In the seeds of cereals, the outer covering of endosperm separates the embryo by a protein-rich layer called :

- (1) Coleoptile (2) Coleorhiza
(3) Integument (4) Aleurone layer

125 Which of the following diagrams is correct with regard to the proximal (P) and distal (D) tubule of the Nephron.



- 126 Identify the part of a bio-reactor which is used as a foam braker from the given figure.



- (1) A (2) B
(3) D (4) C

- 127 Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A typical unfertilised, angiosperm embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R) : The egg apparatus has 2 polar nuclei.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
(2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**
(3) **A** is true but **R** is false
(4) **A** is false but **R** is true

- 128 A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is :

- (1) Mesosome
(2) Chromatophores
(3) Cristae
(4) Endoplasmic Reticulum

- 129 Which of the following are the post-transcriptional events in an eukaryotic cell?

- A. Transport of pre-mRNA to cytoplasm prior to splicing.
B. Removal of introns and joining of exons.
C. Addition of methyl group at 5' end of hnRNA.
D. Addition of adenine residues at 3' end of hnRNA.
E. Base pairing of two complementary RNAs.

Choose the **correct** answer from the options given below :

- (1) A, B, C only
(2) B, C, D only
(3) B, C, E only
(4) C, D, E only

- 130 What is the pattern of inheritance for polygenic trait?

- (1) Mendelian inheritance pattern
(2) Non-mendelian inheritance pattern
(3) Autosomal dominant pattern
(4) X-linked recessive inheritance pattern

- 131 Which one of the following enzymes contains 'Haem' as the prosthetic group?

- (1) RuBisCo
(2) Carbonic anhydrase
(3) Succinate dehydrogenase
(4) Catalase

132 Each of the following characteristics represent a Kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organization.

- A. Multicellular heterotrophs with cell wall made of chitin.
- B. Heterotrophs with tissue/organ/organ system level of body organization.
- C. Prokaryotes with cell wall made of polysaccharides and amino acids.
- D. Eukaryotic autotrophs with tissue/organ level of body organization.
- E. Eukaryotes with cellular body organization.

Choose the **correct** answer from the options given below :

- (1) A, C, E, B, D (2) C, E, A, D, B
- (3) A, C, E, D, B (4) C, E, A, B, D

133 Who is known as the father of Ecology in India?

- (1) S. R. Kashyap (2) Ramdeo Misra
- (3) Ram Udar (4) Birbal Sahni

134 Match **List I** with **List II** :

List-I	List-II
A. Alfred Hershey and Martha Chase	I. Streptococcus pneumoniae
B. Euchromatin	II. Densely packed and dark-stained
C. Frederick Griffith	III. Loosely packed and light-stained
D. Heterochromatin	IV. DNA as genetic material confirmation

Choose the **correct** answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-II, C-I, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-II, C-IV, D-I

135 Neoplastic characteristics of cells refer to :

- A. A mass of proliferating cell
- B. Rapid growth of cells
- C. Invasion and damage to the surrounding tissue
- D. Those confined to original location

Choose the **correct** answer from the options given below:

- (1) A, B only (2) A, B, C only
- (3) A, B, D only (4) B, C, D only

136 Given below are two statements :

Statement I : The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

137 Match **List I** with **List II**.

List I	List II
A. Adenosine	I. Nitrogen base
B. Adenylic acid	II. Nucleotide
C. Adenine	III. Nucleoside
D. Alanine	IV. Amino acid

Choose the option with all **correct** matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-III, C-I, D-IV

138 Consider the following :

- A. The reductive division for the human female gametogenesis starts earlier than that of the male gametogenesis.
- B. The gap between the first meiotic division and the second meiotic division is much shorter for males compared to females.
- C. The first polar body is associated with the formation of the primary oocyte.
- D. Luteinizing Hormone (LH) surge leads to disintegration of the endometrium and onset of menstrual bleeding.

Choose the **correct** answer from the options given below :

- (1) A and B are true
- (2) A and C are true
- (3) B and D are true
- (4) B and C are true

139 All living members of the class Cyclostomata are:

- (1) Free living (2) Endoparasite
- (3) Symbiotic (4) Ectoparasite

140 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R) : Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

141 Match **List I** with **List II** :

List I	List II
A. Scutellum	I. Persistent nucellus
B. Non-albuminous seed	II. Cotyledon of Monocot seed
C. Epiblast	III. Groundnut
D. Perisperm	IV. Rudimentary cotyledon

Choose the option with all **correct** matches.

- (1) A-II, B-III, C-IV, D-I
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-II, B-IV, C-III, D-I

142 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : All vertebrates are chordates but all chordates are not vertebrate.

Reason (R) : The members of subphylum vertebrata possess notochord during the embryonic period, the notochord is replaced by a cartilaginous or bony vertebral column in adults.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

143 Identify the statement that is **NOT** correct.

- (1) Each antibody has two light and two heavy chains.
- (2) The heavy and light chains are held together by disulfide bonds.
- (3) Antigen binding site is located at C-terminal region of antibody molecules.
- (4) Constant region of heavy and light chains are located at C-terminus of antibody molecules.

144 Silencing of specific mRNA is possible via RNAi because of -

- (1) Complementary dsRNA
- (2) Inhibitory ssRNA
- (3) Complementary tRNA
- (4) Non-complementary ssRNA

145 Genes R and Y follow independent assortment. If RRYYY produce round yellow seeds and rryy produce wrinkled green seeds, what will be the phenotypic ratio of the F₂ generation?

- (1) Phenotypic ratio - 1 : 2 : 1
- (2) Phenotypic ratio - 3 : 1
- (3) Phenotypic ratio - 9 : 3 : 3 : 1
- (4) Phenotypic ratio - 9 : 7

146 Histones are enriched with -

- (1) Lysine & Arginine
- (2) Leucine & Lysine
- (3) Phenylalanine & Leucine
- (4) Phenylalanine & Arginine

147 The first menstruation is called :

- (1) Menopause
- (2) Menarche
- (3) Diapause
- (4) Ovulation

148 Match List - I with List - II.

List - I

List - II

- | | |
|----------------------------|--------------------------------|
| A. Heart | I. Erythropoietin |
| B. Kidney | II. Aldosterone |
| C. Gastro-intestinal tract | III. Atrial natriuretic factor |
| D. Adrenal Cortex | IV. Secretin |

Choose the **correct** answer from the options given below :

- (1) A-II, B-I, C-III, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-I, C-IV, D-II

149 The protein portion of an enzyme is called :

- (1) Cofactor
- (2) Coenzyme
- (3) Apoenzyme
- (4) Prosthetic group

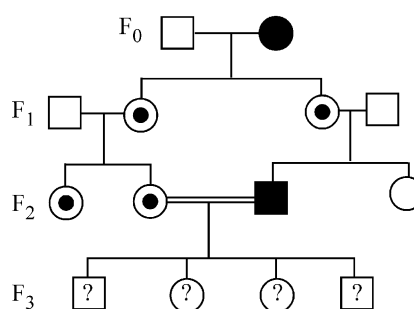
150 Which of the following is the unit of productivity of an Ecosystem?

- (1) gm⁻²
- (2) KCal m⁻²
- (3) KCal m⁻³
- (4) (KCal m⁻²)yr⁻¹

151 Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution.

- (1) Analogy, convergent
- (2) Homology, divergent
- (3) Homology, convergent
- (4) Analogy, divergent

152 With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F₃ generation.



- Unaffected male
 ■ Affected male
 ● Carrier female
 ○ Unaffected female
 ● Affected female

- (1) 1/4
- (2) 1/2
- (3) 1/8
- (4) Zero

- 153** Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

Reason (R) : Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

In light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

- 154** How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?

- (1) 2 Meiosis and 3 Mitosis
- (2) 1 Meiosis and 2 Mitosis
- (3) 1 Meiosis and 3 Mitosis
- (4) No Meiosis and 2 Mitosis

- 155** Which of the following is an example of a zygomorphic flower?

- (1) Petunia (2) Datura
- (3) Pea (4) Chilli

- 156** After maturation, in primary lymphoid organs, the lymphocytes migrate for interaction with antigens to secondary lymphoid organ(s) / tissue(s) like:

- A. thymus B. bone marrow
- C. spleen D. lymph nodes
- E. Peyer's patches

Choose the *correct* answer from the options given below:

- (1) B, C, D only (2) A, B, C only
- (3) E, A, B only (4) C, D, E only

- 157** Given below are two statements :

Statement I : Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.

Statement II : Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

- 158** What is the main function of the spindle fibers during mitosis ?

- (1) To separate the chromosomes
- (2) To synthesize new DNA
- (3) To repair damaged DNA
- (4) To regulate cell growth

- 159** Which one of the following is the characteristic feature of gymnosperms?

- (1) Seeds are enclosed in fruits.
- (2) Seeds are naked.
- (3) Seeds are absent.
- (4) Gymnosperms have flowers for reproduction.

160 Consider the following statements regarding function of adrenal medullary hormones :

- A. It causes pupillary constriction
- B. It is a hyperglycemic hormone
- C. It causes piloerection
- D. It increases strength of heart contraction

Choose the **correct** answer from the options given below :

- (1) C and D Only
- (2) B, C and D Only
- (3) A, C and D Only
- (4) D Only

161 Why can't insulin be given orally to diabetic patients?

- (1) Human body will elicit strong immune response
- (2) It will be digested in Gastro-Intestinal (GI) tract
- (3) Because of structural variation
- (4) Its bioavailability will be increased

162 Match **List I** with **List II**.

List I	List II
A. Pteridophyte	I. <i>Salvia</i>
B. Bryophyte	II. <i>Ginkgo</i>
C. Angiosperm	III. <i>Polytrichum</i>
D. Gymnosperm	IV. <i>Salvinia</i>

Choose the option with all **correct** matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-IV, B-III, C-II, D-I

163 Who proposed that the genetic code for amino acids should be made up of three nucleotides?

- (1) George Gamow
- (2) Francis Crick
- (3) Jacque Monod
- (4) Franklin Stahl

164 Match **List I** with **List II** :

List I	List II
A. The Evil Quartet	I. Cryopreservation
B. Ex situ conservation	II. Alien species invasion
C. <i>Lantana camara</i>	III. Causes of biodiversity losses
D. Dodo	IV. Extinction

Choose the option with all correct matches.

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-IV, D-I

165 Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus ?

- (1) Luteinizing hormone (LH)
- (2) Anti-diuretic hormone (ADH)
- (3) Follicle-stimulating hormone (FSH)
- (4) Adenocorticotrophic hormone (ACTH)

166 Role of the water vascular system in Echinoderms is :

- A. Respiration and Locomotion
- B. Excretion and Locomotion
- C. Capture and transport of food
- D. Digestion and Respiration
- E. Digestion and Excretion

Choose the **correct** answer from the options given below :

- (1) A and B Only
- (2) A and C Only
- (3) B and C Only
- (4) B, D and E Only

167 Which of the following type of immunity is present at the time of birth and is a non-specific type of defence in the human body?

- (1) Acquired Immunity
- (2) Innate Immunity
- (3) Cell-mediated Immunity
- (4) Humoral Immunity

- 168** In bryophytes, the gemmae help in which one of the following?
- (1) Sexual reproduction
 - (2) Asexual reproduction
 - (3) Nutrient absorption
 - (4) Gaseous exchange
- 169** In frog, the Renal portal system is a special venous connection that acts to link :
- (1) Liver and intestine
 - (2) Liver and kidney
 - (3) Kidney and intestine
 - (4) Kidney and lower part of body
- 170** Given below are two statements :
- Statement I :** In ecosystem, there is unidirectional flow of energy of sun from producers to consumers.
- Statement II :** Ecosystems are exempted from 2nd law of thermodynamics.
- In the light of the above statements, choose the **most appropriate** answer from the options given below :
- (1) Both statement I and statement II are correct
 - (2) Both statement I and statement II are incorrect
 - (3) Statement I is correct but statement II is incorrect
 - (4) Statement I is incorrect but statement II is correct
- 171** Which of the following statements about RuBisCO is true?
- (1) It is active only in the dark.
 - (2) It has higher affinity for oxygen than carbon dioxide.
 - (3) It is an enzyme involved in the photolysis of water.
 - (4) It catalyzes the carboxylation of RuBP.
- 172** Which of the following enzyme(s) are **NOT** essential for gene cloning?
- A. Restriction enzymes
 - B. DNA ligase
 - C. DNA mutase
 - D. DNA recombinase
 - E. DNA polymerase
- Choose the **correct** answer from the options given below :
- (1) C and D only (2) A and B only
 - (3) D and E only (4) B and C only
- 173** Read the following statements on plant growth and development.
- A. Parthenocarpy can be induced by auxins.
 - B. Plant growth regulators can be involved in promotion as well as inhibition of growth.
 - C. Dedifferentiation is a pre-requisite for re-differentiation.
 - D. Abscissic acid is a plant growth promoter.
 - E. Apical dominance promotes the growth of lateral buds.
- Choose the option with all correct statements.
- (1) A, B, C only (2) A, C, E only
 - (3) A, D, E only (4) B, D, E only
- 174** Which factor is important for termination of transcription?
- (1) α (alpha) (2) σ (sigma)
 - (3) ρ (rho) (4) γ (gamma)
- 175** Frogs respire in water by skin and buccal cavity and on land by skin, buccal cavity and lungs.
- Choose the **correct** answer from the following :
- (1) The statement is true for water but false for land
 - (2) The statement is true for both the environment
 - (3) The statement is false for water but true for land
 - (4) The statement is false for both the environment

176 Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

- (1) They are monozygotic twins.
- (2) They are fraternal twins.
- (3) They were conceived through in vitro fertilization.
- (4) They have 75% identical genetic content.

177 Which of the following microbes is **NOT** involved in the preparation of household products?

- A. *Aspergillus niger*
- B. *Lactobacillus*
- C. *Trichoderma polysporum*
- D. *Saccharomyces cerevisiae*
- E. *Propionibacterium sharmanii*

Choose the **correct** answer from the options given below:

- (1) A and B only
- (2) A and C only
- (3) C and D only
- (4) C and E only

178 Match List - I with List - II.

List - I	List - II
A. Progesterone	I. Pars intermedia
B. Relaxin	II. Ovary
C. Melanocyte stimulating hormone	III. Adrenal Medulla
D. Catecholamines	IV. Corpus luteum

Choose the **correct** answer from the options given below :

- (1) A-IV, B-II, C-I, D-III
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

179 The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method:

Statement I : The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II : The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

180 Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

- (1) $\frac{dN}{dt} = r \left(\frac{K - N}{K} \right)$
- (2) $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$
- (3) $\frac{dN}{dt} = rN \left(\frac{N - K}{N} \right)$
- (4) $\frac{dN}{dt} = N \left(\frac{r - K}{K} \right)$

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

This Booklet contains **32** pages, including Rough Page.

Do not open this Test Booklet until you are asked to do so.

Important Instructions:

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below:
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos – 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is T1. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
17. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

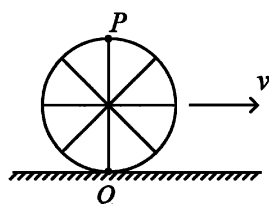
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Candidate's Signature: _____ Invigilator's Signature: _____

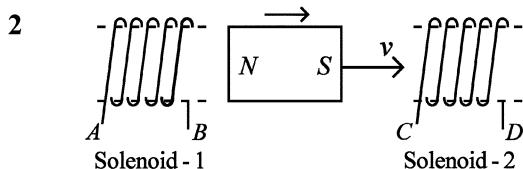
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Physics : Section-A (Q. No. 1 to 35)

- 1 A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?



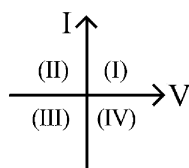
- (1) Both the points P and Q move with equal speed.
- (2) Point P has zero speed.
- (3) Point P moves slower than point Q .
- (4) Point P moves faster than point Q .



In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

- (1) AB and CD
- (2) BA and DC
- (3) AB and DC
- (4) BA and CD

- 3 Consider the following statements A and B and identify the correct answer :

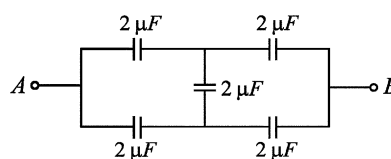


- A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
 - B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.
- (1) Both A and B are correct.
 - (2) Both A and B are incorrect.
 - (3) A is correct but B is incorrect.
 - (4) A is incorrect but B is correct.

- 4 A wire of length ' l ' and resistance 100Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

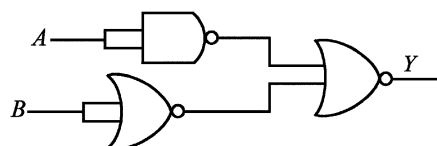
- (1) 55Ω
- (2) 60Ω
- (3) 26Ω
- (4) 52Ω

- 5 In the following circuit, the equivalent capacitance between terminal A and terminal B is :



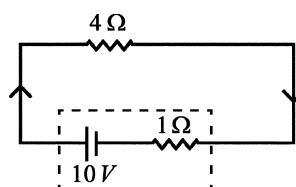
- (1) $0.5 \mu F$
- (2) $4 \mu F$
- (3) $2 \mu F$
- (4) $1 \mu F$

- 6 The output (Y) of the given logic gate is similar to the output of an/a :



- (1) OR gate
- (2) AND gate
- (3) NAND gate
- (4) NOR gate

- 7 The terminal voltage of the battery, whose emf is $10V$ and internal resistance 1Ω , when connected through an external resistance of 4Ω as shown in the figure is :



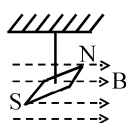
- (1) $8 V$
- (2) $10 V$
- (3) $4 V$
- (4) $6 V$

- 8 The mass of a planet is $\frac{1}{10}$ th that of the earth and

its diameter is half that of the earth. The acceleration due to gravity on that planet is :

- (1) 4.9 m s^{-2} (2) 3.92 m s^{-2}
(3) 19.6 m s^{-2} (4) 9.8 m s^{-2}

- 9 In a uniform magnetic field of 0.049 T , a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is $9.8 \times 10^{-6} \text{ kg m}^2$. If the magnitude of magnetic moment of the needle is $x \times 10^{-5} \text{ Am}^2$; then the value of 'x' is :



- (1) $50 \pi^2$ (2) $1280 \pi^2$
(3) $5 \pi^2$ (4) $128 \pi^2$

- 10 If c is the velocity of light in free space, the correct statements about photon among the following are :

- A. The energy of a photon is $E = h\nu$.
B. The velocity of a photon is c .
C. The momentum of a photon, $p = \frac{h\nu}{c}$.
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses positive charge.

Choose the correct answer from the options given below :

- (1) A, C and D only
(2) A, B, D and E only
(3) A and B only
(4) A, B, C and D only

- 11 The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are $8 \times 10^8 \text{ N m}^{-2}$ and $2 \times 10^{11} \text{ N m}^{-2}$, is :

- (1) 40 mm (2) 8 mm
(3) 4 mm (4) 0.4 mm

- 12 A logic circuit provides the output Y as per the following truth table :

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is :

- (1) $\overline{B}$ (2) B
(3) $A.B + \overline{A}$ (4) $A.\overline{B} + \overline{A}$

- 13 The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is 2400 g cm^2 . The length of the 400 g rod is nearly :

- (1) 20.7 cm (2) 72.0 cm
(3) 8.5 cm (4) 17.5 cm

- 14 In a vernier calipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm , the vernier constant (in cm) is :

- (1) $100N$ (2) $10(N + 1)$
(3) $\frac{1}{10N}$ (4) $\frac{1}{100(N + 1)}$

- 15 Match List I with List II.

List I (Spectral Lines of Hydrogen for transitions from)	List II (Wavelengths (nm))
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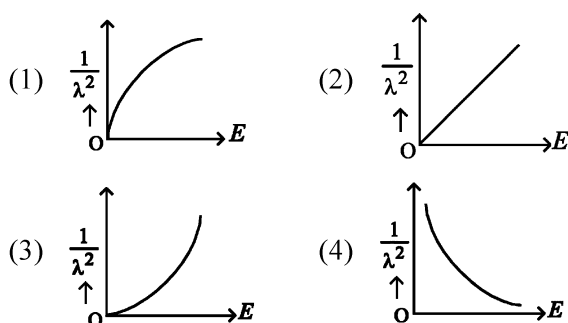
- | | |
|---------------------------|------------|
| A. $n_2 = 3$ to $n_1 = 2$ | I. 410.2 |
| B. $n_2 = 4$ to $n_1 = 2$ | II. 434.1 |
| C. $n_2 = 5$ to $n_1 = 2$ | III. 656.3 |
| D. $n_2 = 6$ to $n_1 = 2$ | IV. 486.1 |

Choose the correct answer from the options given below :

- (1) A-IV, B-III, C-I, D-II
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-II, D-I

- 16 If the monochromatic source in Young's double slit experiment is replaced by white light, then
- (1) there will be a central bright white fringe surrounded by a few coloured fringes.
 - (2) all bright fringes will be of equal width.
 - (3) interference pattern will disappear.
 - (4) there will be a central dark fringe surrounded by a few coloured fringes.

- 17 The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle) :



- 18 Match List-I with List-II.

List-I (Material)	List-II (Susceptibility (χ))
A. Diamagnetic	I. $\chi = 0$
B. Ferromagnetic	II. $0 > \chi \geq -1$
C. Paramagnetic	III. $\chi \gg 1$
D. Non-magnetic	IV. $0 < \chi < \epsilon$ (a small positive number)

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
 - (2) A-IV, B-III, C-II, D-I
 - (3) A-II, B-III, C-IV, D-I
 - (4) A-II, B-I, C-III, D-IV
- 19 An unpolarised light beam strikes a glass surface at Brewster's angle. Then
- (1) both the reflected and refracted light will be completely polarised.
 - (2) the reflected light will be completely polarised but the refracted light will be partially polarised.
 - (3) the reflected light will be partially polarised.
 - (4) the refracted light will be completely polarised.

- 20 A particle moving with uniform speed in a circular path maintains :

- (1) constant velocity but varying acceleration.
- (2) varying velocity and varying acceleration.
- (3) constant velocity.
- (4) constant acceleration.

- 21 The quantities which have the same dimensions as those of solid angle are :

- (1) strain and arc
- (2) angular speed and stress
- (3) strain and angle
- (4) stress and angle

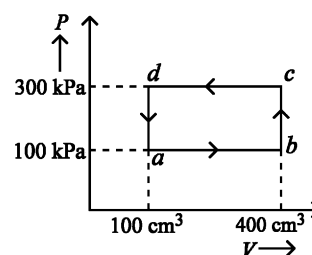
- 22 A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T . If speed becomes 2ω while keeping the same radius, the tension in the string becomes :

- (1) $\frac{T}{4}$
- (2) $\sqrt{2}T$
- (3) T
- (4) $4T$

- 23 If $x = 5 \sin\left(\pi t + \frac{\pi}{3}\right)m$ represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are :

- (1) 5 cm, 1 s
- (2) 5 m, 1 s
- (3) 5 cm, 2 s
- (4) 5 m, 2 s

- 24 A thermodynamic system is taken through the cycle $abcd$. The work done by the gas along the path bc is :



- (1) $-90 J$
- (2) $-60 J$
- (3) zero
- (4) $30 J$

25 Given below are two statements :

Statement I : Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II : Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the *most appropriate* answer from the options given below :

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

26 In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{2}$.

The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning) :

- (1) 1 : 1 (2) 1 : 4
- (3) 1 : 2 (4) 2 : 1

27 Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2 . The ratio $v_1 : v_2$ is :

- (1) 4:1 (2) 1:4
- (3) 1:2 (4) 2:1

28 ${}^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$

In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are :

- (1) 288, 82 (2) 286, 81
- (3) 280, 81 (4) 286, 80

29 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The potential (V) at any axial point, at 2 m distance(r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude, $4 \times 10^{-6} \text{ C m}$, is $\pm 9 \times 10^3 \text{ V}$.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}$)

Reason R : $V = \pm \frac{2P}{4\pi\epsilon_0 r^2}$, where r is the

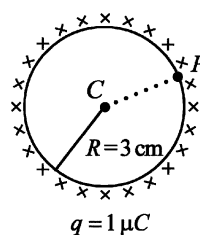
distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the *correct* answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

30 A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is:

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}$)

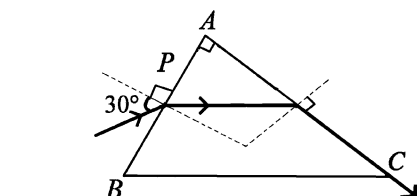


- (1) 0.5×10^5 (2) zero
- (3) 3×10^5 (4) 1×10^5

31 A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 Nm^{-1} , then the excess force required to take it away from the surface is :

- (1) 1.98 mN (2) 99 N
- (3) 19.8 mN (4) 198 N

- 32 A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC . The refractive index of the prism is:



- (1) $\frac{\sqrt{3}}{4}$ (2) $\frac{\sqrt{3}}{2}$
(3) $\frac{\sqrt{5}}{4}$ (4) $\frac{\sqrt{5}}{2}$

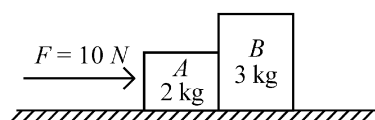
- 33 At any instant of time t , the displacement of any particle is given by $2t - 1$ (SI unit) under the influence of force of $5N$. The value of instantaneous power is (in SI unit):

- (1) 7 (2) 6
(3) 10 (4) 5

- 34 A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

- (1) 4.4 mT (2) 44 T
(3) 44 mT (4) 4.4 T

- 35 A horizontal force $10 N$ is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is :



- (1) 6 N (2) 10 N
(3) zero (4) 4 N

- 36 A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is:

- (1) 17 (2) 32
(3) 34 (4) 28

- 37 Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

- (1) 1 : 2 (2) 2 : 3
(3) 1 : 1 (4) 2 : 9

- 38 If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

- (1) $2\sqrt{3}$ (2) 4
(3) $\sqrt{3}$ (4) $\sqrt{2}$

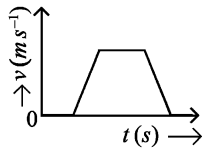
- 39 A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates :

- (1) displacement current of magnitude equal to I flows in a direction opposite to that of I .
(2) displacement current of magnitude greater than I flows but can be in any direction.
(3) there is no current.
(4) displacement current of magnitude equal to I flows in the same direction as I .

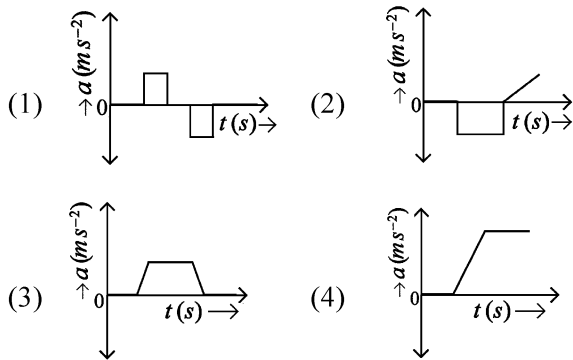
- 40 A metallic bar of Young's modulus, $0.5 \times 10^{11} \text{ N m}^{-2}$ and coefficient of linear thermal expansion $10^{-5} \text{ }^\circ\text{C}^{-1}$, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is:

- (1) $100 \times 10^3 \text{ N}$ (2) $2 \times 10^3 \text{ N}$
(3) $5 \times 10^3 \text{ N}$ (4) $50 \times 10^3 \text{ N}$

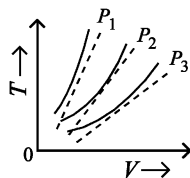
- 41 The velocity (v) – time (t) plot of the motion of a body is shown below :



The acceleration (a) – time (t) graph that best suits this motion is :

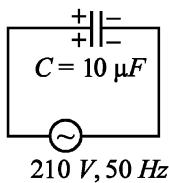


- 42 The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1 , P_2 and P_3 compared with those of Charles's law represented as dotted lines.



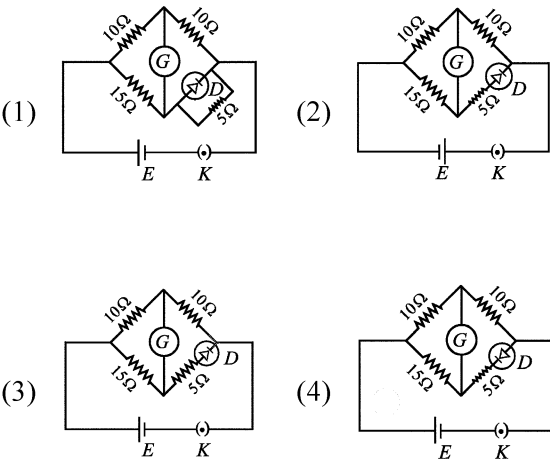
Then the correct relation is:

- (1) $P_2 > P_1 > P_3$ (2) $P_1 > P_2 > P_3$
 (3) $P_3 > P_2 > P_1$ (4) $P_1 > P_3 > P_2$
- 43 A $10\ \mu\text{F}$ capacitor is connected to a $210\ \text{V}$, $50\ \text{Hz}$ source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$):



- (1) $1.20\ \text{A}$ (2) $0.35\ \text{A}$
 (3) $0.58\ \text{A}$ (4) $0.93\ \text{A}$

- 44 Choose the correct circuit which can achieve the bridge balance.



- 45 The property which is not of an electromagnetic wave travelling in free space is that :

- (1) they travel with a speed equal to $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$.
 (2) they originate from charges moving with uniform speed.
 (3) they are transverse in nature.
 (4) the energy density in electric field is equal to energy density in magnetic field.

- 46 The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is:

- (1) $\frac{GmM}{2R}$ (2) $\frac{GmM}{3R}$
 (3) $\frac{5GmM}{6R}$ (4) $\frac{2GmM}{3R}$

- 47 If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then
- the charge stored in it, increases.
 - the energy stored in it, decreases.
 - its capacitance increases.
 - the ratio of charge to its potential remains the same.
 - the product of charge and voltage increases.
- Choose the most appropriate answer from the options given below:

(1) B, D and E only (2) A, B and C only
(3) A, B and E only (4) A, C and E only

- 48 A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t . The factor which is dimensionless, if α and β are constants, is:

(1) $\alpha\beta t$ (2) $\frac{\alpha\beta}{t}$
(3) $\frac{\beta t}{\alpha}$ (4) $\frac{\alpha t}{\beta}$

- 49 A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to :
- hold the sheet there if it is magnetic.
 - hold the sheet there if it is non-magnetic.
 - move the sheet away from the pole with uniform velocity if it is conducting.
 - move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

(1) A, C and D only
(2) C only
(3) B and D only
(4) A and C only

- 50 An iron bar of length L has magnetic moment M . It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is:

(1) $2M$ (2) $\frac{M}{\sqrt{3}}$
(3) M (4) $\frac{M}{2}$

- 51 Match List I with List II.

List I (Process)	List II (Conditions)
A. Isothermal process	I. No heat exchange
B. Isochoric process	II. Carried out at constant temperature
C. Isobaric process	III. Carried out at constant volume
D. Adiabatic process	IV. Carried out at constant pressure

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-IV, B-II, C-III, D-I

- 52 Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order

$n\text{-pentane} > \text{isopentane} > \text{neopentane}$

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

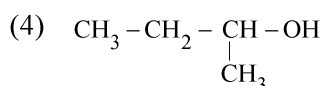
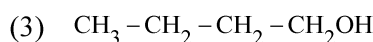
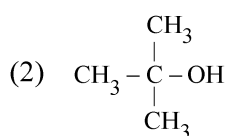
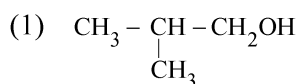
53 Match List I with List II.

List I (Molecule)	List II (Number and types of bond/s between two carbon atoms)
A. ethane	I. one σ -bond and two π -bonds
B. ethene	II. two π -bonds
C. carbon molecule, C_2	III. one σ -bond
D. ethyne	IV. one σ -bond and one π -bond

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-IV, C-II, D-III
- (4) A-IV, B-III, C-II, D-I

54 Which one of the following alcohols reacts instantaneously with Lucas reagent?



55 Match List I with List II.

List I Quantum Number	List II Information provided
A. m_l	I. shape of orbital
B. m_s	II. size of orbital
C. l	III. orientation of orbital
D. n	IV. orientation of spin of electron

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-IV, C-I, D-II

56 'Spin only' magnetic moment is same for which of the following ions?

- | | |
|---------------------|---------------------|
| A. Ti^{3+} | B. Cr^{2+} |
| C. Mn^{2+} | D. Fe^{2+} |
| E. Sc^{3+} | |

Choose the most appropriate answer from the options given below:

- (1) B and C only
- (2) A and D only
- (3) B and D only
- (4) A and E only

57 On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as

- (1) Distillation
- (2) Chromatography
- (3) Crystallization
- (4) Sublimation

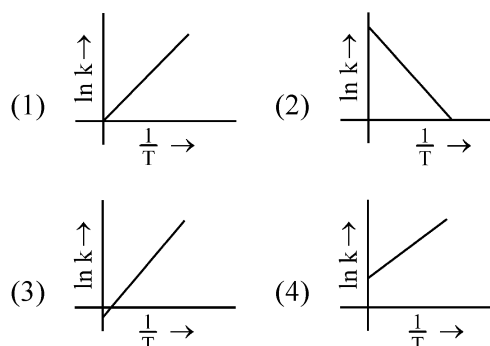
58 The energy of an electron in the ground state ($n = 1$) for He^+ ion is $-x$ J, then that for an electron in $n = 2$ state for Be^{3+} ion in J is :

- | | |
|-----------|---------------------|
| (1) $-4x$ | (2) $-\frac{4}{9}x$ |
| (3) $-x$ | (4) $-\frac{x}{9}$ |

59 Activation energy of any chemical reaction can be calculated if one knows the value of

- (1) orientation of reactant molecules during collision.
- (2) rate constant at two different temperatures.
- (3) rate constant at standard temperature.
- (4) probability of collision.

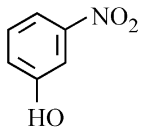
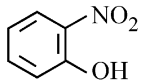
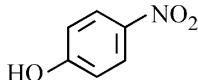
60 Which plot of $\ln k$ vs $\frac{1}{T}$ is consistent with Arrhenius equation?



- 61** Given below are two statements:
Statement I : The boiling point of hydrides of Group 16 elements follow the order $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$.
Statement II : On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O , it has higher boiling point.
 In the light of the above statements, choose the *correct* answer from the options given below:
 (1) Statement I is true but Statement II is false.
 (2) Statement I is false but Statement II is true.
 (3) Both Statement I and Statement II are true.
 (4) Both Statement I and Statement II are false.
- 62** The reagents with which glucose does **not** react to give the corresponding tests/products are
 A. Tollen's reagent
 B. Schiff's reagent
 C. HCN
 D. NH_2OH
 E. NaHSO_3
 Choose the correct options from the given below:
 (1) B and E (2) E and D
 (3) B and C (4) A and D
- 63** Arrange the following elements in increasing order of first ionization enthalpy:
 Li, Be, B, C, N
 Choose the correct answer from the options given below:
 (1) $\text{Li} < \text{Be} < \text{C} < \text{B} < \text{N}$
 (2) $\text{Li} < \text{Be} < \text{N} < \text{B} < \text{C}$
 (3) $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{N}$
 (4) $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N}$
- 64** Arrange the following elements in increasing order of electronegativity:
 N, O, F, C, Si
 Choose the correct answer from the options given below:
 (1) $\text{O} < \text{F} < \text{N} < \text{C} < \text{Si}$
 (2) $\text{F} < \text{O} < \text{N} < \text{C} < \text{Si}$
 (3) $\text{Si} < \text{C} < \text{N} < \text{O} < \text{F}$
 (4) $\text{Si} < \text{C} < \text{O} < \text{N} < \text{F}$

- 65** The E° value for the $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is more positive than that of $\text{Cr}^{3+}/\text{Cr}^{2+}$ or $\text{Fe}^{3+}/\text{Fe}^{2+}$ due to change of
 (1) d^4 to d^5 configuration
 (2) d^3 to d^5 configuration
 (3) d^5 to d^4 configuration
 (4) d^5 to d^2 configuration
- 66** Match List I with List II.
- | List I
(Compound) | List II
(Shape/geometry) |
|------------------------------------|---|
| A. NH_3 | I. Trigonal Pyramidal |
| B. BrF_5 | II. Square Planar |
| C. XeF_4 | III. Octahedral |
| D. SF_6 | IV. Square Pyramidal |
- Choose the correct answer from the options given below:
 (1) A-III, B-IV, C-I, D-II
 (2) A-II, B-III, C-IV, D-I
 (3) A-I, B-IV, C-II, D-III
 (4) A-II, B-IV, C-III, D-I
- 67** Match List I with List II.
- | List I (Complex) | List II (Type of isomerism) |
|--|------------------------------------|
| A. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ | I. Solvate isomerism |
| B. $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$ | II. Linkage isomerism |
| C. $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$ | III. Ionization isomerism |
| D. $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ | IV. Coordination isomerism |
- Choose the correct answer from the options given below:
 (1) A-I, B-IV, C-III, D-II
 (2) A-II, B-IV, C-III, D-I
 (3) A-II, B-III, C-IV, D-I
 (4) A-I, B-III, C-IV, D-II

68 Intramolecular hydrogen bonding is present in

- (1) 
- (2) HF
- (3) 
- (4) 

69 Given below are two statements:

Statement I : Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the *correct* answer from the options given below:

- (1) Statement I is correct but Statement II is false.
- (2) Statement I is incorrect but Statement II is true.
- (3) Both Statement I and Statement II are true.
- (4) Both Statement I and Statement II are false.

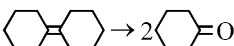
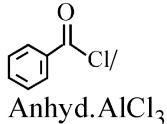
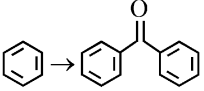
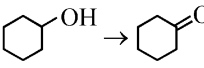
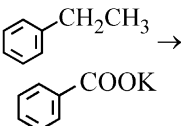
70 The highest number of helium atoms is in

- (1) 4 g of helium
- (2) 2.271098 L of helium at STP
- (3) 4 mol of helium
- (4) 4 u of helium

71 Match List I with List II.

List I (Reaction)

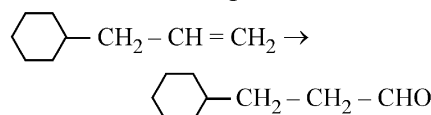
List II (Reagents/Condition)

- A.  I. 
- B.  II. CrO₃
- C.  III. KMnO₄/
KOH, Δ
- D.  IV. (i) O₃
(ii) Zn-H₂O

Choose the correct answer from the options given below:

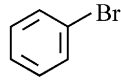
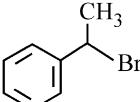
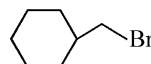
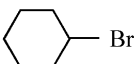
- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-II, D-IV

72 Identify the correct reagents that would bring about the following transformation.



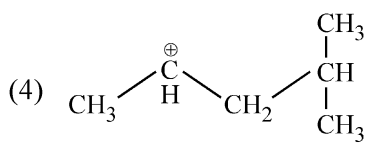
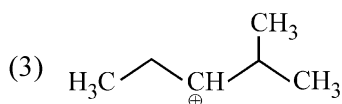
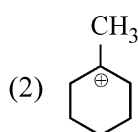
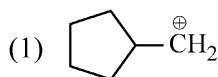
- (1) (i) BH₃
(ii) H₂O₂/[⊖]OH
(iii) alk. KMnO₄
(iv) H₃O[⊕]
- (2) (i) H₂O/H⁺
(ii) PCC
- (3) (i) H₂O/H⁺
(ii) CrO₃
- (4) (i) BH₃
(ii) H₂O₂/[⊖]OH
(iii) PCC

73 The compound that will undergo S_N¹ reaction with the fastest rate is

- (1)  (2) 
- (3)  (4) 

- 74 Fehling's solution 'A' is
- (1) alkaline solution of sodium potassium tartrate (Rochelle's salt)
 - (2) aqueous sodium citrate
 - (3) aqueous copper sulphate
 - (4) alkaline copper sulphate

- 75 The most stable carbocation among the following is:



- 76 For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is : $[A] = [B] = [C] = 2 \times 10^{-3} \text{ M}$.

Then, which of the following is correct?

- (1) Reaction has a tendency to go in backward direction.
 - (2) Reaction has gone to completion in forward direction.
 - (3) Reaction is at equilibrium.
 - (4) Reaction has a tendency to go in forward direction.
- 77 The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order:
- (1) $A > C > B$
 - (2) $A > B > C$
 - (3) $B > A > C$
 - (4) $B > C > A$

- 78 A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is:

- (1) 2,3-dimethylbutane
- (2) 2,2-dimethylbutane
- (3) n-hexane
- (4) 2-methylpentane

- 79 In which of the following equilibria, K_p and K_c are **NOT** equal?

- (1) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$
- (2) $2 BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$
- (3) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$
- (4) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2 HI_{(g)}$

- 80 Which reaction is **NOT** a redox reaction?

- (1) $H_2 + Cl_2 \rightarrow 2 HCl$
- (2) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2 NaCl$
- (3) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
- (4) $2 KClO_3 + I_2 \rightarrow 2 KIO_3 + Cl_2$

- 81 Given below are two statements :

Statement I : Both $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.

Statement II : $[Co(NH_3)_6]^{3+}$ is diamagnetic whereas $[CoF_6]^{3-}$ is paramagnetic.

In the light of the above statements, choose the *correct* answer from the options given below:

- (1) Statement I is true but Statement II is false.
 - (2) Statement I is false but Statement II is true.
 - (3) Both Statement I and Statement II are true.
 - (4) Both Statement I and Statement II are false.
- 82 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to
- (1) Zero mg
 - (2) 200 mg
 - (3) 750 mg
 - (4) 250 mg

83 Among Group 16 elements, which one does **NOT** show -2 oxidation state?

- (1) Te (2) Po
(3) O (4) Se

84 In which of the following processes entropy increases?

- A. A liquid evaporates to vapour.
B. Temperature of a crystalline solid lowered from 130 K to 0 K.
C. $2\text{NaHCO}_{3(s)} \rightarrow \text{Na}_2\text{CO}_{3(s)} + \text{CO}_{2(g)} + \text{H}_2\text{O}_{(g)}$
D. $\text{Cl}_{2(g)} \rightarrow 2\text{Cl}_{(g)}$

Choose the correct answer from the options given below:

- (1) A, C and D (2) C and D
(3) A and C (4) A, B and D

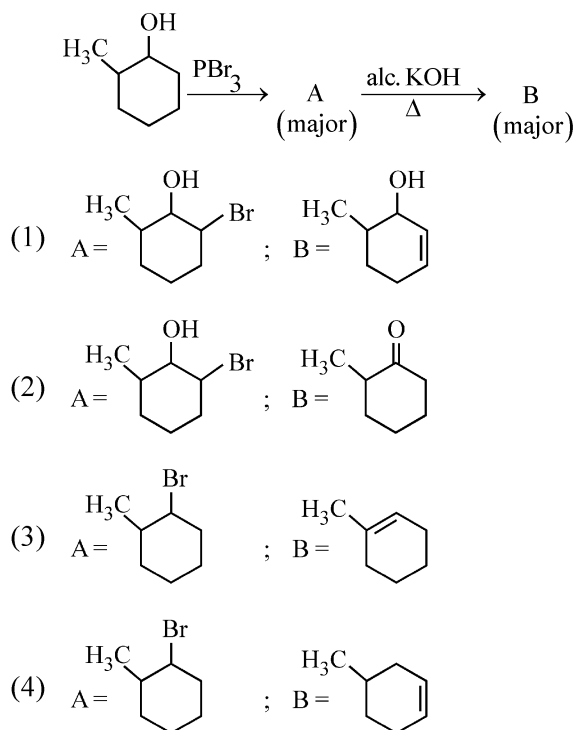
85 Match List I with List II.

List I (Conversion)	List II (Number of Faraday required)
A. 1 mol of H_2O to O_2	I. 3F
B. 1 mol of MnO_4^- to Mn^{2+}	II. 2F
C. 1.5 mol of Ca from molten CaCl_2	III. 1F
D. 1 mol of FeO to Fe_2O_3	IV. 5F

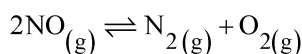
Choose the correct answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-IV, C-I, D-II

86 Major products A and B formed in the following reaction sequence, are



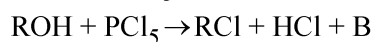
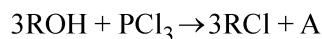
87 Consider the following reaction in a sealed vessel at equilibrium with concentrations of $\text{N}_2 = 3.0 \times 10^{-3} \text{ M}$, $\text{O}_2 = 4.2 \times 10^{-3} \text{ M}$ and $\text{NO} = 2.8 \times 10^{-3} \text{ M}$.



If 0.1 mol L^{-1} of $\text{NO}_{(g)}$ is taken in a closed vessel, what will be degree of dissociation (α) of $\text{NO}_{(g)}$ at equilibrium?

- (1) 0.8889 (2) 0.717
(3) 0.00889 (4) 0.0889

88 The products A and B obtained in the following reactions, respectively, are



- (1) H_3PO_4 and POCl_3
(2) H_3PO_3 and POCl_3
(3) POCl_3 and H_3PO_3
(4) POCl_3 and H_3PO_4

- 89 Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.

A. Al^{3+} B. Cu^{2+}
 C. Ba^{2+} D. Co^{2+}
 E. Mg^{2+}

Choose the correct answer from the options given below:

- (1) E, C, D, B, A
 (2) E, A, B, C, D
 (3) B, A, D, C, E
 (4) B, C, A, D, E

- 90 During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

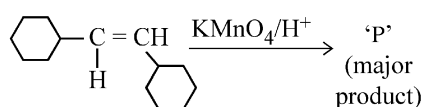
- (1) dilute nitric acid
 (2) dilute sulphuric acid
 (3) dilute hydrochloric acid
 (4) concentrated sulphuric acid

- 91 The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is:

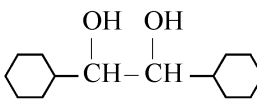
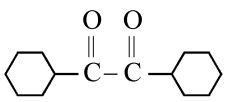
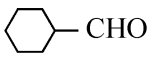
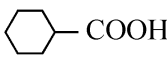
(Given $R = 2.0 \text{ cal K}^{-1} \text{ mol}^{-1}$)

- (1) 413.14 calories
 (2) 100 calories
 (3) 0 calorie
 (4) -413.14 calories

- 92 For the given reaction:



'P' is

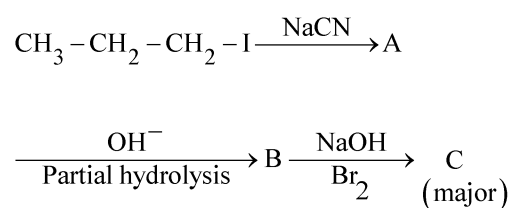
- (1) 
 (2) 
 (3) 
 (4) 

- 93 The plot of osmotic pressure (Π) vs concentration (mol L^{-1}) for a solution gives a straight line with slope $25.73 \text{ L bar mol}^{-1}$. The temperature at which the osmotic pressure measurement is done is:

(Use $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$)

- (1) 25.73°C (2) 12.05°C
 (3) 37°C (4) 310°C

- 94 Identify the major product C formed in the following reaction sequence :



- (1) butanamide
 (2) α -bromobutanoic acid
 (3) propylamine
 (4) butylamine

- 95 A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :

(Given atomic masses of $A = 64$; $B = 40$; $C = 32 \text{ u}$)

- (1) AB_2C_2 (2) ABC_4
 (3) A_2BC_2 (4) ABC_3

- 96 The rate of a reaction quadruples when temperature changes from 27°C to 57°C . Calculate the energy of activation.

Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 4 = 0.6021$

- (1) 3.80 kJ/mol
 (2) 3804 kJ/mol
 (3) 38.04 kJ/mol
 (4) 380.4 kJ/mol

- 97 Identify the **correct** answer.
- (1) Dipole moment of NF_3 is greater than that of NH_3 .
 - (2) Three canonical forms can be drawn for CO_3^{2-} ion.
 - (3) Three resonance structures can be drawn for ozone.
 - (4) BF_3 has non-zero dipole moment.

- 98 The pair of lanthanoid ions which are diamagnetic is
- (1) Gd^{3+} and Eu^{3+}
 - (2) Pm^{3+} and Sm^{3+}
 - (3) Ce^{4+} and Yb^{2+}
 - (4) Ce^{3+} and Eu^{2+}

- 99 Given below are two statements :

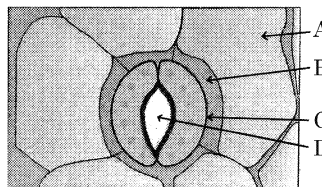
Statement I : $[\text{Co}(\text{NH}_3)_6]^{3+}$ is a homoleptic complex whereas $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is a heteroleptic complex.

Statement II : Complex $[\text{Co}(\text{NH}_3)_6]^{3+}$ has only one kind of ligands but $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ has more than one kind of ligands.

In the light of the above statements, choose the *correct* answer from the options given below:

- (1) Statement I is true but Statement II is false.
 - (2) Statement I is false but Statement II is true.
 - (3) Both Statement I and Statement II are true.
 - (4) Both Statement I and Statement II are false.
- 100 Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is:
(Given : Molar mass of Cu : 63 g mol^{-1} , $1\text{F} = 96487 \text{ C}$)
- (1) 31.5 g
 - (2) 0.0315 g
 - (3) 3.15 g
 - (4) 0.315 g

- 101 In the given figure, which component has thin outer walls and highly thickened inner walls?



- (1) A
 - (2) B
 - (3) C
 - (4) D
- 102 List of endangered species was released by-
- (1) FOAM
 - (2) IUCN
 - (3) GEAC
 - (4) WWF
- 103 The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called;
- (1) Semi-conservative method
 - (2) Sustainable development
 - (3) *in-situ* conservation
 - (4) Biodiversity conservation
- 104 Which one of the following can be explained on the basis of Mendel's Law of Dominance?
- A. Out of one pair of factors one is dominant and the other is recessive.
 - B. Alleles do not show any expression and both the characters appear as such in F_2 generation.
 - C. Factors occur in pairs in normal diploid plants.
 - D. The discrete unit controlling a particular character is called factor.
 - E. The expression of only one of the parental characters is found in a monohybrid cross.
- Choose the correct answer from the options given below:
- (1) B, C and D only
 - (2) A, B, C, D and E
 - (3) A, B and C only
 - (4) A, C, D and E only

105 The lactose present in the growth medium of bacteria is transported to the cell by the action of:

- (1) Permease
- (2) Polymerase
- (3) Beta-galactosidase
- (4) Acetylase

106 Match List I with List II

List I	List II
A. <i>Clostridium butylicum</i>	I. Ethanol
B. <i>Saccharomyces cerevisiae</i>	II. Streptokinase
C. <i>Trichoderma polysporum</i>	III. Butyric acid
D. <i>Streptococcus</i> sp.	IV. Cyclosporin-A

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-IV, B-I, C-III, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-IV, C-III, D-I

107 The equation of Verhulst-Pearl logistic growth is

$$\frac{dN}{dt} = rN \left[\frac{K-N}{K} \right].$$

From this equation, *K* indicates:

- (1) Carrying capacity
- (2) Population density
- (3) Intrinsic rate of natural increase
- (4) Biotic potential

108 A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;

- (1) Inducer, Repressor, Structural gene
- (2) Promotor, Structural gene, Terminator
- (3) Repressor, Operator gene, Structural gene
- (4) Structural gene, Transposons, Operator gene

109 Formation of interfascicular cambium from fully developed parenchyma cells is an example for

- (1) Dedifferentiation
- (2) Maturation
- (3) Differentiation
- (4) Redifferentiation

110 Match List I with List II

List I	List II
A. <i>Rhizopus</i>	I. Mushroom
B. <i>Ustilago</i>	II. Smut fungus
C. <i>Puccinia</i>	III. Bread mould
D. <i>Agaricus</i>	IV. Rust fungus

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-II, C-IV, D-I
- (4) A-I, B-III, C-II, D-IV

111 What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?

- A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.
- B. It may get integrated into the genome of the recipient.
- C. It may multiply and be inherited along with the host DNA.
- D. The alien piece of DNA is not an integral part of chromosome.
- E. It shows ability to replicate.

Choose the correct answer from the options given below:

- (1) B and C only
- (2) A and E only
- (3) A and B only
- (4) D and E only

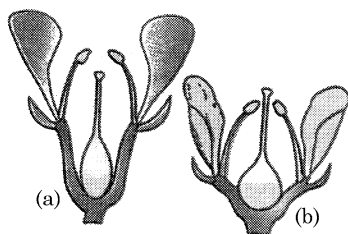
112 Match List I with List II

List I	List II
A. Nucleolus	I. Site of formation of glycolipid
B. Centriole	II. Organization like the cartwheel
C. Leucoplasts	III. Site for active ribosomal RNA synthesis
D. Golgi apparatus	IV. For storing nutrients

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-III, C-I, D-IV

- 113** Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (a) and (b)

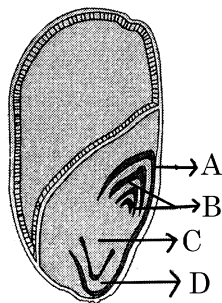


- (1) (a) Perigynous; (b) Epigynous
 (2) (a) Perigynous; (b) Perigynous
 (3) (a) Epigynous; (b) Hypogynous
 (4) (a) Hypogynous; (b) Epigynous
- 114** Spindle fibers attach to kinetochores of chromosomes during
- (1) Anaphase (2) Telophase
 (3) Prophase (4) Metaphase
- 115** These are regarded as major causes of biodiversity loss:
- A. Over exploitation
 B. Co-extinction
 C. Mutation
 D. Habitat loss and fragmentation
 E. Migration
- Choose the correct option:
- (1) A, B and E only
 (2) A, B and D only
 (3) A, C and D only
 (4) A, B, C and D only
- 116** Lecithin, a small molecular weight organic compound found in living tissues, is an example of:
- (1) Glycerides
 (2) Carbohydrates
 (3) Amino acids
 (4) Phospholipids

- 117** Identify the set of correct statements:
- A. The flowers of *Vallisneria* are colourful and produce nectar.
 B. The flowers of waterlily are not pollinated by water.
 C. In most of water-pollinated species, the pollen grains are protected from wetting.
 D. Pollen grains of some hydrophytes are long and ribbon like.
 E. In some hydrophytes, the pollen grains are carried passively inside water.
- Choose the correct answer from the options given below:
- (1) A, C, D and E only
 (2) B, C, D and E only
 (3) C, D and E only
 (4) A, B, C and D only
- 118** The cofactor of the enzyme carboxypeptidase is:
- (1) Flavin (2) Haem
 (3) Zinc (4) Niacin
- 119** Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of:
- (1) Competitive inhibition
 (2) Enzyme activation
 (3) Cofactor inhibition
 (4) Feedback inhibition
- 120** Which of the following is an example of actinomorphic flower?
- (1) *Pisum* (2) *Sesbania*
 (3) *Datura* (4) *Cassia*
- 121** Given below are two statements:
- Statement I :** Chromosomes become gradually visible under light microscope during leptotene stage.
- Statement II :** The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.
- In the light of the above statements, choose the correct answer from the options given below:
- (1) Statement I is true but Statement II is false
 (2) Statement I is false but Statement II is true
 (3) Both Statement I and Statement II are true
 (4) Both Statement I and Statement II are false

- 122** A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?
- (1) Only pink flowered plants
 - (2) Red, Pink as well as white flowered plants
 - (3) Only red flowered plants
 - (4) Red flowered as well as pink flowered plants
- 123** Given below are two statements:
- Statement I :** Parenchyma is living but collenchyma is dead tissue.
- Statement II :** Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperms.
- In the light of the above statements, choose the correct answer from the options given below:
- (1) Statement I is true but Statement II is false
 - (2) Statement I is false but Statement II is true
 - (3) Both Statement I and Statement II are true
 - (4) Both Statement I and Statement II are false
- 124** Given below are two statements:
- Statement I :** Bt toxins are insect group specific and coded by a gene *cry* IAc.
- Statement II :** Bt toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.
- In the light of the above statements, choose the correct answer from the options given below:
- (1) Statement I is true but Statement II is false
 - (2) Statement I is false but Statement II is true
 - (3) Both Statement I and Statement II are true
 - (4) Both Statement I and Statement II are false
- 125** Which one of the following is not a criterion for classification of fungi?
- (1) Mode of spore formation
 - (2) Fruiting body
 - (3) Morphology of mycelium
 - (4) Mode of nutrition
- 126** The capacity to generate a whole plant from any cell of the plant is called:
- (1) Differentiation
 - (2) Somatic hybridization
 - (3) Totipotency
 - (4) Micropropagation
- 127** Which of the following are required for the dark reaction of photosynthesis?
- A. Light
 - B. Chlorophyll
 - C. CO₂
 - D. ATP
 - E. NADPH
- Choose the correct answer from the options given below:
- (1) C, D and E only
 - (2) D and E only
 - (3) A, B and C only
 - (4) B, C and D only
- 128** Tropical regions show greatest level of species richness because
- A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
 - B. Tropical environments are more seasonal.
 - C. More solar energy is available in tropics.
 - D. Constant environments promote niche specialization.
 - E. Tropical environments are constant and predictable.
- Choose the correct answer from the options given below:
- (1) A, B and E only
 - (2) A, B and D only
 - (3) A, C, D and E only
 - (4) A and B only

129 Identify the part of the seed from the given figure which is destined to form root when the seed germinates.



- (1) C (2) D
(3) A (4) B

130 How many molecules of ATP and NADPH are required for every molecule of CO₂ fixed in the Calvin cycle?

- (1) 3 molecules of ATP and 3 molecules of NADPH
(2) 3 molecules of ATP and 2 molecules of NADPH
(3) 2 molecules of ATP and 3 molecules of NADPH
(4) 2 molecules of ATP and 2 molecules of NADPH

131 In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

- (1) Bb (2) BB/Bb
(3) BB (4) bb

132 Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of:

- (1) 4 bp (2) 10 bp
(3) 8 bp (4) 6 bp

133 Bulliform cells are responsible for

- (1) Increased photosynthesis in monocots.
- (2) Providing large spaces for storage of sugars.
- (3) Inward curling of leaves in monocots.
- (4) Protecting the plant from salt stress.

134 Match List I with List II

List I	List II
A. Two or more alternative forms of a gene	I. Back cross
B. Cross of F ₁ progeny with homozygous recessive parent	II. Ploidy
C. Cross of F ₁ progeny with any of the parents	III. Allele
D. Number of chromosome sets in plant	IV. Test cross

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-III, D-IV

135 Auxin is used by gardeners to prepare weed-free lawns. But no damage is caused to grass as auxin

- (1) does not affect mature monocotyledonous plants.
- (2) can help in cell division in grasses, to produce growth.
- (3) promotes apical dominance.
- (4) promotes abscission of mature leaves only.

Botany : Section-B (Q. No. 136 to 150)

136 Match List I with List II

List I	List II
A. Rose	I. Twisted aestivation
B. Pea	II. Perigynous flower
C. Cotton	III. Drupe
D. Mango	IV. Marginal placentation

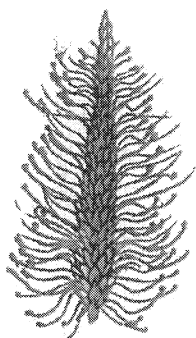
Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-II, B-III, C-IV, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-I, B-II, C-III, D-IV

137 The DNA present in chloroplast is:

- (1) Linear, single stranded
- (2) Circular, single stranded
- (3) Linear, double stranded
- (4) Circular, double stranded

138 Identify the correct description about the given figure:



- (1) Cleistogamous flowers showing autogamy.
- (2) Compact inflorescence showing complete autogamy.
- (3) Wind pollinated plant inflorescence showing flowers with well exposed stamens.
- (4) Water pollinated flowers showing stamens with mucilaginous covering.

139 Match List I with List II

List I	List II
A. Robert May	I. Species-Area relationship
B. Alexander von Humboldt	II. Long term ecosystem experiment using out door plots
C. Paul Ehrlich	III. Global species diversity at about 7 million
D. David Tilman	IV. Rivet popper hypothesis

Choose the correct answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-I, C-IV, D-II

140 Given below are two statements:

Statement I : In C_3 plants, some O_2 binds to RuBisCO, hence CO_2 fixation is decreased.

Statement II : In C_4 plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the *correct* answer from the options given below:

- (1) Statement I is true but Statement II is false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

141 Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.

- (1) Succinyl-CoA $\rightarrow$ Succinic acid
- (2) Isocitrate $\rightarrow$ α -ketoglutaric acid
- (3) Malic acid $\rightarrow$ Oxaloacetic acid
- (4) Succinic acid $\rightarrow$ Malic acid

142 Match List I with List II

List I	List II
A. Citric acid cycle	I. Cytoplasm
B. Glycolysis	II. Mitochondrial matrix
C. Electron transport system	III. Intermembrane space of mitochondria
D. Proton gradient	IV. Inner mitochondrial membrane

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-IV, D-III

143 Which of the following are fused in somatic hybridization involving two varieties of plants?

- (1) Protoplasts
- (2) Pollens
- (3) Callus
- (4) Somatic embryos

144 Match List I with List II

List I	List II
A. GLUT-4	I. Hormone
B. Insulin	II. Enzyme
C. Trypsin	III. Intercellular ground substance
D. Collagen	IV. Enables glucose transport into cells

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-II, C-III, D-IV

145 Match List I with List II

List I (Types of Stamens)	List II (Example)
A. Monoadelphous	I. Citrus
B. Diadelphous	II. Pea
C. Polyadelphous	III. Lily
D. Epiphylous	IV. China-rose

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-I, C-II, D-III

146 Read the following statements and choose the set of correct statements:

- In the members of Phaeophyceae,
- A. Asexual reproduction occurs usually by biflagellate zoospores.
 - B. Sexual reproduction is by oogamous method only.
 - C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.
 - D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.
 - E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

- (1) A, C, D and E only
- (2) A, B, C and E only
- (3) A, B, C and D only
- (4) B, C, D and E only

147 Match List I with List II

List I	List II
A. Frederick Griffith	I. Genetic code
B. Francois Jacob & Jacques Monod	II. Semi-conservative mode of DNA replication
C. Har Gobind Khorana	III. Transformation
D. Meselson & Stahl	IV. Lac operon

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-III, B-II, C-I, D-IV
- (4) A-III, B-IV, C-I, D-II

148 Which of the following statement is correct regarding the process of replication in *E.coli*?

- (1) The DNA dependent DNA polymerase catalyses polymerization in 5' → 3' as well as 3' → 5' direction.
- (2) The DNA dependent DNA polymerase catalyses polymerization in 5' → 3' direction.
- (3) The DNA dependent DNA polymerase catalyses polymerization in one direction that is 3' → 5'.
- (4) The DNA dependent RNA polymerase catalyses polymerization in one direction, that is 5' → 3'.

149 Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield?

- (1) Cytokinin
- (2) Absciscic acid
- (3) Auxin
- (4) Gibberellin

150 In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is

$100x \text{ (kcal m}^{-2}\text{) yr}^{-1}$, what would be the GPP

(Gross Primary Productivity) of the third trophic level of the same ecosystem?

- (1) $10x \text{ (kcal m}^{-2}\text{) yr}^{-1}$
- (2) $\frac{100x}{3x} \text{ (kcal m}^{-2}\text{) yr}^{-1}$
- (3) $\frac{x}{10} \text{ (kcal m}^{-2}\text{) yr}^{-1}$
- (4) $x \text{ (kcal m}^{-2}\text{) yr}^{-1}$

151 Match List I with List II :

List I	List II
A. Down's syndrome	I. 11 th chromosome
B. α-Thalassemia	II. 'X' chromosome
C. β-Thalassemia	III. 21 st chromosome
D. Klinefelter's syndrome	IV. 16 th chromosome

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-I, C-II, D-III
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

152 Match List I with List II :

List I (Sub Phases of Prophase I)	List II (Specific characters)
A. Diakinesis	I. Synaptonemal complex formation
B. Pachytene	II. Completion of terminalisation of chiasmata
C. Zygotene	III. Chromosomes look like thin threads
D. Leptotene	IV. Appearance of recombination nodules

Choose the correct answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-II, C-IV, D-III

153 Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

- (1) Low pCO₂ and High H⁺ concentration
- (2) Low pCO₂ and High temperature
- (3) High pO₂ and High pCO₂
- (4) High pO₂ and Lesser H⁺ concentration

154 Match List I with List II :

List I	List II
A. Typhoid	I. Fungus
B. Leishmaniasis	II. Nematode
C. Ringworm	III. Protozoa
D. Filariasis	IV. Bacteria

Choose the correct answer from the options given below :

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-IV, C-III, D-I
- (3) A-I, B-III, C-II, D-IV
- (4) A-IV, B-III, C-I, D-II

155 Match List I with List II :

List I	List II
A. Expiratory capacity	I. Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
B. Functional residual capacity	II. Tidal volume + Expiratory reserve volume
C. Vital capacity	III. Tidal volume + Inspiratory reserve volume
D. Inspiratory capacity	IV. Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

156 Which of the following are Autoimmune disorders?

- A. Myasthenia gravis
- B. Rheumatoid arthritis
- C. Gout
- D. Muscular dystrophy
- E. Systemic Lupus Erythematosus (SLE)

Choose the most appropriate answer from the options given below :

- (1) B, C & E only
- (2) C, D & E only
- (3) A, B & D only
- (4) A, B & E only

157 Given below are two statements :

Statement I : In the nephron, the descending limb of loop of Henle is impermeable to water and permeable to electrolytes.

Statement II : The proximal convoluted tubule is lined by simple columnar brush border epithelium and increases the surface area for reabsorption.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Statement I is true but Statement II is false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

158 Which of the following is not a steroid hormone?

- (1) Progesterone
- (2) Glucagon
- (3) Cortisol
- (4) Testosterone

159 Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : FSH acts upon ovarian follicles in female and Leydig cells in male.

Reason R : Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in male human being.

In the light of the above statements, choose the correct answer from the options given below :

- (1) A is true but R is false
- (2) A is false but R is true
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

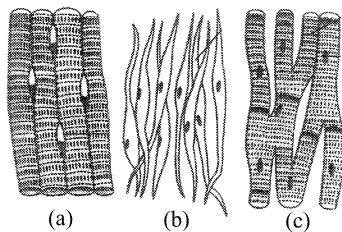
160 Given below are some stages of human evolution. Arrange them in correct sequence. (Past to Recent)

- A. *Homo habilis*
- B. *Homo sapiens*
- C. *Homo neanderthalensis*
- D. *Homo erectus*

Choose the correct sequence of human evolution from the options given below :

- (1) C-B-D-A
- (2) A-D-C-B
- (3) D-A-C-B
- (4) B-A-D-C

161 Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body :



Name of muscle/location

- (1) (a) Skeletal - Biceps
(b) Involuntary - Intestine
(c) Smooth - Heart.
- (2) (a) Involuntary - Nose tip
(b) Skeletal - Bone
(c) Cardiac - Heart.
- (3) (a) Smooth - Toes
(b) Skeletal - Legs
(c) Cardiac - Heart.
- (4) (a) Skeletal - Triceps
(b) Smooth - Stomach
(c) Cardiac - Heart.

162 Match List I with List II :

List I	List II
A. Cocaine	I. Effective sedative in surgery
B. Heroin	II. <i>Cannabis sativa</i>
C. Morphine	III. <i>Erythroxylum</i>
D. Marijuana	IV. <i>Papaver somniferum</i>

Choose the correct answer from the options given below :

- (1) A-II, B-I, C-III, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-III, C-I, D-II
- (4) A-I, B-III, C-II, D-IV

163 The “Ti plasmid” of *Agrobacterium tumefaciens* stands for

- (1) Tumor inducing plasmid
- (2) Temperature independent plasmid
- (3) Tumour inhibiting plasmid
- (4) Tumor independent plasmid

164 Which one is the correct product of DNA dependent RNA polymerase to the given template?

3’TACATGGCAAATATCCATTCA5’

- (1) 5’AUGUACCGUUUAUAGGGAAGU3’
- (2) 5’ATGTACCGTTTATAGGTAAGT3’
- (3) 5’AUGUACCGUUUAUAGGUAAGU3’
- (4) 5’AUGUAAAGUUUAUAGGUAAGU3’

165 Match List I with List II :

List I	List II
A. <i>Pterophyllum</i>	I. Hag fish
B. <i>Myxine</i>	II. Saw fish
C. <i>Pristis</i>	III. Angel fish
D. <i>Exocoetus</i>	IV. Flying fish

Choose the correct answer from the options given below :

- (1) A-IV, B-I, C-II, D-III
- (2) A-III, B-II, C-I, D-IV
- (3) A-II, B-I, C-III, D-IV
- (4) A-III, B-I, C-II, D-IV

166 Match List I with List II :

List I	List II
A. Fibrous joints	I. Adjacent vertebrae, limited movement
B. Cartilaginous joints	II. Humerus and Pectoral girdle, rotational movement
C. Hinge joints	III. Skull, don’t allow any movement
D. Ball and socket joints	IV. Knee, help in locomotion

Choose the correct answer from the options given below :

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-III, C-II, D-IV

- 167** The flippers of the Penguins and Dolphins are the example of the
- (1) Convergent evolution
 - (2) Divergent evolution
 - (3) Adaptive radiation
 - (4) Natural selection

168 Match List I with List II :

List I	List II
A. α -1 antitrypsin	I. Cotton bollworm
B. Cry IAb	II. ADA deficiency
C. Cry IAc	III. Emphysema
D. Enzyme replacement therapy	IV. Corn borer

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-I, C-II, D-IV

169 Match List I with List II :

List I	List II
A. Axoneme	I. Centriole
B. Cartwheel pattern	II. Cilia and flagella
C. Crista	III. Chromosome
D. Satellite	IV. Mitochondria

Choose the correct answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-IV, B-II, C-III, D-I

- 170** Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby.

Reason R : Colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) A is correct but R is not correct.
- (2) A is not correct but R is correct.
- (3) Both A and R are correct and R is the correct explanation of A.
- (4) Both A and R are correct but R is NOT the correct explanation of A.

- 171** Which of the following is not a component of Fallopian tube?

- (1) Infundibulum
- (2) Ampulla
- (3) Uterine fundus
- (4) Isthmus

- 172** Which of the following statements is incorrect?

- (1) Bio-reactors are used to produce small scale bacterial cultures.
- (2) Bio-reactors have an agitator system, an oxygen delivery system and foam control system.
- (3) A bio-reactor provides optimal growth conditions for achieving the desired product.
- (4) Most commonly used bio-reactors are of stirring type.

173 Match List I with List II :

List I	List II
A. Pons	I. Provides additional space for Neurons, regulates posture and balance.
B. Hypothalamus	II. Controls respiration and gastric secretions.
C. Medulla	III. Connects different regions of the brain.
D. Cerebellum	IV. Neuro secretory cells

Choose the correct answer from the options given below :

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-I, C-III, D-IV
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-IV, C-II, D-I

174 Match List I with List II :

List I	List II
A. Lipase	I. Peptide bond
B. Nuclease	II. Ester bond
C. Protease	III. Glycosidic bond
D. Amylase	IV. Phosphodiester bond

Choose the correct answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-I, C-III, D-II
- (3) A-IV, B-II, C-III, D-I
- (4) A-III, B-II, C-I, D-IV

175 Which of the following is not a natural/traditional contraceptive method?

- (1) Lactational amenorrhea
- (2) Vaults
- (3) Coitus interruptus
- (4) Periodic abstinence

176 Following are the stages of pathway for conduction of an action potential through the heart:

- A. AV bundle
- B. Purkinje fibres
- C. AV node
- D. Bundle branches
- E. SA node

Choose the correct sequence of pathway from the options given below :

- (1) B-D-E-C-A (2) E-A-D-B-C
- (3) E-C-A-D-B (4) A-E-C-B-D

177 Match List I with List II :

List I	List II
A. Pleurobrachia	I. Mollusca
B. Radula	II. Ctenophora
C. Stomochord	III. Osteichthyes
D. Air bladder	IV. Hemichordata

Choose the correct answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-II, B-I, C-IV, D-III

178 Match List I with List II :

List I	List II
A. Common cold	I. <i>Plasmodium</i>
B. Haemozoin	II. Typhoid
C. Widal test	III. Rhinoviruses
D. Allergy	IV. Dust mites

Choose the correct answer from the options given below :

- (1) A-III, B-I, C-II, D-IV
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-III, C-II, D-IV

- 179** Consider the following statements :
- Annelids are true coelomates
 - Poriferans are pseudocoelomates
 - Aschelminthes are acoelomates
 - Platyhelminthes are pseudocoelomates
- Choose the correct answer from the options given below :

- (1) C only
- (2) D only
- (3) B only
- (4) A only

- 180** In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on :

- (1) 8th and 9th segment
- (2) 11th segment
- (3) 5th segment
- (4) 10th segment

- 181** Given below are two statements :

Statement I : The presence or absence of hymen is not a reliable indicator of virginity.

Statement II : The hymen is torn during the first coitus only.

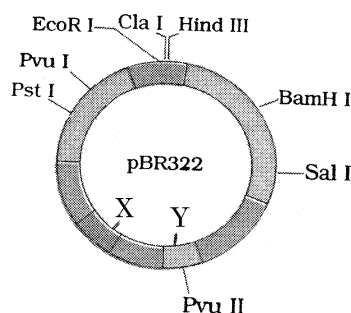
In the light of the above statements, choose the correct answer from the options given below :

- (1) Statement I is true but Statement II is false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

- 182** Which one of the following factors will not affect the Hardy-Weinberg equilibrium?

- (1) Gene migration
- (2) Constant gene pool
- (3) Genetic recombination
- (4) Genetic drift

- 183** The following diagram showing restriction sites in *E. coli* cloning vector pBR322. Find the role of 'X' and 'Y' genes :



- (1) The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.
- (2) Gene 'X' is responsible for recognition sites and 'Y' is responsible for antibiotic resistance.
- (3) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.
- (4) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.

- 184** Match List I with List II :

List I	List II
A. Non-medicated IUD	I. Multiload 375
B. Copper releasing IUD	II. Progestogens
C. Hormone releasing IUD	III. Lippes loop
D. Implants	IV. LNG-20

Choose the correct answer from the options given below :

- (1) A-IV, B-I, C-II, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-I, B-III, C-IV, D-II

- 185** Following are the stages of cell division :

- Gap 2 phase
- Cytokinesis
- Synthesis phase
- Karyokinesis
- Gap 1 phase

Choose the correct sequence of stages from the options given below :

- (1) B-D-E-A-C
- (2) E-C-A-D-B
- (3) C-E-D-A-B
- (4) E-B-D-A-C

Zoology : Section-B (Q. No. 186 to 200)

186 Choose the correct statement given below regarding juxta medullary nephron.

- (1) Loop of Henle of juxta medullary nephron runs deep into medulla.
- (2) Juxta medullary nephrons outnumber the cortical nephrons.
- (3) Juxta medullary nephrons are located in the columns of Bertini.
- (4) Renal corpuscle of juxta medullary nephron lies in the outer portion of the renal medulla.

187 The following are the statements about non-chordates :

- A. Pharynx is perforated by gill slits.
- B. Notochord is absent.
- C. Central nervous system is dorsal.
- D. Heart is dorsal if present.
- E. Post anal tail is absent.

Choose the most appropriate answer from the options given below :

- (1) B, D & E only
- (2) B, C & D only
- (3) A & C only
- (4) A, B & D only

188 Given below are two statements :

Statement I : Mitochondria and chloroplasts are both double membrane bound organelles.

Statement II : Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

189 Regarding catalytic cycle of an enzyme action, select the correct sequential steps :

- A. Substrate enzyme complex formation.
- B. Free enzyme ready to bind with another substrate.
- C. Release of products.
- D. Chemical bonds of the substrate broken.
- E. Substrate binding to active site.

Choose the correct answer from the options given below :

- (1) B, A, C, D, E
- (2) E, D, C, B, A
- (3) E, A, D, C, B
- (4) A, E, B, D, C

190 Match List I with List II :

List I	List II
A. P wave	I. Heart muscles are electrically silent.
B. QRS complex	II. Depolarisation of ventricles.
C. T wave	III. Depolarisation of atria.
D. T-P gap	IV. Repolarisation of ventricles.

Choose the correct answer from the options given below :

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-II, C-I, D-III
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-II, C-IV, D-I

191 Given below are two statements :

Statement I : Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.

Statement II : Both bone marrow and thymus provide micro environments for the development and maturation of T-lymphocytes.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

192 Match List I with List II :

List I	List II
A. Unicellular glandular epithelium	I. Salivary glands
B. Compound epithelium	II. Pancreas
C. Multicellular glandular epithelium	III. Goblet cells of alimentary canal
D. Endocrine glandular epithelium	IV. Moist surface of buccal cavity

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-IV, B-III, C-I, D-II

193 Given below are two statements :

Statement I : The cerebral hemispheres are connected by nerve tract known as corpus callosum.

Statement II : The brain stem consists of the medulla oblongata, pons and cerebrum.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

194 Match List I with List II :

List I	List II
A. RNA polymerase III	I. snRNPs
B. Termination of transcription	II. Promotor
C. Splicing of Exons	III. Rho factor
D. TATA box	IV. SnRNAs, tRNA

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

195 Match List I with List II :

List I	List II
A. Mesozoic Era	I. Lower invertebrates
B. Proterozoic Era	II. Fish & Amphibia
C. Cenozoic Era	III. Birds & Reptiles
D. Paleozoic Era	IV. Mammals

Choose the correct answer from the options given below :

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-I, C-III, D-IV
- (4) A-III, B-I, C-II, D-IV

196 Given below are two statements :

Statement I : Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.

Statement II : According to Gause's principle, during competition, the inferior will be eliminated. This may be true if resources are limiting.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true.
- (3) Both Statement I and Statement II are true.
- (4) Both Statement I and Statement II are false.

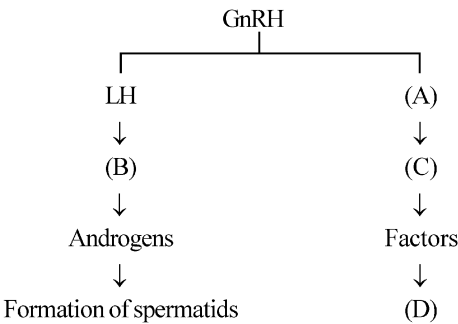
197 Match List I with List II :

List I	List II
A. Exophthalmic goiter	I. Excess secretion of cortisol, moon face & hyperglycemia
B. Acromegaly	II. Hypo-secretion of thyroid hormone and stunted growth.
C. Cushing's syndrome	III. Hyper secretion of thyroid hormone & protruding eye balls.
D. Cretinism	IV. Excessive secretion of growth hormone.

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-III, C-II, D-IV
- (4) A-IV, B-II, C-I, D-III

198 Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.



- (1) FSH, Sertoli cells, Leydig cells, spermatogenesis.
- (2) ICSH, Leydig cells, Sertoli cells, spermatogenesis.
- (3) FSH, Leydig cells, Sertoli cells, spermiogenesis
- (4) ICSH, Interstitial cells, Leydig cells, spermiogenesis.

199 As per ABO blood grouping system, the blood group of father is B⁺, mother is A⁺ and child is O⁺. Their respective genotype can be

- A. I^Bi / I^Ai / ii
- B. I^BI^B / I^AI^A / ii
- C. I^AI^B / ii^A / I^Bi
- D. I^Ai / I^Bi / I^Ai
- E. ii^B / ii^A / I^AI^B

Choose the most appropriate answer from the options given below :

- (1) C & B only
- (2) D & E only
- (3) A only
- (4) B only

200 Match List I with List II related to digestive system of cockroach.

List I	List II
A. The structures used for storing of food.	I. Gizzard
B. Ring of 6-8 blind tubules at junction of foregut and midgut.	II. Gastric Caeca
C. Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	III. Malpighian tubules
D. The structures used for grinding the food.	IV. Crop

Choose the correct answer from the options given below :

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-II, C-III, D-IV

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

Test Booklet No. **ENGLISH**

Test Booklet Code

E1

ADI

This Booklet contains **32** pages, including Rough Page.

Do not open this Test Booklet until you are asked to do so.

Important Instructions:

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below:
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos – 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is E1. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
17. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

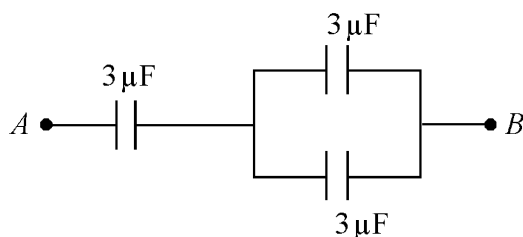
- 1 Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is :

(1) $2W/A$ (2) W/A
(3) $W/2A$ (4) Zero

- 2 The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is :

(1) 3 : 5 (2) 5 : 3
(3) 2 : 5 (4) 5 : 2

- 3 The equivalent capacitance of the system shown in the following circuit is :



(1) $2 \mu\text{F}$ (2) $3 \mu\text{F}$
(3) $6 \mu\text{F}$ (4) $9 \mu\text{F}$

- 4 A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is :

(1) along eastward
(2) along northward
(3) along north-east
(4) along south-west

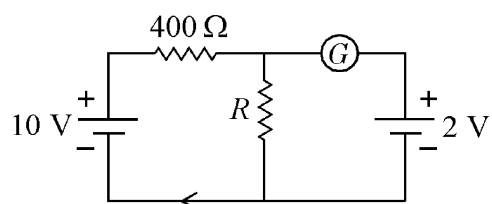
- 5 If $\oint_S \vec{E} \cdot d\vec{S} = 0$ over a surface, then :

(1) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.
(2) the magnitude of electric field on the surface is constant.
(3) all the charges must necessarily be inside the surface.
(4) the electric field inside the surface is necessarily uniform.

- 6 The potential energy of a long spring when stretched by 2 cm is U . If the spring is stretched by 8 cm, potential energy stored in it will be :

(1) $2U$ (2) $4U$
(3) $8U$ (4) $16U$

- 7 If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by :



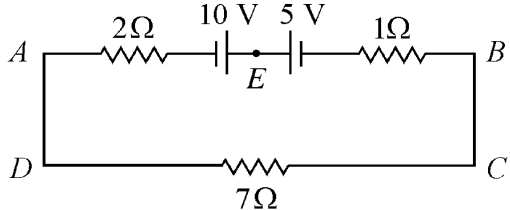
(1) 200Ω (2) 50Ω
(3) 100Ω (4) 400Ω

- 8 A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?

(1) 0.27 A (2) 2.7 A
(3) 3.7 A (4) 0.37 A

- 9 A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output?
- A centre-tapped transformer
 - p-n junction diodes
 - Capacitor
 - Load resistance
- 10 Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?
- $\sin^{-1}\left(\frac{t_2}{t_1}\right)$
 - $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$
 - $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$
 - $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$
- 11 Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%) \Omega$. The colour of third band must be :
- Red
 - Green
 - Orange
 - Yellow
- 12 Given below are two statements:
Statement I : Photovoltaic devices can convert optical radiation into electricity.
Statement II : Zener diode is designed to operate under reverse bias in breakdown region.
 In the light of the above statements, choose the **most appropriate** answer from the options given below :
- Both **Statement I** and **Statement II** are correct.
 - Both **Statement I** and **Statement II** are incorrect.
 - Statement I** is correct but **Statement II** is incorrect.
 - Statement I** is incorrect but **Statement II** is correct.
- 13 The magnetic energy stored in an inductor of inductance $4 \mu\text{H}$ carrying a current of 2 A is :
- $4 \mu\text{J}$
 - 4 mJ
 - 8 mJ
 - $8 \mu\text{J}$
- 14 The angular acceleration of a body, moving along the circumference of a circle, is :
- along the radius, away from centre
 - along the radius towards the centre
 - along the tangent to its position
 - along the axis of rotation
- 15 A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C . The temperature of the sink is :
- 27°C
 - 15°C
 - 100°C
 - 200°C
- 16 Two bodies of mass m and $9m$ are placed at a distance R . The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant) :
- $-\frac{8Gm}{R}$
 - $-\frac{12Gm}{R}$
 - $-\frac{16Gm}{R}$
 - $-\frac{20Gm}{R}$
- 17 A vehicle travels half the distance with speed ϑ and the remaining distance with speed 2ϑ . Its average speed is:
- $\frac{\vartheta}{3}$
 - $\frac{2\vartheta}{3}$
 - $\frac{4\vartheta}{3}$
 - $\frac{3\vartheta}{4}$

- 18 The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly : (surface tension of soap solution = 0.03 N m^{-1})
 (1) $30.16 \times 10^{-4} \text{ J}$ (2) $5.06 \times 10^{-4} \text{ J}$
 (3) $3.01 \times 10^{-4} \text{ J}$ (4) $50.1 \times 10^{-4} \text{ J}$
- 19 The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:
 (1) $\sqrt{V}$ (2) $\frac{1}{V}$
 (3) $\frac{1}{\sqrt{V}}$ (4) V^2
- 20 The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{\text{th}}$ of its initial value?
 (1) 20 minutes (2) 40 minutes
 (3) 60 minutes (4) 80 minutes
- 21 A metal wire has mass $(0.4 \pm 0.002) \text{ g}$, radius $(0.3 \pm 0.001) \text{ mm}$ and length $(5 \pm 0.02) \text{ cm}$. The maximum possible percentage error in the measurement of density will nearly be:
 (1) 1.2% (2) 1.3%
 (3) 1.6% (4) 1.4%
- 22 In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of $2.0 \times 10^{10} \text{ Hz}$ and amplitude 48 V m^{-1} . Then the amplitude of oscillating magnetic field is : (Speed of light in free space = $3 \times 10^8 \text{ m s}^{-1}$)
 (1) $1.6 \times 10^{-9} \text{ T}$ (2) $1.6 \times 10^{-8} \text{ T}$
 (3) $1.6 \times 10^{-7} \text{ T}$ (4) $1.6 \times 10^{-6} \text{ T}$
- 23 The temperature of a gas is -50° C . To what temperature the gas should be heated so that the rms speed is increased by 3 times?
 (1) 669° C (2) 3295° C
 (3) 3097 K (4) 223 K
- 24 An ac source is connected to a capacitor C. Due to decrease in its operating frequency :
 (1) capacitive reactance decreases.
 (2) displacement current increases.
 (3) displacement current decreases.
 (4) capacitive reactance remains constant
- 25 For Young's double slit experiment, two statements are given below:
Statement I : If screen is moved away from the plane of slits, angular separation of the fringes remains constant.
Statement II : If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.
 In the light of the above statements, choose the *correct* answer from the options given below:
 (1) Both **Statement I** and **Statement II** are true.
 (2) Both **Statement I** and **Statement II** are false.
 (3) **Statement I** is true but **Statement II** is false.
 (4) **Statement I** is false but **Statement II** is true.
- 26 In hydrogen spectrum, the shortest wavelength in the Balmer series is λ . The shortest wavelength in the Bracket series is :
 (1) 2λ (2) 4λ
 (3) 9λ (4) 16λ

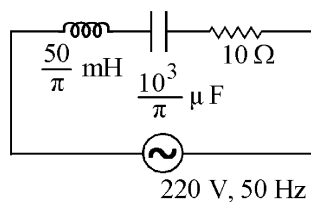
- 27** The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?
- Cs only
 - Both Na and K
 - K only
 - Na only
- 28** The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are :
- Instrumental errors
 - Personal errors
 - Least count errors
 - Random errors
- 29** In a series *LCR* circuit, the inductance L is 10 mH, capacitance C is $1\text{ }\mu\text{F}$ and resistance R is $100\text{ }\Omega$. The frequency at which resonance occurs is :
- 15.9 rad/s
 - 15.9 kHz
 - 1.59 rad/s
 - 1.59 kHz
- 30** The venturi-meter works on :
- Huygen's principle
 - Bernoulli's principle
 - The principle of parallel axes
 - The principle of perpendicular axes
- 31** The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is :
- 1 : 2
 - 2 : 1
 - 1 : 3
 - 3 : 1
- 32** An electric dipole is placed at an angle of 30° with an electric field of intensity $2 \times 10^5 \text{ N C}^{-1}$. It experiences a torque equal to 4 N m. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.
- 8 mC
 - 6 mC
 - 4 mC
 - 2 mC
- 33** The magnitude and direction of the current in the following circuit is
- 
- 0.2 A from B to A through E
 - 0.5 A from A to B through E
 - $\frac{5}{9}$ A from A to B through E
 - 1.5 A from B to A through E
- 34** The net magnetic flux through any closed surface is :
- Zero
 - Positive
 - Infinity
 - Negative
- 35** A bullet is fired from a gun at the speed of 280 m s^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is ($g = 9.8 \text{ m s}^{-2}$, $\sin 30^\circ = 0.5$) :
- 2800 m
 - 2000 m
 - 1000 m
 - 3000 m

Physics : Section-B (Q. No. 36 to 50)

- 36** Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be :

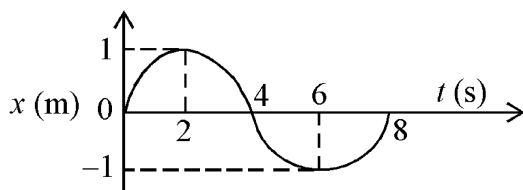
(1) Zero (2) $f/4$
(3) $f/2$ (4) Infinite

- 37** The net impedance of circuit (as shown in figure) will be :



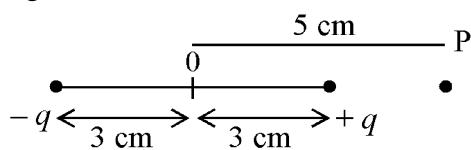
(1) $10\sqrt{2} \Omega$ (2) 15Ω
(3) $5\sqrt{5} \Omega$ (4) 25Ω

- 38** The $x-t$ graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t=2$ s is :



(1) $\frac{\pi^2}{8} \text{ m s}^{-2}$ (2) $-\frac{\pi^2}{8} \text{ m s}^{-2}$
(3) $\frac{\pi^2}{16} \text{ m s}^{-2}$ (4) $-\frac{\pi^2}{16} \text{ m s}^{-2}$

- 39** An electric dipole is placed as shown in the figure.



The electric potential (in 10^2 V) at point P due to the dipole is (ϵ_0 = permittivity of free space and $\frac{1}{4\pi\epsilon_0} = K$):

(1) $\left(\frac{3}{8}\right) qK$ (2) $\left(\frac{5}{8}\right) qK$
(3) $\left(\frac{8}{5}\right) qK$ (4) $\left(\frac{8}{3}\right) qK$

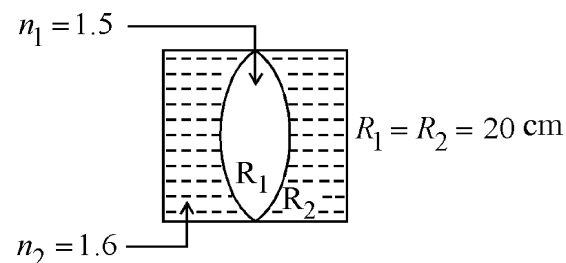
- 40** A bullet from a gun is fired on a rectangular wooden block with velocity u . When bullet travels 24 cm through the block along its length horizontally, velocity of bullet

becomes $\frac{u}{3}$. Then it further penetrates into

the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is :

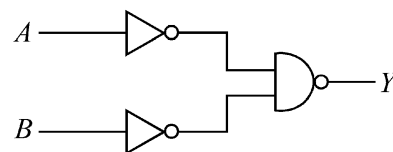
(1) 27 cm (2) 24 cm
(3) 28 cm (4) 30 cm

- 41** In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



(1) 40 cm (2) -40 cm
(3) -100 cm (4) -50 cm

- 42** For the following logic circuit, the truth table is:



(1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

 (2)

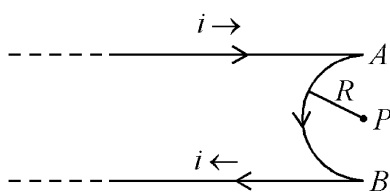
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

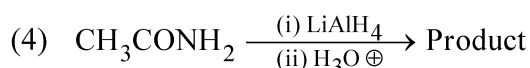
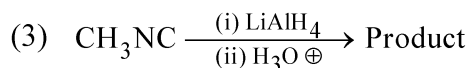
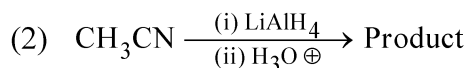
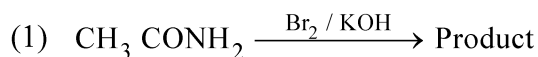
 (4)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

- 43 A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity 4 m s^{-1} . The ball strikes the water surface after 4 s. The height of bridge above water surface is (Take $g = 10 \text{ m s}^{-2}$) :
- (1) 56 m (2) 60 m
(3) 64 m (4) 68 m
- 44 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is :
- (1) 10 (2) 100
(3) 1 (4) 1000
- 45 A wire carrying a current I along the positive x -axis has length L . It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k}) \text{ T}$. The magnitude of the magnetic force acting on the wire is :
- (1) $3 IL$ (2) $\sqrt{5} IL$
(3) $5 IL$ (4) $\sqrt{3} IL$
- 46 A satellite is orbiting just above the surface of the earth with period T . If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents :
- (1) T (2) T^2
(3) T^3 (4) $\sqrt{T}$
- 47 Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10 \text{ m s}^{-2}$).
- (1) 1.2 m s^{-2} (2) 150 m s^{-2}
(3) 1.5 m s^{-2} (4) 50 m s^{-2}
- 48 The resistance of platinum wire at 0°C is 2Ω and 6.8Ω at 80°C . The temperature coefficient of resistance of the wire is :
- (1) $3 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$ (2) $3 \times 10^{-3} \text{ }^\circ\text{C}^{-1}$
(3) $3 \times 10^{-2} \text{ }^\circ\text{C}^{-1}$ (4) $3 \times 10^{-1} \text{ }^\circ\text{C}^{-1}$
- 49 The radius of inner most orbit of hydrogen atom is $5.3 \times 10^{-11} \text{ m}$. What is the radius of third allowed orbit of hydrogen atom?
- (1) $0.53 \text{ \AA}$ (2) $1.06 \text{ \AA}$
(3) $1.59 \text{ \AA}$ (4) $4.77 \text{ \AA}$
- 50 A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by :
- 
- (1) $\frac{\mu_0 i}{4R}$ pointed into the page
(2) $\frac{\mu_0 i}{4R}$ pointed away from the page
(3) $\frac{\mu_0 i}{4R} \left[1 - \frac{2}{\pi} \right]$ pointed away from page
(4) $\frac{\mu_0 i}{4R} \left[1 - \frac{2}{\pi} \right]$ pointed into the page

Chemistry : Section-A (Q. No. 51 to 85)

51 Which of the following reactions will NOT give primary amine as the product?



52 Match **List - I** with **List - II** :

List - I	List - II
A. Coke	I. Carbon atoms are sp^3 hybridised.
B. Diamond	II. Used as a dry lubricant
C. Fullerene	III. Used as a reducing agent
D. Graphite	IV. Cage like molecules

Choose the **correct** answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-I, C-II, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B-IV, C-I, D-II

53 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

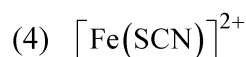
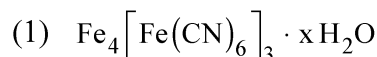
Assertion A : Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic.

Reasons R : The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

54 In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of -



55 The conductivity of centimolar solution of KCl at 25°C is $0.0210 \text{ ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is -

(1) 1.34 cm^{-1} (2) 3.28 cm^{-1}

(3) 1.26 cm^{-1} (4) 3.34 cm^{-1}

56 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : A reaction can have zero activation energy.

Reasons R : The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

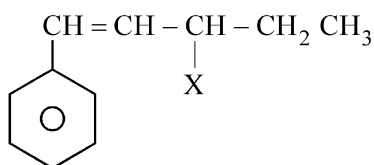
In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

57 Which one is an example of heterogenous catalysis?

- (1) Oxidation of sulphur dioxide into sulphur trioxide in the presence of oxides of nitrogen.
- (2) Hydrolysis of sugar catalysed by H^+ ions.
- (3) Decomposition of ozone in presence of nitrogen monoxide.
- (4) Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.

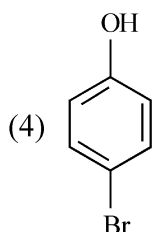
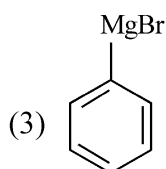
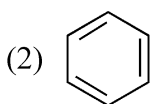
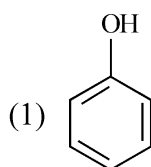
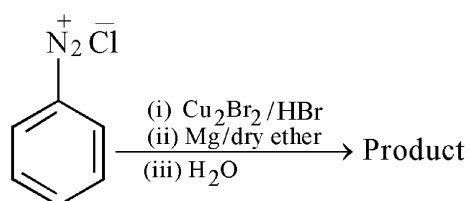
58 The given compound



is an example of _____.

- (1) benzylic halide
- (2) aryl halide
- (3) allylic halide
- (4) vinylic halide

59 Identify the product in the following reaction:



60 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : Helium is used to dilute oxygen in diving apparatus.

Reasons R : Helium has high solubility in O_2 .

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

61 A compound is formed by two elements A and B. The element B forms cubic close packed structure and atoms of A occupy $1/3$ of tetrahedral voids. If the formula of the compound is A_xB_y , then the value of $x + y$ is in option

- (1) 5
- (2) 4
- (3) 3
- (4) 2

62 Given below are two statements :

Statement I : A unit formed by the attachment of a base to $1'$ position of sugar is known as nucleoside

Statement II : When nucleoside is linked to phosphorous acid at $5'$ -position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statement **I** and Statement **II** are true.
- (2) Both Statement **I** and Statement **II** are false.
- (3) Statement **I** is true but Statement **II** is false.
- (4) Statement **I** is false but Statement **II** is true.

- 63 The relation between n_m , (n_m = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

$$(1) \quad l = \frac{n_m - 1}{2} \quad (2) \quad l = 2n_m + 1$$

$$(3) \quad n_m = 2l^2 + 1 \quad (4) \quad n_m = l + 2$$

- 64 Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is



$$(1) \quad 3 \quad (2) \quad 2$$

$$(3) \quad 4 \quad (4) \quad 1$$

- 65 The **correct** order of energies of molecular orbitals of N_2 molecule, is :

$$(1) \quad \sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) <$$

$$\sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

$$(2) \quad \sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z <$$

$$(\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

$$(3) \quad \sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z <$$

$$\sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$$

$$(4) \quad \sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) <$$

$$(\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$$

- 66 The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are:

$$(1) \quad 11, 2, 0 \quad (2) \quad 12, 3, 0$$

$$(3) \quad 11, 3, 1 \quad (4) \quad 12, 2, 1$$

- 67 Intermolecular forces are forces of attraction and repulsion between interacting particles that will include :

- A. dipole - dipole forces.
- B. dipole - induced dipole forces.
- C. hydrogen bonding.
- D. covalent bonding.
- E. dispersion forces.

Choose the **most appropriate** answer from the options given below :

$$(1) \quad \text{B, C, D, E are correct.}$$

$$(2) \quad \text{A, B, C, D are correct.}$$

$$(3) \quad \text{A, B, C, E are correct.}$$

$$(4) \quad \text{A, C, D, E are correct.}$$

- 68 Which of the following statements are **NOT** correct?

- A. Hydrogen is used to reduce heavy metal oxides to metals.
- B. Heavy water is used to study reaction mechanism.
- C. Hydrogen is used to make saturated fats from oils.
- D. The H-H bond dissociation enthalpy is lowest as compared to a single bond between two atoms of any element.
- E. Hydrogen reduces oxides of metals that are more active than iron.

Choose the **most appropriate** answer from the options given below :

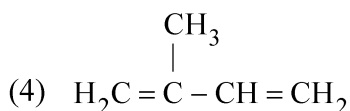
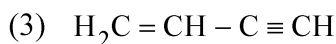
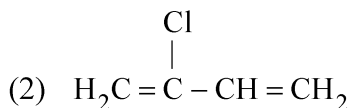
$$(1) \quad \text{B, C, D, E only}$$

$$(2) \quad \text{B, D only}$$

$$(3) \quad \text{D, E only}$$

$$(4) \quad \text{A, B, C only}$$

69 Which amongst the following molecules on polymerization produces neoprene?



70 Some tranquilizers are listed below. Which one from the following belongs to barbiturates?

- (1) Chlordiazepoxide
- (2) Meprobamate
- (3) Valium
- (4) Veronal

71 The element expected to form largest ion to achieve the nearest noble gas configuration is :

- | | |
|-------|--------|
| (1) O | (2) F |
| (3) N | (4) Na |

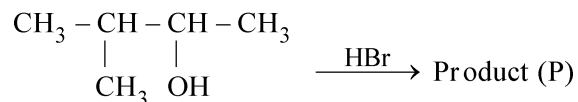
72 Select the **correct** statements from the following :

- A. Atoms of all elements are composed of two fundamental particles.
- B. The mass of the electron is 9.10939×10^{-31} kg.
- C. All the isotopes of a given element show same chemical properties.
- D. Protons and electrons are collectively known as nucleons.
- E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

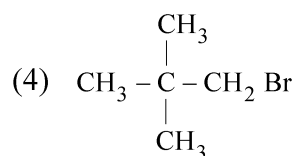
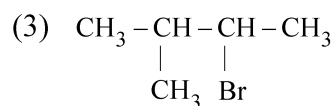
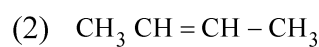
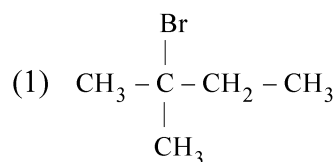
Choose the **correct** answer from the options given below :

- (1) A, B and C only
- (2) C, D and E only
- (3) A and E only
- (4) B, C and E only

73 Consider the following reaction and identify the product (P).



3 - Methylbutan-2 - ol



74 The stability of Cu^{2+} is more than Cu^+ salts in aqueous solution due to -

- (1) first ionisation enthalpy.
- (2) enthalpy of atomization.
- (3) hydration energy.
- (4) second ionisation enthalpy.

75 Which one of the following statements is **correct**?

- (1) The daily requirement of Mg and Ca in the human body is estimated to be 0.2 - 0.3 g.
- (2) All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor.
- (3) The bone in human body is an inert and unchanging substance.
- (4) Mg plays roles in neuromuscular function and interneuronal transmission.

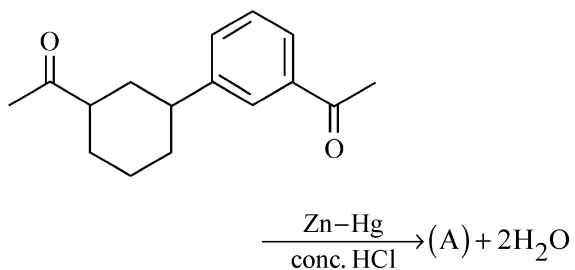
76 Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is :

- (1) 16 (2) 32
(3) 30 (4) 18

77 Amongst the given options which of the following molecules / ion acts as a Lewis acid?

- (1) NH_3 (2) H_2O
(3) BF_3 (4) OH^-

78 Identify product (A) in the following reaction:



- (1)
- (2)
- (3)
- (4)

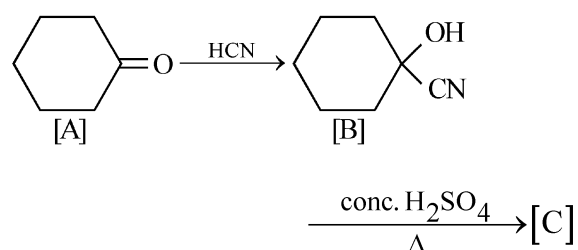
79 Taking stability as the factor, which one of the following represents **correct** relationship?

- (1) $\text{TiCl}_3 > \text{TiCl}$
(2) $\text{InI}_3 > \text{InI}$
(3) $\text{AlCl} > \text{AlCl}_3$
(4) $\text{TlI} > \text{TlI}_3$

80 Homoleptic complex from the following complexes is :

- (1) Potassium trioxalatoaluminate (III)
(2) Diamminechloridonitrito - N - platinum (II)
(3) Pentaamminecarbonatocobalt (III) chloride
(4) Triamminetriaquachromium (III) chloride

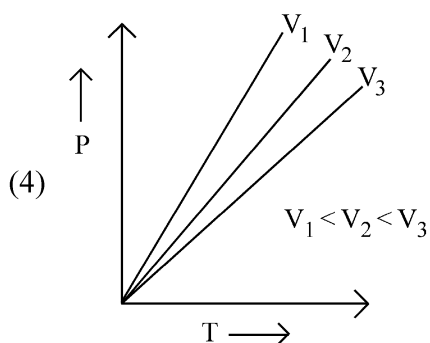
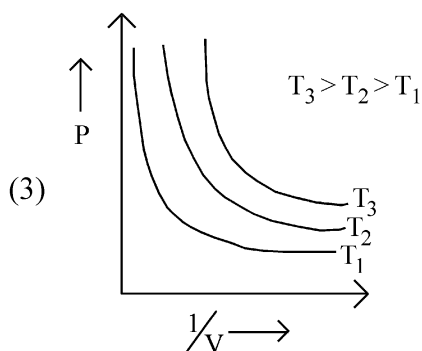
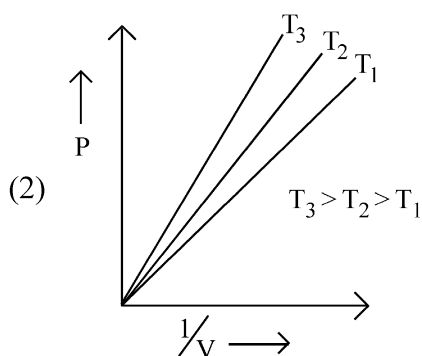
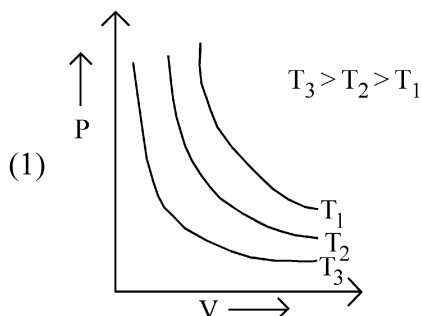
81 Complete the following reaction :



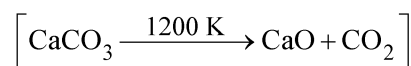
[C] is _____.

- (1)
- (2)
- (3)
- (4)

- 82 Which amongst the following options is **correct** graphical representation of Boyle's Law?



- 83 The **right** option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)



- (1) 1.12 g (2) 1.76 g
(3) 2.64 g (4) 1.32 g

- 84 For a certain reaction, the rate = $k[\text{A}]^2[\text{B}]$, when the initial concentration of A is tripled keeping concentration of B constant, the initial rate would

- (1) decrease by a factor of nine.
(2) increase by a factor of six.
(3) increase by a factor of nine.
(4) increase by a factor of three.

- 85 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : In equation $\Delta_r G = -nFE_{\text{cell}}$, value of $\Delta_r G$ depends on n.

Reasons R : E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

86 Match List - I with List - II :

List - I (Oxoacids of Sulphur) **List - II (Bonds)**

- | | |
|---------------------------|--------------------------------------|
| A. Peroxodisulphuric acid | I. Two S-OH, Four S=O, One S-O-S |
| B. Sulphuric acid | II. Two S-OH, One S=O |
| C. Pyrosulphuric acid | III. Two S-OH, Four S=O, One S-O-O-S |
| D. Sulphurous acid | IV. Two S-OH, Two S=O |

Choose the **correct** answer from the options given below :

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-IV, C-II, D-I

87 Which of the following statements are **INCORRECT**?

- A. All the transition metals except scandium form MO oxides which are ionic.
- B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7 .
- C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5 .
- D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.
- E. CrO is basic but Cr_2O_3 is amphoteric.

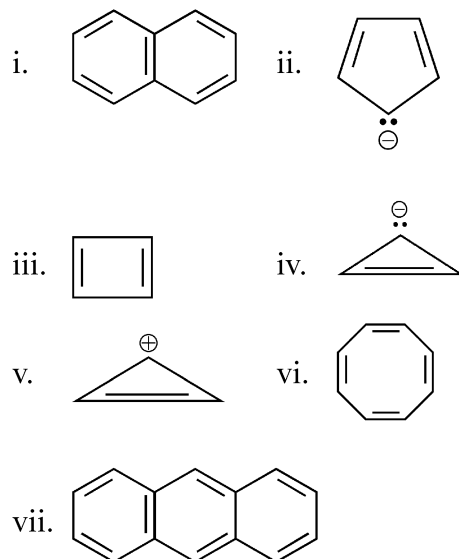
Choose the **correct** answer from the options given below :

- (1) A and E only
- (2) B and D only
- (3) C and D only
- (4) B and C only

88 Which complex compound is most stable?

- (1) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$
- (2) $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$
- (3) $[\text{CoCl}_2(\text{en})_2]\text{NO}_3$
- (4) $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$

89 Consider the following compounds/species:



The number of compounds/species which obey Huckel's rule is _____.

- | | |
|-------|-------|
| (1) 4 | (2) 6 |
| (3) 2 | (4) 5 |

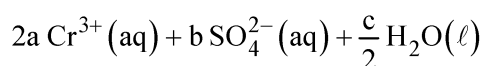
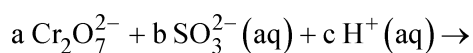
90 What fraction of one edge centred octahedral void lies in one unit cell of fcc?

- | | |
|-------------------|--------------------|
| (1) $\frac{1}{2}$ | (2) $\frac{1}{3}$ |
| (3) $\frac{1}{4}$ | (4) $\frac{1}{12}$ |

91 Which amongst the following options is the **correct** relation between change in enthalpy and change in internal energy?

- (1) $\Delta H = \Delta U - \Delta n_g RT$
 (2) $\Delta H = \Delta U + \Delta n_g RT$
 (3) $\Delta H - \Delta U = - \Delta n RT$
 (4) $\Delta H + \Delta U = \Delta n R$

92 On balancing the given redox reaction,



the coefficients a, b and c are found to be, respectively -

- (1) 1, 3, 8 (2) 3, 8, 1
 (3) 1, 8, 3 (4) 8, 1, 3

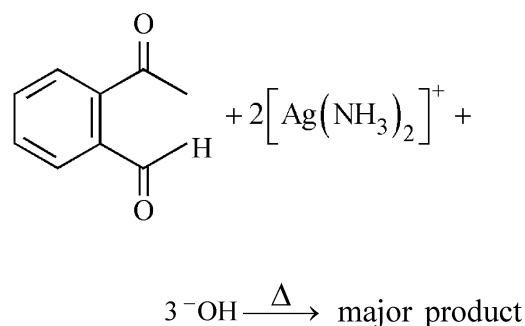
93 The equilibrium concentrations of the species in the reaction $A + B \rightleftharpoons C + D$ are 2, 3, 10 and 6 mol L⁻¹, respectively at 300 K. ΔG° for the reaction is (R = 2 cal / mol K)

- (1) 1372.60 cal (2) -137.26 cal
 (3) -1381.80 cal (4) -13.73 cal

94 Pumice stone is an example of -

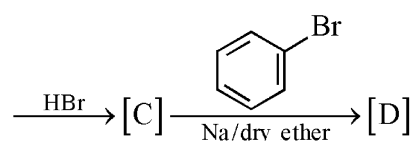
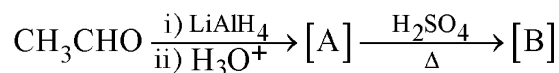
- (1) sol (2) gel
 (3) solid sol (4) foam

95 Identify the major product obtained in the following reaction :



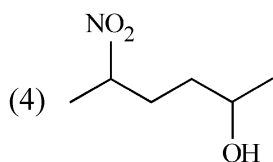
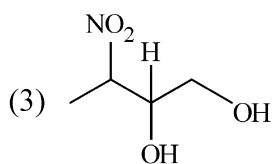
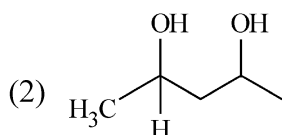
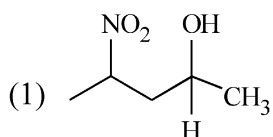
- (1) (2)
 (3) (4)

96 Identify the final product [D] obtained in the following sequence of reactions.



- (1)
 (2)
 (3) C₄H₁₀
 (4) HC $\equiv$ C[⊖] Na⁺

97 Which amongst the following will be most readily dehydrated under acidic conditions ?



98 Given below are two statements :

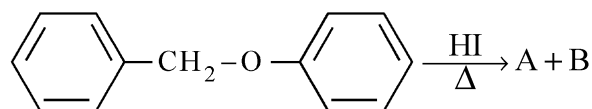
Statement I : The nutrient deficient water bodies lead to eutrophication.

Statement II : Eutrophication leads to decrease in the level of oxygen in the water bodies.

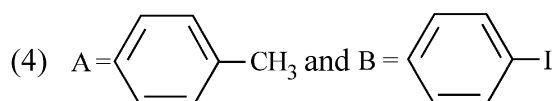
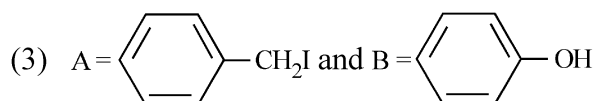
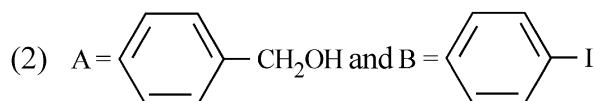
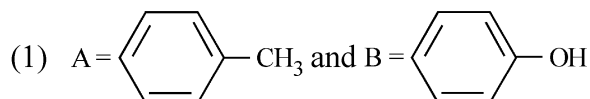
In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** is incorrect but **Statement II** is true.

99 Consider the following reaction :



Identify products A and B.



100 The reaction that does **NOT** take place in a blast furnace between 900 K to 1500 K temperature range during extraction of iron is :

- (1) $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{FeO} + \text{CO}_2$
- (2) $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
- (3) $\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$
- (4) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

Botany : Section-A (Q. No. 101 to 135)

- 101** Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :
Assertion A : The first stage of gametophyte in the life cycle of moss is protonema stage.
Reason R : Protonema develops directly from spores produced in capsule.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
(3) **A** is correct but **R** is not correct.
(4) **A** is not correct but **R** is correct.
- 102** In angiosperm, the haploid, diploid and triploid structures of a fertilized embryo sac sequentially are :
(1) Synergids, Primary endosperm nucleus and zygote
(2) Antipodals, synergids, and primary endosperm nucleus
(3) Synergids, Zygote and Primary endosperm nucleus
(4) Synergids, antipodals and Polar nuclei
- 103** Movement and accumulation of ions across a membrane against their concentration gradient can be explained by
(1) Osmosis
(2) Facilitated Diffusion
(3) Passive Transport
(4) Active Transport
- 104** Large, colourful, fragrant flowers with nectar are seen in :
(1) insect pollinated plants
(2) bird pollinated plants
(3) bat pollinated plants
(4) wind pollinated plants
- 105** The phenomenon of pleiotropism refers to
(1) presence of several alleles of a single gene controlling a single crossover.
(2) presence of two alleles, each of the two genes controlling a single trait.
(3) a single gene affecting multiple phenotypic expression.
(4) more than two genes affecting a single character.
- 106** Which hormone promotes internode/petiole elongation in deep water rice?
(1) GA₃ (2) Kinetin
(3) Ethylene (4) 2, 4-D
- 107** Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species?
(1) Habitat loss and fragmentation
(2) Over exploitation for economic gain
(3) Alien species invasions
(4) Co-extinctions
- 108** Upon exposure to UV radiation, DNA stained with ethidium bromide will show
(1) Bright red colour
(2) Bright blue colour
(3) Bright yellow colour
(4) Bright orange colour
- 109** Which micronutrient is required for splitting of water molecule during photosynthesis?
(1) manganese (2) molybdenum
(3) magnesium (4) copper
- 110** Axile placentation is observed in
(1) Mustard, Cucumber and Primrose
(2) China rose, Beans and Lupin
(3) Tomato, Dianthus and Pea
(4) China rose, Petunia and Lemon
- 111** The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?
(1) Zygotene (2) Pachytene
(3) Diplotene (4) Diakinesis
- 112** The reaction centre in PS II has an absorption maxima at
(1) 680 nm (2) 700 nm
(3) 660 nm (4) 780 nm
- 113** Unequivocal proof that DNA is the genetic material was first proposed by
(1) Frederick Griffith
(2) Alfred Hershey and Martha Chase
(3) Avery, Macleoid and McCarthy
(4) Wilkins and Franklin

- 114** Among eukaryotes, replication of DNA takes place in -
 (1) M phase (2) S phase
 (3) G₁ phase (4) G₂ phase
- 115** In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as -
 (1) Differentiation
 (2) Dedifferentiation
 (3) Development
 (4) Senescence
- 116** Cellulose does not form blue colour with Iodine because
 (1) It is a disaccharide.
 (2) It is a helical molecule.
 (3) It does not contain complex helices and hence cannot hold iodine molecules.
 (4) It breaks down when iodine reacts with it.
- 117** Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production?
 (1) Indole-3-butyric Acid
 (2) Gibberellic Acid
 (3) Zeatin
 (4) Absciscic Acid
- 118** Given below are two statements :
Statement I : The forces generated by transpiration can lift a xylem-sized column of water over 130 meters height.
Statement II : Transpiration cools leaf surfaces sometimes 10 to 15 degrees, by evaporative cooling.
 In the light of the above statements, choose the **most appropriate** answer from the options given below :
 (1) Both **Statement I** and **Statement II** are correct.
 (2) Both **Statement I** and **Statement II** are incorrect.
 (3) **Statement I** is correct but **Statement II** is incorrect.
 (4) **Statement I** is incorrect but **Statement II** is correct.
- 119** Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens, pick out the characteristics specific to family Fabaceae but not found in Solanaceae or Liliaceae.
 (1) Diadelphous and Ditheous anthers
 (2) Polyadelphous and epipetalous stamens
 (3) Monoadelphous and Monotheous anthers
 (4) Epiphyllous and Ditheous anthers
- 120** Expressed Sequence Tags (ESTs) refers to
 (1) All genes that are expressed as RNA.
 (2) All genes that are expressed as proteins.
 (3) All genes whether expressed or unexpressed.
 (4) Certain important expressed genes.
- 121** Identify the **correct** statements :
 A. Detrivores perform fragmentation.
 B. The humus is further degraded by some microbes during mineralization.
 C. Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.
 D. The detritus food chain begins with living organisms.
 E. Earthworms break down detritus into smaller particles by a process called catabolism.
 Choose the **correct** answer from the options given below :
 (1) A, B, C only (2) B, C, D only
 (3) C, D, E only (4) D, E, A only
- 122** The thickness of ozone in a column of air in the atmosphere is measured in terms of :
 (1) Dobson units (2) Decibels
 (3) Decameter (4) Kilobase

- 123** Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :
Assertion A : Late wood has fewer xylary elements with narrow vessels.
Reason R : Cambium is less active in winters.
 In the light of the above statements, choose the **correct** answer from the options given below :
- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
 - (2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
 - (3) **A** is true but **R** is false.
 - (4) **A** is false but **R** is true.
- 124** Which of the following stages of meiosis involves division of centromere?
 (1) Metaphase I (2) Metaphase II
 (3) Anaphase II (4) Telophase
- 125** The historic Convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year :
 (1) 1985 (2) 1992
 (3) 1986 (4) 2002
- 126** How many ATP and NADPH_2 are required for the synthesis of one molecule of Glucose during Calvin cycle?
 (1) 12 ATP and 12 NADPH_2
 (2) 18 ATP and 12 NADPH_2
 (3) 12 ATP and 16 NADPH_2
 (4) 18 ATP and 16 NADPH_2
- 127** In the equation

$$\boxed{\text{GPP} - \text{R} = \text{NPP}}$$
 GPP is Gross Primary Productivity
 NPP is Net Primary Productivity
 R here is _____.
 (1) Photosynthetically active radiation
 (2) Respiratory quotient
 (3) Respiratory loss
 (4) Reproductive allocation
- 128** During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out
 (1) RNA (2) DNA
 (3) Histones (4) Polysaccharides
- 129** What is the role of RNA polymerase III in the process of transcription in Eukaryotes?
 (1) Transcription of rRNAs (28S, 18S and 5.8S)
 (2) Transcription of tRNA, 5 srRNA and snRNA
 (3) Transcription of precursor of mRNA
 (4) Transcription of only snRNAs
- 130** What is the function of tassels in the corn cob?
 (1) To attract insects
 (2) To trap pollen grains
 (3) To disperse pollen grains
 (4) To protect seeds
- 131** Identify the pair of heterosporous pteridophytes among the following :
 (1) *Lycopodium* and *Selaginella*
 (2) *Selaginella* and *Salvinia*
 (3) *Psilotum* and *Salvinia*
 (4) *Equisetum* and *Salvinia*
- 132** In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.
 (1) Copper
 (2) Zinc
 (3) Tungsten or gold
 (4) Silver

- 133** Given below are two statements :
Statement I : Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.
Statement II : Exarch condition is the most common feature of the root system.
 In the light of the above statements, choose the **correct** answer from the options given below :
- (1) Both **Statement I** and **Statement II** are true.
 - (2) Both **Statement I** and **Statement II** are false.
 - (3) **Statement I** is correct but **Statement II** is false.
 - (4) **Statement I** is incorrect but **Statement II** is true.
- 134** Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by
- (1) Thomas Hunt Morgan
 - (2) Sutton and Boveri
 - (3) Alfred Sturtevant
 - (4) Henking
- 135** Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :
Assertion A : ATP is used at two steps in glycolysis.
Reason R : First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1-6-diphosphate.
 In the light of the above statements, choose the **correct** answer from the options given below :
- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
 - (2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
 - (3) **A** is true but **R** is false.
 - (4) **A** is false but **R** is true.

- 136** Which one of the following statements is **NOT** correct?
- (1) The micro-organisms involved in biodegradation of organic matter in a sewage polluted water body consume a lot of oxygen causing the death of aquatic organisms.
 - (2) Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries.
 - (3) Water hyacinth grows abundantly in eutrophic water bodies and leads to an imbalance in the ecosystem dynamics of the water body.
 - (4) The amount of some toxic substances of industrial waste water increases in the organisms at successive trophic levels.
- 137** How many different proteins does the ribosome consist of?
- | | |
|--------|--------|
| (1) 80 | (2) 60 |
| (3) 40 | (4) 20 |
- 138** Which of the following statements are correct about Klinefelter's Syndrome?
- A. This disorder was first described by Langdon Down (1866).
 - B. Such an individual has overall masculine development. However, the feminine development is also expressed.
 - C. The affected individual is short statured.
 - D. Physical, psychomotor and mental development is retarded.
 - E. Such individuals are sterile.
- Choose the **correct** answer from the options given below :
- | | |
|------------------|------------------|
| (1) A and B only | (2) C and D only |
| (3) B and E only | (4) A and E only |
- 139** Match **List I** with **List II** :
- | List I | List II |
|------------------------------|--------------------------------|
| A. Oxidative decarboxylation | I. Citrate synthase |
| B. Glycolysis | II. Pyruvate dehydrogenase |
| C. Oxidative phosphorylation | III. Electron transport system |
| D. Tricarboxylic acid cycle | IV. EMP pathway |
- Choose the correct answer from the options given below :
- (1) A-III, B-IV, C-II, D-I
 - (2) A-II, B-IV, C-I, D-III
 - (3) A-III, B-I, C-II, D-IV
 - (4) A-II, B-IV, C-III, D-I

- 140** Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason R : Internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

- 141** Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.

Reason R : Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

- 142** Match **List I** with **List II** :

List I	List II
A. Cohesion	I. More attraction in liquid phase
B. Adhesion	II. Mutual attraction among water molecules
C. Surface tension	III. Water loss in liquid phase
D. Guttation	IV. Attraction towards polar surfaces

Choose the **correct** answer from the options given below :

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-I, C-IV, D-III

- 143** Which of the following combinations is required for chemiosmosis?

- (1) membrane, proton pump, proton gradient, ATP synthase
- (2) membrane, proton pump, proton gradient, NADP synthase
- (3) proton pump, electron gradient, ATP synthase
- (4) proton pump, electron gradient, NADP synthase

- 144** Melonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

- (1) Succinic dehydrogenase
- (2) Amylase
- (3) Lipase
- (4) Dinitrogenase

- 145** Identify the **correct** statements :

- A. Lenticels are the lens-shaped openings permitting the exchange of gases.
- B. Bark formed early in the season is called hard bark.
- C. Bark is a technical term that refers to all tissues exterior to vascular cambium.
- D. Bark refers to periderm and secondary phloem.
- E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below :

- (1) B, C and E only
- (2) A and D only
- (3) A, B and D only
- (4) B and C only

146 Match List I with List II :

List I	List II
A. M Phase	I. Proteins are synthesized
B. G ₂ Phase	II. Inactive phase
C. Quiescent stage	III. Interval between mitosis and initiation of DNA replication
D. G ₁ Phase	IV. Equational division

Choose the correct answer from the options given below :

- (1) A-III, B-II, C-IV, D-I
- (2) A-IV, B-II, C-I, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-II, B-IV, C-I, D-III

147 Match List I with List II :

List I	List II
(Interaction)	(Species A and B)
A. Mutualism	I. + (A), O(B)
B. Commensalism	II. – (A), O(B)
C. Amensalism	III. + (A), – (B)
D. Parasitism	IV. + (A), + (B)

Choose the **correct** answer from the options given below :

- (1) A-IV, B-II, C-I, D-III
- (2) A-IV, B-I, C-II, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-I, C-IV, D-II

148 Given below are two statements :

Statement I : Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II : In general, carnivores are more adversely affected by competition than herbivores.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** is incorrect but **Statement II** is true.

149 Match List I with List II :

List I	List II
A. Iron	I. Synthesis of auxin
B. Zinc	II. Component of nitrate reductase
C. Boron	III. Activator of catalase
D. Molybdenum	IV. Cell elongation and differentiation

Choose the correct answer from the options given below :

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-IV, C-I, D-III

150 Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.

- A. Insertion of recombinant DNA into the host cell.
- B. Cutting of DNA at specific location by restriction enzyme.
- C. Isolation of desired DNA fragment.
- D. Amplification of gene of interest using PCR.

Choose the correct answer from the options given below :

- (1) B, C, D, A
- (2) C, A, B, D
- (3) C, B, D, A
- (4) B, D, A, C

151 Match List I with List II.

List I	List II
A. Vasectomy	I. Oral method
B. Coitus interruptus	II. Barrier method
C. Cervical caps	III. Surgical method
D. Saheli	IV. Natural method

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-IV, B-II, C-I, D-III

152 Given below are two statements:

Statement I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Statement II: The cavity of the cervix is called cervical canal which along with vagina forms birth canal.

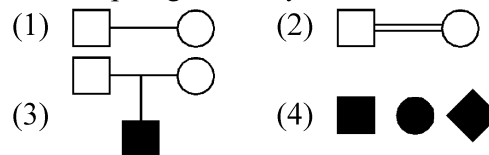
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** incorrect but **Statement II** is true.

153 Which of the following statements is correct?

- (1) Eutrophication refers to increase in domestic sewage and waste water in lakes.
- (2) Biomagnification refers to increase in concentration of the toxicant at successive trophic levels.
- (3) Presence of large amount of nutrients in water restricts 'Algal Bloom'
- (4) Algal Bloom decreases fish mortality

154 Which one of the following symbols represents mating between relatives in human pedigree analysis?



155 Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly?

- (1) Genital herpes
- (2) Gonorrhoea
- (3) Hepatitis-B
- (4) HIV Infection

156 Match List I with List II.

List I	List II
A. Heroin	I. Effect on cardiovascular system
B. Marijuana	II. Slow down body function
C. Cocaine	III. Painkiller
D. Morphine	IV. Interfere with transport of dopamine

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-IV, B-III, C-II, D-I
- (4) A-III, B-IV, C-I, D-II

157 Match List I with List II.

List I (Type of Joint)	List II (Found between)
A. Cartilaginous Joint	I. Between flat skull bones
B. Ball and Socket Joint	II. Between adjacent vertebrae in vertebral column
C. Fibrous Joint	III. Between carpal and metacarpal of thumb
D. Saddle Joint	IV. Between Humerus and Pectoral girdle

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-I, B-IV, C-III, D-II
- (4) A-II, B-IV, C-III, D-I

- 158** Given below are two statements:
Statement I: A protein is imagined as a line, the left end represented by first amino acid (C-terminal) and the right end represented by last amino acid (N-terminal)
Statement II: Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type.)
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **Statement I** and **Statement II** are true.
 - (2) Both **Statement I** and **Statement II** are false.
 - (3) **Statement I** is true but **Statement II** is false.
 - (4) **Statement I** is false but **Statement II** is true.
- 159** Which of the following are NOT considered as the part of endomembrane system?
 A. Mitochondria B. Endoplasmic Reticulum
 C. Chloroplasts D. Golgi complex
 E. Peroxisomes
 Choose the **most appropriate** answer from the options given below:
- (1) B and D only
 - (2) A, C and E only
 - (3) A and D only
 - (4) A, D and E only
- 160** Given below are two statements:
Statement I: RNA mutates at a faster rate.
Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **Statement I** and **Statement II** are true.
 - (2) Both **Statement I** and **Statement II** are false.
 - (3) **Statement I** is true but **Statement II** is false.
 - (4) **Statement I** false but **Statement II** is true.

- 161** Match **List I** with **List II**.
- | List I | List II |
|---------------|--------------------|
| A. CCK | I. Kidney |
| B. GIP | II. Heart |
| C. ANF | III. Gastric gland |
| D. ADH | IV. Pancreas |
- Choose the **correct** answer from the options given below:
- (1) A-IV, B-III, C-II, D-I
 - (2) A-III, B-II, C-IV, D-I
 - (3) A-II, B-IV, C-I, D-III
 - (4) A-IV, B-II, C-III, D-I
- 162** Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Endometrium is necessary for implantation of blastocyst.
Reason R: In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
 - (2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
 - (3) **A** is true but **R** is false.
 - (4) **A** is false but **R** is true.
- 163** Match **List I** with **List II**.
- | List I | List II |
|---------------|----------------------------------|
| A. Ringworm | I. <i>Haemophilus influenzae</i> |
| B. Filariasis | II. <i>Trichophyton</i> |
| C. Malaria | III. <i>Wuchereria bancrofti</i> |
| D. Pneumonia | IV. <i>Plasmodium vivax</i> |
- Choose the **correct** answer from the options given below:
- (1) A-II, B-III, C-IV, D-I
 - (2) A-II, B-III, C-I, D-IV
 - (3) A-III, B-II, C-I, D-IV
 - (4) A-III, B-II, C-IV, D-I

164 Given below are two statements:

Statement I: Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement II: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is true but **Statement II** is false.
- (4) **Statement I** is false but **Statement II** is true.

165 Match **List I** with **List II**.

List I	List II
A. <i>Taenia</i>	I. Nephridia
B. <i>Paramoecium</i>	II. Contractile vacuole
C. <i>Periplaneta</i>	III. Flame cells
D. <i>Pheretima</i>	IV. Urecose gland

Choose the **correct** answer from the options give below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-I, C-IV, D-III

166 Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment?

- (1) Recombinant DNA Technology
- (2) Serum and Urine analysis
- (3) Polymerase Chain Reaction (PCR) technique
- (4) Enzyme Linked Immuno-Sorbent Assay (ELISA) technique

167 Match **List I** with **List II**.

List I (Interacting species)	List II (Name of Interaction)
A. A Leopard and a Lion in a forest/ grassland	I. Competition
B. A Cuckoo laying egg in a Crow's nest	II. Brood parasitism
C. Fungi and root of a higher plant in Mycorrhizae	III. Mutualism
D. A cattle egret and a Cattle in a field	IV. Commensalism

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-I, D-IV

168 Given below are two statements:

Statement I: Ligaments are dense irregular tissue.

Statement II: Cartilage is dense regular tissue.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is true but **Statement II** is false.
- (4) **Statement I** is false but **Statement II** is true.

169 Given below are two statements:

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** incorrect but **Statement II** is true.

170 Match **List I** with **List II** with respect to human eye.

List I	List II
A. Fovea	I. Visible coloured portion of eye that regulates diameter of pupil.
B. Iris	II. External layer of eye formed of dense connective tissue.
C. Blind spot	III. Point of greatest visual acuity or resolution.
D. Sclera	IV. Point where optic nerve leaves the eyeball and photoreceptor cells are absent.

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-IV, C-III, D-II
- (4) A-II, B-I, C-III, D-IV

171 Select the correct group/set of Australian Marsupials exhibiting adaptive radiation.

- (1) Tasmanian wolf, Bobcat, Marsupial mole
- (2) Numbat, Spotted cuscus, Flying phalanger
- (3) Mole, Flying squirrel, Tasmanian tiger cat
- (4) Lemur, Anteater, Wolf

172 Which of the following statements are correct regarding female reproductive cycle?

- A. In non-primate mammals cyclical changes during reproduction are called oestrus cycle.
- B. First menstrual cycle begins at puberty and is called menopause.
- C. Lack of menstruation may be indicative of pregnancy.
- D. Cyclic menstruation extends between menarche and menopause.

Choose the **most appropriate** answer from the options given below:

- (1) A and D only
- (2) A and B only
- (3) A, B and C only
- (4) A, C and D only

173 Vital capacity of lung is _____.

- (1) IRV + ERV
- (2) IRV + ERV + TV + RV
- (3) IRV + ERV + TV – RV
- (4) IRV + ERV + TV

174 Match **List I** with **List II**.

List I	List II
A. P - wave	I. Beginning of systole
B. Q - wave	II. Repolarisation of ventricles
C. QRS complex	III. Depolarisation of atria
D. T - wave	IV. Depolarisation of ventricles

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-I, B-II, C-III, D-IV

- 175** Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Amniocentesis for sex determination is one of the strategies of Reproductive and Child Health Care Programme.

Reason R: Ban on amniocentesis checks increasing menace of female foeticide.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true and **R** is NOT the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

- 176** Once the undigested and unabsorbed substances enter the caecum, their backflow is prevented by-

- (1) Sphincter of Oddi
- (2) Ileo - caecal valve
- (3) Gastro - oesophageal sphincter
- (4) Pyloric sphincter

- 177** Match **List I** with **List II**.

List I	List II
A. Gene 'a'	I. β -galactosidase
B. Gene 'y'	II. Transacetylase
C. Gene 'i'	III. Permease
D. Gene 'z'	IV. Repressor protein

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-I, C-IV, D-II

- 178** Match **List I** with **List II**.

List I (Cells)	List II (Secretion)
A. Peptic cells	I. Mucus
B. Goblet cells	II. Bile juice
C. Oxyntic cells	III. Proenzyme pepsinogen
D. Hepatic cells	IV. HCl and intrinsic factor for absorption of vitamin B ₁₂

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-II, B-I, C-III, D-IV
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-IV, C-I, D-III

- 179** Which of the following functions is carried out by cytoskeleton in a cell?

- (1) Nuclear division
- (2) Protein synthesis
- (3) Motility
- (4) Transportation

- 180** Given below are statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in cortex and medulla.

Reason R: Juxta medullary nephrons have short loop of Henle whereas, cortical nephrons have longer loop of Henle.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

181 Given below are two statements:

Statement I: Electrostatic precipitator is most widely used in thermal power plant.

Statement II: Electrostatic precipitator in thermal power plant removes ionising radiations

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are correct.
- (2) Both **Statement I** and **Statement II** are incorrect.
- (3) **Statement I** is correct but **Statement II** is incorrect.
- (4) **Statement I** incorrect but **Statement II** is correct.

182 Broad palm with single palm crease is visible in a person suffering from-

- (1) Down's syndrome
- (2) Turner's syndrome
- (3) Klinefelter's syndrome
- (4) Thalassemia

183 Radial symmetry is NOT found in adults of phylum _____.

- (1) Ctenophora (2) Hemichordata
- (3) Coelenterata (4) Echinodermata

184 In which blood corpuscles, the HIV undergoes replication and produces progeny viruses?

- (1) T_H cells (2) B-lymphocytes
- (3) Basophils (4) Eosinophils

185 Which of the following is not a cloning vector?

- (1) BAC (2) YAC
- (3) pBR322 (4) Probe

186 Match **List I** with **List II**.

List I	List II
A. Logistic growth	I. Unlimited resource availability condition
B. Exponential growth	II. Limited resource availability condition
C. Expanding age pyramid	III. The percent individuals of pre-reproductive age is largest followed by reproductive and post reproductive age groups
D. Stable age pyramid	IV. The percent individuals of pre-reproductives and reproductive age group are same

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-II, B-IV, C-I, D-III
- (4) A-II, B-IV, C-III, D-I

187 Select the correct statements with reference to chordates.

- A. Presence of a mid-dorsal, solid and double nerve cord.
- B. Presence of closed circulatory system.
- C. Presence of paired pharyngeal gillslits.
- D. Presence of dorsal heart
- E. Triploblastic pseudocoelomate animals.

Choose the **correct** answer from the options given below:

- (1) A, C and D only
- (2) B and C only
- (3) B, D and E only
- (4) C, D and E only

- 188** The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc. are :
- (1) Limbic system & hypothalamus
 - (2) Corpora quadrigemina & hippocampus
 - (3) Brain stem & epithalamus
 - (4) Corpus callosum and thalamus

- 189** The unique mammalian characteristics are:
- (1) hairs, tympanic membrane and mammary glands
 - (2) hairs, pinna and mammary glands
 - (3) hairs, pinna and indirect development
 - (4) pinna, monocondylic skull and mammary glands

- 190** Which of the following are NOT under the control of thyroid hormone?
- A. Maintenance of water and electrolyte balance
 - B. Regulation of basal metabolic rate
 - C. Normal rhythm of sleep-wake cycle
 - D. Development of immune system
 - E. Support the process of R.B.Cs formation
- Choose the **correct** answer from the options given below:
- (1) A and D only (2) B and C only
 - (3) C and D only (4) D and E only

- 191** Select the correct statements.
- A. Tetrad formation is seen during Leptotene.
 - B. During Anaphase, the centromeres split and chromatids separate.
 - C. Terminalization takes place during Pachytene.
 - D. Nucleolus, Golgi complex and ER are reformed during Telophase.
 - E. Crossing over takes place between sister chromatids of homologous chromosome.
- Choose the **correct** answer from the options given below:
- (1) A and C only
 - (2) B and D only
 - (3) A, C and E only
 - (4) B and E only

- 192** Match **List I** with **List II**.

List I	List II
A. Mast cells	I. Ciliated epithelium
B. Inner surface of bronchiole	II. Areolar connective tissue
C. Blood	III. Cuboidal epithelium
D. Tubular parts of nephron	IV. specialised connective tissue

Choose the **correct** answer from the options give below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-II, B-III, C-I, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-IV, C-II, D-I

- 193** Which of the following is characteristic feature of cockroach regarding sexual dimorphism ?
- (1) Dark brown body colour and anal cerci
 - (2) Presence of anal styles
 - (3) Presence of sclerites
 - (4) Presence of anal cerci

- 194** Which one of the following is the sequence on corresponding coding strand, if the sequence on mRNA formed is as follows 5' AUCGAUCGAUCGAUCG AUCG AUCG 3'?
- (1) 5' UAGCUAGCUAGCUAGCUA GCUAGC UAGC 3'
 - (2) 3' UAGCUAGCUAGCUAGCUA GCUAGCUAGC 5'
 - (3) 5' ATCGATCGATCGATCGATCG ATCGATCG 3'
 - (4) 3' ATCGATCGATCGATCGATCG ATCGATCG 5'

- 195** In cockroach, excretion is brought about by-
- A. Phallic gland B. Ureose gland
 - C. Nephrocytes D. Fat body
 - E. Collateral glands
- Choose the **correct** answer from the options given below:
- (1) A and E only
 - (2) A, B and E only
 - (3) B, C and D only
 - (4) B and D only

196 Given below are two statements:
Statement I: During G_0 phase of cell cycle, the cell is metabolically inactive.
Statement II: The centrosome undergoes duplication during S phase of interphase.
 In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are correct.
- (2) Both **Statement I** and **Statement II** are incorrect.
- (3) **Statement I** is correct but **Statement II** is incorrect.
- (4) **Statement I** is incorrect but **Statement II** is correct.

197 Which one of the following is NOT an advantage of inbreeding?

- (1) It decreases homozygosity.
- (2) It exposes harmful recessive genes that are eliminated by selection.
- (3) Elimination of less desirable genes and accumulation of superior genes takes place due to it.
- (4) It decreases the productivity of inbred population, after continuous inbreeding.

198 Which of the following statements are correct?

- A. An excessive loss of body fluid from the body switches off osmoreceptors.
- B. ADH facilitates water reabsorption to prevent diuresis.
- C. ANF causes vasodilation.
- D. ADH causes increase in blood pressure.
- E. ADH is responsible for decrease in GFR.

Choose the **correct** answer from the options given below:

- (1) A and B only
- (2) B, C and D only
- (3) A, B and E only
- (4) C, D and E only

199 Which of the following statements are correct regarding skeletal muscle?

- A. Muscle bundles are held together by collagenous connective tissue layer called fascicle.
- B. Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.
- C. Striated appearance of skeletal muscle fibre is due to distribution pattern of actin and myosin proteins.
- D. M line is considered as functional unit of contraction called sarcomere.

Choose the **most appropriate** answer from the options given below:

- (1) A, B and C only
- (2) B and C only
- (3) A, C and D only
- (4) C and D only

200 Which of the following statements are correct ?

- A. Basophils are most abundant cells of the total WBCs
- B. Basophils secrete histamine, serotonin and heparin
- C. Basophils are involved in inflammatory response
- D. Basophils have kidney shaped nucleus
- E. Basophils are agranulocytes

Choose the **correct** answer from the options given below:

- (1) D and E only
- (2) C and E only
- (3) B and C only
- (4) A and B only

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

Test Booklet Code

M5

JAHAGA

No. :

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on OFFICE Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos – 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is M5. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

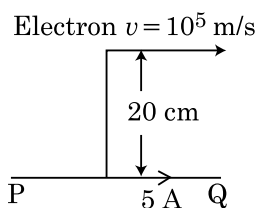
Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

Section - A (Physics)

1. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.

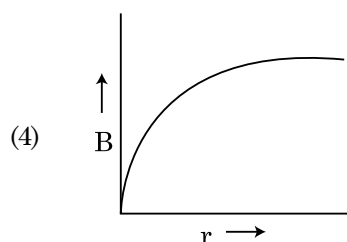
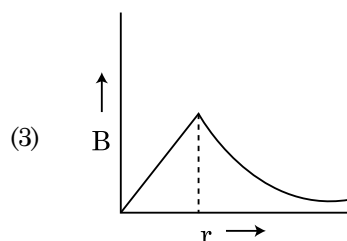
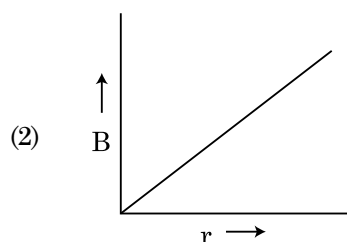
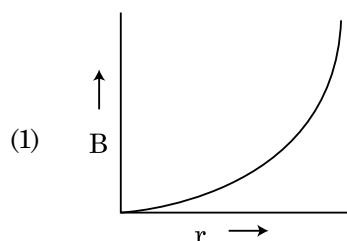


- (1) 4×10^{-20} N
 (2) $8\pi \times 10^{-20}$ N
 (3) $4\pi \times 10^{-20}$ N
 (4) 8×10^{-20} N
2. A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is :
- (1) n
 (2) 2n
 (3) 3n
 (4) 4n
3. A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence ${}^A_ZX \rightarrow {}^{Z-1}_B \rightarrow {}^{Z-3}_C \rightarrow {}^{Z-2}_D$, where Z is the atomic number of element X. The possible decay particles in the sequence are :
- (1) α , β^- , β^+
 (2) α , β^+ , β^-
 (3) β^+ , α , β^-
 (4) β^- , α , β^+
4. The escape velocity from the Earth's surface is v . The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is :
- (1) v
 (2) $2v$
 (3) $3v$
 (4) $4v$

5. The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be :
- (1) $1/2$
 (2) $\frac{1}{2\sqrt{2}}$
 (3) $\frac{2}{3}$
 (4) $\frac{2}{3\sqrt{2}}$
6. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be :
- (1) 25
 (2) 15
 (3) 50
 (4) 30
7. A capacitor of capacitance 'C', is connected across an ac source of voltage V, given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would then be given by :
- (1) $I_d = V_0 \omega C \cos \omega t$
 (2) $I_d = \frac{V_0}{\omega C} \cos \omega t$
 (3) $I_d = \frac{V_0}{\omega C} \sin \omega t$
 (4) $I_d = V_0 \omega C \sin \omega t$
8. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$ is :
- (1) $\frac{2n-1}{2n}$
 (2) $\frac{2n-1}{2n+1}$
 (3) $\frac{2n+1}{2n-1}$
 (4) $\frac{2n}{2n-1}$

9. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively :
- (1) $\frac{S}{4}, \frac{3gS}{2}$
 - (2) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
 - (3) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$
 - (4) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$
10. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs ?
- (1) 60 cm
 - (2) 21.6 cm
 - (3) 64 cm
 - (4) 62 cm
11. For a plane electromagnetic wave propagating in x -direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively ?
- (1) $\hat{j} + \hat{k}, \hat{j} + \hat{k}$
 - (2) $-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$
 - (3) $\hat{j} + \hat{k}, -\hat{j} - \hat{k}$
 - (4) $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$
12. Polar molecules are the molecules :
- (1) having zero dipole moment.
 - (2) acquire a dipole moment only in the presence of electric field due to displacement of charges.
 - (3) acquire a dipole moment only when magnetic field is absent.
 - (4) having a permanent electric dipole moment.
13. The velocity of a small ball of mass M and density d , when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be :
- (1) $\frac{Mg}{2}$
 - (2) Mg
 - (3) $\frac{3}{2}Mg$
 - (4) $2Mg$
14. Match **Column - I** and **Column - II** and choose the **correct** match from the given choices.
- | Column - I | Column - II |
|---|------------------------------|
| (A) Root mean square speed of gas molecules | (P) $\frac{1}{3}nm\bar{v}^2$ |
| (B) Pressure exerted by ideal gas | (Q) $\sqrt{\frac{3RT}{M}}$ |
| (C) Average kinetic energy of a molecule | (R) $\frac{5}{2}RT$ |
| (D) Total internal energy of 1 mole of a diatomic gas | (S) $\frac{3}{2}k_B T$ |
- (1) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)
 - (2) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)
 - (3) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)
 - (4) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)
15. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine ? ($g = 10 \text{ m/s}^2$)
- (1) 10.2 kW
 - (2) 8.1 kW
 - (3) 12.3 kW
 - (4) 7.0 kW
16. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since :
- (1) a large aperture contributes to the quality and visibility of the images.
 - (2) a large area of the objective ensures better light gathering power.
 - (3) a large aperture provides a better resolution.
 - (4) all of the above.

17. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.
- current in n-type = current in p-type.
 - current in p-type > current in n-type.
 - current in n-type > current in p-type.
 - No current will flow in p-type, current will only flow in n-type.
18. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is :
- 0.9 MeV
 - 9.4 MeV
 - 804 MeV
 - 216 MeV
19. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field B(r) due to the cable with the distance 'r' from the axis of the cable is represented by :



20. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :

- $\frac{R_1}{R_2}$
- $\frac{R_2}{R_1}$
- $\sqrt{\frac{R_1}{R_2}}$
- $\frac{R_1^2}{R_2^2}$

21. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of :

- $[M^2] [L^{-1}] [T^0]$
- $[M] [L^{-1}] [T^{-1}]$
- $[M] [L^0] [T^0]$
- $[M^2] [L^{-2}] [T^{-1}]$

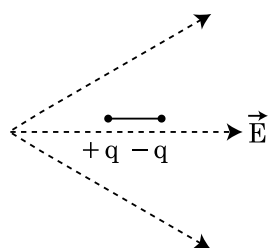
22. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is :

- 0.0628 s
- 6.28 s
- 3.14 s
- 0.628 s

23. **Column - I** gives certain physical terms associated with flow of current through a metallic conductor. **Column - II** gives some mathematical relations involving electrical quantities. Match **Column - I** and **Column - II** with appropriate relations.

Column - I	Column - II
(A) Drift Velocity	(P) $\frac{m}{ne^2\rho}$
(B) Electrical Resistivity	(Q) nev_d
(C) Relaxation Period	(R) $\frac{eE}{m}\tau$
(D) Current Density	(S) $\frac{E}{J}$
(1) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)	
(2) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)	
(3) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)	
(4) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)	

24. A dipole is placed in an electric field as shown. In which direction will it move ?



- (1) towards the left as its potential energy will increase.
- (2) towards the right as its potential energy will decrease.
- (3) towards the left as its potential energy will decrease.
- (4) towards the right as its potential energy will increase.

25. Consider the following **statements (A)** and **(B)** and identify the **correct** answer.

(A) A zener diode is connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

- (1) **(A)** and **(B)** both are correct.
- (2) **(A)** and **(B)** both are incorrect.
- (3) **(A)** is correct and **(B)** is incorrect.
- (4) **(A)** is incorrect but **(B)** is correct.

26. A screw gauge gives the following readings when used to measure the diameter of a wire

Main scale reading : 0 mm

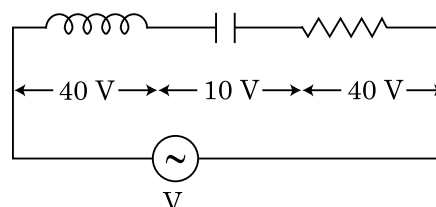
Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is :

- (1) 0.52 cm
- (2) 0.026 cm
- (3) 0.26 cm
- (4) 0.052 cm

27. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.

Potential difference across L , C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is :



- (1) $4\sqrt{2} \Omega$
- (2) $5/\sqrt{2} \Omega$
- (3) 4Ω
- (4) 5Ω

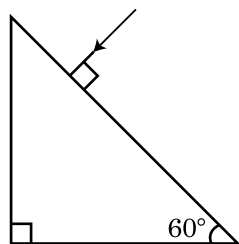
28. A parallel plate capacitor has a uniform electric field ' $\vec{E}$ ' in the space between the plates. If the distance between the plates is ' d ' and the area of each plate is ' A ', the energy stored in the capacitor is : (ϵ_0 = permittivity of free space)

- (1) $\frac{1}{2} \epsilon_0 E^2$
- (2) $\epsilon_0 E A d$
- (3) $\frac{1}{2} \epsilon_0 E^2 A d$
- (4) $\frac{E^2 A d}{\epsilon_0}$

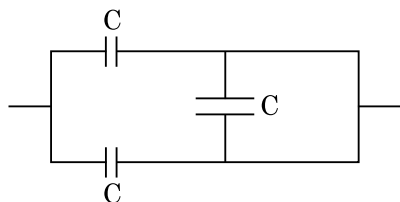
29. An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of negligible work function. If ' m ' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d , then :

- (1) $\lambda = \left(\frac{2m}{hc} \right) \lambda_d^2$
- (2) $\lambda_d = \left(\frac{2mc}{h} \right) \lambda^2$
- (3) $\lambda = \left(\frac{2mc}{h} \right) \lambda_d^2$
- (4) $\lambda = \left(\frac{2h}{mc} \right) \lambda_d^2$

30. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



- (1) 60°
 (2) 30°
 (3) 45°
 (4) 90°
31. The equivalent capacitance of the combination shown in the figure is :



- (1) $3C$
 (2) $2C$
 (3) $C/2$
 (4) $3C/2$
32. If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.
- (1) [F] [A] [T]
 (2) [F] [A] [T²]
 (3) [F] [A] [T⁻¹]
 (4) [F] [A⁻¹] [T]
33. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C . The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is :

- (1) $\frac{13}{10}t$
 (2) $\frac{13}{5}t$
 (3) $\frac{10}{13}t$
 (4) $\frac{5}{13}t$

34. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is $0.25\ \Omega$. What will be the effective resistance if they are connected in series ?

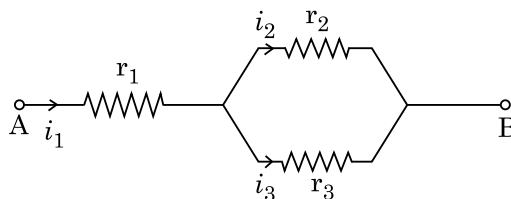
- (1) $0.25\ \Omega$
 (2) $0.5\ \Omega$
 (3) $1\ \Omega$
 (4) $4\ \Omega$

35. The number of photons per second on an average emitted by the source of monochromatic light of wavelength $600\ \text{nm}$, when it delivers the power of $3.3 \times 10^{-3}\ \text{watt}$ will be : ($h = 6.6 \times 10^{-34}\ \text{Js}$)

- (1) 10^{18}
 (2) 10^{17}
 (3) 10^{16}
 (4) 10^{15}

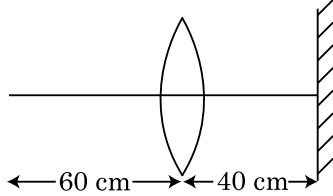
Section - B (Physics)

36. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is :

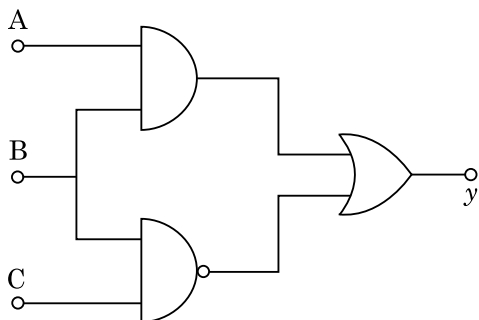
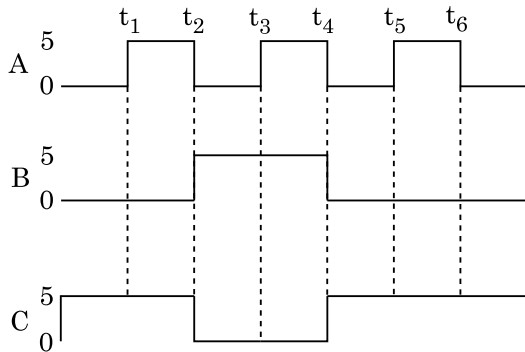


- (1) $\frac{r_1}{r_2 + r_3}$
 (2) $\frac{r_2}{r_2 + r_3}$
 (3) $\frac{r_1}{r_1 + r_2}$
 (4) $\frac{r_2}{r_1 + r_3}$

37. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of :



- (1) 20 cm from the lens, it would be a real image.
 (2) 30 cm from the lens, it would be a real image.
 (3) 30 cm from the plane mirror, it would be a virtual image.
 (4) 20 cm from the plane mirror, it would be a virtual image.
38. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y?



- (1) 0 V
 (2) 5 V
 (3) 5 V
 (4) 5 V
 (4) 0 V

39. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit ?

- (1) 0.2 A
 (2) 0.4 A
 (3) 2 A
 (4) 4 A

40. A uniform conducting wire of length $12a$ and resistance 'R' is wound up as a current carrying coil in the shape of,

- (i) an equilateral triangle of side 'a'.
 (ii) a square of side 'a'.

The magnetic dipole moments of the coil in each case respectively are :

- (1) $\sqrt{3} Ia^2$ and $3 Ia^2$
 (2) $3 Ia^2$ and Ia^2
 (3) $3 Ia^2$ and $4 Ia^2$
 (4) $4 Ia^2$ and $3 Ia^2$

41. In the product

$$\vec{F} = q \left(\vec{v} \times \vec{B} \right)$$

$$= q \vec{v} \times \left(B \hat{i} + B \hat{j} + B_0 \hat{k} \right)$$

For $q = 1$ and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and

$$\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$$

What will be the complete expression for $\vec{B}$?

- (1) $-8\hat{i} - 8\hat{j} - 6\hat{k}$
 (2) $-6\hat{i} - 6\hat{j} - 8\hat{k}$
 (3) $8\hat{i} + 8\hat{j} - 6\hat{k}$
 (4) $6\hat{i} + 6\hat{j} - 8\hat{k}$

42. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ , is then given by :

$$(1) \quad \theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{1/2}$$

$$(2) \quad \theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$$

$$(3) \quad \theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$$

$$(4) \quad \theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{1/2}$$

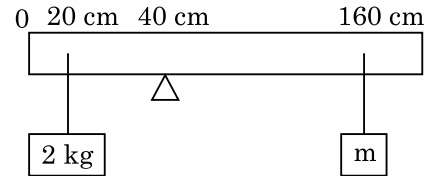
43. A series LCR circuit containing 5.0 H inductor, $80 \mu\text{F}$ capacitor and 40Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be :

- (1) 25 rad/s and 75 rad/s
- (2) 50 rad/s and 25 rad/s
- (3) 46 rad/s and 54 rad/s
- (4) 42 rad/s and 58 rad/s

44. From a circular ring of mass ' M ' and radius ' R ' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is ' K ' times ' MR^2 '. Then the value of ' K ' is :

- (1) $\frac{3}{4}$
- (2) $\frac{7}{8}$
- (3) $\frac{1}{4}$
- (4) $\frac{1}{8}$

45. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass ' m ' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of ' m ' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$)



- (1) $\frac{1}{2} \text{ kg}$
 - (2) $\frac{1}{3} \text{ kg}$
 - (3) $\frac{1}{6} \text{ kg}$
 - (4) $\frac{1}{12} \text{ kg}$
46. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.
- (1) 660 V
 - (2) 1320 V
 - (3) 1520 V
 - (4) 1980 V
47. A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$? (Take $g = 10 \text{ m/s}^2$)
- (1) 20 m/s , 5 m/s^2
 - (2) 20 m/s , 0
 - (3) $20\sqrt{2} \text{ m/s}$, 0
 - (4) $20\sqrt{2} \text{ m/s}$, 10 m/s^2
48. A particle of mass ' m ' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth. (V_e = escape velocity)
The maximum height above the surface reached by the particle is :
- (1) $R \left(\frac{k}{1-k} \right)^2$
 - (2) $R \left(\frac{k}{1+k} \right)^2$
 - (3) $\frac{R^2 k}{1+k}$
 - (4) $\frac{Rk^2}{1-k^2}$

49. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly :

- (1) 0 kg m/s
- (2) 4.2 kg m/s
- (3) 2.1 kg m/s
- (4) 1.4 kg m/s

50. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to :

- (1) $\frac{R_1}{R_2}$
- (2) $\frac{R_2}{R_1}$
- (3) $\frac{R_1^2}{R_2}$
- (4) $\frac{R_2^2}{R_1}$

Section - A (Chemistry)

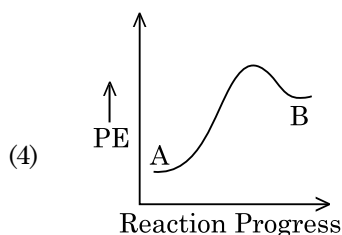
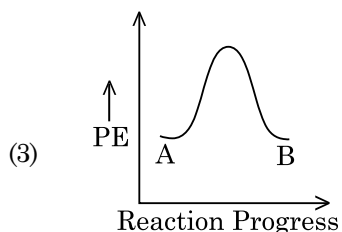
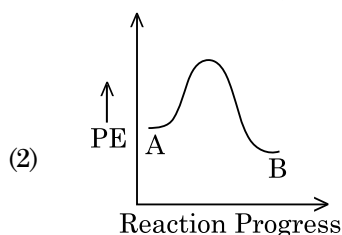
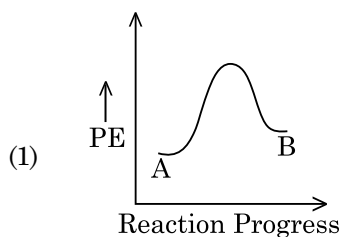
51. Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are :

- (1) 8, 4
- (2) 6, 12
- (3) 2, 1
- (4) 12, 6

52. Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of :

- (1) belonging to same group
- (2) diagonal relationship
- (3) lanthanoid contraction
- (4) having similar chemical properties

53. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.



54. Tritium, a radioactive isotope of hydrogen, emits which of the following particles ?

- (1) Beta (β^-)
- (2) Alpha (α)
- (3) Gamma (γ)
- (4) Neutron (n)

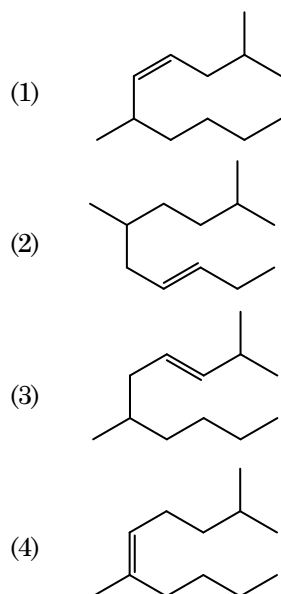
55. The RBC deficiency is deficiency disease of :

- (1) Vitamin B_{12}
- (2) Vitamin B_6
- (3) Vitamin B_1
- (4) Vitamin B_2

56. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.

- (1) $201.28 \text{ S cm}^2 \text{ mol}^{-1}$
- (2) $390.71 \text{ S cm}^2 \text{ mol}^{-1}$
- (3) $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
- (4) $540.48 \text{ S cm}^2 \text{ mol}^{-1}$

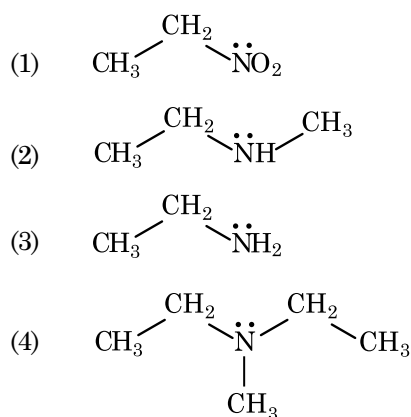
57. The correct structure of 2,6-Dimethyl-dec-4-ene is :



58. The maximum temperature that can be achieved in blast furnace is :

- (1) upto 1200 K
(2) upto 2200 K
(3) upto 1900 K
(4) upto 5000 K

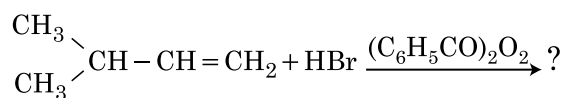
59. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.



60. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :

- (1) $P_2 > P_1 > P_3$
(2) $P_1 > P_2 > P_3$
(3) $P_2 > P_3 > P_1$
(4) $P_3 > P_1 > P_2$

61. The major product of the following chemical reaction is :



- (1)
- (2)
- (3)
- (4)

62. Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true.
(2) Both **Statement I** and **Statement II** are false.
(3) **Statement I** is correct but **Statement II** is false.
(4) **Statement I** is incorrect but **Statement II** is true.

63. The correct sequence of bond enthalpy of 'C-X' bond is :

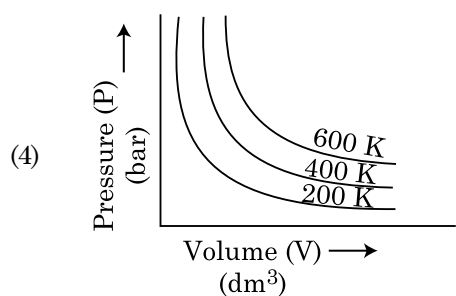
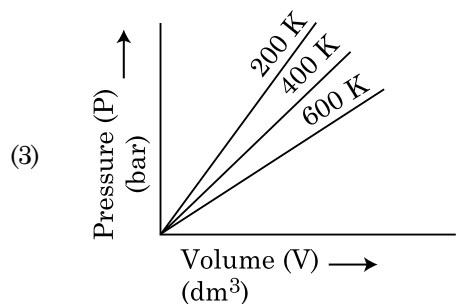
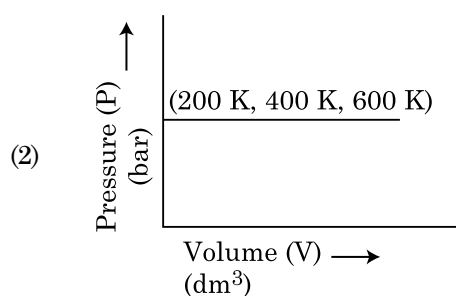
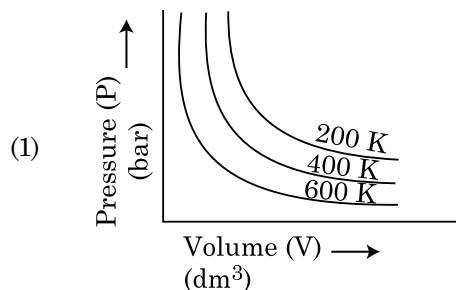
- (1) $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} < \text{CH}_3 - \text{Br} < \text{CH}_3 - \text{I}$
(2) $\text{CH}_3 - \text{F} > \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
(3) $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
(4) $\text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{F} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$

64. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are :

- (1) sp^3 and 4
(2) sp^3 and 6
(3) sp^2 and 6
(4) sp^2 and 8

65. Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas ?
- $C_P + C_V = R$
 - $C_P - C_V = R$
 - $C_P = RC_V$
 - $C_V = RC_P$
66. Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :
- Calcium chloride
 - Strontium chloride
 - Magnesium chloride
 - Beryllium chloride
67. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]
- CH
 - CH₂
 - CH₃
 - CH₄
68. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on ?
- Saytzeff's Rule
 - Hund's Rule
 - Hofmann Rule
 - Huckel's Rule
69. What is the IUPAC name of the organic compound formed in the following chemical reaction ?
- Acetone $\xrightarrow[\text{(ii) H}_2\text{O, H}^+]{\text{(i) C}_2\text{H}_5\text{MgBr, dry Ether}}$ Product
- 2-methyl propan-2-ol
 - pentan-2-ol
 - pentan-3-ol
 - 2-methyl butan-2-ol
70. Noble gases are named because of their inertness towards reactivity. Identify an **incorrect** statement about them.
- Noble gases are sparingly soluble in water.
 - Noble gases have very high melting and boiling points.
 - Noble gases have weak dispersion forces.
 - Noble gases have large positive values of electron gain enthalpy.
71. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is :
- 8.50
 - 5.50
 - 7.75
 - 6.25
72. The right option for the statement "Tyndall effect is exhibited by", is :
- NaCl solution
 - Glucose solution
 - Starch solution
 - Urea solution
73. **Statement I :**
- Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.
- Statement II :**
- As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.
- In the light of the above statements, choose the **correct** answer from the options given below.
- Both **Statement I** and **Statement II** are true.
 - Both **Statement I** and **Statement II** are false.
 - Statement I** is correct but **Statement II** is false.
 - Statement I** is incorrect but **Statement II** is true.
74. Ethylene diaminetetraacetate (EDTA) ion is :
- Hexadentate ligand with four "O" and two "N" donor atoms
 - Unidentate ligand
 - Bidentate ligand with two "N" donor atoms
 - Tridentate ligand with three "N" donor atoms

75. Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures :



76. The structures of beryllium chloride in solid state and vapour phase, are :

- (1) Chain and dimer, respectively
- (2) Linear in both
- (3) Dimer and Linear, respectively
- (4) Chain in both

77. Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature ?

- (1) Electrolysis
- (2) Chromatography
- (3) Distillation
- (4) Zone refining

78. The compound which shows metamerism is :

- (1) C_5H_{12}
- (2) C_3H_8O
- (3) C_3H_6O
- (4) $C_4H_{10}O$

79. The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is :

- (1) 7
- (2) 5
- (3) 2
- (4) 3

80. Which one of the following polymers is prepared by addition polymerisation ?

- (1) Teflon
- (2) Nylon-66
- (3) Novolac
- (4) Dacron

81. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light, $c = 3.0 \times 10^8 \text{ ms}^{-1}$]

- (1) 219.3 m
- (2) 219.2 m
- (3) 2192 m
- (4) 21.92 cm

82. Which of the following reactions is the metal displacement reaction? Choose the right option.

- (1) $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$
- (2) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
- (3) $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2\uparrow$
- (4) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2\uparrow$

83. The **incorrect** statement among the following is :

- (1) Actinoid contraction is greater for element to element than Lanthanoid contraction.
- (2) Most of the trivalent Lanthanoid ions are colorless in the solid state.
- (3) Lanthanoids are good conductors of heat and electricity.
- (4) Actinoids are highly reactive metals, especially when finely divided.

84. Dihedral angle of least stable conformer of ethane is :

- (1) 120°
- (2) 180°
- (3) 60°
- (4) 0°

85. Match List - I with List - II.

List - I	List - II
(a) PCl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

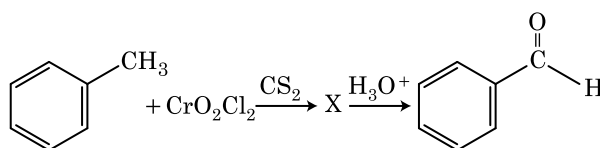
Section - B (Chemistry)

86. $\text{CH}_3\text{CH}_2\text{COO}^- \text{Na}^+ \xrightarrow[\text{Heat}]{\text{NaOH, + ?}} \text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$.

Consider the above reaction and identify the missing reagent/chemical.

- (1) B_2H_6
- (2) Red Phosphorus
- (3) CaO
- (4) DIBAL-H

87. The intermediate compound 'X' in the following chemical reaction is :



- (1) $\text{C}_6\text{H}_5\text{CH}(\text{OCrOHCl}_2)_2$
- (2) $\text{C}_6\text{H}_5\text{CH}(\text{OCOCH}_3)_2$
- (3) $\text{C}_6\text{H}_5\text{CH}(\text{Cl})_2$
- (4) $\text{C}_6\text{H}_5\text{CH}(\text{Cl})\text{H}$

88. Match List - I with List - II.

List - I	List - II
(a) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$	(i) Acid rain
(b) $\text{HOCl}(\text{g}) \xrightarrow{h\nu} \cdot\text{OH} + \cdot\text{Cl}$	(ii) Smog
(c) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$	(iii) Ozone depletion
(d) $\text{NO}_2(\text{g}) \xrightarrow{h\nu} \text{NO}(\text{g}) + \text{O}(\text{g})$	(iv) Tropospheric pollution

Choose the **correct** answer from the options given below.

- (1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

89. Match List - I with List - II.

- | List - I | List - II |
|--|------------------------------------|
| (a)  $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$ | (i) Hell-Volhard-Zelinsky reaction |
| (b) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$ | (ii) Gattermann-Koch reaction |
| (c) $\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$ | (iii) Haloform reaction |
| (d) $\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$ | (iv) Esterification |

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
 (3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
 (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

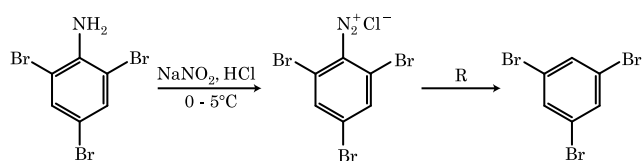
90. Match List - I with List - II.

- | List - I | List - II |
|--|---------------|
| (a) $[\text{Fe}(\text{CN})_6]^{3-}$ | (i) 5.92 BM |
| (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ | (ii) 0 BM |
| (c) $[\text{Fe}(\text{CN})_6]^{4-}$ | (iii) 4.90 BM |
| (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ | (iv) 1.73 BM |

Choose the **correct** answer from the options given below.

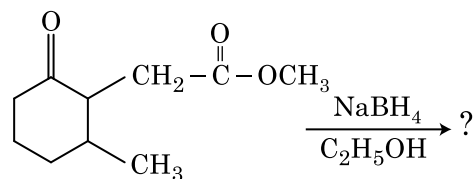
- (1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
 (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

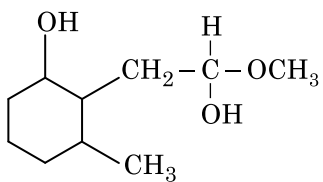
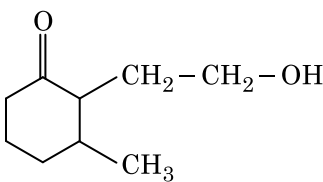
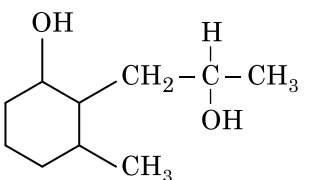
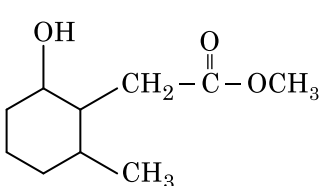
91. The reagent 'R' in the given sequence of chemical reaction is :



- (1) H_2O
 (2) $\text{CH}_3\text{CH}_2\text{OH}$
 (3) HI
 (4) CuCN/KCN

92. The product formed in the following chemical reaction is :



- (1) 
 (2) 
 (3) 
 (4) 

93. From the following pairs of ions which one is not an iso-electronic pair ?

- (1) O^{2-} , F^-
 (2) Na^+ , Mg^{2+}
 (3) Mn^{2+} , Fe^{3+}
 (4) Fe^{2+} , Mn^{2+}

94. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid ? Choose the correct option.

$$\left[\begin{array}{l} \Lambda_{\text{H}^+}^\circ = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda_{\text{CH}_3\text{COO}^-}^\circ = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$

- (1) $1.75 \times 10^{-4} \text{ mol L}^{-1}$
 (2) $2.50 \times 10^{-4} \text{ mol L}^{-1}$
 (3) $1.75 \times 10^{-5} \text{ mol L}^{-1}$
 (4) $2.50 \times 10^{-5} \text{ mol L}^{-1}$

95. Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O₂ and 2 g H₂ confined in a total volume of one litre at 0°C is :
[Given R = 0.082 L atm mol⁻¹K⁻¹, T = 273 K]
(1) 2.518
(2) 2.602
(3) 25.18
(4) 26.02
96. The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is :
[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]
(1) 160 mm of Hg
(2) 168 mm of Hg
(3) 336 mm of Hg
(4) 350 mm of Hg
97. For irreversible expansion of an ideal gas under isothermal condition, the correct option is :
(1) $\Delta U = 0, \Delta S_{\text{total}} = 0$
(2) $\Delta U \neq 0, \Delta S_{\text{total}} \neq 0$
(3) $\Delta U = 0, \Delta S_{\text{total}} \neq 0$
(4) $\Delta U \neq 0, \Delta S_{\text{total}} = 0$
98. Which of the following molecules is non-polar in nature ?
(1) POCl₃
(2) CH₂O
(3) SbCl₅
(4) NO₂
99. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?
(1) HF < HCl : Increasing acidic strength
< HBr < HI
(2) H₂O < H₂S : Increasing pK_a
< H₂Se < H₂Te values
(3) NH₃ < PH₃ : Increasing acidic character
< AsH₃ < SbH₃
(4) CO₂ < SiO₂ : Increasing oxidizing power
< SnO₂ < PbO₂
100. The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T} \right)$ of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is. Choose the correct option for your answer.
[Given R = 8.314 JK⁻¹mol⁻¹]
(1) 41.5 kJ mol⁻¹
(2) 83.0 kJ mol⁻¹
(3) 166 kJ mol⁻¹
(4) -83 kJ mol⁻¹

Section - A (Biology : Botany)

101. Mutations in plant cells can be induced by :
(1) Kinetin
(2) Infrared rays
(3) Gamma rays
(4) Zeatin

102. Match List - I with List - II.

List - I		List - II	
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the **correct** answer from the options given below.

- (a) (b) (c) (d)
(1) (ii) (iv) (i) (iii)
(2) (iv) (iii) (ii) (i)
(3) (i) (ii) (iii) (iv)
(4) (iii) (ii) (iv) (i)
103. Which of the following is a **correct** sequence of steps in a PCR (Polymerase Chain Reaction) ?
(1) Denaturation, Annealing, Extension
(2) Denaturation, Extension, Annealing
(3) Extension, Denaturation, Annealing
(4) Annealing, Denaturation, Extension

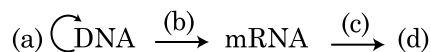
104. Match List - I with List - II.

List - I		List - II	
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
(1) (iv) (i) (iii) (ii)
(2) (iii) (i) (iv) (ii)
(3) (ii) (iii) (iv) (i)
(4) (iv) (ii) (i) (iii)

105. Complete the flow chart on central dogma.



- (1) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein
- (2) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Transduction
- (3) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein
- (4) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein

106. The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is :

- (1) Xenogamy
- (2) Geitonogamy
- (3) Chasmogamy
- (4) Cleistogamy

107. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as :

- (1) Yellow bands
- (2) Bright orange bands
- (3) Dark red bands
- (4) Bright blue bands

108. Which of the following is an **incorrect** statement ?

- (1) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
- (2) Microbodies are present both in plant and animal cells.
- (3) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
- (4) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.

109. In spite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival ?

- (1) Resource partitioning
- (2) Competitive release
- (3) Mutualism
- (4) Predation

110. Gemmae are present in :

- (1) Mosses
- (2) Pteridophytes
- (3) Some Gymnosperms
- (4) Some Liverworts

111. Match **List - I** with **List - II**.

List - I		List - II	
(a)	Protoplast fusion	(i)	Totipotency
(b)	Plant tissue culture	(ii)	Pomato
(c)	Meristem culture	(iii)	Somaclones
(d)	Micropropagation	(iv)	Virus free plants

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

112. The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called :

- (1) Bullet square
- (2) Punch square
- (3) Punnett square
- (4) Net square

113. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as :

- (1) Homosorus
- (2) Heterosorus
- (3) Homosporous
- (4) Heterosporous

114. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as :

- (1) Climax
- (2) Climax community
- (3) Standing state
- (4) Standing crop

115. Amensalism can be represented as :

- (1) Species A (−) ; Species B (0)
- (2) Species A (+) ; Species B (+)
- (3) Species A (−) ; Species B (−)
- (4) Species A (+) ; Species B (0)

116. Match List - I with List - II.

List - I		List - II	
(a)	Cohesion	(i)	More attraction in liquid phase
(b)	Adhesion	(ii)	Mutual attraction among water molecules
(c)	Surface tension	(iii)	Water loss in liquid phase
(d)	Guttation	(iv)	Attraction towards polar surfaces

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (ii) (iv) (i) (iii)
- (2) (iv) (iii) (ii) (i)
- (3) (iii) (i) (iv) (ii)
- (4) (ii) (i) (iv) (iii)

117. Which of the following is **not** an application of PCR (Polymerase Chain Reaction) ?

- (1) Molecular diagnosis
- (2) Gene amplification
- (3) Purification of isolated protein
- (4) Detection of gene mutation

118. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out :

- (1) RNA
- (2) DNA
- (3) Histones
- (4) Polysaccharides

119. In the equation $GPP - R = NPP$
R represents :

- (1) Radiant energy
- (2) Retardation factor
- (3) Environment factor
- (4) Respiration losses

120. The first stable product of CO_2 fixation in sorghum is :

- (1) Pyruvic acid
- (2) Oxaloacetic acid
- (3) Succinic acid
- (4) Phosphoglyceric acid

121. Which of the following algae produce Carrageen ?

- (1) Green algae
- (2) Brown algae
- (3) Red algae
- (4) Blue-green algae

122. Which of the following statements is **not** correct ?

- (1) Pyramid of biomass in sea is generally inverted.
- (2) Pyramid of biomass in sea is generally upright.
- (3) Pyramid of energy is always upright.
- (4) Pyramid of numbers in a grassland ecosystem is upright.

123. Match List - I with List - II.

List - I		List - II	
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (iv) (iii) (ii) (i)
- (2) (i) (iv) (iii) (ii)
- (3) (iii) (iv) (i) (ii)
- (4) (ii) (iii) (iv) (i)

124. Which of the following are **not** secondary metabolites in plants ?

- (1) Morphine, codeine
- (2) Amino acids, glucose
- (3) Vinblastin, curcumin
- (4) Rubber, gums

125. Which of the following algae contains mannitol as reserve food material ?

- (1) *Ectocarpus*
- (2) *Gracilaria*
- (3) *Volvox*
- (4) *Ulothrix*

126. A typical angiosperm embryo sac at maturity is :

- (1) 8-nucleate and 7-celled
- (2) 7-nucleate and 8-celled
- (3) 7-nucleate and 7-celled
- (4) 8-nucleate and 8-celled

- 127.** Diadelphous stamens are found in :
- (1) China rose
 - (2) Citrus
 - (3) Pea
 - (4) China rose and citrus
- 128.** When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :
- (1) Biopiracy
 - (2) Gene therapy
 - (3) Molecular diagnosis
 - (4) Safety testing
- 129.** Which of the following stages of meiosis involves division of centromere ?
- (1) Metaphase I
 - (2) Metaphase II
 - (3) Anaphase II
 - (4) Telophase II
- 130.** Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called :
- (1) Elasticity
 - (2) Flexibility
 - (3) Plasticity
 - (4) Maturity
- 131.** Which of the following plants is monoecious ?
- (1) *Carica papaya*
 - (2) Chara
 - (3) *Marchantia polymorpha*
 - (4) *Cycas circinalis*
- 132.** When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as :
- (1) Metacentric
 - (2) Telocentric
 - (3) Sub-metacentric
 - (4) Acrocentric

- 133.** The site of perception of light in plants during photoperiodism is :
- (1) Shoot apex
 - (2) Stem
 - (3) Axillary bud
 - (4) Leaf
- 134.** The factor that leads to Founder effect in a population is :
- (1) Natural selection
 - (2) Genetic recombination
 - (3) Mutation
 - (4) Genetic drift
- 135.** The plant hormone used to destroy weeds in a field is :
- (1) IAA
 - (2) NAA
 - (3) 2, 4-D
 - (4) IBA

Section - B (Biology : Botany)

- 136.** Match Column - I with Column - II.

Column - I		Column - II	
(a)	<i>Nitrococcus</i>	(i)	Denitrification
(b)	<i>Rhizobium</i>	(ii)	Conversion of ammonia to nitrite
(c)	<i>Thiobacillus</i>	(iii)	Conversion of nitrite to nitrate
(d)	<i>Nitrobacter</i>	(iv)	Conversion of atmospheric nitrogen to ammonia

Choose the **correct** answer from options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iv) | (i) | (iii) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (iii) | (i) | (iv) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |
- 137.** Select the **correct** pair.
- (1) Large colorless empty cells in the epidermis of grass leaves - Subsidiary cells
 - (2) In dicot leaves, vascular bundles are surrounded by large thick-walled cells - Conjunctive tissue
 - (3) Cells of medullary rays that form part of cambial ring - Interfascicular cambium
 - (4) Loose parenchyma cells rupturing the epidermis and forming a lens-shaped opening in bark - Spongy parenchyma

138. Which of the following statements is **correct** ?

- (1) Fusion of two cells is called Karyogamy.
- (2) Fusion of protoplasts between two motile on non-motile gametes is called plasmogamy.
- (3) Organisms that depend on living plants are called saprophytes.
- (4) Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.

139. Identify the **correct** statement.

- (1) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.
- (2) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.
- (3) The coding strand in a transcription unit is copied to an mRNA.
- (4) Split gene arrangement is characteristic of prokaryotes.

140. In the exponential growth equation

$$N_t = N_0 e^{rt}, \text{ e represents :}$$

- (1) The base of number logarithms
- (2) The base of exponential logarithms
- (3) The base of natural logarithms
- (4) The base of geometric logarithms

141. Which of the following statements is **incorrect** ?

- (1) During aerobic respiration, role of oxygen is limited to the terminal stage.
- (2) In ETC (Electron Transport Chain), one molecule of $\text{NADH} + \text{H}^+$ gives rise to 2 ATP molecules, and one FADH_2 gives rise to 3 ATP molecules.
- (3) ATP is synthesized through complex V.
- (4) Oxidation-reduction reactions produce proton gradient in respiration.

142. What is the role of RNA polymerase III in the process of transcription in eukaryotes ?

- (1) Transcribes rRNAs (28S, 18S and 5.8S)
- (2) Transcribes tRNA, 5s rRNA and snRNA
- (3) Transcribes precursor of mRNA
- (4) Transcribes only snRNAs

143. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

- (1) Poaceae ; Rosaceae
- (2) Poaceae ; Leguminosae
- (3) Poaceae ; Solanaceae
- (4) Rosaceae ; Leguminosae

144. Match **List - I** with **List - II**.

List - I		List - II	
(a)	S phase	(i)	Proteins are synthesized
(b)	G ₂ phase	(ii)	Inactive phase
(c)	Quiescent stage	(iii)	Interval between mitosis and initiation of DNA replication
(d)	G ₁ phase	(iv)	DNA replication

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (ii) | (i) | (iv) |
| (2) | (iv) | (ii) | (iii) | (i) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (ii) | (iv) | (iii) | (i) |

145. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because :

- (1) mutated gene partially appears on a photographic film.
- (2) mutated gene completely and clearly appears on a photographic film.
- (3) mutated gene does not appear on a photographic film as the probe has no complementarity with it.
- (4) mutated gene does not appear on photographic film as the probe has complementarity with it.

146. Plasmid pBR322 has PstI restriction enzyme site within gene amp^R that confers ampicillin resistance. If this enzyme is used for inserting a gene for β -galactoside production and the recombinant plasmid is inserted in an *E.coli* strain

- (1) it will not be able to confer ampicillin resistance to the host cell.
- (2) the transformed cells will have the ability to resist ampicillin as well as produce β -galactoside.
- (3) it will lead to lysis of host cell.
- (4) it will be able to produce a novel protein with dual ability.

147. Match Column - I with Column - II.

Column - I	Column - II
(a) $\% \frac{\phi}{\phi} K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$	(i) Brassicaceae
(b) $\oplus \frac{\phi}{\phi} K_{(5)} \widehat{C_{(5)} A_5} \underline{G}_2$	(ii) Liliaceae
(c) $\oplus \frac{\phi}{\phi} \widehat{P_{(3+3)} A_{3+3}} \underline{G}_{(3)}$	(iii) Fabaceae
(d) $\oplus \frac{\phi}{\phi} K_{2+2} C_4 A_{2-4} \underline{G}_{(2)}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (iv) | (ii) | (i) | (iii) |

148. Which of the following statements is **incorrect** ?

- (1) Both ATP and $NADPH + H^+$ are synthesized during non-cyclic photophosphorylation.
- (2) Stroma lamellae have PS I only and lack NADP reductase.
- (3) Grana lamellae have both PS I and PS II.
- (4) Cyclic photophosphorylation involves both PS I and PS II.

149. Match List - I with List - II.

List - I		List - II	
(a)	Protein	(i)	C = C double bonds
(b)	Unsaturated fatty acid	(ii)	Phosphodiester bonds
(c)	Nucleic acid	(iii)	Glycosidic bonds
(d)	Polysaccharide	(iv)	Peptide bonds

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (ii) | (iii) |
| (2) | (i) | (iv) | (iii) | (ii) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (iv) | (iii) | (i) | (ii) |

150. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as :

- (1) Satellite DNA
- (2) Repetitive DNA
- (3) Single nucleotides
- (4) Polymorphic DNA

Section - A (Biology : Zoology)

151. Which of the following statements wrongly represents the nature of smooth muscle ?

- (1) These muscle have no striations
- (2) They are involuntary muscles
- (3) Communication among the cells is performed by intercalated discs
- (4) These muscles are present in the wall of blood vessels

152. The organelles that are included in the endomembrane system are :

- (1) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
- (2) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles
- (3) Golgi complex, Mitochondria, Ribosomes and Lysosomes
- (4) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes

153. With regard to insulin choose correct options.

- (a) C-peptide is not present in mature insulin.
- (b) The insulin produced by rDNA technology has C-peptide.
- (c) The pro-insulin has C-peptide.
- (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

- (1) (b) and (d) only
- (2) (b) and (c) only
- (3) (a), (c) and (d) only
- (4) (a) and (d) only

154. Venereal diseases can spread through :

- (a) Using sterile needles
- (b) Transfusion of blood from infected person
- (c) Infected mother to foetus
- (d) Kissing
- (e) Inheritance

Choose the **correct** answer from the options given below.

- (1) (a), (b) and (c) only
- (2) (b), (c) and (d) only
- (3) (b) and (c) only
- (4) (a) and (c) only

155. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

- (1) High pO_2 , low pCO_2 , less H^+ , lower temperature
- (2) Low pO_2 , high pCO_2 , more H^+ , higher temperature
- (3) High pO_2 , high pCO_2 , less H^+ , higher temperature
- (4) Low pO_2 , low pCO_2 , more H^+ , higher temperature

156. Match the following :

List - I		List - II	
(a)	<i>Physalia</i>	(i)	Pearl oyster
(b)	<i>Limulus</i>	(ii)	Portuguese Man of War
(c)	<i>Ancylostoma</i>	(iii)	Living fossil
(d)	<i>Pinctada</i>	(iv)	Hookworm

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (i) | (iv) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (i) | (iv) | (iii) | (ii) |

157. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?

- (1) 50%
- (2) 75%
- (3) 25%
- (4) 100%

158. Match List - I with List - II.

List - I		List - II	
(a)	<i>Aspergillus niger</i>	(i)	Acetic Acid
(b)	<i>Acetobacter aceti</i>	(ii)	Lactic Acid
(c)	<i>Clostridium butylicum</i>	(iii)	Citric Acid
(d)	<i>Lactobacillus</i>	(iv)	Butyric Acid

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (i) | (iv) | (ii) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (ii) | (iii) | (i) | (iv) |
| (4) | (iv) | (ii) | (i) | (iii) |

159. Sphincter of oddi is present at :

- (1) Ileo-caecal junction
- (2) Junction of hepato-pancreatic duct and duodenum
- (3) Gastro-oesophageal junction
- (4) Junction of jejunum and duodenum

160. Which of the following is **not** an objective of Biofortification in crops ?

- (1) Improve protein content
- (2) Improve resistance to diseases
- (3) Improve vitamin content
- (4) Improve micronutrient and mineral content

161. Erythropoietin hormone which stimulates R.B.C. formation is produced by :

- (1) Alpha cells of pancreas
- (2) The cells of rostral adenohypophysis
- (3) The cells of bone marrow
- (4) Juxtaglomerular cells of the kidney

162. Which one of the following organisms bears hollow and pneumatic long bones ?

- (1) *Neophron*
- (2) *Hemidactylus*
- (3) *Macropus*
- (4) *Ornithorhynchus*

163. The centriole undergoes duplication during :

- (1) S-phase
- (2) Prophase
- (3) Metaphase
- (4) G₂ phase

164. Dobson units are used to measure thickness of :

- (1) CFCs
- (2) Stratosphere
- (3) Ozone
- (4) Troposphere

165. Which one of the following belongs to the family Muscidae ?

- (1) Fire fly
- (2) Grasshopper
- (3) Cockroach
- (4) House fly

166. Read the following statements.

- (a) Metagenesis is observed in Helminths.
- (b) Echinoderms are triploblastic and coelomate animals.
- (c) Round worms have organ-system level of body organization.
- (d) Comb plates present in ctenophores help in digestion.
- (e) Water vascular system is characteristic of Echinoderms.

Choose the **correct** answer from the options given below.

- (1) (c), (d) and (e) are correct
- (2) (a), (b) and (c) are correct
- (3) (a), (d) and (e) are correct
- (4) (b), (c) and (e) are correct

167. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection ?

- (1) Western Blotting Technique
- (2) Southern Blotting Technique
- (3) ELISA Technique
- (4) Hybridization Technique

168. Receptors for sperm binding in mammals are present on :

- (1) Corona radiata
- (2) Vitelline membrane
- (3) Perivitelline space
- (4) Zona pellucida

169. Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :

- (1) Arthritis
- (2) Muscular dystrophy
- (3) Myasthenia gravis
- (4) Gout

170. Which one of the following is an example of Hormone releasing IUD ?

- (1) CuT
- (2) LNG 20
- (3) Cu 7
- (4) Multiload 375

171. Match List - I with List - II.

List - I		List - II	
(a)	Vaults	(i)	Entry of sperm through Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of fallopian tube

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (ii) | (i) | (iii) |
| (2) | (i) | (iii) | (ii) | (iv) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (iii) | (i) | (iv) | (ii) |

172. Match List - I with List - II.

List - I		List - II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (i) | (ii) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (iv) | (i) | (ii) | (iii) |

173. The fruit fly has 8 chromosomes ($2n$) in each cell. During interphase of Mitosis if the number of chromosomes at G_1 phase is 8, what would be the number of chromosomes after S phase ?
- (1) 8
 - (2) 16
 - (3) 4
 - (4) 32
174. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?
- (1) Annealing
 - (2) Extension
 - (3) Denaturation
 - (4) Ligation
175. If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it ?
- (1) T : 20 ; G : 30 ; C : 20
 - (2) T : 20 ; G : 20 ; C : 30
 - (3) T : 30 ; G : 20 ; C : 20
 - (4) T : 20 ; G : 25 ; C : 25
176. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?
- (1) Thrombin
 - (2) Renin
 - (3) Epinephrine
 - (4) Thrombokinas
177. Succus entericus is referred to as :
- (1) Pancreatic juice
 - (2) Intestinal juice
 - (3) Gastric juice
 - (4) Chyme
178. Identify the **incorrect** pair.
- (1) Alkaloids - Codeine
 - (2) Toxin - Abrin
 - (3) Lectins - Concanavalin A
 - (4) Drugs - Ricin
179. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature ?
- (1) Leptotene
 - (2) Zygotene
 - (3) Diakinesis
 - (4) Pachytene
180. Persons with 'AB' blood group are called as "Universal recipients". This is due to :
- (1) Absence of antigens A and B on the surface of RBCs
 - (2) Absence of antigens A and B in plasma
 - (3) Presence of antibodies, anti-A and anti-B, on RBCs
 - (4) Absence of antibodies, anti-A and anti-B, in plasma
181. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :
- (1) Degenerate primer sequence
 - (2) Okazaki sequences
 - (3) Palindromic Nucleotide sequences
 - (4) Poly(A) tail sequences
182. Which of the following characteristics is **incorrect** with respect to cockroach ?
- (1) A ring of gastric caeca is present at the junction of midgut and hind gut.
 - (2) Hypopharynx lies within the cavity enclosed by the mouth parts.
 - (3) In females, 7th-9th sterna together form a genital pouch.
 - (4) 10th abdominal segment in both sexes, bears a pair of anal cerci.
183. Which of the following RNAs is not required for the synthesis of protein ?
- (1) mRNA
 - (2) tRNA
 - (3) rRNA
 - (4) siRNA
184. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?
- (1) DNA dependent DNA polymerase
 - (2) DNA dependent RNA polymerase
 - (3) DNA Ligase
 - (4) DNase
185. The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are :
- (1) $pO_2 = 104$ and $pCO_2 = 40$
 - (2) $pO_2 = 40$ and $pCO_2 = 45$
 - (3) $pO_2 = 95$ and $pCO_2 = 40$
 - (4) $pO_2 = 159$ and $pCO_2 = 0.3$

Section - B (Biology : Zoology)

186. Which of the following is **not** a step in Multiple Ovulation Embryo Transfer Technology (MOET) ?
- (1) Cow is administered hormone having LH like activity for super ovulation
 - (2) Cow yields about 6-8 eggs at a time
 - (3) Cow is fertilized by artificial insemination
 - (4) Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage
187. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?
- (1) Graafian follicle
 - (2) Corpus luteum
 - (3) Foetus
 - (4) Uterus
188. During muscular contraction which of the following events occur ?
- (a) 'H' zone disappears
 - (b) 'A' band widens
 - (c) 'I' band reduces in width
 - (d) Myosine hydrolyzes ATP, releasing the ADP and Pi
 - (e) Z-lines attached to actins are pulled inwards
- Choose the **correct** answer from the options given below.
- (1) (a), (c), (d), (e) only
 - (2) (a), (b), (c), (d) only
 - (3) (b), (c), (d), (e) only
 - (4) (b), (d), (e), (a) only
189. Following are the statements with reference to 'lipids'.
- (a) Lipids having only single bonds are called unsaturated fatty acids.
 - (b) Lecithin is a phospholipid.
 - (c) Trihydroxy propane is glycerol.
 - (d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
 - (e) Arachidonic acid has 16 carbon atoms.
- Choose the **correct** answer from the options given below.
- (1) (a) and (b) only
 - (2) (c) and (d) only
 - (3) (b) and (c) only
 - (4) (b) and (e) only
190. Which one of the following statements about Histones is **wrong** ?
- (1) Histones are organized to form a unit of 8 molecules.
 - (2) The pH of histones is slightly acidic.
 - (3) Histones are rich in amino acids - Lysine and Arginine.
 - (4) Histones carry positive charge in the side chain.

191. Match List - I with List - II.

List - I		List - II	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (ii) | (i) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (i) | (iv) | (iii) | (ii) |
192. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.
- (1) Gap junctions and Adhering junctions, respectively.
 - (2) Tight junctions and Gap junctions, respectively.
 - (3) Adhering junctions and Tight junctions, respectively.
 - (4) Adhering junctions and Gap junctions, respectively.
193. Following are the statements about prostomium of earthworm.
- (a) It serves as a covering for mouth.
 - (b) It helps to open cracks in the soil into which it can crawl.
 - (c) It is one of the sensory structures.
 - (d) It is the first body segment.
- Choose the **correct** answer from the options given below.
- (1) (a), (b) and (c) are correct
 - (2) (a), (b) and (d) are correct
 - (3) (a), (b), (c) and (d) are correct
 - (4) (b) and (c) are correct
194. The Adenosine deaminase deficiency results into :
- (1) Dysfunction of Immune system
 - (2) Parkinson's disease
 - (3) Digestive disorder
 - (4) Addison's disease

195. **Statement I :**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true
- (2) Both **Statement I** and **Statement II** are false
- (3) **Statement I** is correct but **Statement II** is false
- (4) **Statement I** is incorrect but **Statement II** is true

196. **Assertion (A) :**

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
- (2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**
- (3) **(A)** is true but **(R)** is false
- (4) **(A)** is false but **(R)** is true

197. Match List - I with List - II.

List - I		List - II	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical adaptation	(iv)	Polar seal

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (ii) | (iii) | (i) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

198. Match List - I with List - II.

List - I		List - II	
(a)	Filariasis	(i)	<i>Haemophilus influenzae</i>
(b)	Amoebiasis	(ii)	<i>Trichophyton</i>
(c)	Pneumonia	(iii)	<i>Wuchereria bancrofti</i>
(d)	Ringworm	(iv)	<i>Entamoeba histolytica</i>

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (iii) | (ii) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (ii) | (iii) | (i) | (iv) |

199. Which of these is not an important component of initiation of parturition in humans ?

- (1) Increase in estrogen and progesterone ratio
- (2) Synthesis of prostaglandins
- (3) Release of Oxytocin
- (4) Release of Prolactin

200. Match List - I with List - II.

List - I		List - II	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (iii) | (ii) | (iv) |
| (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iv) | (ii) | (iii) | (i) |
| (4) | (iv) | (iii) | (ii) | (i) |

Space For Rough Work

Space For Rough Work

Space For Rough Work

Test Booklet Code

M1

GAJAH

No. :

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on OFFICE Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos – 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is M1. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

Section - A (Physics)

1. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :

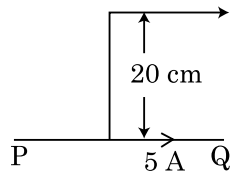
- (1) $\frac{R_1}{R_2}$
 (2) $\frac{R_2}{R_1}$
 (3) $\sqrt{\left(\frac{R_1}{R_2}\right)}$
 (4) $\frac{R_1^2}{R_2^2}$

2. The velocity of a small ball of mass M and density d , when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be :

- (1) $\frac{Mg}{2}$
 (2) Mg
 (3) $\frac{3}{2}Mg$
 (4) $2Mg$

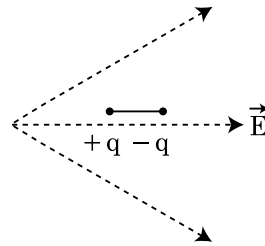
3. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.

Electron $v = 10^5$ m/s



- (1) 4×10^{-20} N
 (2) $8\pi \times 10^{-20}$ N
 (3) $4\pi \times 10^{-20}$ N
 (4) 8×10^{-20} N

4. A dipole is placed in an electric field as shown. In which direction will it move ?



- (1) towards the left as its potential energy will increase.
 (2) towards the right as its potential energy will decrease.
 (3) towards the left as its potential energy will decrease.
 (4) towards the right as its potential energy will increase.
5. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since :
- (1) a large aperture contributes to the quality and visibility of the images.
 (2) a large area of the objective ensures better light gathering power.
 (3) a large aperture provides a better resolution.
 (4) all of the above.
6. A screw gauge gives the following readings when used to measure the diameter of a wire
 Main scale reading : 0 mm
 Circular scale reading : 52 divisions
 Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is :
- (1) 0.52 cm
 (2) 0.026 cm
 (3) 0.26 cm
 (4) 0.052 cm
7. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is :
- (1) 0.9 MeV
 (2) 9.4 MeV
 (3) 804 MeV
 (4) 216 MeV

8. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$ is :

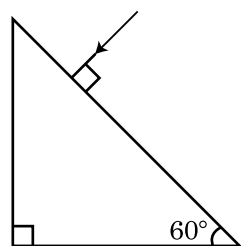
(1) $\frac{2n-1}{2n}$

(2) $\frac{2n-1}{2n+1}$

(3) $\frac{2n+1}{2n-1}$

(4) $\frac{2n}{2n-1}$

9. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



- (1) 60°
 (2) 30°
 (3) 45°
 (4) 90°

10. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is :

- (1) 0.0628 s
 (2) 6.28 s
 (3) 3.14 s
 (4) 0.628 s

11. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be :

- (1) 25
 (2) 15
 (3) 50
 (4) 30

12. **Column - I** gives certain physical terms associated with flow of current through a metallic conductor. **Column - II** gives some mathematical relations involving electrical quantities. Match **Column - I** and **Column - II** with appropriate relations.

Column - I	Column - II
(A) Drift Velocity	(P) $\frac{m}{ne^2 \rho}$
(B) Electrical Resistivity	(Q) nev_d
(C) Relaxation Period	(R) $\frac{eE}{m} \tau$
(D) Current Density	(S) $\frac{E}{J}$

- (1) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)
 (2) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)
 (3) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)
 (4) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)

13. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

- (1) current in n-type = current in p-type.
 (2) current in p-type > current in n-type.
 (3) current in n-type > current in p-type.
 (4) No current will flow in p-type, current will only flow in n-type.

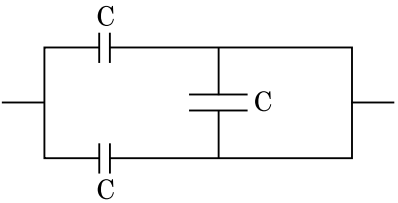
14. A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence

${}^A_ZX \rightarrow {}^{Z-1}_B \rightarrow {}^{Z-3}_C \rightarrow {}^{Z-2}_D$, where Z is the atomic number of element X. The possible decay particles in the sequence are :

- (1) α , β^- , β^+
 (2) α , β^+ , β^-
 (3) β^+ , α , β^-
 (4) β^- , α , β^+

15. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine ? ($g = 10 \text{ m/s}^2$)

- (1) 10.2 kW
 (2) 8.1 kW
 (3) 12.3 kW
 (4) 7.0 kW

16. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be : ($h = 6.6 \times 10^{-34}$ Js)
- 10^{18}
 - 10^{17}
 - 10^{16}
 - 10^{15}
17. For a plane electromagnetic wave propagating in x -direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively ?
- $\hat{j} + \hat{k}, \hat{j} + \hat{k}$
 - $-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$
 - $\hat{j} + \hat{k}, -\hat{j} - \hat{k}$
 - $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$
18. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25Ω . What will be the effective resistance if they are connected in series ?
- 0.25Ω
 - 0.5Ω
 - 1Ω
 - 4Ω
19. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs ?
- 60 cm
 - 21.6 cm
 - 64 cm
 - 62 cm
20. Polar molecules are the molecules :
- having zero dipole moment.
 - acquire a dipole moment only in the presence of electric field due to displacement of charges.
 - acquire a dipole moment only when magnetic field is absent.
 - having a permanent electric dipole moment.
21. A parallel plate capacitor has a uniform electric field ' $\vec{E}$ ' in the space between the plates. If the distance between the plates is ' d ' and the area of each plate is ' A ', the energy stored in the capacitor is : (ϵ_0 = permittivity of free space)
- $\frac{1}{2} \epsilon_0 E^2$
 - $\epsilon_0 E A d$
 - $\frac{1}{2} \epsilon_0 E^2 A d$
 - $\frac{E^2 A d}{\epsilon_0}$
22. The equivalent capacitance of the combination shown in the figure is :
- 
- The diagram shows a circuit with two input terminals on the left and two output terminals on the right. Between the input and output terminals, there are two parallel branches. The top branch contains two capacitors labeled 'C' connected in series. The bottom branch contains two capacitors labeled 'C' connected in parallel.
- $3C$
 - $2C$
 - $C/2$
 - $3C/2$
23. Consider the following **statements (A) and (B)** and identify the **correct** answer.
- (A) A zener diode is connected in reverse bias, when used as a voltage regulator.
- (B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.
- (A) and (B) both are correct.
 - (A) and (B) both are incorrect.
 - (A) is correct and (B) is incorrect.
 - (A) is incorrect but (B) is correct.
24. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively :
- $\frac{S}{4}, \frac{3gS}{2}$
 - $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
 - $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$
 - $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

25. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of :

- (1) $[M^2] [L^{-1}] [T^0]$
- (2) $[M] [L^{-1}] [T^{-1}]$
- (3) $[M] [L^0] [T^0]$
- (4) $[M^2] [L^{-2}] [T^{-1}]$

26. Match **Column - I** and **Column - II** and choose the **correct** match from the given choices.

Column - I	Column - II
(A) Root mean square speed of gas molecules	(P) $\frac{1}{3} n m \bar{v}^2$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3 RT}{M}}$
(C) Average kinetic energy of a molecule	(R) $\frac{5}{2} RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2} k_B T$

- (1) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)
- (2) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)
- (3) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)
- (4) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)

27. A body is executing simple harmonic motion with frequency ' n ', the frequency of its potential energy is :

- (1) n
- (2) $2n$
- (3) $3n$
- (4) $4n$

28. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C . The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is :

- (1) $\frac{13}{10} t$
- (2) $\frac{13}{5} t$
- (3) $\frac{10}{13} t$
- (4) $\frac{5}{13} t$

29. A capacitor of capacitance ' C ', is connected across an ac source of voltage V , given by

$$V = V_0 \sin \omega t$$

The displacement current between the plates of the capacitor, would then be given by :

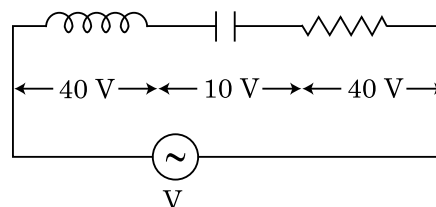
- (1) $I_d = V_0 \omega C \cos \omega t$
- (2) $I_d = \frac{V_0}{\omega C} \cos \omega t$
- (3) $I_d = \frac{V_0}{\omega C} \sin \omega t$
- (4) $I_d = V_0 \omega C \sin \omega t$

30. The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be :

- (1) $1/2$
- (2) $\frac{1}{2\sqrt{2}}$
- (3) $\frac{2}{3}$
- (4) $\frac{2}{3\sqrt{2}}$

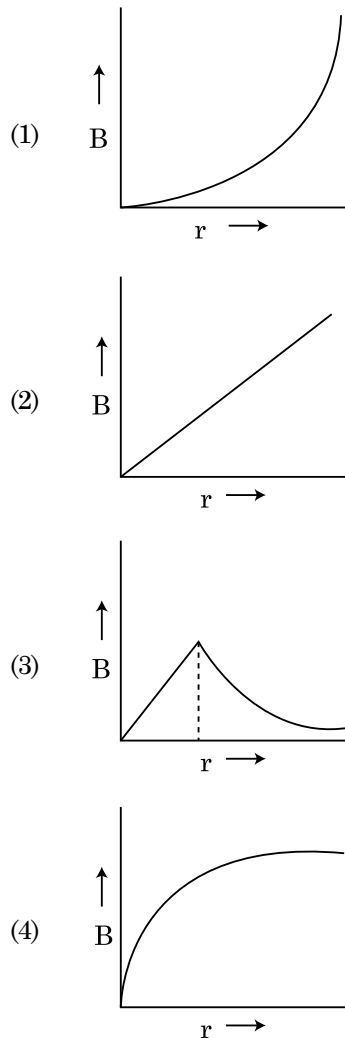
31. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.

Potential difference across L , C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is :



- (1) $4\sqrt{2} \Omega$
- (2) $5/\sqrt{2} \Omega$
- (3) 4Ω
- (4) 5Ω

32. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by :



33. An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d , then :

- (1) $\lambda = \left(\frac{2m}{hc} \right) \lambda_d^2$
- (2) $\lambda_d = \left(\frac{2mc}{h} \right) \lambda^2$
- (3) $\lambda = \left(\frac{2mc}{h} \right) \lambda_d^2$
- (4) $\lambda = \left(\frac{2h}{mc} \right) \lambda_d^2$

34. The escape velocity from the Earth's surface is v . The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is :

- (1) v
- (2) $2v$
- (3) $3v$
- (4) $4v$

35. If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.

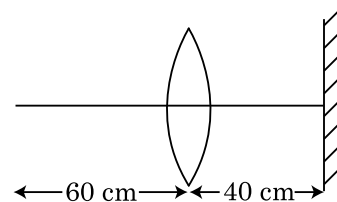
- (1) [F] [A] [T]
- (2) [F] [A] [T²]
- (3) [F] [A] [T⁻¹]
- (4) [F] [A⁻¹] [T]

Section - B (Physics)

36. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to :

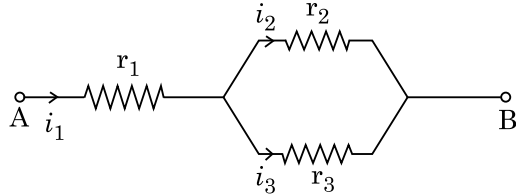
- (1) $\frac{R_1}{R_2}$
- (2) $\frac{R_2}{R_1}$
- (3) $\frac{R_1^2}{R_2}$
- (4) $\frac{R_2^2}{R_1}$

37. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of :

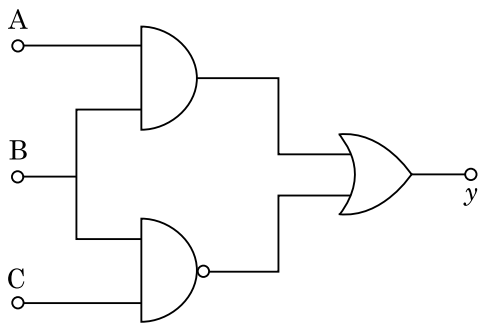
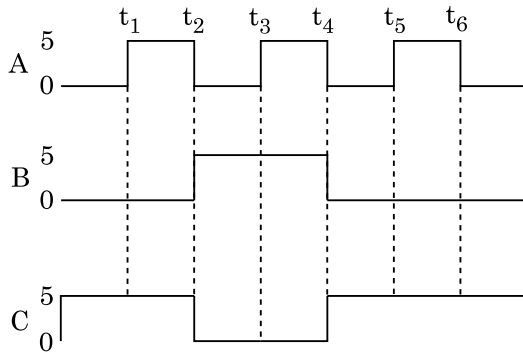


- (1) 20 cm from the lens, it would be a real image.
- (2) 30 cm from the lens, it would be a real image.
- (3) 30 cm from the plane mirror, it would be a virtual image.
- (4) 20 cm from the plane mirror, it would be a virtual image.

38. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is :



- (1) $\frac{r_1}{r_2 + r_3}$
 (2) $\frac{r_2}{r_2 + r_3}$
 (3) $\frac{r_1}{r_1 + r_2}$
 (4) $\frac{r_2}{r_1 + r_3}$
39. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y ?



- (1) y 0 V
 (2) 5 V
 (3) 0 V
 (4) 5 V
 (4) 0 V

40. A series LCR circuit containing 5.0 H inductor, $80 \mu\text{F}$ capacitor and 40Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be :

- (1) 25 rad/s and 75 rad/s
 (2) 50 rad/s and 25 rad/s
 (3) 46 rad/s and 54 rad/s
 (4) 42 rad/s and 58 rad/s

41. A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$? (Take $g = 10 \text{ m/s}^2$)

- (1) 20 m/s, 5 m/s^2
 (2) 20 m/s, 0
 (3) $20\sqrt{2} \text{ m/s}$, 0
 (4) $20\sqrt{2} \text{ m/s}$, 10 m/s^2

42. A particle of mass 'm' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth.

(V_e = escape velocity)

The maximum height above the surface reached by the particle is :

- (1) $R \left(\frac{k}{1-k} \right)^2$
 (2) $R \left(\frac{k}{1+k} \right)^2$
 (3) $\frac{R^2 k}{1+k}$
 (4) $\frac{Rk^2}{1-k^2}$

43. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ , is then given by :

$$(1) \quad \theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{1/2}$$

$$(2) \quad \theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$$

$$(3) \quad \theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$$

$$(4) \quad \theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{1/2}$$

44. In the product

$$\begin{aligned} \vec{F} &= q \left(\vec{v} \times \vec{B} \right) \\ &= q \vec{v} \times \left(B \hat{i} + B_0 \hat{j} + B_0 \hat{k} \right) \end{aligned}$$

$$\begin{aligned} \text{For } q=1 \text{ and } \vec{v} &= 2\hat{i} + 4\hat{j} + 6\hat{k} \text{ and} \\ \vec{F} &= 4\hat{i} - 20\hat{j} + 12\hat{k} \end{aligned}$$

What will be the complete expression for $\vec{B}$?

$$(1) \quad -8\hat{i} - 8\hat{j} - 6\hat{k}$$

$$(2) \quad -6\hat{i} - 6\hat{j} - 8\hat{k}$$

$$(3) \quad 8\hat{i} + 8\hat{j} - 6\hat{k}$$

$$(4) \quad 6\hat{i} + 6\hat{j} - 8\hat{k}$$

45. A ball of mass 0.15 kg is dropped from a height 10 m , strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly :

$$(1) \quad 0 \text{ kg m/s}$$

$$(2) \quad 4.2 \text{ kg m/s}$$

$$(3) \quad 2.1 \text{ kg m/s}$$

$$(4) \quad 1.4 \text{ kg m/s}$$

46. A uniform conducting wire of length $12a$ and resistance ' R ' is wound up as a current carrying coil in the shape of,

(i) an equilateral triangle of side ' a '.

(ii) a square of side ' a '.

The magnetic dipole moments of the coil in each case respectively are :

$$(1) \quad \sqrt{3} Ia^2 \text{ and } 3 Ia^2$$

$$(2) \quad 3 Ia^2 \text{ and } Ia^2$$

$$(3) \quad 3 Ia^2 \text{ and } 4 Ia^2$$

$$(4) \quad 4 Ia^2 \text{ and } 3 Ia^2$$

47. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

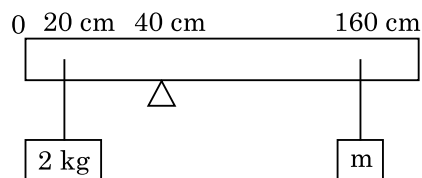
$$(1) \quad 660 \text{ V}$$

$$(2) \quad 1320 \text{ V}$$

$$(3) \quad 1520 \text{ V}$$

$$(4) \quad 1980 \text{ V}$$

48. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$)



- (1) $\frac{1}{2} \text{ kg}$
 (2) $\frac{1}{3} \text{ kg}$
 (3) $\frac{1}{6} \text{ kg}$
 (4) $\frac{1}{12} \text{ kg}$
49. From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times ' MR^2 '. Then the value of 'K' is :
- (1) $\frac{3}{4}$
 (2) $\frac{7}{8}$
 (3) $\frac{1}{4}$
 (4) $\frac{1}{8}$
50. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit ?
- (1) 0.2 A
 (2) 0.4 A
 (3) 2 A
 (4) 4 A

Section - A (Chemistry)

51. Dihedral angle of least stable conformer of ethane is :
- (1) 120°
 (2) 180°
 (3) 60°
 (4) 0°
52. Noble gases are named because of their inertness towards reactivity. Identify an **incorrect** statement about them.
- (1) Noble gases are sparingly soluble in water.
 (2) Noble gases have very high melting and boiling points.
 (3) Noble gases have weak dispersion forces.
 (4) Noble gases have large positive values of electron gain enthalpy.
53. The major product of the following chemical reaction is :
- $$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} = \text{CH}_2 \\ \diagup \\ \text{CH}_3 \end{array} + \text{HBr} \xrightarrow{(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2} ?$$
- (1) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Br} \\ \diagup \\ \text{CH}_3 \end{array}$
 (2) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{COC}_6\text{H}_5 \\ \diagup \\ \text{CH}_3 \end{array}$
 (3) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} - \text{CH}_3 \\ \diagup \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$
 (4) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CBr} - \text{CH}_2 - \text{CH}_3 \\ \diagup \\ \text{CH}_3 \end{array}$
54. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light, $c = 3.0 \times 10^8 \text{ ms}^{-1}$]
- (1) 219.3 m
 (2) 219.2 m
 (3) 2192 m
 (4) 21.92 cm

55. The **incorrect** statement among the following is :

- (1) Actinoid contraction is greater for element to element than Lanthanoid contraction.
- (2) Most of the trivalent Lanthanoid ions are colorless in the solid state.
- (3) Lanthanoids are good conductors of heat and electricity.
- (4) Actinoids are highly reactive metals, especially when finely divided.

56. Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are :

- (1) 8, 4
- (2) 6, 12
- (3) 2, 1
- (4) 12, 6

57. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.

- (1) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{O}_2$
- (2) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{H}-\text{CH}_3$
- (3) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{H}_2$
- (4) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$

58. Which of the following reactions is the metal displacement reaction? Choose the right option.

- (1) $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$
- (2) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
- (3) $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2\uparrow$
- (4) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2\uparrow$

59. Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

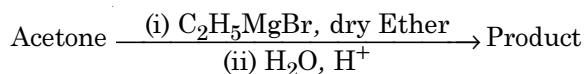
Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** is incorrect but **Statement II** is true.

60. What is the IUPAC name of the organic compound formed in the following chemical reaction?



- (1) 2-methyl propan-2-ol
- (2) pentan-2-ol
- (3) pentan-3-ol
- (4) 2-methyl butan-2-ol

61. **Statement I :**

Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.

Statement II :

As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** is incorrect but **Statement II** is true.

62. Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas?

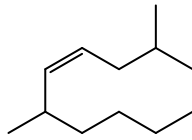
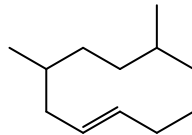
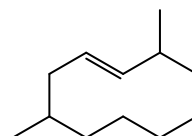
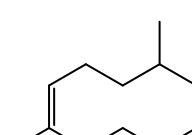
- (1) $C_P + C_V = R$
- (2) $C_P - C_V = R$
- (3) $C_P = RC_V$
- (4) $C_V = RC_P$

63. Match **List - I** with **List - II**.

List - I	List - II
(a) PCl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

64. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :
- $P_2 > P_1 > P_3$
 - $P_1 > P_2 > P_3$
 - $P_2 > P_3 > P_1$
 - $P_3 > P_1 > P_2$
65. Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature ?
- Electrolysis
 - Chromatography
 - Distillation
 - Zone refining
66. Tritium, a radioactive isotope of hydrogen, emits which of the following particles ?
- Beta (β^-)
 - Alpha (α)
 - Gamma (γ)
 - Neutron (n)
67. The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is :
- 7
 - 5
 - 2
 - 3
68. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on ?
- Saytzeff's Rule
 - Hund's Rule
 - Hofmann Rule
 - Huckel's Rule
69. The maximum temperature that can be achieved in blast furnace is :
- upto 1200 K
 - upto 2200 K
 - upto 1900 K
 - upto 5000 K
70. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.
- $201.28 \text{ S cm}^2 \text{ mol}^{-1}$
 - $390.71 \text{ S cm}^2 \text{ mol}^{-1}$
 - $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
 - $540.48 \text{ S cm}^2 \text{ mol}^{-1}$
71. Ethylene diaminetetraacetate (EDTA) ion is :
- Hexadentate ligand with four "O" and two "N" donor atoms
 - Unidentate ligand
 - Bidentate ligand with two "N" donor atoms
 - Tridentate ligand with three "N" donor atoms
72. The structures of beryllium chloride in solid state and vapour phase, are :
- Chain and dimer, respectively
 - Linear in both
 - Dimer and Linear, respectively
 - Chain in both
73. The correct structure of 2,6-Dimethyl-dec-4-ene is :
- 
 - 
 - 
 - 
74. The right option for the statement "Tyndall effect is exhibited by", is :
- NaCl solution
 - Glucose solution
 - Starch solution
 - Urea solution

75. Which one of the following polymers is prepared by addition polymerisation ?

- (1) Teflon
- (2) Nylon-66
- (3) Novolac
- (4) Dacron

76. The RBC deficiency is deficiency disease of :

- (1) Vitamin B₁₂
- (2) Vitamin B₆
- (3) Vitamin B₁
- (4) Vitamin B₂

77. The correct sequence of bond enthalpy of 'C-X' bond is :

- (1) $\text{CH}_3-\text{F} < \text{CH}_3-\text{Cl} < \text{CH}_3-\text{Br} < \text{CH}_3-\text{I}$
- (2) $\text{CH}_3-\text{F} > \text{CH}_3-\text{Cl} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
- (3) $\text{CH}_3-\text{F} < \text{CH}_3-\text{Cl} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
- (4) $\text{CH}_3-\text{Cl} > \text{CH}_3-\text{F} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$

78. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]

- (1) CH
- (2) CH₂
- (3) CH₃
- (4) CH₄

79. Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :

- (1) Calcium chloride
- (2) Strontium chloride
- (3) Magnesium chloride
- (4) Beryllium chloride

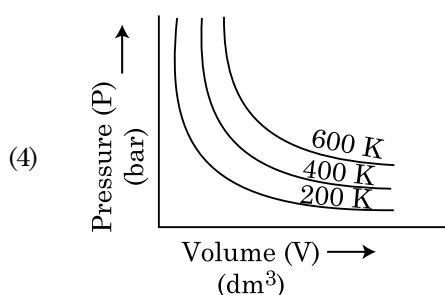
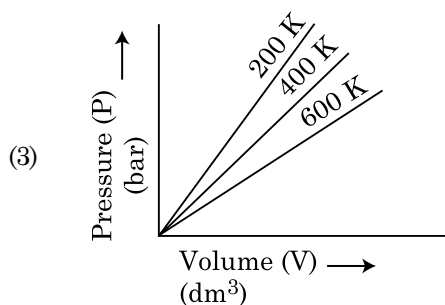
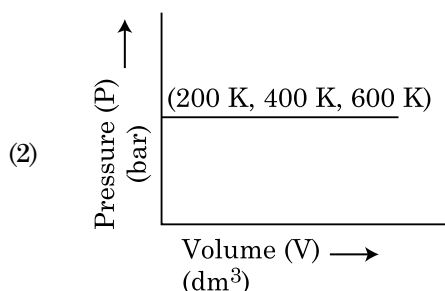
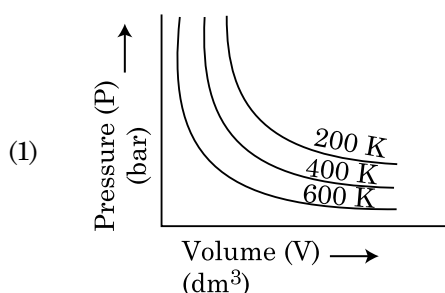
80. BF₃ is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are :

- (1) sp³ and 4
- (2) sp³ and 6
- (3) sp² and 6
- (4) sp² and 8

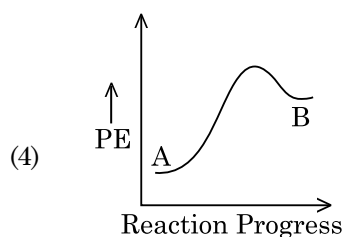
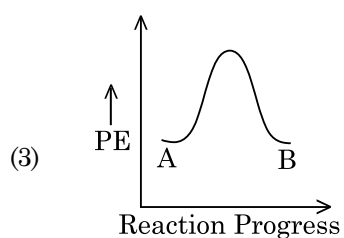
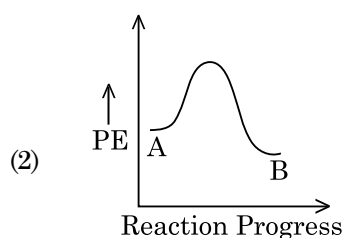
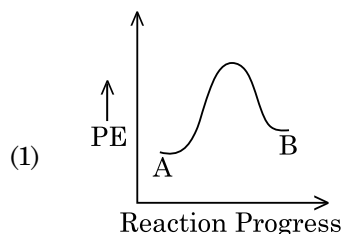
81. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is :

- (1) 8.50
- (2) 5.50
- (3) 7.75
- (4) 6.25

82. Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures :



83. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.



84. The compound which shows metamerism is :

- (1) C_5H_{12}
- (2) C_3H_8O
- (3) C_3H_6O
- (4) $C_4H_{10}O$

85. Zr ($Z=40$) and Hf ($Z=72$) have similar atomic and ionic radii because of :

- (1) belonging to same group
- (2) diagonal relationship
- (3) lanthanoid contraction
- (4) having similar chemical properties

Section - B (Chemistry)

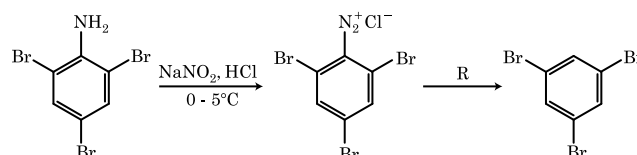
86. Match List - I with List - II.

List - I		List - II	
(a)	$[Fe(CN)_6]^{3-}$	(i)	5.92 BM
(b)	$[Fe(H_2O)_6]^{3+}$	(ii)	0 BM
(c)	$[Fe(CN)_6]^{4-}$	(iii)	4.90 BM
(d)	$[Fe(H_2O)_6]^{2+}$	(iv)	1.73 BM

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
- (3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

87. The reagent 'R' in the given sequence of chemical reaction is :



- (1) H_2O
- (2) CH_3CH_2OH
- (3) HI
- (4) $CuCN/KCN$

88. From the following pairs of ions which one is not an iso-electronic pair ?

- (1) O^{2-} , F^-
- (2) Na^+ , Mg^{2+}
- (3) Mn^{2+} , Fe^{3+}
- (4) Fe^{2+} , Mn^{2+}

89. The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T} \right)$ of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is. Choose the correct option for your answer.

[Given $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$]

- (1) 41.5 kJ mol^{-1}
- (2) 83.0 kJ mol^{-1}
- (3) 166 kJ mol^{-1}
- (4) -83 kJ mol^{-1}

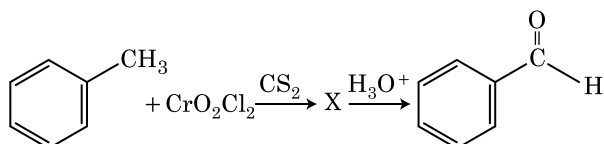
90. For irreversible expansion of an ideal gas under isothermal condition, the correct option is :

- (1) $\Delta U = 0$, $\Delta S_{\text{total}} = 0$
- (2) $\Delta U \neq 0$, $\Delta S_{\text{total}} \neq 0$
- (3) $\Delta U = 0$, $\Delta S_{\text{total}} \neq 0$
- (4) $\Delta U \neq 0$, $\Delta S_{\text{total}} = 0$

91. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?

- (1) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength
- (2) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values
- (3) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character
- (4) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power

92. The intermediate compound 'X' in the following chemical reaction is :



- (1)
- (2)
- (3)
- (4)

93. The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is :

[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]

- (1) 160 mm of Hg
- (2) 168 mm of Hg
- (3) 336 mm of Hg
- (4) 350 mm of Hg

94. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid ? Choose the correct option.

$$\left[\begin{array}{l} \Lambda^\circ_{\text{H}^+} = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda^\circ_{\text{CH}_3\text{COO}^-} = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$

- (1) $1.75 \times 10^{-4} \text{ mol L}^{-1}$
- (2) $2.50 \times 10^{-4} \text{ mol L}^{-1}$
- (3) $1.75 \times 10^{-5} \text{ mol L}^{-1}$
- (4) $2.50 \times 10^{-5} \text{ mol L}^{-1}$

95. $\text{CH}_3\text{CH}_2\text{COO}^- \text{Na}^+ \xrightarrow[\text{Heat}]{\text{NaOH, + ?}} \text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$.

Consider the above reaction and identify the missing reagent/chemical.

- (1) B_2H_6
- (2) Red Phosphorus
- (3) CaO
- (4) DIBAL-H

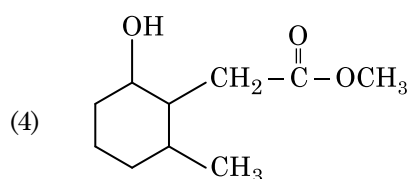
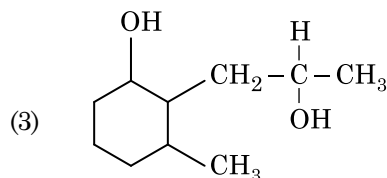
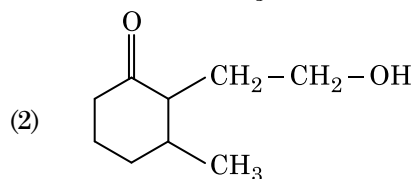
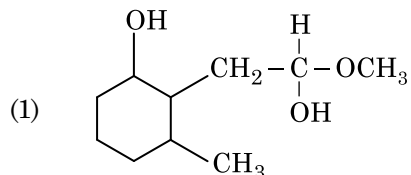
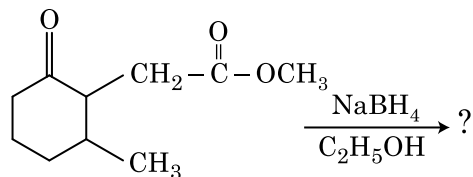
96. Match List - I with List - II.

- | List - I | List - II |
|--|------------------------------------|
| (a) $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$ | (i) Hell-Volhard-Zelinsky reaction |
| (b) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$ | (ii) Gattermann-Koch reaction |
| (c) $\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$ | (iii) Haloform reaction |
| (d) $\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$ | (iv) Esterification |

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
- (3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

97. The product formed in the following chemical reaction is :



98. Which of the following molecules is non-polar in nature ?

- (1) POCl_3
- (2) CH_2O
- (3) SbCl_5
- (4) NO_2

99. Match List - I with List - II.

- | List - I | List - II |
|--|-----------------------------|
| (a) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ | (i) Acid rain |
| (b) $\text{HOCl}(\text{g}) \xrightarrow{h\nu} \cdot\text{OH} + \cdot\text{Cl}$ | (ii) Smog |
| (c) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$ | (iii) Ozone depletion |
| (d) $\text{NO}_2(\text{g}) \xrightarrow{h\nu} \text{NO}(\text{g}) + \text{O}(\text{g})$ | (iv) Tropospheric pollution |

Choose the **correct** answer from the options given below.

- (1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

100. Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one litre at 0°C is :

[Given $R = 0.082 \text{ L atm mol}^{-1} \text{K}^{-1}$, $T = 273 \text{ K}$]

- (1) 2.518
- (2) 2.602
- (3) 25.18
- (4) 26.02

Section - A (Biology : Botany)

101. Which of the following plants is monoecious ?

- (1) *Carica papaya*
- (2) *Chara*
- (3) *Marchantia polymorpha*
- (4) *Cycas circinalis*

102. A typical angiosperm embryo sac at maturity is :

- (1) 8-nucleate and 7-celled
- (2) 7-nucleate and 8-celled
- (3) 7-nucleate and 7-celled
- (4) 8-nucleate and 8-celled

103. Gemmae are present in :

- (1) Mosses
- (2) Pteridophytes
- (3) Some Gymnosperms
- (4) Some Liverworts

104. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as :

- (1) Metacentric
- (2) Telocentric
- (3) Sub-metacentric
- (4) Acrocentric

105. Which of the following stages of meiosis involves division of centromere ?

- (1) Metaphase I
- (2) Metaphase II
- (3) Anaphase II
- (4) Telophase II

106. The first stable product of CO_2 fixation in sorghum is :

- (1) Pyruvic acid
- (2) Oxaloacetic acid
- (3) Succinic acid
- (4) Phosphoglyceric acid

107. The factor that leads to Founder effect in a population is :
- (1) Natural selection
 - (2) Genetic recombination
 - (3) Mutation
 - (4) Genetic drift
108. Which of the following is an **incorrect** statement ?
- (1) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
 - (2) Microbodies are present both in plant and animal cells.
 - (3) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
 - (4) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
109. Amensalism can be represented as :
- (1) Species A (–) ; Species B (0)
 - (2) Species A (+) ; Species B (+)
 - (3) Species A (–) ; Species B (–)
 - (4) Species A (+) ; Species B (0)
110. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as :
- (1) Climax
 - (2) Climax community
 - (3) Standing state
 - (4) Standing crop
111. Which of the following algae contains mannitol as reserve food material ?
- (1) *Ectocarpus*
 - (2) *Gracilaria*
 - (3) *Volvox*
 - (4) *Ulothrix*
112. In spite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival ?
- (1) Resource partitioning
 - (2) Competitive release
 - (3) Mutualism
 - (4) Predation

113. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as :

- (1) Homosorus
- (2) Heterosorus
- (3) Homosporous
- (4) Heterosporous

114. The site of perception of light in plants during photoperiodism is :

- (1) Shoot apex
- (2) Stem
- (3) Axillary bud
- (4) Leaf

115. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called :

- (1) Elasticity
- (2) Flexibility
- (3) Plasticity
- (4) Maturity

116. Match List - I with List - II.

List - I		List - II	
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (ii) | (i) |
| (2) | (i) | (iv) | (iii) | (ii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (ii) | (iii) | (iv) | (i) |

117. Match List - I with List - II.

List - I		List - II	
(a)	Cohesion	(i)	More attraction in liquid phase
(b)	Adhesion	(ii)	Mutual attraction among water molecules
(c)	Surface tension	(iii)	Water loss in liquid phase
(d)	Guttation	(iv)	Attraction towards polar surfaces

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iv) | (i) | (iii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (iii) | (i) | (iv) | (ii) |
| (4) | (ii) | (i) | (iv) | (iii) |

118. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as :

- (1) Yellow bands
- (2) Bright orange bands
- (3) Dark red bands
- (4) Bright blue bands

119. The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is :

- (1) Xenogamy
- (2) Geitonogamy
- (3) Chasmogamy
- (4) Cleistogamy

120. Which of the following algae produce Carrageen ?

- (1) Green algae
- (2) Brown algae
- (3) Red algae
- (4) Blue-green algae

121. Which of the following statements is **not** correct ?

- (1) Pyramid of biomass in sea is generally inverted.
- (2) Pyramid of biomass in sea is generally upright.
- (3) Pyramid of energy is always upright.
- (4) Pyramid of numbers in a grassland ecosystem is upright.

122. In the equation $GPP - R = NPP$

R represents :

- (1) Radiant energy
- (2) Retardation factor
- (3) Environment factor
- (4) Respiration losses

123. Diadelphous stamens are found in :

- (1) China rose
- (2) Citrus
- (3) Pea
- (4) China rose and citrus

124. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :

- (1) Biopiracy
- (2) Gene therapy
- (3) Molecular diagnosis
- (4) Safety testing

125. Match List - I with List - II.

List - I		List - II	
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (iii) | (ii) |
| (2) | (iii) | (i) | (iv) | (ii) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (iv) | (ii) | (i) | (iii) |

126. The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called :

- (1) Bullet square
- (2) Punch square
- (3) Punnett square
- (4) Net square

127. Which of the following is a **correct** sequence of steps in a PCR (Polymerase Chain Reaction) ?

- (1) Denaturation, Annealing, Extension
- (2) Denaturation, Extension, Annealing
- (3) Extension, Denaturation, Annealing
- (4) Annealing, Denaturation, Extension

128. Mutations in plant cells can be induced by :

- (1) Kinetin
- (2) Infrared rays
- (3) Gamma rays
- (4) Zeatin

129. Match List - I with List - II.

List - I		List - II	
(a)	Protoplast fusion	(i)	Totipotency
(b)	Plant tissue culture	(ii)	Pomato
(c)	Meristem culture	(iii)	Somaclones
(d)	Micropropagation	(iv)	Virus free plants

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

130. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out :

- (1) RNA
- (2) DNA
- (3) Histones
- (4) Polysaccharides

131. Complete the flow chart on central dogma.

(a) $\text{DNA} \xrightarrow{(b)} \text{mRNA} \xrightarrow{(c)} (d)$

- (1) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein
- (2) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Transduction
- (3) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein
- (4) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein

132. Which of the following is **not** an application of PCR (Polymerase Chain Reaction) ?

- (1) Molecular diagnosis
- (2) Gene amplification
- (3) Purification of isolated protein
- (4) Detection of gene mutation

133. The plant hormone used to destroy weeds in a field is :

- (1) IAA
- (2) NAA
- (3) 2, 4-D
- (4) IBA

134. Match List - I with List - II.

List - I		List - II	
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iv) | (i) | (iii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (i) | (ii) | (iii) | (iv) |
| (4) | (iii) | (ii) | (iv) | (i) |

135. Which of the following are **not** secondary metabolites in plants ?

- (1) Morphine, codeine
- (2) Amino acids, glucose
- (3) Vinblastin, curcumin
- (4) Rubber, gums

Section - B (Biology : Botany)

136. Which of the following statements is **incorrect** ?

- (1) During aerobic respiration, role of oxygen is limited to the terminal stage.
- (2) In ETC (Electron Transport Chain), one molecule of $\text{NADH} + \text{H}^+$ gives rise to 2 ATP molecules, and one FADH_2 gives rise to 3 ATP molecules.
- (3) ATP is synthesized through complex V.
- (4) Oxidation-reduction reactions produce proton gradient in respiration.

137. Match Column - I with Column - II.

Column - I		Column - II	
(a)	<i>Nitrococcus</i>	(i)	Denitrification
(b)	<i>Rhizobium</i>	(ii)	Conversion of ammonia to nitrite
(c)	<i>Thiobacillus</i>	(iii)	Conversion of nitrite to nitrate
(d)	<i>Nitrobacter</i>	(iv)	Conversion of atmospheric nitrogen to ammonia

Choose the **correct** answer from options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iv) | (i) | (iii) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (iii) | (i) | (iv) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

138. Match Column - I with Column - II.

Column - I	Column - II
(a) $\% \frac{\phi}{\psi} K_{(5)} C_{1+2+(2)} A_{(9)+1} G_1$	(i) Brassicaceae
(b) $\oplus \frac{\phi}{\psi} K_{(5)} \widehat{C_{(5)} A_5} G_2$	(ii) Liliaceae
(c) $\oplus \frac{\phi}{\psi} P_{(3+3)} \widehat{A_{3+3}} G_{(3)}$	(iii) Fabaceae
(d) $\oplus \frac{\phi}{\psi} K_{2+2} C_4 A_{2-4} G_{(2)}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (iv) | (ii) | (i) | (iii) |

139. Identify the **correct** statement.

- (1) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.
- (2) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.
- (3) The coding strand in a transcription unit is copied to an mRNA.
- (4) Split gene arrangement is characteristic of prokaryotes.

140. Which of the following statements is **correct** ?

- (1) Fusion of two cells is called Karyogamy.
- (2) Fusion of protoplasts between two motile on non-motile gametes is called plasmogamy.
- (3) Organisms that depend on living plants are called saprophytes.
- (4) Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.

141. Select the **correct** pair.

- (1) Large colorless empty cells in the epidermis of grass leaves - Subsidiary cells
- (2) In dicot leaves, vascular bundles are surrounded by large thick-walled cells - Conjunctive tissue
- (3) Cells of medullary rays that form part of cambial ring - Interfascicular cambium
- (4) Loose parenchyma cells rupturing the epidermis and forming a lens-shaped opening in bark - Spongy parenchyma

142. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as :

- (1) Satellite DNA
- (2) Repetitive DNA
- (3) Single nucleotides
- (4) Polymorphic DNA

143. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

- (1) Poaceae ; Rosaceae
- (2) Poaceae ; Leguminosae
- (3) Poaceae ; Solanaceae
- (4) Rosaceae ; Leguminosae

144. Which of the following statements is **incorrect** ?

- (1) Both ATP and NADPH + H⁺ are synthesized during non-cyclic photophosphorylation.
- (2) Stroma lamellae have PS I only and lack NADP reductase.
- (3) Grana lamellae have both PS I and PS II.
- (4) Cyclic photophosphorylation involves both PS I and PS II.

145. What is the role of RNA polymerase III in the process of transcription in eukaryotes ?

- (1) Transcribes rRNAs (28S, 18S and 5.8S)
- (2) Transcribes tRNA, 5s rRNA and snRNA
- (3) Transcribes precursor of mRNA
- (4) Transcribes only snRNAs

146. In the exponential growth equation

$$N_t = N_0 e^{rt}, e \text{ represents :}$$

- (1) The base of number logarithms
- (2) The base of exponential logarithms
- (3) The base of natural logarithms
- (4) The base of geometric logarithms

147. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because :

- (1) mutated gene partially appears on a photographic film.
- (2) mutated gene completely and clearly appears on a photographic film.
- (3) mutated gene does not appear on a photographic film as the probe has no complimentarity with it.
- (4) mutated gene does not appear on photographic film as the probe has complimentarity with it.

148. Match List - I with List - II.

List - I		List - II	
(a)	Protein	(i)	C = C double bonds
(b)	Unsaturated fatty acid	(ii)	Phosphodiester bonds
(c)	Nucleic acid	(iii)	Glycosidic bonds
(d)	Polysaccharide	(iv)	Peptide bonds

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (ii) | (iii) |
| (2) | (i) | (iv) | (iii) | (ii) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (iv) | (iii) | (i) | (ii) |

149. Match List - I with List - II.

List - I		List - II	
(a)	S phase	(i)	Proteins are synthesized
(b)	G ₂ phase	(ii)	Inactive phase
(c)	Quiescent stage	(iii)	Interval between mitosis and initiation of DNA replication
(d)	G ₁ phase	(iv)	DNA replication

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (ii) | (i) | (iv) |
| (2) | (iv) | (ii) | (iii) | (i) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (ii) | (iv) | (iii) | (i) |

150. Plasmid pBR322 has PstI restriction enzyme site within gene *amp^R* that confers ampicillin resistance. If this enzyme is used for inserting a gene for β -galactoside production and the recombinant plasmid is inserted in an *E.coli* strain

- (1) it will not be able to confer ampicillin resistance to the host cell.
- (2) the transformed cells will have the ability to resist ampicillin as well as produce β -galactoside.
- (3) it will lead to lysis of host cell.
- (4) it will be able to produce a novel protein with dual ability.

Section - A (Biology : Zoology)

151. Identify the **incorrect** pair.

- | | | | |
|-----|-----------|---|----------------|
| (1) | Alkaloids | - | Codeine |
| (2) | Toxin | - | Abrin |
| (3) | Lectins | - | Concanavalin A |
| (4) | Drugs | - | Ricin |

152. The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G₁ phase is 8, what would be the number of chromosomes after S phase ?

- (1) 8
- (2) 16
- (3) 4
- (4) 32

153. Match List - I with List - II.

List - I		List - II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (i) | (ii) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (iv) | (i) | (ii) | (iii) |

154. Match List - I with List - II.

List - I		List - II	
(a)	Vaults	(i)	Entry of sperm through Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of fallopian tube

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (ii) | (i) | (iii) |
| (2) | (i) | (iii) | (ii) | (iv) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (iii) | (i) | (iv) | (ii) |

155. If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it ?

- (1) T : 20 ; G : 30 ; C : 20
- (2) T : 20 ; G : 20 ; C : 30
- (3) T : 30 ; G : 20 ; C : 20
- (4) T : 20 ; G : 25 ; C : 25

156. Which of the following RNAs is not required for the synthesis of protein ?

- (1) mRNA
- (2) tRNA
- (3) rRNA
- (4) siRNA

157. Which one of the following is an example of Hormone releasing IUD ?

- (1) CuT
- (2) LNG 20
- (3) Cu 7
- (4) Multiload 375

158. Succus entericus is referred to as :

- (1) Pancreatic juice
- (2) Intestinal juice
- (3) Gastric juice
- (4) Chyme

159. Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :

- (1) Arthritis
- (2) Muscular dystrophy
- (3) Myasthenia gravis
- (4) Gout

160. With regard to insulin choose correct options.

- (a) C-peptide is not present in mature insulin.
- (b) The insulin produced by rDNA technology has C-peptide.
- (c) The pro-insulin has C-peptide.
- (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

- (1) (b) and (d) only
- (2) (b) and (c) only
- (3) (a), (c) and (d) only
- (4) (a) and (d) only

161. Which one of the following belongs to the family Muscidae ?

- (1) Fire fly
- (2) Grasshopper
- (3) Cockroach
- (4) House fly

162. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?

- (1) DNA dependent DNA polymerase
- (2) DNA dependent RNA polymerase
- (3) DNA Ligase
- (4) DNase

163. Receptors for sperm binding in mammals are present on :

- (1) Corona radiata
- (2) Vitelline membrane
- (3) Perivitelline space
- (4) Zona pellucida

164. The centriole undergoes duplication during :

- (1) S-phase
- (2) Prophase
- (3) Metaphase
- (4) G₂ phase

165. Which one of the following organisms bears hollow and pneumatic long bones ?

- (1) *Neophron*
- (2) *Hemidactylus*
- (3) *Macropus*
- (4) *Ornithorhynchus*

166. Erythropoietin hormone which stimulates R.B.C. formation is produced by :

- (1) Alpha cells of pancreas
- (2) The cells of rostral adenohypophysis
- (3) The cells of bone marrow
- (4) Juxtaglomerular cells of the kidney

167. Match the following :

List - I		List - II	
(a)	<i>Physalia</i>	(i)	Pearl oyster
(b)	<i>Limulus</i>	(ii)	Portuguese Man of War
(c)	<i>Ancylostoma</i>	(iii)	Living fossil
(d)	<i>Pinctada</i>	(iv)	Hookworm

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (i) | (iv) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (i) | (iv) | (iii) | (ii) |

168. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?

- (1) Annealing
- (2) Extension
- (3) Denaturation
- (4) Ligation

169. Which of the following statements wrongly represents the nature of smooth muscle ?

- (1) These muscle have no striations
- (2) They are involuntary muscles
- (3) Communication among the cells is performed by intercalated discs
- (4) These muscles are present in the wall of blood vessels

170. The organelles that are included in the endomembrane system are :

- (1) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
- (2) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles
- (3) Golgi complex, Mitochondria, Ribosomes and Lysosomes
- (4) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes

171. Match List - I with List - II.

List - I		List - II	
(a)	<i>Aspergillus niger</i>	(i)	Acetic Acid
(b)	<i>Acetobacter aceti</i>	(ii)	Lactic Acid
(c)	<i>Clostridium butylicum</i>	(iii)	Citric Acid
(d)	<i>Lactobacillus</i>	(iv)	Butyric Acid

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (i) | (iv) | (ii) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (ii) | (iii) | (i) | (iv) |
| (4) | (iv) | (ii) | (i) | (iii) |

172. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection ?

- (1) Western Blotting Technique
- (2) Southern Blotting Technique
- (3) ELISA Technique
- (4) Hybridization Technique

173. Which of the following characteristics is **incorrect** with respect to cockroach ?

- (1) A ring of gastric caeca is present at the junction of midgut and hind gut.
- (2) Hypopharynx lies within the cavity enclosed by the mouth parts.
- (3) In females, 7th-9th sterna together form a genital pouch.
- (4) 10th abdominal segment in both sexes, bears a pair of anal cerci.

174. Persons with 'AB' blood group are called as "Universal recipients". This is due to :

- (1) Absence of antigens A and B on the surface of RBCs
- (2) Absence of antigens A and B in plasma
- (3) Presence of antibodies, anti-A and anti-B, on RBCs
- (4) Absence of antibodies, anti-A and anti-B, in plasma

- 175.** Dobson units are used to measure thickness of :
- (1) CFCs
 - (2) Stratosphere
 - (3) Ozone
 - (4) Troposphere
- 176.** Read the following statements.
- (a) Metagenesis is observed in Helminths.
 - (b) Echinoderms are triploblastic and coelomate animals.
 - (c) Round worms have organ-system level of body organization.
 - (d) Comb plates present in ctenophores help in digestion.
 - (e) Water vascular system is characteristic of Echinoderms.
- Choose the **correct** answer from the options given below.
- (1) (c), (d) and (e) are correct
 - (2) (a), (b) and (c) are correct
 - (3) (a), (d) and (e) are correct
 - (4) (b), (c) and (e) are correct
- 177.** In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?
- (1) 50%
 - (2) 75%
 - (3) 25%
 - (4) 100%
- 178.** Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?
- (1) Thrombin
 - (2) Renin
 - (3) Epinephrine
 - (4) Thrombokinase
- 179.** Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.
- (1) High pO_2 , low pCO_2 , less H^+ , lower temperature
 - (2) Low pO_2 , high pCO_2 , more H^+ , higher temperature
 - (3) High pO_2 , high pCO_2 , less H^+ , higher temperature
 - (4) Low pO_2 , low pCO_2 , more H^+ , higher temperature
- 180.** Sphincter of oddi is present at :
- (1) Ileo-caecal junction
 - (2) Junction of hepato-pancreatic duct and duodenum
 - (3) Gastro-oesophageal junction
 - (4) Junction of jejunum and duodenum
- 181.** Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature ?
- (1) Leptotene
 - (2) Zygotene
 - (3) Diakinesis
 - (4) Pachytene
- 182.** Which of the following is **not** an objective of Biofortification in crops ?
- (1) Improve protein content
 - (2) Improve resistance to diseases
 - (3) Improve vitamin content
 - (4) Improve micronutrient and mineral content
- 183.** The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are :
- (1) $pO_2 = 104$ and $pCO_2 = 40$
 - (2) $pO_2 = 40$ and $pCO_2 = 45$
 - (3) $pO_2 = 95$ and $pCO_2 = 40$
 - (4) $pO_2 = 159$ and $pCO_2 = 0.3$
- 184.** Veneral diseases can spread through :
- (a) Using sterile needles
 - (b) Transfusion of blood from infected person
 - (c) Infected mother to foetus
 - (d) Kissing
 - (e) Inheritance
- Choose the **correct** answer from the options given below.
- (1) (a), (b) and (c) only
 - (2) (b), (c) and (d) only
 - (3) (b) and (c) only
 - (4) (a) and (c) only
- 185.** A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :
- (1) Degenerate primer sequence
 - (2) Okazaki sequences
 - (3) Palindromic Nucleotide sequences
 - (4) Poly(A) tail sequences

Section - B (Biology : Zoology)

186. Which one of the following statements about Histones is **wrong** ?

- (1) Histones are organized to form a unit of 8 molecules.
- (2) The pH of histones is slightly acidic.
- (3) Histones are rich in amino acids - Lysine and Arginine.
- (4) Histones carry positive charge in the side chain.

187. During muscular contraction which of the following events occur ?

- (a) 'H' zone disappears
- (b) 'A' band widens
- (c) 'T' band reduces in width
- (d) Myosine hydrolyzes ATP, releasing the ADP and Pi
- (e) Z-lines attached to actins are pulled inwards

Choose the **correct** answer from the options given below.

- (1) (a), (c), (d), (e) only
- (2) (a), (b), (c), (d) only
- (3) (b), (c), (d), (e) only
- (4) (b), (d), (e), (a) only

188. Match List - I with List - II.

List - I		List - II	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (iii) | (ii) | (iv) |
| (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iv) | (ii) | (iii) | (i) |
| (4) | (iv) | (iii) | (ii) | (i) |

189. Which of these is not an important component of initiation of parturition in humans ?

- (1) Increase in estrogen and progesterone ratio
- (2) Synthesis of prostaglandins
- (3) Release of Oxytocin
- (4) Release of Prolactin

190. Assertion (A) :

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

191. Following are the statements with reference to 'lipids'.

- (a) Lipids having only single bonds are called unsaturated fatty acids.
- (b) Lecithin is a phospholipid.
- (c) Trihydroxy propane is glycerol.
- (d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
- (e) Arachidonic acid has 16 carbon atoms.

Choose the **correct** answer from the options given below.

- (1) (a) and (b) only
- (2) (c) and (d) only
- (3) (b) and (c) only
- (4) (b) and (e) only

192. The Adenosine deaminase deficiency results into :

- (1) Dysfunction of Immune system
- (2) Parkinson's disease
- (3) Digestive disorder
- (4) Addison's disease

193. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?

- (1) Graafian follicle
- (2) Corpus luteum
- (3) Foetus
- (4) Uterus

194. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.

- (1) Gap junctions and Adhering junctions, respectively.
- (2) Tight junctions and Gap junctions, respectively.
- (3) Adhering junctions and Tight junctions, respectively.
- (4) Adhering junctions and Gap junctions, respectively.

195. Which of the following is **not** a step in Multiple Ovulation Embryo Transfer Technology (MOET)?

- (1) Cow is administered hormone having LH like activity for super ovulation
- (2) Cow yields about 6-8 eggs at a time
- (3) Cow is fertilized by artificial insemination
- (4) Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage

196. Match List - I with List - II.

List - I		List - II	
(a)	Filariasis	(i)	<i>Haemophilus influenzae</i>
(b)	Amoebiasis	(ii)	<i>Trichophyton</i>
(c)	Pneumonia	(iii)	<i>Wuchereria bancrofti</i>
(d)	Ringworm	(iv)	<i>Entamoeba histolytica</i>

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (iii) | (ii) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (ii) | (iii) | (i) | (iv) |

197. Match List - I with List - II.

List - I		List - II	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical adaptation	(iv)	Polar seal

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (ii) | (iii) | (i) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

198. Match List - I with List - II.

List - I		List - II	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (ii) | (i) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (i) | (iv) | (iii) | (ii) |

199. **Statement I :**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true
- (2) Both **Statement I** and **Statement II** are false
- (3) **Statement I** is correct but **Statement II** is false
- (4) **Statement I** is incorrect but **Statement II** is true

200. Following are the statements about prostomium of earthworm.

- (a) It serves as a covering for mouth.
- (b) It helps to open cracks in the soil into which it can crawl.
- (c) It is one of the sensory structures.
- (d) It is the first body segment.

Choose the **correct** answer from the options given below.

- (1) (a), (b) and (c) are correct
- (2) (a), (b) and (d) are correct
- (3) (a), (b), (c) and (d) are correct
- (4) (b) and (c) are correct

M1

26

Space For Rough Work

Space For Rough Work

M1

28

Space For Rough Work

Test Booklet Code

M6

AJHGAA

No. :

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on OFFICE Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos – 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is M6. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

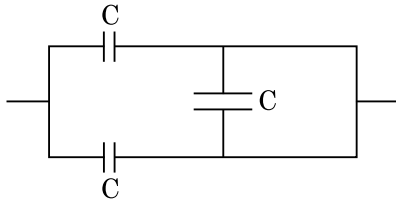
Candidate's Signature : _____ Invigilator's Signature : _____

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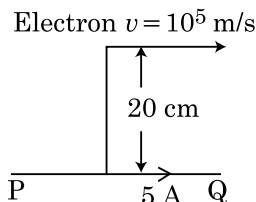
Centre Superintendent : _____

Section - A (Physics)

1. The equivalent capacitance of the combination shown in the figure is :

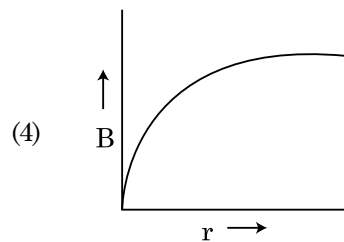
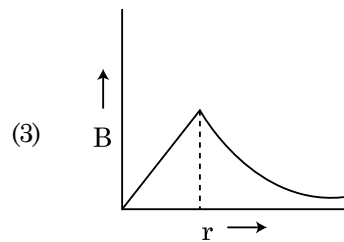
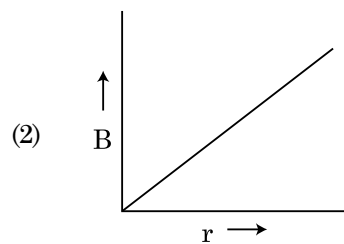
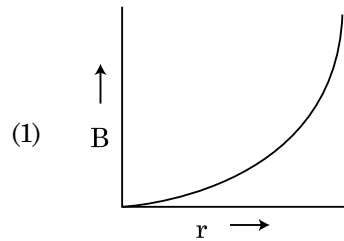


- (1) $3C$
 (2) $2C$
 (3) $C/2$
 (4) $3C/2$
2. Polar molecules are the molecules :
- (1) having zero dipole moment.
 (2) acquire a dipole moment only in the presence of electric field due to displacement of charges.
 (3) acquire a dipole moment only when magnetic field is absent.
 (4) having a permanent electric dipole moment.
3. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



- (1) 4×10^{-20} N
 (2) $8\pi \times 10^{-20}$ N
 (3) $4\pi \times 10^{-20}$ N
 (4) 8×10^{-20} N

4. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by :



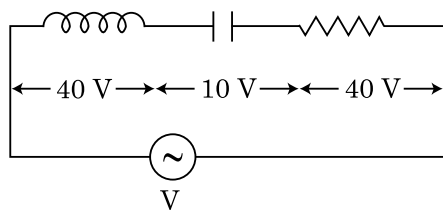
5. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs ?
- (1) 60 cm
 (2) 21.6 cm
 (3) 64 cm
 (4) 62 cm

6. For a plane electromagnetic wave propagating in x -direction, which one of the following combination gives the correct possible directions for electric field ($\vec{E}$) and magnetic field ($\vec{B}$) respectively ?

- (1) $\hat{j} + \hat{k}, \hat{j} + \hat{k}$
 (2) $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$
 (3) $\hat{j} + \hat{k}, -\hat{j} - \hat{k}$
 (4) $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$

7. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.

Potential difference across L , C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is :



- (1) $4\sqrt{2} \Omega$
 (2) $5/\sqrt{2} \Omega$
 (3) 4Ω
 (4) 5Ω
8. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be : ($h = 6.6 \times 10^{-34}$ Js)

- (1) 10^{18}
 (2) 10^{17}
 (3) 10^{16}
 (4) 10^{15}

9. An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of negligible work function. If ' m ' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d , then :

- (1) $\lambda = \left(\frac{2m}{hc} \right) \lambda_d^2$
 (2) $\lambda_d = \left(\frac{2mc}{h} \right) \lambda^2$
 (3) $\lambda = \left(\frac{2mc}{h} \right) \lambda_d^2$
 (4) $\lambda = \left(\frac{2h}{mc} \right) \lambda_d^2$

10. **Column - I** gives certain physical terms associated with flow of current through a metallic conductor. **Column - II** gives some mathematical relations involving electrical quantities. Match **Column - I** and **Column - II** with appropriate relations.

Column - I	Column - II
(A) Drift Velocity	(P) $\frac{m}{ne^2 \rho}$
(B) Electrical Resistivity	(Q) nev_d
(C) Relaxation Period	(R) $\frac{eE}{m} \tau$
(D) Current Density	(S) $\frac{E}{J}$
(1) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)	
(2) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)	
(3) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)	
(4) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)	

11. The escape velocity from the Earth's surface is v . The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is :

- (1) v
 (2) $2v$
 (3) $3v$
 (4) $4v$

12. The velocity of a small ball of mass M and density d , when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be :

- (1) $\frac{Mg}{2}$
- (2) Mg
- (3) $\frac{3}{2}Mg$
- (4) $2Mg$

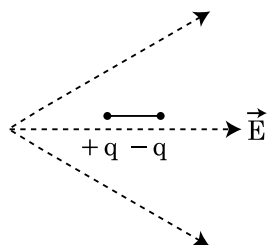
13. A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is :

- (1) n
- (2) $2n$
- (3) $3n$
- (4) $4n$

14. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine ? ($g = 10 \text{ m/s}^2$)

- (1) 10.2 kW
- (2) 8.1 kW
- (3) 12.3 kW
- (4) 7.0 kW

15. A dipole is placed in an electric field as shown. In which direction will it move ?



- (1) towards the left as its potential energy will increase.
- (2) towards the right as its potential energy will decrease.
- (3) towards the left as its potential energy will decrease.
- (4) towards the right as its potential energy will increase.

16. A capacitor of capacitance 'C', is connected across an ac source of voltage V , given by

$$V = V_0 \sin \omega t$$

The displacement current between the plates of the capacitor, would then be given by :

- (1) $I_d = V_0 \omega C \cos \omega t$
- (2) $I_d = \frac{V_0}{\omega C} \cos \omega t$
- (3) $I_d = \frac{V_0}{\omega C} \sin \omega t$
- (4) $I_d = V_0 \omega C \sin \omega t$

17. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C . The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is :

- (1) $\frac{13}{10}t$
- (2) $\frac{13}{5}t$
- (3) $\frac{10}{13}t$
- (4) $\frac{5}{13}t$

18. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25Ω . What will be the effective resistance if they are connected in series ?

- (1) 0.25Ω
- (2) 0.5Ω
- (3) 1Ω
- (4) 4Ω

19. Match **Column - I** and **Column - II** and choose the **correct** match from the given choices.

Column - I	Column - II
(A) Root mean square speed of gas molecules	(P) $\frac{1}{3}nm\bar{v}^2$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3RT}{M}}$
(C) Average kinetic energy of a molecule	(R) $\frac{5}{2}RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2}k_B T$
(1) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)	
(2) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)	
(3) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)	
(4) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)	

20. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$ is :

- (1) $\frac{2n-1}{2n}$
- (2) $\frac{2n-1}{2n+1}$
- (3) $\frac{2n+1}{2n-1}$
- (4) $\frac{2n}{2n-1}$

21. A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence

${}^A_ZX \rightarrow {}^{Z-1}_B \rightarrow {}^{Z-3}_C \rightarrow {}^{Z-2}_D$, where Z is the atomic number of element X . The possible decay particles in the sequence are :

- (1) α, β^-, β^+
- (2) α, β^+, β^-
- (3) β^+, α, β^-
- (4) β^-, α, β^+

22. A screw gauge gives the following readings when used to measure the diameter of a wire

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is :

- (1) 0.52 cm
- (2) 0.026 cm
- (3) 0.26 cm
- (4) 0.052 cm

23. A parallel plate capacitor has a uniform electric field ' $\vec{E}$ ' in the space between the plates. If the distance between the plates is ' d ' and the area of each plate is ' A ', the energy stored in the capacitor is : (ϵ_0 = permittivity of free space)

- (1) $\frac{1}{2}\epsilon_0 E^2$
- (2) $\epsilon_0 EAd$
- (3) $\frac{1}{2}\epsilon_0 E^2 Ad$
- (4) $\frac{E^2 Ad}{\epsilon_0}$

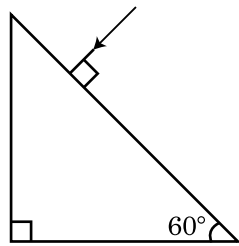
24. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

- (1) current in n-type = current in p-type.
- (2) current in p-type > current in n-type.
- (3) current in n-type > current in p-type.
- (4) No current will flow in p-type, current will only flow in n-type.

25. A convex lens ' A ' of focal length 20 cm and a concave lens ' B ' of focal length 5 cm are kept along the same axis with a distance ' d ' between them. If a parallel beam of light falling on ' A ' leaves ' B ' as a parallel beam, then the distance ' d ' in cm will be :

- (1) 25
- (2) 15
- (3) 50
- (4) 30

26. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



- (1) 60°
 (2) 30°
 (3) 45°
 (4) 90°
27. Consider the following **statements (A) and (B)** and identify the **correct** answer.
- (A)** A zener diode is connected in reverse bias, when used as a voltage regulator.
(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.
- (1) **(A)** and **(B)** both are correct.
 (2) **(A)** and **(B)** both are incorrect.
 (3) **(A)** is correct and **(B)** is incorrect.
 (4) **(A)** is incorrect but **(B)** is correct.
28. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :
- (1) $\frac{R_1}{R_2}$
 (2) $\frac{R_2}{R_1}$
 (3) $\sqrt{\left(\frac{R_1}{R_2}\right)}$
 (4) $\frac{R_1^2}{R_2^2}$
29. If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.
- (1) [F] [A] [T]
 (2) [F] [A] [T²]
 (3) [F] [A] [T⁻¹]
 (4) [F] [A⁻¹] [T]

30. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of :

- (1) [M²] [L⁻¹] [T⁰]
 (2) [M] [L⁻¹] [T⁻¹]
 (3) [M] [L⁰] [T⁰]
 (4) [M²] [L⁻²] [T⁻¹]

31. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since :

- (1) a large aperture contributes to the quality and visibility of the images.
 (2) a large area of the objective ensures better light gathering power.
 (3) a large aperture provides a better resolution.
 (4) all of the above.

32. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is :

- (1) 0.9 MeV
 (2) 9.4 MeV
 (3) 804 MeV
 (4) 216 MeV

33. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively :

- (1) $\frac{S}{4}, \frac{3gS}{2}$
 (2) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
 (3) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$
 (4) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

34. The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be :

- (1) $\frac{1}{2}$
 (2) $\frac{1}{2\sqrt{2}}$
 (3) $\frac{2}{3}$
 (4) $\frac{2}{3\sqrt{2}}$

35. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is :

- (1) 0.0628 s
 (2) 6.28 s
 (3) 3.14 s
 (4) 0.628 s

Section - B (Physics)

36. A series LCR circuit containing 5.0 H inductor, $80 \mu\text{F}$ capacitor and 40Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be :

- (1) 25 rad/s and 75 rad/s
 (2) 50 rad/s and 25 rad/s
 (3) 46 rad/s and 54 rad/s
 (4) 42 rad/s and 58 rad/s

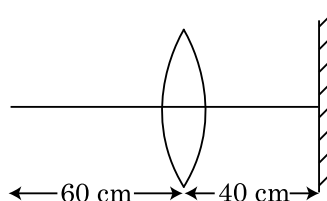
37. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 > R_2$, the mutual inductance M between them will be directly proportional to :

- (1) $\frac{R_1}{R_2}$
 (2) $\frac{R_2}{R_1}$
 (3) $\frac{R_1^2}{R_2}$
 (4) $\frac{R_2^2}{R_1}$

38. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly :

- (1) 0 kg m/s
 (2) 4.2 kg m/s
 (3) 2.1 kg m/s
 (4) 1.4 kg m/s

39. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of :



- (1) 20 cm from the lens, it would be a real image.
 (2) 30 cm from the lens, it would be a real image.
 (3) 30 cm from the plane mirror, it would be a virtual image.
 (4) 20 cm from the plane mirror, it would be a virtual image.

40. A uniform conducting wire of length $12a$ and resistance ' R ' is wound up as a current carrying coil in the shape of,

- (i) an equilateral triangle of side ' a '.
 (ii) a square of side ' a '.

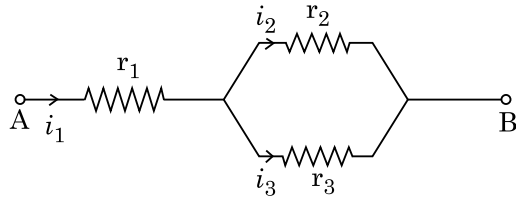
The magnetic dipole moments of the coil in each case respectively are :

- (1) $\sqrt{3} Ia^2$ and $3 Ia^2$
 (2) $3 Ia^2$ and Ia^2
 (3) $3 Ia^2$ and $4 Ia^2$
 (4) $4 Ia^2$ and $3 Ia^2$

41. A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$? (Take $g = 10 \text{ m/s}^2$)

- (1) 20 m/s, 5 m/s^2
 (2) 20 m/s, 0
 (3) $20\sqrt{2} \text{ m/s}$, 0
 (4) $20\sqrt{2} \text{ m/s}$, 10 m/s^2

42. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is :

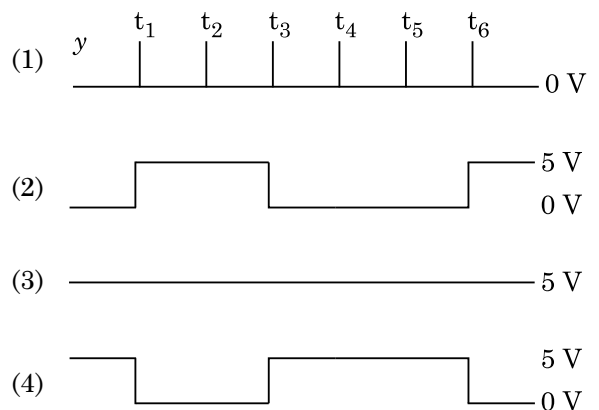
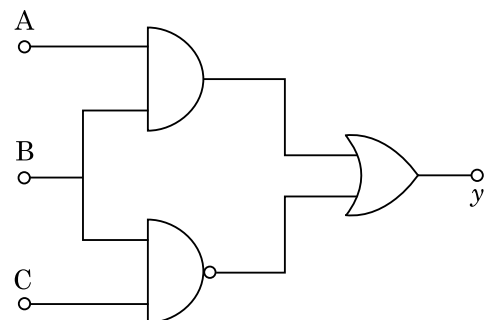
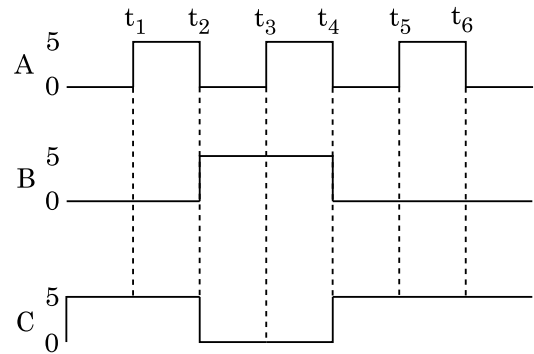


- (1) $\frac{r_1}{r_2 + r_3}$
- (2) $\frac{r_2}{r_2 + r_3}$
- (3) $\frac{r_1}{r_1 + r_2}$
- (4) $\frac{r_2}{r_1 + r_3}$
43. A particle of mass 'm' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth. (V_e = escape velocity)
- The maximum height above the surface reached by the particle is :
- (1) $R \left(\frac{k}{1-k} \right)^2$
- (2) $R \left(\frac{k}{1+k} \right)^2$
- (3) $\frac{R^2 k}{1+k}$
- (4) $\frac{Rk^2}{1-k^2}$
44. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit ?
- (1) 0.2 A
- (2) 0.4 A
- (3) 2 A
- (4) 4 A

45. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

- (1) 660 V
- (2) 1320 V
- (3) 1520 V
- (4) 1980 V

46. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y ?



47. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ , is then given by :

(1) $\theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{1/2}$

(2) $\theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$

(3) $\theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$

(4) $\theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{1/2}$

48. From a circular ring of mass ' M ' and radius ' R ' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is ' K ' times ' MR^2 '. Then the value of ' K ' is :

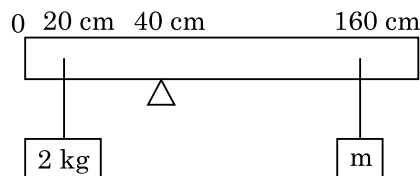
(1) $\frac{3}{4}$

(2) $\frac{7}{8}$

(3) $\frac{1}{4}$

(4) $\frac{1}{8}$

49. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass ' m ' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of ' m ' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$)



(1) $\frac{1}{2} \text{ kg}$

(2) $\frac{1}{3} \text{ kg}$

(3) $\frac{1}{6} \text{ kg}$

(4) $\frac{1}{12} \text{ kg}$

50. In the product

$$\begin{aligned} \vec{F} &= q \left(\vec{v} \times \vec{B} \right) \\ &= q \vec{v} \times \left(B \hat{i} + B \hat{j} + B_0 \hat{k} \right) \end{aligned}$$

For $q = 1$ and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and $\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$

What will be the complete expression for $\vec{B}$?

(1) $-8\hat{i} - 8\hat{j} - 6\hat{k}$

(2) $-6\hat{i} - 6\hat{j} - 8\hat{k}$

(3) $8\hat{i} + 8\hat{j} - 6\hat{k}$

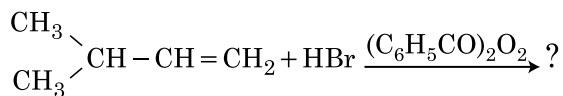
(4) $6\hat{i} + 6\hat{j} - 8\hat{k}$

Section - A (Chemistry)

51. The correct sequence of bond enthalpy of 'C-X' bond is :
- (1) $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} < \text{CH}_3 - \text{Br} < \text{CH}_3 - \text{I}$
 - (2) $\text{CH}_3 - \text{F} > \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
 - (3) $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
 - (4) $\text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{F} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
52. Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature ?
- (1) Electrolysis
 - (2) Chromatography
 - (3) Distillation
 - (4) Zone refining
53. The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is :
- (1) 7
 - (2) 5
 - (3) 2
 - (4) 3
54. Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :
- (1) Calcium chloride
 - (2) Strontium chloride
 - (3) Magnesium chloride
 - (4) Beryllium chloride
55. Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of :
- (1) belonging to same group
 - (2) diagonal relationship
 - (3) lanthanoid contraction
 - (4) having similar chemical properties
56. The maximum temperature that can be achieved in blast furnace is :
- (1) upto 1200 K
 - (2) upto 2200 K
 - (3) upto 1900 K
 - (4) upto 5000 K
57. What is the IUPAC name of the organic compound formed in the following chemical reaction ?
- Acetone $\xrightarrow[\text{(ii) } \text{H}_2\text{O, H}^+]{\text{(i) } \text{C}_2\text{H}_5\text{MgBr, dry Ether}}$ Product
- (1) 2-methyl propan-2-ol
 - (2) pentan-2-ol
 - (3) pentan-3-ol
 - (4) 2-methyl butan-2-ol

58. Which one of the following polymers is prepared by addition polymerisation ?
- (1) Teflon
 - (2) Nylon-66
 - (3) Novolac
 - (4) Dacron
59. Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are :
- (1) 8, 4
 - (2) 6, 12
 - (3) 2, 1
 - (4) 12, 6
60. **Statement I :**
- Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.
- Statement II :**
- As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.
- In the light of the above statements, choose the **correct** answer from the options given below.
- (1) Both **Statement I** and **Statement II** are true.
 - (2) Both **Statement I** and **Statement II** are false.
 - (3) **Statement I** is correct but **Statement II** is false.
 - (4) **Statement I** is incorrect but **Statement II** is true.
61. The **incorrect** statement among the following is :
- (1) Actinoid contraction is greater for element to element than Lanthanoid contraction.
 - (2) Most of the trivalent Lanthanoid ions are colorless in the solid state.
 - (3) Lanthanoids are good conductors of heat and electricity.
 - (4) Actinoids are highly reactive metals, especially when finely divided.

62. The major product of the following chemical reaction is :



- (1) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Br} \\ \diagup \\ \text{CH}_3 \end{array}$
- (2) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{COC}_6\text{H}_5 \\ \diagup \\ \text{CH}_3 \end{array}$
- (3) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} - \text{CH}_3 \\ \diagup \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$
- (4) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CBr} - \text{CH}_2 - \text{CH}_3 \\ \diagup \\ \text{CH}_3 \end{array}$

63. The structures of beryllium chloride in solid state and vapour phase, are :

- (1) Chain and dimer, respectively
- (2) Linear in both
- (3) Dimer and Linear, respectively
- (4) Chain in both

64. Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true.
- (2) Both **Statement I** and **Statement II** are false.
- (3) **Statement I** is correct but **Statement II** is false.
- (4) **Statement I** is incorrect but **Statement II** is true.

65. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]

- (1) CH
- (2) CH₂
- (3) CH₃
- (4) CH₄

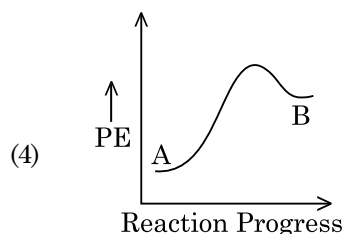
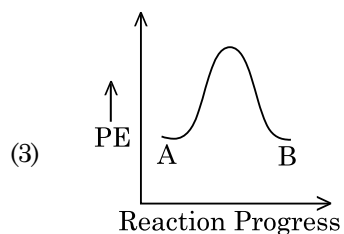
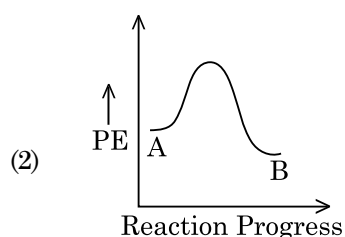
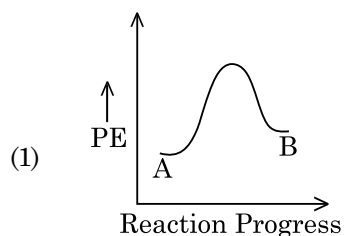
66. The correct structure of 2,6-Dimethyl-dec-4-ene is :

- (1)
- (2)
- (3)
- (4)

67. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on ?

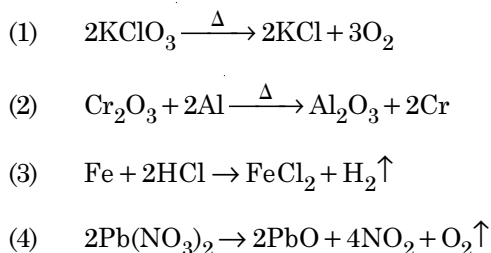
- (1) Saytzeff's Rule
- (2) Hund's Rule
- (3) Hofmann Rule
- (4) Huckel's Rule

68. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.

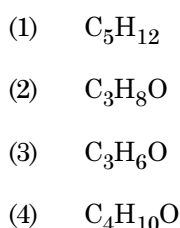


69. Ethylene diaminetetraacetate (EDTA) ion is :
- (1) Hexadentate ligand with four "O" and two "N" donor atoms
 - (2) Unidentate ligand
 - (3) Bidentate ligand with two "N" donor atoms
 - (4) Tridentate ligand with three "N" donor atoms
70. Noble gases are named because of their inertness towards reactivity. Identify an **incorrect** statement about them.
- (1) Noble gases are sparingly soluble in water.
 - (2) Noble gases have very high melting and boiling points.
 - (3) Noble gases have weak dispersion forces.
 - (4) Noble gases have large positive values of electron gain enthalpy.

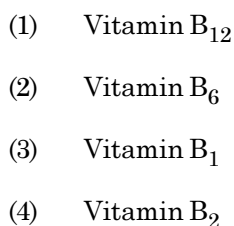
71. Which of the following reactions is the metal displacement reaction ? Choose the right option.



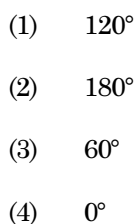
72. The compound which shows metamerism is :



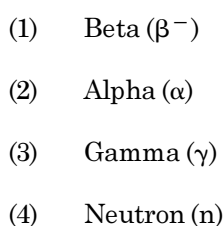
73. The RBC deficiency is deficiency disease of :



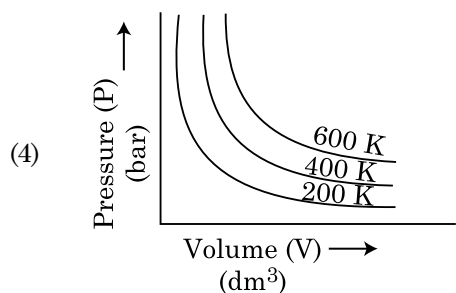
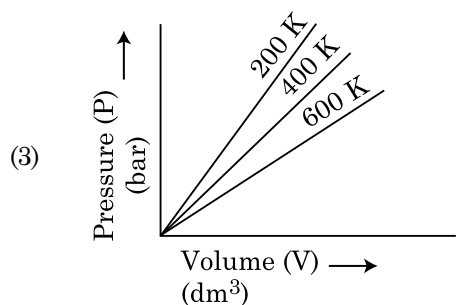
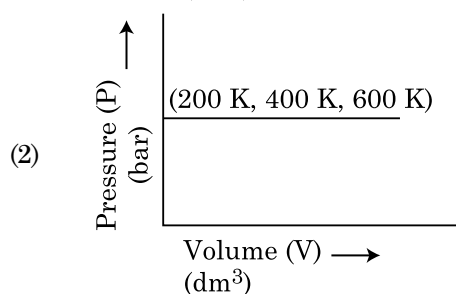
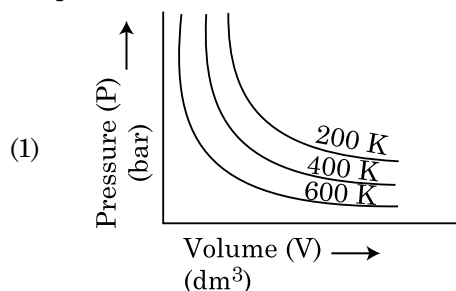
74. Dihedral angle of least stable conformer of ethane is :



75. Tritium, a radioactive isotope of hydrogen, emits which of the following particles ?



76. Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures :



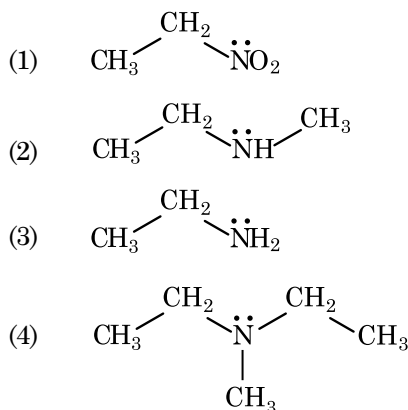
77. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.
- (1) $201.28 \text{ S cm}^2 \text{ mol}^{-1}$
 - (2) $390.71 \text{ S cm}^2 \text{ mol}^{-1}$
 - (3) $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
 - (4) $540.48 \text{ S cm}^2 \text{ mol}^{-1}$
78. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is :
- (1) 8.50
 - (2) 5.50
 - (3) 7.75
 - (4) 6.25

79. Match List - I with List - II.

List - I	List - II
(a) PCl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 - (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 - (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 - (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
80. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.



81. The right option for the statement "Tyndall effect is exhibited by", is :
- (1) NaCl solution
 - (2) Glucose solution
 - (3) Starch solution
 - (4) Urea solution
82. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are :
- (1) sp^3 and 4
 - (2) sp^3 and 6
 - (3) sp^2 and 6
 - (4) sp^2 and 8
83. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light, $c = 3.0 \times 10^8 \text{ ms}^{-1}$]
- (1) 219.3 m
 - (2) 219.2 m
 - (3) 2192 m
 - (4) 21.92 cm

84. Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas ?

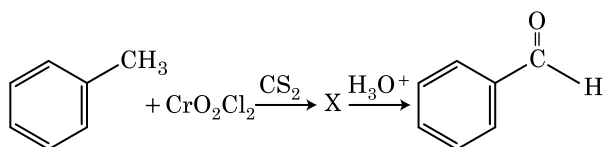
- (1) $C_P + C_V = R$
- (2) $C_P - C_V = R$
- (3) $C_P = RC_V$
- (4) $C_V = RC_P$

85. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :

- (1) $P_2 > P_1 > P_3$
- (2) $P_1 > P_2 > P_3$
- (3) $P_2 > P_3 > P_1$
- (4) $P_3 > P_1 > P_2$

Section - B (Chemistry)

86. The intermediate compound 'X' in the following chemical reaction is :



- (1)
- (2)
- (3)
- (4)

87. For irreversible expansion of an ideal gas under isothermal condition, the correct option is :

- (1) $\Delta U = 0, \Delta S_{\text{total}} = 0$
- (2) $\Delta U \neq 0, \Delta S_{\text{total}} \neq 0$
- (3) $\Delta U = 0, \Delta S_{\text{total}} \neq 0$
- (4) $\Delta U \neq 0, \Delta S_{\text{total}} = 0$

88. Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one litre at 0°C is :

[Given $R = 0.082 \text{ L atm mol}^{-1}\text{K}^{-1}$, $T = 273 \text{ K}$]

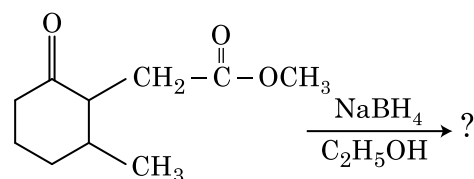
- (1) 2.518
- (2) 2.602
- (3) 25.18
- (4) 26.02

89. The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is :

[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]

- (1) 160 mm of Hg
- (2) 168 mm of Hg
- (3) 336 mm of Hg
- (4) 350 mm of Hg

90. The product formed in the following chemical reaction is :



- (1)
- (2)
- (3)
- (4)

91. Which of the following molecules is non-polar in nature ?

- (1) POCl_3
- (2) CH_2O
- (3) SbCl_5
- (4) NO_2

92. From the following pairs of ions which one is not an iso-electronic pair ?

- (1) O^{2-} , F^-
- (2) Na^+ , Mg^{2+}
- (3) Mn^{2+} , Fe^{3+}
- (4) Fe^{2+} , Mn^{2+}

93. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid ? Choose the correct option.

$$\left[\begin{array}{l} \Lambda_{\text{H}^+}^\circ = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda_{\text{CH}_3\text{COO}^-}^\circ = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$

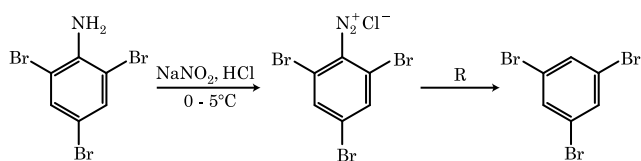
- (1) $1.75 \times 10^{-4} \text{ mol L}^{-1}$
- (2) $2.50 \times 10^{-4} \text{ mol L}^{-1}$
- (3) $1.75 \times 10^{-5} \text{ mol L}^{-1}$
- (4) $2.50 \times 10^{-5} \text{ mol L}^{-1}$

94. The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T} \right)$ of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is. Choose the correct option for your answer.

[Given $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$]

- (1) 41.5 kJ mol^{-1}
- (2) 83.0 kJ mol^{-1}
- (3) 166 kJ mol^{-1}
- (4) -83 kJ mol^{-1}

95. The reagent 'R' in the given sequence of chemical reaction is :



- (1) H_2O
- (2) $\text{CH}_3\text{CH}_2\text{OH}$
- (3) HI
- (4) CuCN/KCN

96. Match List - I with List - II.

- | List - I | | List - II |
|--|-------|--------------------------------|
| (a) $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$ | (i) | Hell-Volhard-Zelinsky reaction |
| (b) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$ | (ii) | Gattermann-Koch reaction |
| (c) $\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$ | (iii) | Haloform reaction |
| (d) $\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$ | (iv) | Esterification |

Choose the **correct** answer from the options given below.

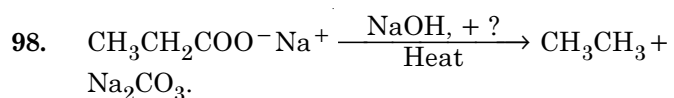
- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
- (3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

97. Match List - I with List - II.

- | List - I | | List - II |
|--|-------|------------------------|
| (a) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ | (i) | Acid rain |
| (b) $\text{HOCl}(\text{g}) \xrightarrow{h\nu} \dot{\text{O}}\text{H} + \dot{\text{Cl}}$ | (ii) | Smog |
| (c) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$ | (iii) | Ozone depletion |
| (d) $\text{NO}_2(\text{g}) \xrightarrow{h\nu} \text{NO}(\text{g}) + \text{O}(\text{g})$ | (iv) | Tropospheric pollution |

Choose the **correct** answer from the options given below.

- (1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)



Consider the above reaction and identify the missing reagent/chemical.

- (1) B_2H_6
- (2) Red Phosphorus
- (3) CaO
- (4) DIBAL-H

99. Match List - I with List - II.

List - I		List - II	
(a)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(b)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(c)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(d)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
- (3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

100. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?

- (1) $\text{HF} < \text{HCl}$: Increasing acidic strength
 $< \text{HBr} < \text{HI}$
- (2) $\text{H}_2\text{O} < \text{H}_2\text{S}$: Increasing pK_a values
 $< \text{H}_2\text{Se} < \text{H}_2\text{Te}$
- (3) $\text{NH}_3 < \text{PH}_3$: Increasing acidic character
 $< \text{AsH}_3 < \text{SbH}_3$
- (4) $\text{CO}_2 < \text{SiO}_2$: Increasing oxidizing power
 $< \text{SnO}_2 < \text{PbO}_2$

Section - A (Biology : Botany)

101. Match List - I with List - II.

List - I		List - II	
(a)	Protoplast fusion	(i)	Totipotency
(b)	Plant tissue culture	(ii)	Pomato
(c)	Meristem culture	(iii)	Somaclones
(d)	Micropropagation	(iv)	Virus free plants

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

102. In the equation $\text{GPP} - \text{R} = \text{NPP}$
R represents :

- (1) Radiant energy
- (2) Retardation factor
- (3) Environment factor
- (4) Respiration losses

103. Which of the following are **not** secondary metabolites in plants ?

- (1) Morphine, codeine
- (2) Amino acids, glucose
- (3) Vinblastin, curcumin
- (4) Rubber, gums

104. The factor that leads to Founder effect in a population is :

- (1) Natural selection
- (2) Genetic recombination
- (3) Mutation
- (4) Genetic drift

105. Amensalism can be represented as :

- (1) Species A (–) ; Species B (0)
- (2) Species A (+) ; Species B (+)
- (3) Species A (–) ; Species B (–)
- (4) Species A (+) ; Species B (0)

106. A typical angiosperm embryo sac at maturity is :

- (1) 8-nucleate and 7-celled
- (2) 7-nucleate and 8-celled
- (3) 7-nucleate and 7-celled
- (4) 8-nucleate and 8-celled

107. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out :

- (1) RNA
- (2) DNA
- (3) Histones
- (4) Polysaccharides

108. Gemmae are present in :

- (1) Mosses
- (2) Pteridophytes
- (3) Some Gymnosperms
- (4) Some Liverworts

109. Which of the following stages of meiosis involves division of centromere ?

- (1) Metaphase I
- (2) Metaphase II
- (3) Anaphase II
- (4) Telophase II

110. Match List - I with List - II.

List - I		List - II	
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (iv) (i) (iii) (ii)
- (2) (iii) (i) (iv) (ii)
- (3) (ii) (iii) (iv) (i)
- (4) (iv) (ii) (i) (iii)

111. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called :

- (1) Elasticity
- (2) Flexibility
- (3) Plasticity
- (4) Maturity

112. The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is :

- (1) Xenogamy
- (2) Geitonogamy
- (3) Chasmogamy
- (4) Cleistogamy

113. Which of the following plants is monoecious ?

- (1) *Carica papaya*
- (2) *Chara*
- (3) *Marchantia polymorpha*
- (4) *Cycas circinalis*

114. Complete the flow chart on central dogma.

(a) $\text{DNA} \xrightarrow{(b)} \text{mRNA} \xrightarrow{(c)} (d)$

- (1) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein
- (2) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Transduction
- (3) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein
- (4) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein

115. Match List - I with List - II.

List - I		List - II	
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (iv) (iii) (ii) (i)
- (2) (i) (iv) (iii) (ii)
- (3) (iii) (iv) (i) (ii)
- (4) (ii) (iii) (iv) (i)

116. Mutations in plant cells can be induced by :

- (1) Kinetin
- (2) Infrared rays
- (3) Gamma rays
- (4) Zeatin

117. Which of the following statements is **not** correct ?

- (1) Pyramid of biomass in sea is generally inverted.
- (2) Pyramid of biomass in sea is generally upright.
- (3) Pyramid of energy is always upright.
- (4) Pyramid of numbers in a grassland ecosystem is upright.

118. Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival ?

- (1) Resource partitioning
- (2) Competitive release
- (3) Mutualism
- (4) Predation

119. Match List - I with List - II.

List - I		List - II	
(a)	Cohesion	(i)	More attraction in liquid phase
(b)	Adhesion	(ii)	Mutual attraction among water molecules
(c)	Surface tension	(iii)	Water loss in liquid phase
(d)	Guttation	(iv)	Attraction towards polar surfaces

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (ii) (iv) (i) (iii)
- (2) (iv) (iii) (ii) (i)
- (3) (iii) (i) (iv) (ii)
- (4) (ii) (i) (iv) (iii)

120. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as :
- (1) Yellow bands
 - (2) Bright orange bands
 - (3) Dark red bands
 - (4) Bright blue bands
121. Which of the following is an **incorrect** statement?
- (1) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
 - (2) Microbodies are present both in plant and animal cells.
 - (3) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
 - (4) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
122. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :
- (1) Biopiracy
 - (2) Gene therapy
 - (3) Molecular diagnosis
 - (4) Safety testing
123. Match List - I with List - II.

List - I		List - II	
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iv) | (i) | (iii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (i) | (ii) | (iii) | (iv) |
| (4) | (iii) | (ii) | (iv) | (i) |
124. Which of the following is a **correct** sequence of steps in a PCR (Polymerase Chain Reaction) ?
- (1) Denaturation, Annealing, Extension
 - (2) Denaturation, Extension, Annealing
 - (3) Extension, Denaturation, Annealing
 - (4) Annealing, Denaturation, Extension

125. Which of the following algae produce Carrageen ?
- (1) Green algae
 - (2) Brown algae
 - (3) Red algae
 - (4) Blue-green algae
126. Which of the following is **not** an application of PCR (Polymerase Chain Reaction) ?
- (1) Molecular diagnosis
 - (2) Gene amplification
 - (3) Purification of isolated protein
 - (4) Detection of gene mutation
127. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as :
- (1) Homosorus
 - (2) Heterosorus
 - (3) Homosporous
 - (4) Heterosporous
128. Diadelphous stamens are found in :
- (1) China rose
 - (2) Citrus
 - (3) Pea
 - (4) China rose and citrus
129. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as :
- (1) Metacentric
 - (2) Telocentric
 - (3) Sub-metacentric
 - (4) Acrocentric
130. Which of the following algae contains mannitol as reserve food material ?
- (1) *Ectocarpus*
 - (2) *Gracilaria*
 - (3) *Volvox*
 - (4) *Ulothrix*
131. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as :
- (1) Climax
 - (2) Climax community
 - (3) Standing state
 - (4) Standing crop
132. The first stable product of CO₂ fixation in sorghum is :
- (1) Pyruvic acid
 - (2) Oxaloacetic acid
 - (3) Succinic acid
 - (4) Phosphoglyceric acid
133. The site of perception of light in plants during photoperiodism is :
- (1) Shoot apex
 - (2) Stem
 - (3) Axillary bud
 - (4) Leaf

134. The plant hormone used to destroy weeds in a field is :

(1) IAA
(2) NAA
(3) 2, 4-D
(4) IBA

135. The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called :

(1) Bullet square
(2) Punch square
(3) Punnett square
(4) Net square

Section - B (Biology : Botany)

136. In the exponential growth equation

$$N_t = N_0 e^{rt}, e \text{ represents :}$$

(1) The base of number logarithms
(2) The base of exponential logarithms
(3) The base of natural logarithms
(4) The base of geometric logarithms

137. Match Column - I with Column - II.

Column - I	Column - II
(a) <i>Nitrococcus</i>	(i) Denitrification
(b) <i>Rhizobium</i>	(ii) Conversion of ammonia to nitrite
(c) <i>Thiobacillus</i>	(iii) Conversion of nitrite to nitrate
(d) <i>Nitrobacter</i>	(iv) Conversion of atmospheric nitrogen to ammonia

Choose the **correct** answer from options given below.

(a) (b) (c) (d)
(1) (ii) (iv) (i) (iii)
(2) (i) (ii) (iii) (iv)
(3) (iii) (i) (iv) (ii)
(4) (iv) (iii) (ii) (i)

138. Match List - I with List - II.

List - I	List - II
(a) S phase	(i) Proteins are synthesized
(b) G ₂ phase	(ii) Inactive phase
(c) Quiescent stage	(iii) Interval between mitosis and initiation of DNA replication
(d) G ₁ phase	(iv) DNA replication

Choose the **correct** answer from the options given below.

(a) (b) (c) (d)
(1) (iii) (ii) (i) (iv)
(2) (iv) (ii) (iii) (i)
(3) (iv) (i) (ii) (iii)
(4) (ii) (iv) (iii) (i)

139. Identify the **correct** statement.

(1) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.
(2) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.
(3) The coding strand in a transcription unit is copied to an mRNA.
(4) Split gene arrangement is characteristic of prokaryotes.

140. Plasmid pBR322 has PstI restriction enzyme site within gene *amp^R* that confers ampicillin resistance. If this enzyme is used for inserting a gene for β -galactoside production and the recombinant plasmid is inserted in an *E.coli* strain

(1) it will not be able to confer ampicillin resistance to the host cell.
(2) the transformed cells will have the ability to resist ampicillin as well as produce β -galactoside.
(3) it will lead to lysis of host cell.
(4) it will be able to produce a novel protein with dual ability.

141. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as :

(1) Satellite DNA
(2) Repetitive DNA
(3) Single nucleotides
(4) Polymorphic DNA

142. Which of the following statements is **correct** ?

(1) Fusion of two cells is called Karyogamy.
(2) Fusion of protoplasts between two motile on non-motile gametes is called plasmogamy.
(3) Organisms that depend on living plants are called saprophytes.
(4) Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.

143. Match Column - I with Column - II.

Column - I	Column - II
(a) $\% \frac{1}{2} K_{(5)} C_{1+2+(2)} A_{(9)+1} G_1$	(i) Brassicaceae
(b) $\oplus \frac{1}{2} K_{(5)} \widehat{C_{(5)} A_5} G_2$	(ii) Liliaceae
(c) $\oplus \frac{1}{2} P_{(3+3)} \widehat{A_{3+3} G_{(3)}}$	(iii) Fabaceae
(d) $\oplus \frac{1}{2} K_{2+2} C_4 A_{2-4} G_{(2)}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

(a) (b) (c) (d)
(1) (iii) (iv) (ii) (i)
(2) (i) (ii) (iii) (iv)
(3) (ii) (iii) (iv) (i)
(4) (iv) (ii) (i) (iii)

144. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because :

- (1) mutated gene partially appears on a photographic film.
- (2) mutated gene completely and clearly appears on a photographic film.
- (3) mutated gene does not appear on a photographic film as the probe has no complementarity with it.
- (4) mutated gene does not appear on photographic film as the probe has complementarity with it.

145. Which of the following statements is **incorrect** ?

- (1) Both ATP and NADPH + H⁺ are synthesized during non-cyclic photophosphorylation.
- (2) Stroma lamellae have PS I only and lack NADP reductase.
- (3) Grana lamellae have both PS I and PS II.
- (4) Cyclic photophosphorylation involves both PS I and PS II.

146. Which of the following statements is **incorrect** ?

- (1) During aerobic respiration, role of oxygen is limited to the terminal stage.
- (2) In ETC (Electron Transport Chain), one molecule of NADH + H⁺ gives rise to 2 ATP molecules, and one FADH₂ gives rise to 3 ATP molecules.
- (3) ATP is synthesized through complex V.
- (4) Oxidation-reduction reactions produce proton gradient in respiration.

147. Match List - I with List - II.

List - I		List - II	
(a)	Protein	(i)	C = C double bonds
(b)	Unsaturated fatty acid	(ii)	Phosphodiester bonds
(c)	Nucleic acid	(iii)	Glycosidic bonds
(d)	Polysaccharide	(iv)	Peptide bonds

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (ii) | (iii) |
| (2) | (i) | (iv) | (iii) | (ii) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (iv) | (iii) | (i) | (ii) |

148. What is the role of RNA polymerase III in the process of transcription in eukaryotes ?

- (1) Transcribes rRNAs (28S, 18S and 5.8S)
- (2) Transcribes tRNA, 5s rRNA and snRNA
- (3) Transcribes precursor of mRNA
- (4) Transcribes only snRNAs

149. Select the **correct** pair.

- (1) Large colorless empty cells in the epidermis of grass leaves - Subsidiary cells
- (2) In dicot leaves, vascular bundles are surrounded by large thick-walled cells - Conjunctive tissue
- (3) Cells of medullary rays that form part of cambial ring - Interfascicular cambium
- (4) Loose parenchyma cells rupturing the epidermis and forming a lens-shaped opening in bark - Spongy parenchyma

150. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

- (1) Poaceae ; Rosaceae
- (2) Poaceae ; Leguminosae
- (3) Poaceae ; Solanaceae
- (4) Rosaceae ; Leguminosae

Section - A (Biology : Zoology)

151. Match List - I with List - II.

List - I		List - II	
(a)	Vaults	(i)	Entry of sperm through Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of fallopian tube

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (ii) | (i) | (iii) |
| (2) | (i) | (iii) | (ii) | (iv) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (iii) | (i) | (iv) | (ii) |

152. Which of the following statements wrongly represents the nature of smooth muscle ?

- (1) These muscle have no striations
- (2) They are involuntary muscles
- (3) Communication among the cells is performed by intercalated discs
- (4) These muscles are present in the wall of blood vessels

153. The organelles that are included in the endomembrane system are :
- (1) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
 - (2) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles
 - (3) Golgi complex, Mitochondria, Ribosomes and Lysosomes
 - (4) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes
154. Succus entericus is referred to as :
- (1) Pancreatic juice
 - (2) Intestinal juice
 - (3) Gastric juice
 - (4) Chyme
155. Which one of the following is an example of Hormone releasing IUD ?
- (1) CuT
 - (2) LNG 20
 - (3) Cu 7
 - (4) Multiload 375
156. Which one of the following belongs to the family Muscidae ?
- (1) Fire fly
 - (2) Grasshopper
 - (3) Cockroach
 - (4) House fly
157. If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it ?
- (1) T : 20 ; G : 30 ; C : 20
 - (2) T : 20 ; G : 20 ; C : 30
 - (3) T : 30 ; G : 20 ; C : 20
 - (4) T : 20 ; G : 25 ; C : 25
158. Receptors for sperm binding in mammals are present on :
- (1) Corona radiata
 - (2) Vitelline membrane
 - (3) Perivitelline space
 - (4) Zona pellucida
159. Which of the following is **not** an objective of Biofortification in crops ?
- (1) Improve protein content
 - (2) Improve resistance to diseases
 - (3) Improve vitamin content
 - (4) Improve micronutrient and mineral content
160. The centriole undergoes duplication during :
- (1) S-phase
 - (2) Prophase
 - (3) Metaphase
 - (4) G₂ phase

161. Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :

- (1) Arthritis
- (2) Muscular dystrophy
- (3) Myasthenia gravis
- (4) Gout

162. Match List - I with List - II.

List - I		List - II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (i) | (ii) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (iv) | (i) | (ii) | (iii) |
163. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?
- (1) Annealing
 - (2) Extension
 - (3) Denaturation
 - (4) Ligation
164. Read the following statements.
- (a) Metagenesis is observed in Helminths.
 - (b) Echinoderms are triploblastic and coelomate animals.
 - (c) Round worms have organ-system level of body organization.
 - (d) Comb plates present in ctenophores help in digestion.
 - (e) Water vascular system is characteristic of Echinoderms.

Choose the **correct** answer from the options given below.

- (1) (c), (d) and (e) are correct
- (2) (a), (b) and (c) are correct
- (3) (a), (d) and (e) are correct
- (4) (b), (c) and (e) are correct

165. Dobson units are used to measure thickness of :
 (1) CFCs
 (2) Stratosphere
 (3) Ozone
 (4) Troposphere
166. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?
 (1) DNA dependent DNA polymerase
 (2) DNA dependent RNA polymerase
 (3) DNA Ligase
 (4) DNase
167. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :
 (1) Degenerate primer sequence
 (2) Okazaki sequences
 (3) Palindromic Nucleotide sequences
 (4) Poly(A) tail sequences
168. The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G₁ phase is 8, what would be the number of chromosomes after S phase ?
 (1) 8
 (2) 16
 (3) 4
 (4) 32
169. Sphincter of oddi is present at :
 (1) Ileo-caecal junction
 (2) Junction of hepato-pancreatic duct and duodenum
 (3) Gastro-oesophageal junction
 (4) Junction of jejunum and duodenum

170. Match List - I with List - II.

List - I		List - II	
(a)	<i>Aspergillus niger</i>	(i)	Acetic Acid
(b)	<i>Acetobacter aceti</i>	(ii)	Lactic Acid
(c)	<i>Clostridium butylicum</i>	(iii)	Citric Acid
(d)	<i>Lactobacillus</i>	(iv)	Butyric Acid

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (i) | (iv) | (ii) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (ii) | (iii) | (i) | (iv) |
| (4) | (iv) | (ii) | (i) | (iii) |

171. Which one of the following organisms bears hollow and pneumatic long bones ?
 (1) *Neophron*
 (2) *Hemidactylus*
 (3) *Macropus*
 (4) *Ornithorhynchus*
172. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?
 (1) 50%
 (2) 75%
 (3) 25%
 (4) 100%
173. The partial pressures (in mm Hg) of oxygen (O₂) and carbon dioxide (CO₂) at alveoli (the site of diffusion) are :
 (1) pO₂ = 104 and pCO₂ = 40
 (2) pO₂ = 40 and pCO₂ = 45
 (3) pO₂ = 95 and pCO₂ = 40
 (4) pO₂ = 159 and pCO₂ = 0.3
174. Veneral diseases can spread through :
 (a) Using sterile needles
 (b) Transfusion of blood from infected person
 (c) Infected mother to foetus
 (d) Kissing
 (e) Inheritance
- Choose the **correct** answer from the options given below.
- (1) (a), (b) and (c) only
 (2) (b), (c) and (d) only
 (3) (b) and (c) only
 (4) (a) and (c) only
175. Which of the following RNAs is not required for the synthesis of protein ?
 (1) mRNA
 (2) tRNA
 (3) rRNA
 (4) siRNA

176. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

- (1) High pO_2 , low pCO_2 , less H^+ , lower temperature
- (2) Low pO_2 , high pCO_2 , more H^+ , higher temperature
- (3) High pO_2 , high pCO_2 , less H^+ , higher temperature
- (4) Low pO_2 , low pCO_2 , more H^+ , higher temperature

177. Match the following :

List - I		List - II	
(a)	<i>Physalia</i>	(i)	Pearl oyster
(b)	<i>Limulus</i>	(ii)	Portuguese Man of War
(c)	<i>Ancylostoma</i>	(iii)	Living fossil
(d)	<i>Pinctada</i>	(iv)	Hookworm

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (i) | (iv) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (i) | (iv) | (iii) | (ii) |

178. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?

- (1) Thrombin
- (2) Renin
- (3) Epinephrine
- (4) Thrombokinas

179. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection ?

- (1) Western Blotting Technique
- (2) Southern Blotting Technique
- (3) ELISA Technique
- (4) Hybridization Technique

180. Identify the **incorrect** pair.

- (1) Alkaloids - Codeine
- (2) Toxin - Abrin
- (3) Lectins - Concanavalin A
- (4) Drugs - Ricin

181. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature ?

- (1) Leptotene
- (2) Zygotene
- (3) Diakinesis
- (4) Pachytene

182. With regard to insulin choose correct options.

- (a) C-peptide is not present in mature insulin.
- (b) The insulin produced by rDNA technology has C-peptide.
- (c) The pro-insulin has C-peptide.
- (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

- (1) (b) and (d) only
- (2) (b) and (c) only
- (3) (a), (c) and (d) only
- (4) (a) and (d) only

183. Persons with 'AB' blood group are called as "Universal recipients". This is due to :

- (1) Absence of antigens A and B on the surface of RBCs
- (2) Absence of antigens A and B in plasma
- (3) Presence of antibodies, anti-A and anti-B, on RBCs
- (4) Absence of antibodies, anti-A and anti-B, in plasma

184. Erythropoietin hormone which stimulates R.B.C. formation is produced by :

- (1) Alpha cells of pancreas
- (2) The cells of rostral adenohypophysis
- (3) The cells of bone marrow
- (4) Juxtaglomerular cells of the kidney

185. Which of the following characteristics is **incorrect** with respect to cockroach ?

- (1) A ring of gastric caeca is present at the junction of midgut and hind gut.
- (2) Hypopharynx lies within the cavity enclosed by the mouth parts.
- (3) In females, 7th-9th sterna together form a genital pouch.
- (4) 10th abdominal segment in both sexes, bears a pair of anal cerci.

Section - B (Biology : Zoology)

186. The Adenosine deaminase deficiency results into :
- (1) Dysfunction of Immune system
 - (2) Parkinson's disease
 - (3) Digestive disorder
 - (4) Addison's disease
187. Which of the following is **not** a step in Multiple Ovulation Embryo Transfer Technology (MOET) ?
- (1) Cow is administered hormone having LH like activity for super ovulation
 - (2) Cow yields about 6-8 eggs at a time
 - (3) Cow is fertilized by artificial insemination
 - (4) Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage

188. Match List - I with List - II.

List - I		List - II	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (iii) | (ii) | (i) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (i) | (iv) | (iii) | (ii) |
189. Which one of the following statements about Histones is **wrong** ?
- (1) Histones are organized to form a unit of 8 molecules.
 - (2) The pH of histones is slightly acidic.
 - (3) Histones are rich in amino acids - Lysine and Arginine.
 - (4) Histones carry positive charge in the side chain.
190. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?
- (1) Graafian follicle
 - (2) Corpus luteum
 - (3) Foetus
 - (4) Uterus

191. Following are the statements with reference to 'lipids'.

- (a) Lipids having only single bonds are called unsaturated fatty acids.
- (b) Lecithin is a phospholipid.
- (c) Trihydroxy propane is glycerol.
- (d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
- (e) Arachidonic acid has 16 carbon atoms.

Choose the **correct** answer from the options given below.

- (1) (a) and (b) only
- (2) (c) and (d) only
- (3) (b) and (c) only
- (4) (b) and (e) only

192. Match List - I with List - II.

List - I		List - II	
(a)	Filariasis	(i)	<i>Haemophilus influenzae</i>
(b)	Amoebiasis	(ii)	<i>Trichophyton</i>
(c)	Pneumonia	(iii)	<i>Wuchereria bancrofti</i>
(d)	Ringworm	(iv)	<i>Entamoeba histolytica</i>

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (iii) | (ii) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (ii) | (iii) | (i) | (iv) |
193. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.
- (1) Gap junctions and Adhering junctions, respectively.
 - (2) Tight junctions and Gap junctions, respectively.
 - (3) Adhering junctions and Tight junctions, respectively.
 - (4) Adhering junctions and Gap junctions, respectively.

194. During muscular contraction which of the following events occur ?

- (a) 'H' zone disappears
 - (b) 'A' band widens
 - (c) 'I' band reduces in width
 - (d) Myosine hydrolyzes ATP, releasing the ADP and Pi
 - (e) Z-lines attached to actins are pulled inwards
- Choose the **correct** answer from the options given below.

- (1) (a), (c), (d), (e) only
- (2) (a), (b), (c), (d) only
- (3) (b), (c), (d), (e) only
- (4) (b), (d), (e), (a) only

195. Following are the statements about prostomium of earthworm.

- (a) It serves as a covering for mouth.
- (b) It helps to open cracks in the soil into which it can crawl.
- (c) It is one of the sensory structures.
- (d) It is the first body segment.

Choose the **correct** answer from the options given below.

- (1) (a), (b) and (c) are correct
- (2) (a), (b) and (d) are correct
- (3) (a), (b), (c) and (d) are correct
- (4) (b) and (c) are correct

196. **Assertion (A) :**

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
- (2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**
- (3) **(A)** is true but **(R)** is false
- (4) **(A)** is false but **(R)** is true

197. Which of these is not an important component of initiation of parturition in humans ?

- (1) Increase in estrogen and progesterone ratio
- (2) Synthesis of prostaglandins
- (3) Release of Oxytocin
- (4) Release of Prolactin

198. Match **List - I** with **List - II**.

List - I		List - II	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical adaptation	(iv)	Polar seal

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (ii) | (iii) | (i) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

199. Match **List - I** with **List - II**.

List - I		List - II	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (iii) | (ii) | (iv) |
| (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iv) | (ii) | (iii) | (i) |
| (4) | (iv) | (iii) | (ii) | (i) |

200. **Statement I :**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are true
- (2) Both **Statement I** and **Statement II** are false
- (3) **Statement I** is correct but **Statement II** is false
- (4) **Statement I** is incorrect but **Statement II** is true

Space For Rough Work

Space For Rough Work

Space For Rough Work

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on OFFICE Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos – 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is O2. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

Section - A (Physics)

1. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively :

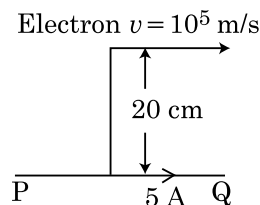
(1) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$

(2) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

(3) $\frac{S}{4}, \frac{3gS}{2}$

(4) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$

2. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



- (1) $4\pi \times 10^{-20}$ N
 (2) 8×10^{-20} N
 (3) 4×10^{-20} N
 (4) $8\pi \times 10^{-20}$ N
3. A screw gauge gives the following readings when used to measure the diameter of a wire

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is :

- (1) 0.26 cm
 (2) 0.052 cm
 (3) 0.52 cm
 (4) 0.026 cm

4. If force $[F]$, acceleration $[A]$ and time $[T]$ are chosen as the fundamental physical quantities. Find the dimensions of energy.

(1) $[F][A][T^{-1}]$

(2) $[F][A^{-1}][T]$

(3) $[F][A][T]$

(4) $[F][A][T^2]$

5. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since :

(1) a large aperture contributes to the quality and visibility of the images.

(2) a large area of the objective ensures better light gathering power.

(3) a large aperture provides a better resolution.

(4) all of the above.

6. An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of negligible work function. If ' m ' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d , then :

(1) $\lambda = \left(\frac{2mc}{h} \right) \lambda_d^2$

(2) $\lambda = \left(\frac{2h}{mc} \right) \lambda_d^2$

(3) $\lambda = \left(\frac{2m}{hc} \right) \lambda_d^2$

(4) $\lambda_d = \left(\frac{2mc}{h} \right) \lambda^2$

7. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :

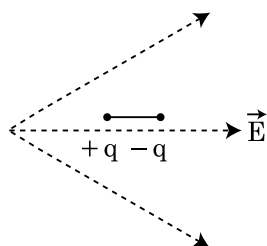
(1) $\sqrt{\left(\frac{R_1}{R_2} \right)}$

(2) $\frac{R_1^2}{R_2^2}$

(3) $\frac{R_1}{R_2}$

(4) $\frac{R_2}{R_1}$

8. A dipole is placed in an electric field as shown. In which direction will it move ?



- (1) towards the left as its potential energy will decrease.
- (2) towards the right as its potential energy will increase.
- (3) towards the left as its potential energy will increase.
- (4) towards the right as its potential energy will decrease.

9. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be : ($h = 6.6 \times 10^{-34}$ Js)

- (1) 10^{16}
- (2) 10^{15}
- (3) 10^{18}
- (4) 10^{17}

10. A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is :

- (1) $3n$
- (2) $4n$
- (3) n
- (4) $2n$

11. The escape velocity from the Earth's surface is v . The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is :

- (1) $3v$
- (2) $4v$
- (3) v
- (4) $2v$

12. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C . The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is :

- (1) $\frac{10}{13}t$
- (2) $\frac{5}{13}t$
- (3) $\frac{13}{10}t$
- (4) $\frac{13}{5}t$

13. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be :

- (1) 50
- (2) 30
- (3) 25
- (4) 15

14. Match **Column - I** and **Column - II** and choose the **correct** match from the given choices.

Column - I	Column - II
(A) Root mean square speed of gas molecules	(P) $\frac{1}{3}nm\bar{v}^2$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3RT}{M}}$
(C) Average kinetic energy of a molecule	(R) $\frac{5}{2}RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2}k_B T$

- (1) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)
- (2) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)
- (3) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)
- (4) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)

15. Polar molecules are the molecules :

- (1) acquire a dipole moment only when magnetic field is absent.
- (2) having a permanent electric dipole moment.
- (3) having zero dipole moment.
- (4) acquire a dipole moment only in the presence of electric field due to displacement of charges.

16. A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence

${}^A_ZX \rightarrow {}^{Z-1}_B \rightarrow {}^{Z-3}_C \rightarrow {}^{Z-2}_D$, where Z is the atomic number of element X. The possible decay particles in the sequence are :

- (1) β^+ , α , β^-
- (2) β^- , α , β^+
- (3) α , β^- , β^+
- (4) α , β^+ , β^-

17. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs ?

- (1) 64 cm
- (2) 62 cm
- (3) 60 cm
- (4) 21.6 cm

18. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$ is :

- (1) $\frac{2n+1}{2n-1}$
- (2) $\frac{2n}{2n-1}$
- (3) $\frac{2n-1}{2n}$
- (4) $\frac{2n-1}{2n+1}$

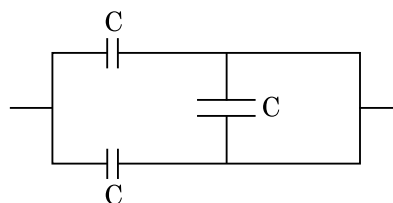
19. Consider the following **statements (A)** and **(B)** and identify the **correct** answer.

(A) A zener diode is connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

- (1) **(A)** is correct and **(B)** is incorrect.
- (2) **(A)** is incorrect but **(B)** is correct.
- (3) **(A)** and **(B)** both are correct.
- (4) **(A)** and **(B)** both are incorrect.

20. The equivalent capacitance of the combination shown in the figure is :



- (1) $C/2$
- (2) $3C/2$
- (3) $3C$
- (4) $2C$

21. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

- (1) current in n-type > current in p-type.
- (2) No current will flow in p-type, current will only flow in n-type.
- (3) current in n-type = current in p-type.
- (4) current in p-type > current in n-type.

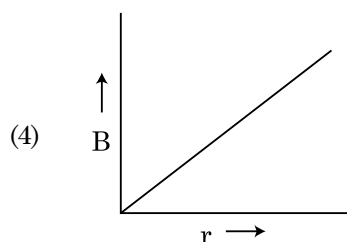
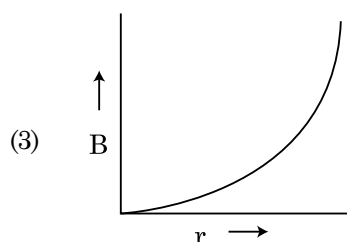
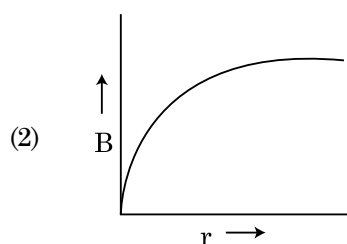
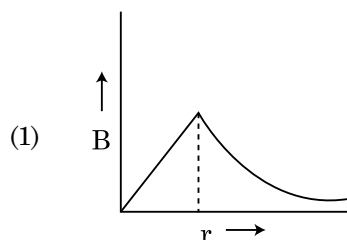
22. A capacitor of capacitance 'C', is connected across an ac source of voltage V, given by

$$V = V_0 \sin \omega t$$

The displacement current between the plates of the capacitor, would then be given by :

- (1) $I_d = \frac{V_0}{\omega C} \sin \omega t$
- (2) $I_d = V_0 \omega C \sin \omega t$
- (3) $I_d = V_0 \omega C \cos \omega t$
- (4) $I_d = \frac{V_0}{\omega C} \cos \omega t$

23. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by :



24. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is :

- (1) 3.14 s
(2) 0.628 s
(3) 0.0628 s
(4) 6.28 s

25. A parallel plate capacitor has a uniform electric field ' $\vec{E}$ ' in the space between the plates. If the distance between the plates is 'd' and the area of each plate is 'A', the energy stored in the capacitor is : (ϵ_0 = permittivity of free space)

(1) $\frac{1}{2} \epsilon_0 E^2 Ad$

(2) $\frac{E^2 Ad}{\epsilon_0}$

(3) $\frac{1}{2} \epsilon_0 E^2$

(4) $\epsilon_0 EAd$

26. The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be :

(1) $\frac{3}{2} Mg$

(2) $2Mg$

(3) $\frac{Mg}{2}$

(4) Mg

27. For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively ?

(1) $\hat{j} + \hat{k}, -\hat{j} - \hat{k}$

(2) $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$

(3) $\hat{j} + \hat{k}, \hat{j} + \hat{k}$

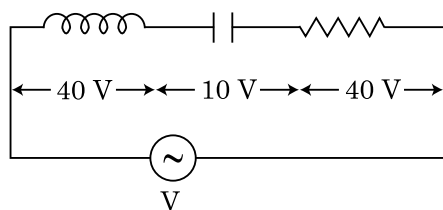
(4) $-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$

28. **Column - I** gives certain physical terms associated with flow of current through a metallic conductor. **Column - II** gives some mathematical relations involving electrical quantities. Match **Column - I** and **Column - II** with appropriate relations.

Column - I	Column - II
(A) Drift Velocity	(P) $\frac{m}{ne^2 \rho}$
(B) Electrical Resistivity	(Q) nev_d
(C) Relaxation Period	(R) $\frac{eE}{m} \tau$
(D) Current Density	(S) $\frac{E}{J}$
(1) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)	
(2) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)	
(3) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)	
(4) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)	

29. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.

Potential difference across L , C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is :



- (1) 4Ω
 (2) 5Ω
 (3) $4\sqrt{2} \Omega$
 (4) $5/\sqrt{2} \Omega$
30. The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be :

- (1) $\frac{2}{3}$
 (2) $\frac{2}{3\sqrt{2}}$
 (3) $\frac{1}{2}$
 (4) $\frac{1}{2\sqrt{2}}$

31. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25Ω . What will be the effective resistance if they are connected in series ?

- (1) 1Ω
 (2) 4Ω
 (3) 0.25Ω
 (4) 0.5Ω

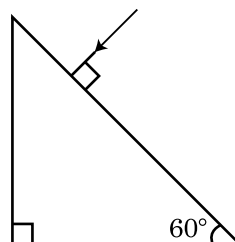
32. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of :

- (1) $[M] [L^0] [T^0]$
 (2) $[M^2] [L^{-2}] [T^{-1}]$
 (3) $[M^2] [L^{-1}] [T^0]$
 (4) $[M] [L^{-1}] [T^{-1}]$

33. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine ? ($g = 10 \text{ m/s}^2$)

- (1) 12.3 kW
 (2) 7.0 kW
 (3) 10.2 kW
 (4) 8.1 kW

34. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



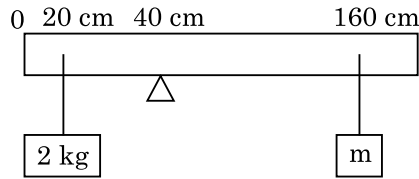
- (1) 45°
 (2) 90°
 (3) 60°
 (4) 30°

35. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is :

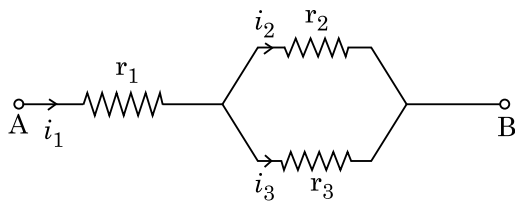
- (1) 804 MeV
 (2) 216 MeV
 (3) 0.9 MeV
 (4) 9.4 MeV

Section - B (Physics)

36. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$)

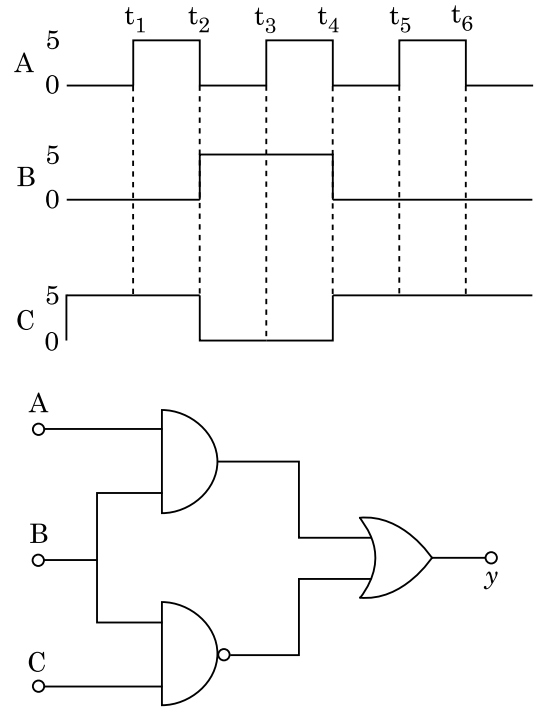


- (1) $\frac{1}{6} \text{ kg}$
 (2) $\frac{1}{12} \text{ kg}$
 (3) $\frac{1}{2} \text{ kg}$
 (4) $\frac{1}{3} \text{ kg}$
37. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is :



- (1) $\frac{r_1}{r_1 + r_2}$
 (2) $\frac{r_2}{r_1 + r_3}$
 (3) $\frac{r_1}{r_2 + r_3}$
 (4) $\frac{r_2}{r_2 + r_3}$

38. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y?



- (1) y waveform: 0 V at $t_1, t_2, t_3, t_4, t_5, t_6$
 (2) y waveform: 5 V at $t_1, t_2, t_3, t_4, t_5, t_6$
 (3) y waveform: 0 V at $t_1, t_2, t_3, t_4, t_5, t_6$
 (4) y waveform: 5 V at $t_1, t_2, t_3, t_4, t_5, t_6$

39. In the product

$$\vec{F} = q \left(\vec{v} \times \vec{B} \right)$$

$$= q \vec{v} \times \left(B \hat{i} + B \hat{j} + B_0 \hat{k} \right)$$

For $q = 1$ and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and

$$\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$$

What will be the complete expression for $\vec{B}$?

- (1) $8\hat{i} + 8\hat{j} - 6\hat{k}$
 (2) $6\hat{i} + 6\hat{j} - 8\hat{k}$
 (3) $-8\hat{i} - 8\hat{j} - 6\hat{k}$
 (4) $-6\hat{i} - 6\hat{j} - 8\hat{k}$

40. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit ?

(1) 2 A
(2) 4 A
(3) 0.2 A
(4) 0.4 A

41. From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times 'MR²'. Then the value of 'K' is :

(1) $\frac{1}{4}$
(2) $\frac{1}{8}$
(3) $\frac{3}{4}$
(4) $\frac{7}{8}$

42. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection, θ , is then given by :

(1) $\theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$
(2) $\theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{1/2}$
(3) $\theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{1/2}$
(4) $\theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$

43. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

(1) 1520 V
(2) 1980 V
(3) 660 V
(4) 1320 V

44. A particle of mass 'm' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth.

(V_e = escape velocity)

The maximum height above the surface reached by the particle is :

(1) $\frac{R^2 k}{1+k}$
(2) $\frac{Rk^2}{1-k^2}$
(3) $R \left(\frac{k}{1-k} \right)^2$
(4) $R \left(\frac{k}{1+k} \right)^2$

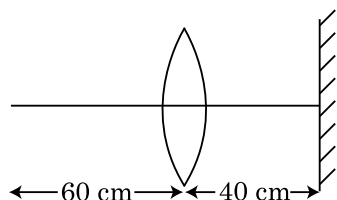
45. A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$?
(Take $g = 10 \text{ m/s}^2$)

(1) $20\sqrt{2} \text{ m/s}$, 0
(2) $20\sqrt{2} \text{ m/s}$, 10 m/s^2
(3) 20 m/s , 5 m/s^2
(4) 20 m/s , 0

46. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to :

(1) $\frac{R_1^2}{R_2}$
(2) $\frac{R_2^2}{R_1}$
(3) $\frac{R_1}{R_2}$
(4) $\frac{R_2}{R_1}$

47. A series LCR circuit containing 5.0 H inductor, 80 μF capacitor and 40 Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be :
- 46 rad/s and 54 rad/s
 - 42 rad/s and 58 rad/s
 - 25 rad/s and 75 rad/s
 - 50 rad/s and 25 rad/s
48. A uniform conducting wire of length 12a and resistance 'R' is wound up as a current carrying coil in the shape of,
- an equilateral triangle of side 'a'.
 - a square of side 'a'.
- The magnetic dipole moments of the coil in each case respectively are :
- 3 Ia^2 and 4 Ia^2
 - 4 Ia^2 and 3 Ia^2
 - $\sqrt{3} \text{Ia}^2$ and 3 Ia^2
 - 3 Ia^2 and Ia^2
49. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly :
- 2.1 kg m/s
 - 1.4 kg m/s
 - 0 kg m/s
 - 4.2 kg m/s
50. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of :



- 30 cm from the plane mirror, it would be a virtual image.
- 20 cm from the plane mirror, it would be a virtual image.
- 20 cm from the lens, it would be a real image.
- 30 cm from the lens, it would be a real image.

Section - A (Chemistry)

51. Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the **correct** answer from the options given below.

- Statement I** is correct but **Statement II** is false.
 - Statement I** is incorrect but **Statement II** is true.
 - Both **Statement I** and **Statement II** are true.
 - Both **Statement I** and **Statement II** are false.
52. Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of :
- lanthanoid contraction
 - having similar chemical properties
 - belonging to same group
 - diagonal relationship
53. The correct sequence of bond enthalpy of 'C-X' bond is :
- $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
 - $\text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{F} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
 - $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} < \text{CH}_3 - \text{Br} < \text{CH}_3 - \text{I}$
 - $\text{CH}_3 - \text{F} > \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$

54. Ethylene diaminetetraacetate (EDTA) ion is :

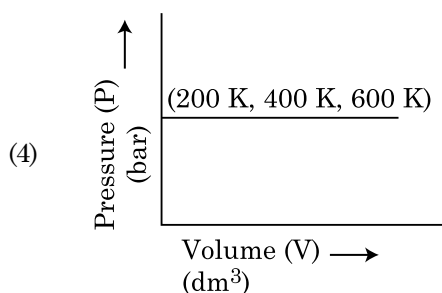
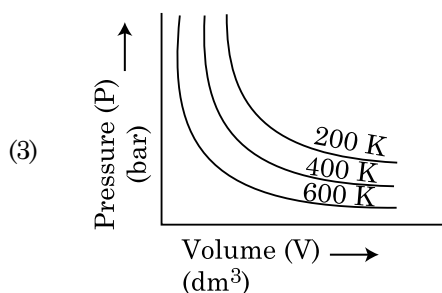
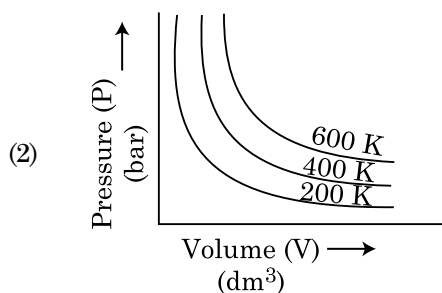
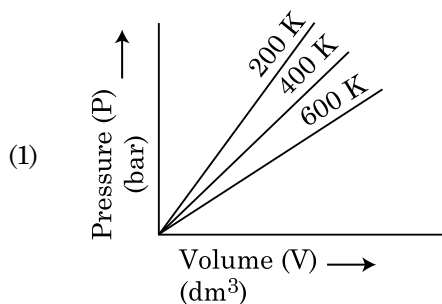
- Bidentate ligand with two "N" donor atoms
- Tridentate ligand with three "N" donor atoms
- Hexadentate ligand with four "O" and two "N" donor atoms
- Unidentate ligand

55. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are :

- sp^2 and 6
- sp^2 and 8
- sp^3 and 4
- sp^3 and 6

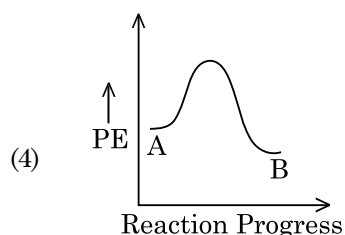
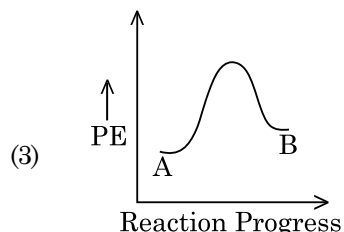
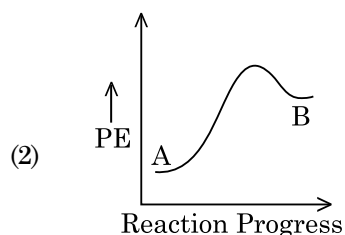
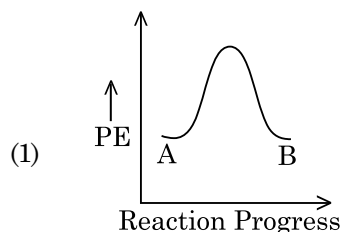
56. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light, $c = 3.0 \times 10^8 \text{ ms}^{-1}$]
- 2192 m
 - 21.92 cm
 - 219.3 m
 - 219.2 m
57. Match List - I with List - II.
- | List - I | List - II |
|--------------------|---------------------------|
| (a) PCl_5 | (i) Square pyramidal |
| (b) SF_6 | (ii) Trigonal planar |
| (c) BrF_5 | (iii) Octahedral |
| (d) BF_3 | (iv) Trigonal bipyramidal |
- Choose the **correct** answer from the options given below.
- (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 - (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 - (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 - (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
58. Dihedral angle of least stable conformer of ethane is :
- 60°
 - 0°
 - 120°
 - 180°
59. Which of the following reactions is the metal displacement reaction ? Choose the right option.
- $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2 \uparrow$
 - $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2 \uparrow$
 - $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$
 - $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
60. The compound which shows metamerism is :
- $\text{C}_3\text{H}_6\text{O}$
 - $\text{C}_4\text{H}_{10}\text{O}$
 - C_5H_{12}
 - $\text{C}_3\text{H}_8\text{O}$
61. Which one among the following is the correct option for right relationship between C_p and C_v for one mole of ideal gas ?
- $C_p = RC_v$
 - $C_v = RC_p$
 - $C_p + C_v = R$
 - $C_p - C_v = R$
62. Which one of the following polymers is prepared by addition polymerisation ?
- Novolac
 - Dacron
 - Teflon
 - Nylon-66

63. The right option for the statement "Tyndall effect is exhibited by", is :
- Starch solution
 - Urea solution
 - NaCl solution
 - Glucose solution
64. The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is :
- 2
 - 3
 - 7
 - 5
65. Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures :

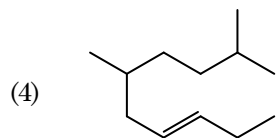
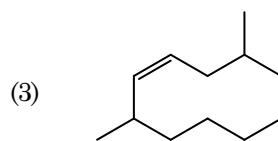
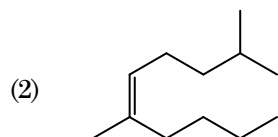
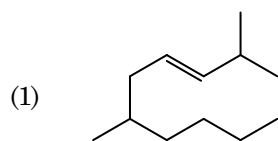


66. Noble gases are named because of their inertness towards reactivity. Identify an **incorrect** statement about them.
- (1) Noble gases have weak dispersion forces.
 - (2) Noble gases have large positive values of electron gain enthalpy.
 - (3) Noble gases are sparingly soluble in water.
 - (4) Noble gases have very high melting and boiling points.
67. Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature ?
- (1) Distillation
 - (2) Zone refining
 - (3) Electrolysis
 - (4) Chromatography
68. Tritium, a radioactive isotope of hydrogen, emits which of the following particles ?
- (1) Gamma (γ)
 - (2) Neutron (n)
 - (3) Beta (β^-)
 - (4) Alpha (α)
69. Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :
- (1) Magnesium chloride
 - (2) Beryllium chloride
 - (3) Calcium chloride
 - (4) Strontium chloride
70. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is :
- (1) 7.75
 - (2) 6.25
 - (3) 8.50
 - (4) 5.50
71. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.
- (1) $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
 - (2) $540.48 \text{ S cm}^2 \text{ mol}^{-1}$
 - (3) $201.28 \text{ S cm}^2 \text{ mol}^{-1}$
 - (4) $390.71 \text{ S cm}^2 \text{ mol}^{-1}$
72. What is the IUPAC name of the organic compound formed in the following chemical reaction ?
- Acetone $\xrightarrow[\text{(ii) H}_2\text{O, H}^+]{\text{(i) C}_2\text{H}_5\text{MgBr, dry Ether}}$ Product
- (1) pentan-3-ol
 - (2) 2-methyl butan-2-ol
 - (3) 2-methyl propan-2-ol
 - (4) pentan-2-ol
73. The major product of the following chemical reaction is :
- $$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} = \text{CH}_2 \\ \diagup \\ \text{CH}_3 \end{array} + \text{HBr} \xrightarrow{(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2} ?$$
- (1)
$$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} - \text{CH}_3 \\ \diagup \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$$
 - (2)
$$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CBr} - \text{CH}_2 - \text{CH}_3 \\ \diagup \\ \text{CH}_3 \end{array}$$
 - (3)
$$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Br} \\ \diagup \\ \text{CH}_3 \end{array}$$
 - (4)
$$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{COC}_6\text{H}_5 \\ \diagup \\ \text{CH}_3 \end{array}$$

74. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.



75. The correct structure of 2,6-Dimethyl-dec-4-ene is :



76. **Statement I :**

Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.

Statement II :

As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) **Statement I** is correct but **Statement II** is false.
- (2) **Statement I** is incorrect but **Statement II** is true.
- (3) Both **Statement I** and **Statement II** are true.
- (4) Both **Statement I** and **Statement II** are false.
77. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on ?
- (1) Hofmann Rule
- (2) Huckel's Rule
- (3) Saytzeff's Rule
- (4) Hund's Rule
78. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]
- (1) CH_3
- (2) CH_4
- (3) CH
- (4) CH_2

79. The RBC deficiency is deficiency disease of :

- (1) Vitamin B_1
- (2) Vitamin B_2
- (3) Vitamin B_{12}
- (4) Vitamin B_6

80. The maximum temperature that can be achieved in blast furnace is :

- (1) upto 1900 K
- (2) upto 5000 K
- (3) upto 1200 K
- (4) upto 2200 K

81. The **incorrect** statement among the following is :

- (1) Lanthanoids are good conductors of heat and electricity.
- (2) Actinoids are highly reactive metals, especially when finely divided.
- (3) Actinoid contraction is greater for element to element than Lanthanoid contraction.
- (4) Most of the trivalent Lanthanoid ions are colorless in the solid state.

82. The structures of beryllium chloride in solid state and vapour phase, are :

- (1) Dimer and Linear, respectively
- (2) Chain in both
- (3) Chain and dimer, respectively
- (4) Linear in both

83. Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are :

- (1) 2, 1
- (2) 12, 6
- (3) 8, 4
- (4) 6, 12

84. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :

- (1) $P_2 > P_3 > P_1$
- (2) $P_3 > P_1 > P_2$
- (3) $P_2 > P_1 > P_3$
- (4) $P_1 > P_2 > P_3$

85. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.

- (1) $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2 \\ \quad \quad \diagdown \quad \diagup \\ \quad \quad \text{N} \quad \text{H}_2 \\ \quad \quad \cdot \cdot \end{array}$
- (2) $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2 \quad \text{CH}_2 \quad \text{CH}_3 \\ \quad \quad \diagdown \quad \diagup \\ \quad \quad \text{N} \\ \quad \quad | \\ \quad \quad \text{CH}_3 \end{array}$
- (3) $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2 \\ \quad \quad \diagdown \quad \diagup \\ \quad \quad \text{N} \quad \text{O}_2 \\ \quad \quad \cdot \cdot \end{array}$
- (4) $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2 \quad \text{CH}_3 \\ \quad \quad \diagdown \quad \diagup \\ \quad \quad \text{N} \quad \text{H} \end{array}$

Section - B (Chemistry)

86. The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is :

[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]

- (1) 336 mm of Hg
- (2) 350 mm of Hg
- (3) 160 mm of Hg
- (4) 168 mm of Hg

87. Match List - I with List - II.

List - I		List - II	
(a)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(b)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(c)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(d)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the **correct** answer from the options given below.

- (1) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

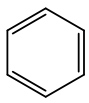
88. From the following pairs of ions which one is not an iso-electronic pair ?

- (1) Mn^{2+} , Fe^{3+}
- (2) Fe^{2+} , Mn^{2+}
- (3) O^{2-} , F^-
- (4) Na^+ , Mg^{2+}

89. Which of the following molecules is non-polar in nature ?

- (1) SbCl_5
- (2) NO_2
- (3) POCl_3
- (4) CH_2O

90. Match List - I with List - II.

List - I	List - II
(a)  $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$	(i) Hell-Volhard-Zelinsky reaction
(b) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$	(ii) Gattermann-Koch reaction
(c) $\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$	(iii) Haloform reaction
(d) $\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$	(iv) Esterification

Choose the **correct** answer from the options given below.

- (1) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
 (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

91. Match List - I with List - II.

List - I	List - II
(a) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$	(i) Acid rain
(b) $\text{HOCl}(\text{g}) \xrightarrow{h\nu} \dot{\text{O}}\text{H} + \dot{\text{Cl}}$	(ii) Smog
(c) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$	(iii) Ozone depletion
(d) $\text{NO}_2(\text{g}) \xrightarrow{h\nu} \text{NO}(\text{g}) + \text{O}(\text{g})$	(iv) Tropospheric pollution

Choose the **correct** answer from the options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
 (3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

92. The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T} \right)$ of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is. Choose the correct option for your answer.

[Given $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$]

- (1) 166 kJ mol^{-1}
 (2) -83 kJ mol^{-1}
 (3) 41.5 kJ mol^{-1}
 (4) 83.0 kJ mol^{-1}

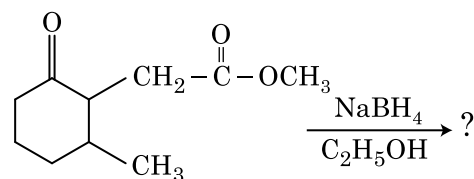
93. For irreversible expansion of an ideal gas under isothermal condition, the correct option is :

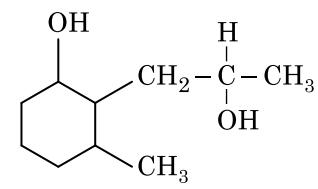
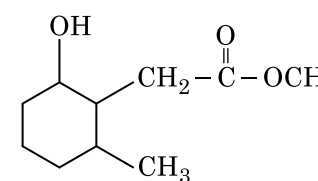
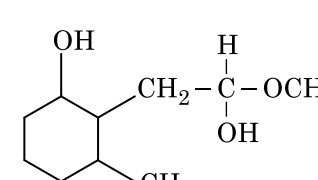
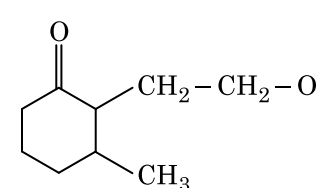
- (1) $\Delta U = 0, \Delta S_{\text{total}} \neq 0$
 (2) $\Delta U \neq 0, \Delta S_{\text{total}} = 0$
 (3) $\Delta U = 0, \Delta S_{\text{total}} = 0$
 (4) $\Delta U \neq 0, \Delta S_{\text{total}} \neq 0$

94. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?

- (1) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character
 (2) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power
 (3) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength
 (4) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values

95. The product formed in the following chemical reaction is :



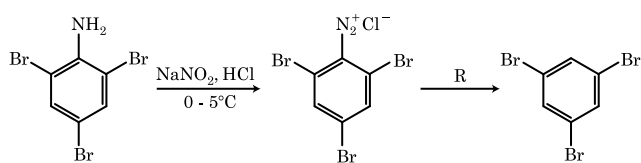
- (1) 
 (2) 
 (3) 
 (4) 

96. Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one litre at $0^\circ C$ is :

[Given $R = 0.082 \text{ L atm mol}^{-1} K^{-1}$, $T = 273 \text{ K}$]

- (1) 25.18
- (2) 26.02
- (3) 2.518
- (4) 2.602

97. The reagent 'R' in the given sequence of chemical reaction is :



- (1) HI
- (2) $CuCN/KCN$
- (3) H_2O
- (4) CH_3CH_2OH

98. $CH_3CH_2COO^- Na^+ + \xrightarrow[\text{Heat}]{NaOH, + ?} CH_3CH_3 + Na_2CO_3$.

Consider the above reaction and identify the missing reagent/chemical.

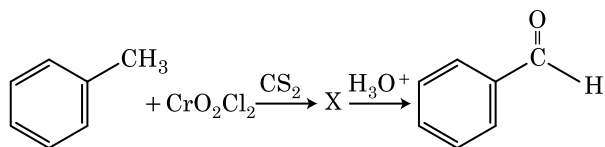
- (1) CaO
- (2) DIBAL-H
- (3) B_2H_6
- (4) Red Phosphorus

99. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid ? Choose the correct option.

$$\left[\begin{array}{l} \Lambda_{H^+}^\circ = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda_{CH_3COO^-}^\circ = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$

- (1) $1.75 \times 10^{-5} \text{ mol L}^{-1}$
- (2) $2.50 \times 10^{-5} \text{ mol L}^{-1}$
- (3) $1.75 \times 10^{-4} \text{ mol L}^{-1}$
- (4) $2.50 \times 10^{-4} \text{ mol L}^{-1}$

100. The intermediate compound 'X' in the following chemical reaction is :



- (1)
- (2)
- (3)
- (4)

Section - A (Biology : Botany)

101. Match List - I with List - II.

List - I		List - II	
(a)	Protoplast fusion	(i)	Totipotency
(b)	Plant tissue culture	(ii)	Pomato
(c)	Meristem culture	(iii)	Somaclones
(d)	Micropropagation	(iv)	Virus free plants

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (ii) | (i) | (iv) | (iii) |

102. The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is :

- (1) Chasmogamy
- (2) Cleistogamy
- (3) Xenogamy
- (4) Geitonogamy

103. The factor that leads to Founder effect in a population is :

- (1) Mutation
- (2) Genetic drift
- (3) Natural selection
- (4) Genetic recombination

104. Match List - I with List - II.

List - I		List - II	
(a)	Cohesion	(i)	More attraction in liquid phase
(b)	Adhesion	(ii)	Mutual attraction among water molecules
(c)	Surface tension	(iii)	Water loss in liquid phase
(d)	Guttation	(iv)	Attraction towards polar surfaces

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (i) | (iv) | (ii) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (ii) | (iv) | (i) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

105. A typical angiosperm embryo sac at maturity is :

- (1) 7-nucleate and 7-celled
- (2) 8-nucleate and 8-celled
- (3) 8-nucleate and 7-celled
- (4) 7-nucleate and 8-celled

106. Amensalism can be represented as :

- (1) Species A (–) ; Species B (–)
- (2) Species A (+) ; Species B (0)
- (3) Species A (–) ; Species B (0)
- (4) Species A (+) ; Species B (+)

107. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as :

- (1) Dark red bands
- (2) Bright blue bands
- (3) Yellow bands
- (4) Bright orange bands

108. Which of the following stages of meiosis involves division of centromere ?

- (1) Anaphase II
- (2) Telophase II
- (3) Metaphase I
- (4) Metaphase II

109. Which of the following plants is monoecious ?

- (1) *Marchantia polymorpha*
- (2) *Cycas circinalis*
- (3) *Carica papaya*
- (4) Chara

110. The site of perception of light in plants during photoperiodism is :

- (1) Axillary bud
- (2) Leaf
- (3) Shoot apex
- (4) Stem

111. Which of the following are **not** secondary metabolites in plants ?

- (1) Vinblastin, curcumin
- (2) Rubber, gums
- (3) Morphine, codeine
- (4) Amino acids, glucose

112. In the equation $GPP - R = NPP$

R represents :

- (1) Environment factor
- (2) Respiration losses
- (3) Radiant energy
- (4) Retardation factor

113. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called :

- (1) Plasticity
- (2) Maturity
- (3) Elasticity
- (4) Flexibility

114. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as :

- (1) Standing state
- (2) Standing crop
- (3) Climax
- (4) Climax community

115. Which of the following is an **incorrect** statement ?

- (1) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
- (2) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
- (3) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
- (4) Microbodies are present both in plant and animal cells.

116. Match List - I with List - II.

List - I		List - II	
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (iii) | (ii) | (iv) | (i) |
| (3) | (ii) | (iv) | (i) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

117. The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called :

- (1) Punnett square
- (2) Net square
- (3) Bullet square
- (4) Punch square

118. Mutations in plant cells can be induced by :

- (1) Gamma rays
- (2) Zeatin
- (3) Kinetin
- (4) Infrared rays

119. The first stable product of CO_2 fixation in sorghum is :

- (1) Succinic acid
- (2) Phosphoglyceric acid
- (3) Pyruvic acid
- (4) Oxaloacetic acid

120. Match List - I with List - II.

List - I		List - II	
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iv) | (iii) | (ii) | (i) |
| (4) | (i) | (iv) | (iii) | (ii) |

121. Which of the following statements is **not** correct ?

- (1) Pyramid of energy is always upright.
- (2) Pyramid of numbers in a grassland ecosystem is upright.
- (3) Pyramid of biomass in sea is generally inverted.
- (4) Pyramid of biomass in sea is generally upright.

122. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as :

- (1) Sub-metacentric
- (2) Acrocentric
- (3) Metacentric
- (4) Telocentric

123. Which of the following is **not** an application of PCR (Polymerase Chain Reaction) ?

- (1) Purification of isolated protein
- (2) Detection of gene mutation
- (3) Molecular diagnosis
- (4) Gene amplification

124. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as :

- (1) Homosporous
- (2) Heterosporous
- (3) Homosorus
- (4) Heterosorus

125. Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival ?

- (1) Mutualism
- (2) Predation
- (3) Resource partitioning
- (4) Competitive release

126. Gemmae are present in :

- (1) Some Gymnosperms
- (2) Some Liverworts
- (3) Mosses
- (4) Pteridophytes

127. Which of the following algae produce Carrageen ?

- (1) Red algae
- (2) Blue-green algae
- (3) Green algae
- (4) Brown algae

128. Which of the following is a **correct** sequence of steps in a PCR (Polymerase Chain Reaction) ?

- (1) Extension, Denaturation, Annealing
- (2) Annealing, Denaturation, Extension
- (3) Denaturation, Annealing, Extension
- (4) Denaturation, Extension, Annealing

129. Which of the following algae contains mannitol as reserve food material ?

- (1) *Volvox*
- (2) *Ulothrix*
- (3) *Ectocarpus*
- (4) *Gracilaria*

130. Diadelphous stamens are found in :

- (1) Pea
- (2) China rose and citrus
- (3) China rose
- (4) Citrus

131. Complete the flow chart on central dogma.

(a) $\text{DNA} \xrightarrow{(b)} \text{mRNA} \xrightarrow{(c)} (d)$

- (1) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein
- (2) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein
- (3) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein
- (4) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Transduction

132. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :

- (1) Molecular diagnosis
- (2) Safety testing
- (3) Biopiracy
- (4) Gene therapy

133. The plant hormone used to destroy weeds in a field is :

- (1) 2, 4-D
- (2) IBA
- (3) IAA
- (4) NAA

134. Match List - I with List - II.

List - I		List - II	
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (iv) | (ii) | (i) | (iii) |
| (3) | (iv) | (i) | (iii) | (ii) |
| (4) | (iii) | (i) | (iv) | (ii) |

135. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out :

- (1) Histones
- (2) Polysaccharides
- (3) RNA
- (4) DNA

Section - B (Biology : Botany)

136. Match List - I with List - II.

List - I		List - II	
(a)	Protein	(i)	C = C double bonds
(b)	Unsaturated fatty acid	(ii)	Phosphodiester bonds
(c)	Nucleic acid	(iii)	Glycosidic bonds
(d)	Polysaccharide	(iv)	Peptide bonds

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (i) | (iv) | (iii) |
| (2) | (iv) | (iii) | (i) | (ii) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (i) | (iv) | (iii) | (ii) |

137. Match Column - I with Column - II.

Column - I	Column - II
(a) $\% \frac{1}{5} K_{(5)} C_{1+2+(2)} A_{(9)+1} G_1$	(i) Brassicaceae
(b) $\frac{1}{5} K_{(5)} \widehat{C_{(5)} A_5} G_2$	(ii) Liliaceae
(c) $\frac{1}{5} P_{(3+3)} \widehat{A_{3+3} G_{(3)}}$	(iii) Fabaceae
(d) $\frac{1}{5} K_{2+2} C_4 A_{2-4} G_{(2)}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (iv) | (ii) | (i) | (iii) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (i) | (ii) | (iii) | (iv) |

138. Plasmid pBR322 has PstI restriction enzyme site within gene *amp^R* that confers ampicillin resistance. If this enzyme is used for inserting a gene for β -galactoside production and the recombinant plasmid is inserted in an *E.coli* strain

- (1) it will lead to lysis of host cell.
- (2) it will be able to produce a novel protein with dual ability.
- (3) it will not be able to confer ampicillin resistance to the host cell.
- (4) the transformed cells will have the ability to resist ampicillin as well as produce β -galactoside.

139. Which of the following statements is **correct** ?

- (1) Organisms that depend on living plants are called saprophytes.
- (2) Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.
- (3) Fusion of two cells is called Karyogamy.
- (4) Fusion of protoplasts between two motile on non-motile gametes is called plasmogamy.

140. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as :

- (1) Single nucleotides
- (2) Polymorphic DNA
- (3) Satellite DNA
- (4) Repetitive DNA

141. Select the **correct** pair.

- (1) Cells of medullary rays that form part of cambial ring - Interfascicular cambium
- (2) Loose parenchyma cells rupturing the epidermis and forming a lens-shaped opening in bark - Spongy parenchyma
- (3) Large colorless empty cells in the epidermis of grass leaves - Subsidiary cells
- (4) In dicot leaves, vascular bundles are surrounded by large thick-walled cells - Conjunctive tissue

142. In the exponential growth equation

$$N_t = N_0 e^{rt}, \text{ e represents :}$$

- (1) The base of natural logarithms
- (2) The base of geometric logarithms
- (3) The base of number logarithms
- (4) The base of exponential logarithms

143. Which of the following statements is **incorrect** ?

- (1) ATP is synthesized through complex V.
- (2) Oxidation-reduction reactions produce proton gradient in respiration.
- (3) During aerobic respiration, role of oxygen is limited to the terminal stage.
- (4) In ETC (Electron Transport Chain), one molecule of $NADH + H^+$ gives rise to 2 ATP molecules, and one $FADH_2$ gives rise to 3 ATP molecules.

144. Identify the **correct** statement.

- (1) The coding strand in a transcription unit is copied to an mRNA.
- (2) Split gene arrangement is characteristic of prokaryotes.
- (3) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.
- (4) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

145. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

- (1) Poaceae ; Solanaceae
- (2) Rosaceae ; Leguminosae
- (3) Poaceae ; Rosaceae
- (4) Poaceae ; Leguminosae

146. Which of the following statements is **incorrect** ?

- (1) Grana lamellae have both PS I and PS II.
- (2) Cyclic photophosphorylation involves both PS I and PS II.
- (3) Both ATP and NADPH + H⁺ are synthesized during non-cyclic photophosphorylation.
- (4) Stroma lamellae have PS I only and lack NADP reductase.

147. Match List - I with List - II.

List - I		List - II	
(a)	S phase	(i)	Proteins are synthesized
(b)	G ₂ phase	(ii)	Inactive phase
(c)	Quiescent stage	(iii)	Interval between mitosis and initiation of DNA replication
(d)	G ₁ phase	(iv)	DNA replication

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iv) | (i) | (ii) | (iii) |
| (2) | (ii) | (iv) | (iii) | (i) |
| (3) | (iii) | (ii) | (i) | (iv) |
| (4) | (iv) | (ii) | (iii) | (i) |

148. What is the role of RNA polymerase III in the process of transcription in eukaryotes ?

- (1) Transcribes precursor of mRNA
- (2) Transcribes only snRNAs
- (3) Transcribes rRNAs (28S, 18S and 5.8S)
- (4) Transcribes tRNA, 5s rRNA and snRNA

149. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complementary DNA in a clone of cells, followed by its detection using autoradiography because :

- (1) mutated gene does not appear on a photographic film as the probe has no complementarity with it.
- (2) mutated gene does not appear on photographic film as the probe has complementarity with it.
- (3) mutated gene partially appears on a photographic film.
- (4) mutated gene completely and clearly appears on a photographic film.

150. Match Column - I with Column - II.

Column - I		Column - II	
(a)	<i>Nitrococcus</i>	(i)	Denitrification
(b)	<i>Rhizobium</i>	(ii)	Conversion of ammonia to nitrite
(c)	<i>Thiobacillus</i>	(iii)	Conversion of nitrite to nitrate
(d)	<i>Nitrobacter</i>	(iv)	Conversion of atmospheric nitrogen to ammonia

Choose the **correct** answer from options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (i) | (iv) | (ii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (ii) | (iv) | (i) | (iii) |
| (4) | (i) | (ii) | (iii) | (iv) |

Section - A (Biology : Zoology)

151. The centriole undergoes duplication during :

- (1) Metaphase
- (2) G₂ phase
- (3) S-phase
- (4) Prophase

152. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?

- (1) Denaturation
- (2) Ligation
- (3) Annealing
- (4) Extension

153. Which one of the following belongs to the family Muscidae ?

- (1) Cockroach
- (2) House fly
- (3) Fire fly
- (4) Grasshopper

154. Which of the following statements wrongly represents the nature of smooth muscle ?
- (1) Communication among the cells is performed by intercalated discs
 - (2) These muscles are present in the wall of blood vessels
 - (3) These muscle have no striations
 - (4) They are involuntary muscles

155. Dobson units are used to measure thickness of :

- (1) Ozone
- (2) Troposphere
- (3) CFCs
- (4) Stratosphere

156. Match List - I with List - II.

List - I		List - II	
(a)	<i>Aspergillus niger</i>	(i)	Acetic Acid
(b)	<i>Acetobacter aceti</i>	(ii)	Lactic Acid
(c)	<i>Clostridium butylicum</i>	(iii)	Citric Acid
(d)	<i>Lactobacillus</i>	(iv)	Butyric Acid

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (i) | (iv) |
| (2) | (iv) | (ii) | (i) | (iii) |
| (3) | (iii) | (i) | (iv) | (ii) |
| (4) | (i) | (ii) | (iii) | (iv) |

157. Succus entericus is referred to as :

- (1) Gastric juice
- (2) Chyme
- (3) Pancreatic juice
- (4) Intestinal juice

158. Match List - I with List - II.

List - I		List - II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (iv) | (i) | (ii) | (iii) |
| (3) | (iv) | (iii) | (i) | (ii) |
| (4) | (iii) | (iv) | (i) | (ii) |

159. Receptors for sperm binding in mammals are present on :

- (1) Perivitelline space
- (2) Zona pellucida
- (3) Corona radiata
- (4) Vitelline membrane

160. Which one of the following is an example of Hormone releasing IUD ?

- (1) Cu 7
- (2) Multiload 375
- (3) CuT
- (4) LNG 20

161. Venereal diseases can spread through :

- (a) Using sterile needles
- (b) Transfusion of blood from infected person
- (c) Infected mother to foetus
- (d) Kissing
- (e) Inheritance

Choose the **correct** answer from the options given below.

- (1) (b) and (c) only
- (2) (a) and (c) only
- (3) (a), (b) and (c) only
- (4) (b), (c) and (d) only

162. Which one of the following organisms bears hollow and pneumatic long bones ?

- (1) *Macropus*
- (2) *Ornithorhynchus*
- (3) *Neophron*
- (4) *Hemidactylus*

163. The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are :

- (1) $pO_2 = 95$ and $pCO_2 = 40$
- (2) $pO_2 = 159$ and $pCO_2 = 0.3$
- (3) $pO_2 = 104$ and $pCO_2 = 40$
- (4) $pO_2 = 40$ and $pCO_2 = 45$

164. If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it ?

- (1) T : 30 ; G : 20 ; C : 20
- (2) T : 20 ; G : 25 ; C : 25
- (3) T : 20 ; G : 30 ; C : 20
- (4) T : 20 ; G : 20 ; C : 30

165. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?

- (1) 25%
- (2) 100%
- (3) 50%
- (4) 75%

166. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection ?

- (1) ELISA Technique
- (2) Hybridization Technique
- (3) Western Blotting Technique
- (4) Southern Blotting Technique

167. Read the following statements.

- (a) Metagenesis is observed in Helminths.
- (b) Echinoderms are triploblastic and coelomate animals.
- (c) Round worms have organ-system level of body organization.
- (d) Comb plates present in ctenophores help in digestion.
- (e) Water vascular system is characteristic of Echinoderms.

Choose the **correct** answer from the options given below.

- (1) (a), (d) and (e) are correct
- (2) (b), (c) and (e) are correct
- (3) (c), (d) and (e) are correct
- (4) (a), (b) and (c) are correct

168. Match List - I with List - II.

List - I		List - II	
(a)	Vaults	(i)	Entry of sperm through Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of fallopian tube

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iv) | (iii) | (i) |
| (2) | (iii) | (i) | (iv) | (ii) |
| (3) | (iv) | (ii) | (i) | (iii) |
| (4) | (i) | (iii) | (ii) | (iv) |

169. Persons with 'AB' blood group are called as "Universal recipients". This is due to :

- (1) Presence of antibodies, anti-A and anti-B, on RBCs
- (2) Absence of antibodies, anti-A and anti-B, in plasma
- (3) Absence of antigens A and B on the surface of RBCs
- (4) Absence of antigens A and B in plasma

170. With regard to insulin choose correct options.

- (a) C-peptide is not present in mature insulin.
- (b) The insulin produced by rDNA technology has C-peptide.
- (c) The pro-insulin has C-peptide.
- (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

- (1) (a), (c) and (d) only
- (2) (a) and (d) only
- (3) (b) and (d) only
- (4) (b) and (c) only

171. Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :

- (1) Myasthenia gravis
- (2) Gout
- (3) Arthritis
- (4) Muscular dystrophy

172. The organelles that are included in the endomembrane system are :

- (1) Golgi complex, Mitochondria, Ribosomes and Lysosomes
- (2) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes
- (3) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
- (4) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

173. The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G₁ phase is 8, what would be the number of chromosomes after S phase ?

- (1) 4
- (2) 32
- (3) 8
- (4) 16

174. Which is the “Only enzyme” that has “Capability” to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?

- (1) DNA Ligase
- (2) DNase
- (3) DNA dependent DNA polymerase
- (4) DNA dependent RNA polymerase

175. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

- (1) High pO_2 , high pCO_2 , less H^+ , higher temperature
- (2) Low pO_2 , low pCO_2 , more H^+ , higher temperature
- (3) High pO_2 , low pCO_2 , less H^+ , lower temperature
- (4) Low pO_2 , high pCO_2 , more H^+ , higher temperature

176. Match the following :

List - I		List - II	
(a)	<i>Physalia</i>	(i)	Pearl oyster
(b)	<i>Limulus</i>	(ii)	Portuguese Man of War
(c)	<i>Ancylostoma</i>	(iii)	Living fossil
(d)	<i>Pinctada</i>	(iv)	Hookworm

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|------|-------|-------|------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (i) | (iv) | (iii) | (ii) |
| (3) | (ii) | (iii) | (i) | (iv) |
| (4) | (iv) | (i) | (iii) | (ii) |

177. Which of the following RNAs is not required for the synthesis of protein ?

- (1) rRNA
- (2) siRNA
- (3) mRNA
- (4) tRNA

178. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?

- (1) Epinephrine
- (2) Thrombokinas
- (3) Thrombin
- (4) Renin

179. Erythropoietin hormone which stimulates R.B.C. formation is produced by :

- (1) The cells of bone marrow
- (2) Juxtaglomerular cells of the kidney
- (3) Alpha cells of pancreas
- (4) The cells of rostral adenohypophysis

180. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature ?

- (1) Diakinesis
- (2) Pachytene
- (3) Leptotene
- (4) Zygotene

181. Sphincter of oddi is present at :

- (1) Gastro-oesophageal junction
- (2) Junction of jejunum and duodenum
- (3) Ileo-caecal junction
- (4) Junction of hepato-pancreatic duct and duodenum

182. Which of the following characteristics is **incorrect** with respect to cockroach ?

- (1) In females, 7th-9th sterna together form a genital pouch.
- (2) 10th abdominal segment in both sexes, bears a pair of anal cerci.
- (3) A ring of gastric caeca is present at the junction of midgut and hind gut.
- (4) Hypopharynx lies within the cavity enclosed by the mouth parts.

183. Which of the following is **not** an objective of Biofortification in crops ?

- (1) Improve vitamin content
- (2) Improve micronutrient and mineral content
- (3) Improve protein content
- (4) Improve resistance to diseases

184. Identify the **incorrect** pair.

- | | | | |
|-----|-----------|---|----------------|
| (1) | Lectins | - | Concanavalin A |
| (2) | Drugs | - | Ricin |
| (3) | Alkaloids | - | Codeine |
| (4) | Toxin | - | Abrin |

185. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :

- (1) Palindromic Nucleotide sequences
- (2) Poly(A) tail sequences
- (3) Degenerate primer sequence
- (4) Okazaki sequences

Section - B (Biology : Zoology)

186. Match List - I with List - II.

List - I		List - II	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical adaptation	(iv)	Polar seal

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (iv) (i) (ii) (iii)
- (2) (iv) (iii) (ii) (i)
- (3) (iv) (ii) (iii) (i)
- (4) (iv) (i) (iii) (ii)

187. **Statement I :**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) **Statement I** is correct but **Statement II** is false
- (2) **Statement I** is incorrect but **Statement II** is true
- (3) Both **Statement I** and **Statement II** are true
- (4) Both **Statement I** and **Statement II** are false

188. Match List - I with List - II.

List - I		List - II	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

- (a) (b) (c) (d)
- (1) (iv) (ii) (iii) (i)
- (2) (iv) (iii) (ii) (i)
- (3) (i) (iii) (ii) (iv)
- (4) (ii) (iii) (iv) (i)

189. **Assertion (A) :**

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) **(A)** is true but **(R)** is false
- (2) **(A)** is false but **(R)** is true
- (3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
- (4) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

190. Following are the statements about prostomium of earthworm.

- (a) It serves as a covering for mouth.
- (b) It helps to open cracks in the soil into which it can crawl.
- (c) It is one of the sensory structures.
- (d) It is the first body segment.

Choose the **correct** answer from the options given below.

- (1) (a), (b), (c) and (d) are correct
- (2) (b) and (c) are correct
- (3) (a), (b) and (c) are correct
- (4) (a), (b) and (d) are correct

191. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.

- (1) Adhering junctions and Tight junctions, respectively.
- (2) Adhering junctions and Gap junctions, respectively.
- (3) Gap junctions and Adhering junctions, respectively.
- (4) Tight junctions and Gap junctions, respectively.

192. Which of these is not an important component of initiation of parturition in humans ?

- (1) Release of Oxytocin
- (2) Release of Prolactin
- (3) Increase in estrogen and progesterone ratio
- (4) Synthesis of prostaglandins

193. Match List - I with List - II.

List - I		List - II	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (i) | (iv) | (iii) |
| (2) | (i) | (iv) | (iii) | (ii) |
| (3) | (iv) | (iii) | (ii) | (i) |
| (4) | (iii) | (ii) | (i) | (iv) |

194. The Adenosine deaminase deficiency results into :

- (1) Digestive disorder
- (2) Addison's disease
- (3) Dysfunction of Immune system
- (4) Parkinson's disease

195. Match List - I with List - II.

List - I		List - II	
(a)	Filariasis	(i)	<i>Haemophilus influenzae</i>
(b)	Amoebiasis	(ii)	<i>Trichophyton</i>
(c)	Pneumonia	(iii)	<i>Wuchereria bancrofti</i>
(d)	Ringworm	(iv)	<i>Entamoeba histolytica</i>

Choose the **correct** answer from the options given below.

- | | | | | |
|-----|-------|-------|-------|-------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (ii) | (iv) | (iii) |
| (2) | (ii) | (iii) | (i) | (iv) |
| (3) | (iv) | (i) | (iii) | (ii) |
| (4) | (iii) | (iv) | (i) | (ii) |

196. Which of the following is **not** a step in Multiple Ovulation Embryo Transfer Technology (MOET) ?

- (1) Cow is fertilized by artificial insemination
- (2) Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage
- (3) Cow is administered hormone having LH like activity for super ovulation
- (4) Cow yields about 6-8 eggs at a time

197. During muscular contraction which of the following events occur ?

- (a) 'H' zone disappears
- (b) 'A' band widens
- (c) 'T' band reduces in width
- (d) Myosine hydrolyzes ATP, releasing the ADP and Pi
- (e) Z-lines attached to actins are pulled inwards

Choose the **correct** answer from the options given below.

- (1) (b), (c), (d), (e) only
- (2) (b), (d), (e), (a) only
- (3) (a), (c), (d), (e) only
- (4) (a), (b), (c), (d) only

198. Following are the statements with reference to 'lipids'.

- (a) Lipids having only single bonds are called unsaturated fatty acids.
- (b) Lecithin is a phospholipid.
- (c) Trihydroxy propane is glycerol.
- (d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
- (e) Arachidonic acid has 16 carbon atoms.

Choose the **correct** answer from the options given below.

- (1) (b) and (c) only
- (2) (b) and (e) only
- (3) (a) and (b) only
- (4) (c) and (d) only

199. Which one of the following statements about Histones is **wrong** ?

- (1) Histones are rich in amino acids - Lysine and Arginine.
- (2) Histones carry positive charge in the side chain.
- (3) Histones are organized to form a unit of 8 molecules.
- (4) The pH of histones is slightly acidic.

200. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?

- (1) Foetus
- (2) Uterus
- (3) Graafian follicle
- (4) Corpus luteum

Space For Rough Work

Space For Rough Work

Test Booklet Code

KHANA

No. :

E6

This Booklet contains 24 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on **side-1** and **side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The CODE for this Booklet is **E6**. Make sure that the CODE printed on **Side-2** of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. **Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

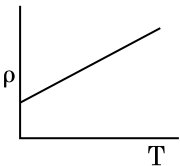
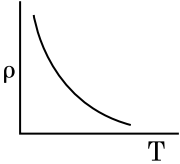
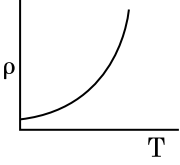
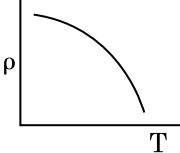
1. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth ?
 - (1) 48 N
 - (2) 32 N
 - (3) 30 N
 - (4) 24 N
2. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be :
 - (1) 523 Hz
 - (2) 524 Hz
 - (3) 536 Hz
 - (4) 537 Hz
3. The capacitance of a parallel plate capacitor with air as medium is $6 \mu\text{F}$. With the introduction of a dielectric medium, the capacitance becomes $30 \mu\text{F}$. The permittivity of the medium is :

$$(\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2})$$
 - (1) $0.44 \times 10^{-13} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - (2) $1.77 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - (3) $0.44 \times 10^{-10} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - (4) $5.00 \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
4. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is :
 - (1) 0.01 mm
 - (2) 0.25 mm
 - (3) 0.5 mm
 - (4) 1.0 mm
5. A short electric dipole has a dipole moment of $16 \times 10^{-9} \text{ C m}$. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is :

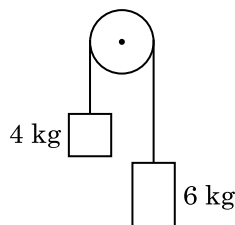
$$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$$
 - (1) 50 V
 - (2) 200 V
 - (3) 400 V
 - (4) zero
6. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , then the angle of incidence is nearly equal to :
 - (1) $\frac{A}{2\mu}$
 - (2) $\frac{2A}{\mu}$
 - (3) μA
 - (4) $\frac{\mu A}{2}$
7. A spherical conductor of radius 10 cm has a charge of $3.2 \times 10^{-7} \text{ C}$ distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere ?

$$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$$
 - (1) $1.28 \times 10^4 \text{ N/C}$
 - (2) $1.28 \times 10^5 \text{ N/C}$
 - (3) $1.28 \times 10^6 \text{ N/C}$
 - (4) $1.28 \times 10^7 \text{ N/C}$
8. For transistor action, which of the following statements is **correct** ?
 - (1) Base, emitter and collector regions should have same doping concentrations.
 - (2) Base, emitter and collector regions should have same size.
 - (3) Both emitter junction as well as the collector junction are forward biased.
 - (4) The base region must be very thin and lightly doped.
9. Dimensions of stress are :
 - (1) $[\text{MLT}^{-2}]$
 - (2) $[\text{ML}^2\text{T}^{-2}]$
 - (3) $[\text{ML}^0\text{T}^{-2}]$
 - (4) $[\text{ML}^{-1}\text{T}^{-2}]$
10. In a certain region of space with volume 0.2 m^3 , the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is :
 - (1) zero
 - (2) 0.5 N/C
 - (3) 1 N/C
 - (4) 5 N/C

11. When a uranium isotope ${}_{92}^{235}\text{U}$ is bombarded with a neutron, it generates ${}_{36}^{89}\text{Kr}$, three neutrons and :
- ${}_{56}^{144}\text{Ba}$
 - ${}_{40}^{91}\text{Zr}$
 - ${}_{36}^{101}\text{Kr}$
 - ${}_{36}^{103}\text{Kr}$
12. The increase in the width of the depletion region in a p-n junction diode is due to :
- forward bias only
 - reverse bias only
 - both forward bias and reverse bias
 - increase in forward current
13. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is : ($g = 10 \text{ m/s}^2$)
- 360 m
 - 340 m
 - 320 m
 - 300 m
14. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5 r_2$) through 1 K are in the ratio :
- $\frac{27}{8}$
 - $\frac{9}{4}$
 - $\frac{3}{2}$
 - $\frac{5}{3}$
15. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C . Its density is : ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)
- 0.5 kg/m^3
 - 0.2 kg/m^3
 - 0.1 kg/m^3
 - 0.02 kg/m^3
16. For which one of the following, Bohr model is **not** valid ?
- Hydrogen atom
 - Singly ionised helium atom (He^+)
 - Deuteron atom
 - Singly ionised neon atom (Ne^+)
17. A wire of length L , area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is :
- $\frac{MgL_1}{AL}$
 - $\frac{Mg(L_1 - L)}{AL}$
 - $\frac{MgL}{AL_1}$
 - $\frac{MgL}{A(L_1 - L)}$
18. The solids which have the negative temperature coefficient of resistance are :
- metals
 - insulators only
 - semiconductors only
 - insulators and semiconductors
19. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :
- $\pi \text{ rad}$
 - $\frac{3\pi}{2} \text{ rad}$
 - $\frac{\pi}{2} \text{ rad}$
 - zero
20. Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2 . The energy received by the surface during time span of 1 minute is :
- $10 \times 10^3 \text{ J}$
 - $12 \times 10^3 \text{ J}$
 - $24 \times 10^3 \text{ J}$
 - $48 \times 10^3 \text{ J}$

21. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves)
- (1) $c : 1$
 - (2) $1 : 1$
 - (3) $1 : c$
 - (4) $1 : c^2$
22. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes :
- (1) double
 - (2) half
 - (3) four times
 - (4) one-fourth
23. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is :
- (1) 10 V
 - (2) 10^2 V
 - (3) 10^3 V
 - (4) 10^4 V
24. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper ?
- (1) 
 - (2) 
 - (3) 
 - (4) 
25. The average thermal energy for a mono-atomic gas is : (k_B is Boltzmann constant and T , absolute temperature)
- (1) $\frac{1}{2} k_B T$
 - (2) $\frac{3}{2} k_B T$
 - (3) $\frac{5}{2} k_B T$
 - (4) $\frac{7}{2} k_B T$
26. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :
- ($\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$)
- (1) $6.28 \times 10^{-4} \text{ T}$
 - (2) $3.14 \times 10^{-4} \text{ T}$
 - (3) $6.28 \times 10^{-5} \text{ T}$
 - (4) $3.14 \times 10^{-5} \text{ T}$
27. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1} . The permeability of the material of the rod is :
- ($\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$)
- (1) $2.4\pi \times 10^{-4} \text{ T m A}^{-1}$
 - (2) $8.0 \times 10^{-5} \text{ T m A}^{-1}$
 - (3) $2.4\pi \times 10^{-5} \text{ T m A}^{-1}$
 - (4) $2.4\pi \times 10^{-7} \text{ T m A}^{-1}$
28. Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$?
- (1) 9.9801 m
 - (2) 9.98 m
 - (3) 9.980 m
 - (4) 9.9 m
29. A charged particle having drift velocity of $7.5 \times 10^{-4} \text{ m s}^{-1}$ in an electric field of $3 \times 10^{-10} \text{ V m}^{-1}$, has a mobility in $\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$ of :
- (1) 2.25×10^{15}
 - (2) 2.5×10^6
 - (3) 2.5×10^{-6}
 - (4) 2.25×10^{-15}

30. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of :
- (1) 33 cm
 - (2) 50 cm
 - (3) 67 cm
 - (4) 80 cm
31. The mean free path for a gas, with molecular diameter d and number density n can be expressed as :
- (1) $\frac{1}{\sqrt{2} n \pi d}$
 - (2) $\frac{1}{\sqrt{2} n \pi d^2}$
 - (3) $\frac{1}{\sqrt{2} n^2 \pi d^2}$
 - (4) $\frac{1}{\sqrt{2} n^2 \pi^2 d^2}$
32. The energy equivalent of 0.5 g of a substance is :
- (1) 4.5×10^{16} J
 - (2) 4.5×10^{13} J
 - (3) 1.5×10^{13} J
 - (4) 0.5×10^{13} J
33. A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of 1Ω of the resistance wire is :
- (1) 1.0×10^{-2} m
 - (2) 1.0×10^{-1} m
 - (3) 1.5×10^{-1} m
 - (4) 1.5×10^{-2} m
34. The Brewsters angle i_b for an interface should be :
- (1) $0^\circ < i_b < 30^\circ$
 - (2) $30^\circ < i_b < 45^\circ$
 - (3) $45^\circ < i_b < 90^\circ$
 - (4) $i_b = 90^\circ$
35. A capillary tube of radius r is immersed in water and water rises in it to a height h . The mass of the water in the capillary is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is :
- (1) 2.5 g
 - (2) 5.0 g
 - (3) 10.0 g
 - (4) 20.0 g
36. Find the torque about the origin when a force of $3\hat{j}$ N acts on a particle whose position vector is $2\hat{k}$ m.
- (1) $6\hat{i}$ N m
 - (2) $6\hat{j}$ N m
 - (3) $-6\hat{i}$ N m
 - (4) $6\hat{k}$ N m
37. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is :
- (1) zero
 - (2) 0.5
 - (3) 1.0
 - (4) -1.0
38. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly :
- (1) 6
 - (2) 0.6
 - (3) 0.06
 - (4) 0.006
39. Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is :

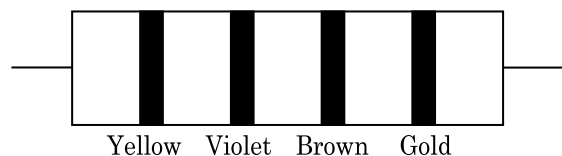


- (1) g
- (2) $g/2$
- (3) $g/5$
- (4) $g/10$

40. A $40\ \mu\text{F}$ capacitor is connected to a $200\ \text{V}$, $50\ \text{Hz}$ ac supply. The rms value of the current in the circuit is, nearly :

- (1) $1.7\ \text{A}$
- (2) $2.05\ \text{A}$
- (3) $2.5\ \text{A}$
- (4) $25.1\ \text{A}$

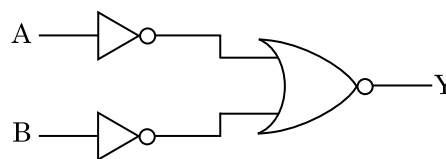
41. The color code of a resistance is given below :



The values of resistance and tolerance, respectively, are :

- (1) $470\ \text{k}\Omega$, 5%
 - (2) $47\ \text{k}\Omega$, 10%
 - (3) $4.7\ \text{k}\Omega$, 5%
 - (4) $470\ \Omega$, 5%
42. Assume that light of wavelength $600\ \text{nm}$ is coming from a star. The limit of resolution of telescope whose objective has a diameter of $2\ \text{m}$ is :
- (1) $3.66 \times 10^{-7}\ \text{rad}$
 - (2) $1.83 \times 10^{-7}\ \text{rad}$
 - (3) $7.32 \times 10^{-7}\ \text{rad}$
 - (4) $6.00 \times 10^{-7}\ \text{rad}$
43. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is :
- (1) isothermal
 - (2) adiabatic
 - (3) isochoric
 - (4) isobaric
44. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled ?
- (1) doubled
 - (2) four times
 - (3) one-fourth
 - (4) zero

45. For the logic circuit shown, the truth table is :



- (1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1
- (2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1
- (3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0
- (4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

46. Match the following :

	Oxide		Nature
(a)	CO	(i)	Basic
(b)	BaO	(ii)	Neutral
(c)	Al_2O_3	(iii)	Acidic
(d)	Cl_2O_7	(iv)	Amphoteric

Which of the following is **correct** option ?

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

47. The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

- (1) Iron
- (2) Copper
- (3) Calcium
- (4) Potassium

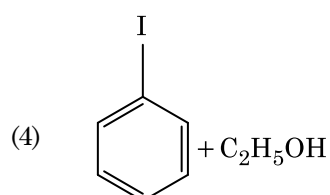
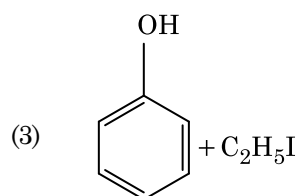
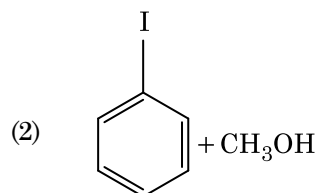
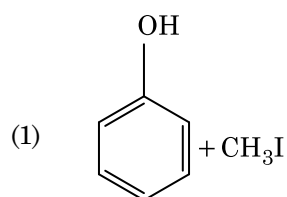
48. The number of Faradays(F) required to produce 20 g of calcium from molten CaCl_2 (Atomic mass of Ca = 40 g mol⁻¹) is :
- 1
 - 2
 - 3
 - 4
49. Which of the following alkane cannot be made in good yield by Wurtz reaction ?
- n-Hexane
 - 2,3-Dimethylbutane
 - n-Heptane
 - n-Butane
50. The freezing point depression constant (K_f) of benzene is 5.12 K kg mol⁻¹. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) :
- 0.20 K
 - 0.80 K
 - 0.40 K
 - 0.60 K
51. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is :
- β -Elimination reaction
 - Follows Zaitsev rule
 - Dehydrohalogenation reaction
 - Dehydration reaction
- (a), (b), (c)
 - (a), (c), (d)
 - (b), (c), (d)
 - (a), (b), (d)
52. Match the following and identify the **correct** option.
- | | | | |
|-----|---------------------------------------|-------|---|
| (a) | $\text{CO(g)} + \text{H}_2\text{(g)}$ | (i) | $\text{Mg(HCO}_3)_2 + \text{Ca(HCO}_3)_2$ |
| (b) | Temporary hardness of water | (ii) | An electron deficient hydride |
| (c) | B_2H_6 | (iii) | Synthesis gas |
| (d) | H_2O_2 | (iv) | Non-planar structure |
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|------|
| (1) | (iii) | (i) | (ii) | (iv) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (i) | (iii) | (ii) | (iv) |
53. Which one of the followings has maximum number of atoms ?
- 1 g of Ag(s) [Atomic mass of Ag = 108]
 - 1 g of Mg(s) [Atomic mass of Mg = 24]
 - 1 g of $\text{O}_2\text{(g)}$ [Atomic mass of O = 16]
 - 1 g of Li(s) [Atomic mass of Li = 7]
54. The rate constant for a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :
- 100 s
 - 200 s
 - 500 s
 - 1000 s
55. Which of the following is the **correct** order of increasing field strength of ligands to form coordination compounds ?
- $\text{SCN}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 - $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$
 - $\text{F}^- < \text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 - $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{SCN}^- < \text{F}^-$
56. A mixture of N_2 and Ar gases in a cylinder contains 7 g of N_2 and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N_2 is :
- [Use atomic masses (in g mol⁻¹) : N = 14, Ar = 40]
- 9 bar
 - 12 bar
 - 15 bar
 - 18 bar
57. Measuring Zeta potential is useful in determining which property of colloidal solution ?
- Viscosity
 - Solubility
 - Stability of the colloidal particles
 - Size of the colloidal particles
58. Sucrose on hydrolysis gives :
- β -D-Glucose + α -D-Fructose
 - α -D-Glucose + β -D-Glucose
 - α -D-Glucose + β -D-Fructose
 - α -D-Fructose + β -D-Fructose

59. Find out the solubility of $\text{Ni}(\text{OH})_2$ in 0.1 M NaOH. Given that the ionic product of $\text{Ni}(\text{OH})_2$ is 2×10^{-15} .
- $2 \times 10^{-13} \text{ M}$
 - $2 \times 10^{-8} \text{ M}$
 - $1 \times 10^{-13} \text{ M}$
 - $1 \times 10^8 \text{ M}$
60. The mixture which shows positive deviation from Raoult's law is :
- Ethanol + Acetone
 - Benzene + Toluene
 - Acetone + Chloroform
 - Chloroethane + Bromoethane
61. Identify a molecule which does **not** exist.
- He_2
 - Li_2
 - C_2
 - O_2
62. Identify the **incorrect** statement.
- $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.
 - The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
 - Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
 - The oxidation states of chromium in CrO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ are not the same.
63. Which of the following is a basic amino acid ?
- Serine
 - Alanine
 - Tyrosine
 - Lysine
64. An element has a body centered cubic (bcc) structure with a cell edge of 288 pm. The atomic radius is :
- $\frac{\sqrt{3}}{4} \times 288 \text{ pm}$
 - $\frac{\sqrt{2}}{4} \times 288 \text{ pm}$
 - $\frac{4}{\sqrt{3}} \times 288 \text{ pm}$
 - $\frac{4}{\sqrt{2}} \times 288 \text{ pm}$
65. Hydrolysis of sucrose is given by the following reaction.
- $$\text{Sucrose} + \text{H}_2\text{O} \rightleftharpoons \text{Glucose} + \text{Fructose}$$
- If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^\ominus$ at the same temperature will be :
- $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(2 \times 10^{13})$
 - $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(2 \times 10^{13})$
 - $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(3 \times 10^{13})$
 - $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(4 \times 10^{13})$
66. Which of the following is a natural polymer ?
- cis*-1,4-polyisoprene
 - poly (Butadiene-styrene)
 - polybutadiene
 - poly (Butadiene-acrylonitrile)
67. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following ?
- I effect of $-\text{CH}_3$ groups
 - + R effect of $-\text{CH}_3$ groups
 - R effect of $-\text{CH}_3$ groups
 - Hyperconjugation
68. An increase in the concentration of the reactants of a reaction leads to change in :
- activation energy
 - heat of reaction
 - threshold energy
 - collision frequency

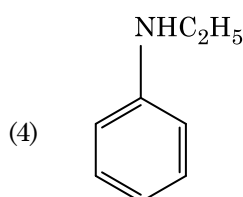
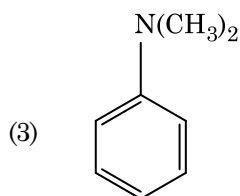
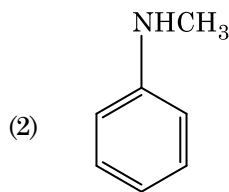
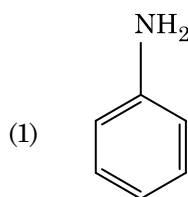
69. Which of the following is **not** correct about carbon monoxide ?

- (1) It forms carboxyhaemoglobin.
- (2) It reduces oxygen carrying ability of blood.
- (3) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
- (4) It is produced due to incomplete combustion.

70. Anisole on cleavage with HI gives :



71. Which of the following amine will give the carbylamine test ?



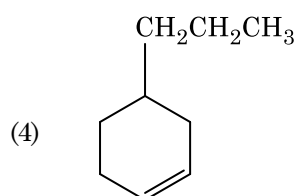
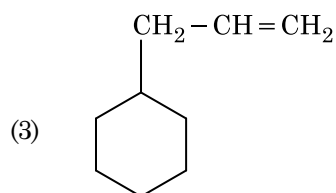
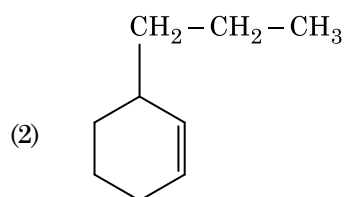
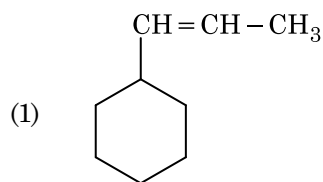
72. Identify the **incorrect** match.

	Name	IUPAC Official Name
(a)	Unnilunium	(i) Mendelevium
(b)	Unniltrium	(ii) Lawrencium
(c)	Unnilhexium	(iii) Seaborgium
(d)	Unununnium	(iv) Darmstadtium
(1)	(a), (i)	
(2)	(b), (ii)	
(3)	(c), (iii)	
(4)	(d), (iv)	

73. On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be :

- (1) Hydrogen gas
- (2) Oxygen gas
- (3) H₂S gas
- (4) SO₂ gas

74. An alkene on ozonolysis gives methanal as one of the product. Its structure is :



75. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

- (1) Isopropyl alcohol
- (2) Sec. butyl alcohol
- (3) Tert. butyl alcohol
- (4) Isobutyl alcohol

76. The calculated spin only magnetic moment of Cr^{2+} ion is :

- (1) 3.87 BM
- (2) 4.90 BM
- (3) 5.92 BM
- (4) 2.84 BM

77. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as :

- (1) Aldol condensation
- (2) Cannizzaro's reaction
- (3) Cross Cannizzaro's reaction
- (4) Cross Aldol condensation

78. Which of the following oxoacid of sulphur has $-\text{O}-\text{O}-$ linkage ?

- (1) H_2SO_3 , sulphurous acid
- (2) H_2SO_4 , sulphuric acid
- (3) $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid
- (4) $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid

79. Which of the following set of molecules will have zero dipole moment ?

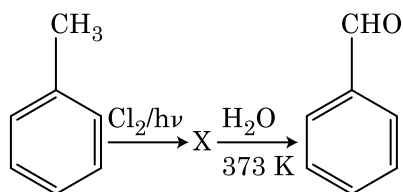
- (1) Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene
- (2) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene
- (3) Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene
- (4) Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene

80. The number of protons, neutrons and electrons in $^{175}_{71}\text{Lu}$, respectively, are :

- (1) 71, 104 and 71
- (2) 104, 71 and 71
- (3) 71, 71 and 104
- (4) 175, 104 and 71

81. Identify the **correct** statements from the following :
- $\text{CO}_2(\text{g})$ is used as refrigerant for ice-cream and frozen food.
 - The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
 - ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
 - CO is colorless and odourless gas.
- (a), (b) and (c) only
 - (a) and (c) only
 - (b) and (c) only
 - (c) and (d) only
82. Urea reacts with water to form **A** which will decompose to form **B**. **B** when passed through $\text{Cu}^{2+}(\text{aq})$, deep blue colour solution **C** is formed. What is the formula of **C** from the following ?
- CuSO_4
 - $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 - $\text{Cu}(\text{OH})_2$
 - $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
83. Which of the following is a cationic detergent ?
- Sodium lauryl sulphate
 - Sodium stearate
 - Cetyltrimethyl ammonium bromide
 - Sodium dodecylbenzene sulphonate
84. For the reaction, $2\text{Cl}(\text{g}) \rightarrow \text{Cl}_2(\text{g})$, the **correct** option is :
- $\Delta_r H > 0$ and $\Delta_r S > 0$
 - $\Delta_r H > 0$ and $\Delta_r S < 0$
 - $\Delta_r H < 0$ and $\Delta_r S > 0$
 - $\Delta_r H < 0$ and $\Delta_r S < 0$
85. Identify the **correct** statement from the following :
- Wrought iron is impure iron with 4% carbon.
 - Blister copper has blistered appearance due to evolution of CO_2 .
 - Vapour phase refining is carried out for Nickel by Van Arkel method.
 - Pig iron can be moulded into a variety of shapes.
86. The correct option for free expansion of an ideal gas under adiabatic condition is :
- $q = 0$, $\Delta T = 0$ and $w = 0$
 - $q = 0$, $\Delta T < 0$ and $w > 0$
 - $q < 0$, $\Delta T = 0$ and $w = 0$
 - $q > 0$, $\Delta T > 0$ and $w > 0$
87. HCl was passed through a solution of CaCl_2 , MgCl_2 and NaCl . Which of the following compound(s) crystallise(s) ?
- Both MgCl_2 and CaCl_2
 - Only NaCl
 - Only MgCl_2
 - NaCl , MgCl_2 and CaCl_2
88. What is the change in oxidation number of carbon in the following reaction ?
- $$\text{CH}_4(\text{g}) + 4\text{Cl}_2(\text{g}) \rightarrow \text{CCl}_4(\text{l}) + 4\text{HCl}(\text{g})$$
- + 4 to + 4
 - 0 to + 4
 - 4 to + 4
 - 0 to - 4

89. Identify compound X in the following sequence of reactions :



- (1)
- (2)
- (3)
- (4)

90. Paper chromatography is an example of :

- (1) Adsorption chromatography
- (2) Partition chromatography
- (3) Thin layer chromatography
- (4) Column chromatography

91. Dissolution of the synaptonemal complex occurs during :

- (1) Pachytene
- (2) Zygotene
- (3) Diplotene
- (4) Leptotene

92. Select the option including all sexually transmitted diseases.

- (1) Gonorrhoea, Syphilis, Genital herpes
- (2) Gonorrhoea, Malaria, Genital herpes
- (3) AIDS, Malaria, Filariasis
- (4) Cancer, AIDS, Syphilis

93. Which of the following would help in prevention of diuresis ?

- (1) More water reabsorption due to undersecretion of ADH
- (2) Reabsorption of Na^+ and water from renal tubules due to aldosterone
- (3) Atrial natriuretic factor causes vasoconstriction
- (4) Decrease in secretion of renin by JG cells

94. Cuboidal epithelium with brush border of microvilli is found in :

- (1) lining of intestine
- (2) ducts of salivary glands
- (3) proximal convoluted tubule of nephron
- (4) eustachian tube

95. Identify the substances having glycosidic bond and peptide bond, respectively in their structure :

- (1) Chitin, cholesterol
- (2) Glycerol, trypsin
- (3) Cellulose, lecithin
- (4) Inulin, insulin

96. Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to :

- (1) Insect pests
- (2) Fungal diseases
- (3) Plant nematodes
- (4) Insect predators

97. The ovary is half inferior in :

- (1) Brinjal
- (2) Mustard
- (3) Sunflower
- (4) Plum

98. Identify the **incorrect** statement.

- (1) Heart wood does not conduct water but gives mechanical support.
- (2) Sapwood is involved in conduction of water and minerals from root to leaf.
- (3) Sapwood is the innermost secondary xylem and is lighter in colour.
- (4) Due to deposition of tannins, resins, oils etc., heart wood is dark in colour.

99. Select the **correct** events that occur during inspiration.

- (a) Contraction of diaphragm
- (b) Contraction of external inter-costal muscles
- (c) Pulmonary volume decreases
- (d) Intra pulmonary pressure increases
- (1) (a) and (b)
- (2) (c) and (d)
- (3) (a), (b) and (d)
- (4) only (d)

100. The process responsible for facilitating loss of water in liquid form from the tip of grass blades at night and in early morning is :

- (1) Transpiration
- (2) Root pressure
- (3) Imbibition
- (4) Plasmolysis

101. Identify the **wrong** statement with reference to immunity.

- (1) When exposed to antigen (living or dead) antibodies are produced in the host's body. It is called "Active immunity".
- (2) When ready-made antibodies are directly given, it is called "Passive immunity".
- (3) Active immunity is quick and gives full response.
- (4) Foetus receives some antibodies from mother, it is an example for passive immunity.

102. Match the following :

- | | |
|-------------------------------------|---------------|
| (a) Inhibitor of catalytic activity | (i) Ricin |
| (b) Possess peptide bonds | (ii) Malonate |
| (c) Cell wall material in fungi | (iii) Chitin |
| (d) Secondary metabolite | (iv) Collagen |

Choose the **correct** option from the following :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (ii) | (iv) | (iii) | (i) |
| (2) | (iii) | (i) | (iv) | (ii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (ii) | (iii) | (i) | (iv) |

103. Identify the **correct** statement with regard to G₁ phase (Gap 1) of interphase.

- (1) DNA synthesis or replication takes place.
- (2) Reorganisation of all cell components takes place.
- (3) Cell is metabolically active, grows but does not replicate its DNA.
- (4) Nuclear Division takes place.

104. Experimental verification of the chromosomal theory of inheritance was done by :

- (1) Mendel
- (2) Sutton
- (3) Boveri
- (4) Morgan

105. Which of the following hormone levels will cause release of ovum (ovulation) from the graffian follicle ?

- (1) High concentration of Estrogen
- (2) High concentration of Progesterone
- (3) Low concentration of LH
- (4) Low concentration of FSH

106. If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is approximately :

- (1) 2.0 meters
- (2) 2.5 meters
- (3) 2.2 meters
- (4) 2.7 meters

107. Which of the following statements are **true** for the phylum-Chordata ?
- In Urochordata notochord extends from head to tail and it is present throughout their life.
 - In Vertebrata notochord is present during the embryonic period only.
 - Central nervous system is dorsal and hollow.
 - Chordata is divided into 3 subphyla : Hemichordata, Tunicata and Cephalochordata.
- (d) and (c)
 - (c) and (a)
 - (a) and (b)
 - (b) and (c)
108. The specific palindromic sequence which is recognized by EcoRI is :
- 5' - GAATTC - 3'
3' - CTTAAG - 5'
 - 5' - GGAACC - 3'
3' - CCTTGG - 5'
 - 5' - CTTAAG - 3'
3' - GAATTC - 5'
 - 5' - GGATCC - 3'
3' - CCTAGG - 5'
109. Ray florets have :
- Inferior ovary
 - Superior ovary
 - Hypogynous ovary
 - Half inferior ovary
110. Select the **correct** match.
- | | | |
|-------------------------|---|--|
| (1) Haemophilia | - | Y linked |
| (2) Phenylketonuria | - | Autosomal dominant trait |
| (3) Sickle cell anaemia | - | Autosomal recessive trait, chromosome-11 |
| (4) Thalassemia | - | X linked |
111. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits ?
- 4
 - 2
 - 14
 - 8
112. In which of the following techniques, the embryos are transferred to assist those females who cannot conceive ?
- ZIFT and IUT
 - GIFT and ZIFT
 - ICSI and ZIFT
 - GIFT and ICSI
113. The enzyme enterokinase helps in conversion of :
- protein into polypeptides
 - trypsinogen into trypsin
 - caseinogen into casein
 - pepsinogen into pepsin
114. The sequence that controls the copy number of the linked DNA in the vector, is termed :
- Selectable marker
 - Ori site
 - Palindromic sequence
 - Recognition site
115. Goblet cells of alimentary canal are modified from :
- Squamous epithelial cells
 - Columnar epithelial cells
 - Chondrocytes
 - Compound epithelial cells
116. Which of the following statements about inclusion bodies is **incorrect** ?
- They are not bound by any membrane.
 - These are involved in ingestion of food particles.
 - They lie free in the cytoplasm.
 - These represent reserve material in cytoplasm.
117. Which of the following statements is **correct** ?
- Adenine pairs with thymine through two H-bonds.
 - Adenine pairs with thymine through one H-bond.
 - Adenine pairs with thymine through three H-bonds.
 - Adenine does not pair with thymine.

118. Identify the **wrong** statement with reference to the gene 'I' that controls ABO blood groups.

- (1) The gene (I) has three alleles.
- (2) A person will have only two of the three alleles.
- (3) When I^A and I^B are present together, they express same type of sugar.
- (4) Allele 'i' does not produce any sugar.

119. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Bt cotton	(i)	Gene therapy
(b)	Adenosine deaminase deficiency	(ii)	Cellular defence
(c)	RNAi	(iii)	Detection of HIV infection
(d)	PCR	(iv)	<i>Bacillus thuringiensis</i>

	(a)	(b)	(c)	(d)
(1)	(iv)	(i)	(ii)	(iii)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(i)	(ii)	(iii)	(iv)

120. Montreal protocol was signed in 1987 for control of :

- (1) Transport of Genetically modified organisms from one country to another
- (2) Emission of ozone depleting substances
- (3) Release of Green House gases
- (4) Disposal of e-wastes

121. Match the following diseases with the causative organism and select the **correct** option.

Column - I		Column - II	
(a)	Typhoid	(i)	<i>Wuchereria</i>
(b)	Pneumonia	(ii)	<i>Plasmodium</i>
(c)	Filariasis	(iii)	<i>Salmonella</i>
(d)	Malaria	(iv)	<i>Haemophilus</i>

	(a)	(b)	(c)	(d)
(1)	(i)	(iii)	(ii)	(iv)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(ii)	(i)	(iii)	(iv)
(4)	(iv)	(i)	(ii)	(iii)

122. Identify the **wrong** statement with regard to Restriction Enzymes.

- (1) Each restriction enzyme functions by inspecting the length of a DNA sequence.
- (2) They cut the strand of DNA at palindromic sites.
- (3) They are useful in genetic engineering.
- (4) Sticky ends can be joined by using DNA ligases.

123. The QRS complex in a standard ECG represents :

- (1) Repolarisation of auricles
- (2) Depolarisation of auricles
- (3) Depolarisation of ventricles
- (4) Repolarisation of ventricles

124. Which one of the following is the most abundant protein in the animals ?

- (1) Haemoglobin
- (2) Collagen
- (3) Lectin
- (4) Insulin

125. In gel electrophoresis, separated DNA fragments can be visualized with the help of :

- (1) Acetocarmine in bright blue light
- (2) Ethidium bromide in UV radiation
- (3) Acetocarmine in UV radiation
- (4) Ethidium bromide in infrared radiation

126. Which of the following is **not** an inhibitory substance governing seed dormancy ?

- (1) Gibberellic acid
- (2) Abscissic acid
- (3) Phenolic acid
- (4) Para-ascorbic acid

127. In water hyacinth and water lily, pollination takes place by :

- (1) insects or wind
- (2) water currents only
- (3) wind and water
- (4) insects and water

128. By which method was a new breed 'Hisardale' of sheep formed by using Bikaneri ewes and Marino rams ?

- (1) Out crossing
- (2) Mutational breeding
- (3) Cross breeding
- (4) Inbreeding

129. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus ?

- (1) Uremia and Ketonuria
- (2) Uremia and Renal Calculi
- (3) Ketonuria and Glycosuria
- (4) Renal calculi and Hyperglycaemia

130. Which of the following refer to **correct** example(s) of organisms which have evolved due to changes in environment brought about by anthropogenic action ?

- (a) Darwin's Finches of Galapagos islands.
- (b) Herbicide resistant weeds.
- (c) Drug resistant eukaryotes.
- (d) Man-created breeds of domesticated animals like dogs.

- (1) only (a)
- (2) (a) and (c)
- (3) (b), (c) and (d)
- (4) only (d)

131. Embryological support for evolution was disapproved by :

- (1) Karl Ernst von Baer
- (2) Alfred Wallace
- (3) Charles Darwin
- (4) Oparin

132. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Floating Ribs	(i)	Located between second and seventh ribs
(b)	Acromion	(ii)	Head of the Humerus
(c)	Scapula	(iii)	Clavicle
(d)	Glenoid cavity	(iv)	Do not connect with the sternum

	(a)	(b)	(c)	(d)
(1)	(ii)	(iv)	(i)	(iii)
(2)	(i)	(iii)	(ii)	(iv)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(iv)	(iii)	(i)	(ii)

133. The body of the ovule is fused within the funicle at :

- (1) Hilum
- (2) Micropyle
- (3) Nucellus
- (4) Chalaza

134. Which of the following is put into Anaerobic sludge digester for further sewage treatment ?

- (1) Primary sludge
- (2) Floating debris
- (3) Effluents of primary treatment
- (4) Activated sludge

135. The product(s) of reaction catalyzed by nitrogenase in root nodules of leguminous plants is/are :

- (1) Ammonia alone
- (2) Nitrate alone
- (3) Ammonia and oxygen
- (4) Ammonia and hydrogen

136. Choose the **correct** pair from the following :

- | | | | |
|-----|--------------|---|--|
| (1) | Ligases | - | Join the two DNA molecules |
| (2) | Polymerases | - | Break the DNA into fragments |
| (3) | Nucleases | - | Separate the two strands of DNA |
| (4) | Exonucleases | - | Make cuts at specific positions within DNA |

137. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop.

- (1) Cytokinin
- (2) Gibberellin
- (3) Ethylene
- (4) Absciscic acid

138. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Gregarious, polyphagous pest	(i)	<i>Asterias</i>
(b)	Adult with radial symmetry and larva with bilateral symmetry	(ii)	Scorpion
(c)	Book lungs	(iii)	<i>Ctenoplanea</i>
(d)	Bioluminescence	(iv)	<i>Locusta</i>

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (i) | (iii) | (ii) | (iv) |
| (2) | (iv) | (i) | (ii) | (iii) |
| (3) | (iii) | (ii) | (i) | (iv) |
| (4) | (ii) | (i) | (iii) | (iv) |

139. Identify the **wrong** statement with reference to transport of oxygen.

- (1) Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2 .
- (2) Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.
- (3) Higher H^+ conc. in alveoli favours the formation of oxyhaemoglobin.
- (4) Low pCO_2 in alveoli favours the formation of oxyhaemoglobin.

140. Match the following concerning essential elements and their functions in plants :

- | | | | |
|-----|-----------|-------|---------------------------------------|
| (a) | Iron | (i) | Photolysis of water |
| (b) | Zinc | (ii) | Pollen germination |
| (c) | Boron | (iii) | Required for chlorophyll biosynthesis |
| (d) | Manganese | (iv) | IAA biosynthesis |

Select the **correct** option :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|-------|
| (1) | (ii) | (i) | (iv) | (iii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (iv) | (i) | (ii) | (iii) |

141. The transverse section of a plant shows following anatomical features :

- (a) Large number of scattered vascular bundles surrounded by bundle sheath.
- (b) Large conspicuous parenchymatous ground tissue.
- (c) Vascular bundles conjoint and closed.
- (d) Phloem parenchyma absent.

Identify the category of plant and its part :

- (1) Monocotyledonous stem
- (2) Monocotyledonous root
- (3) Dicotyledonous stem
- (4) Dicotyledonous root

142. In light reaction, plastoquinone facilitates the transfer of electrons from :

- (1) PS-II to $Cytb_6f$ complex
- (2) $Cytb_6f$ complex to PS-I
- (3) PS-I to $NADP^+$
- (4) PS-I to ATP synthase

143. Which of the following regions of the globe exhibits highest species diversity ?

- (1) Western Ghats of India
- (2) Madagascar
- (3) Himalayas
- (4) Amazon forests

144. The first phase of translation is :
- (1) Binding of mRNA to ribosome
 - (2) Recognition of DNA molecule
 - (3) Aminoacylation of tRNA
 - (4) Recognition of an anti-codon
145. The infectious stage of *Plasmodium* that enters the human body is :
- (1) Trophozoites
 - (2) Sporozoites
 - (3) Female gametocytes
 - (4) Male gametocytes
146. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells ?
- (1) Endoplasmic reticulum
 - (2) Peroxisomes
 - (3) Golgi bodies
 - (4) Polysomes
147. Identify the basic amino acid from the following.
- (1) Tyrosine
 - (2) Glutamic Acid
 - (3) Lysine
 - (4) Valine
148. The number of substrate level phosphorylations in one turn of citric acid cycle is :
- (1) Zero
 - (2) One
 - (3) Two
 - (4) Three
149. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask :
- (1) CH₄, H₂, NH₃ and water vapor at 800°C
 - (2) CH₃, H₂, NH₄ and water vapor at 800°C
 - (3) CH₄, H₂, NH₃ and water vapor at 600°C
 - (4) CH₃, H₂, NH₃ and water vapor at 600°C

150. Strobili or cones are found in :
- (1) *Salvinia*
 - (2) *Pteris*
 - (3) *Marchantia*
 - (4) *Equisetum*
151. Name the enzyme that facilitates opening of DNA helix during transcription.
- (1) DNA ligase
 - (2) DNA helicase
 - (3) DNA polymerase
 - (4) RNA polymerase
152. The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of :
- (1) 2 molecules of 3-C compound
 - (2) 1 molecule of 3-C compound
 - (3) 1 molecule of 6-C compound
 - (4) 1 molecule of 4-C compound and 1 molecule of 2-C compound
153. Floridean starch has structure similar to :
- (1) Starch and cellulose
 - (2) Amylopectin and glycogen
 - (3) Mannitol and algin
 - (4) Laminarin and cellulose
154. The process of growth is maximum during :
- (1) Log phase
 - (2) Lag phase
 - (3) Senescence
 - (4) Dormancy
155. Which of the following statements is **not correct** ?
- (1) In man insulin is synthesised as a proinsulin.
 - (2) The proinsulin has an extra peptide called C-peptide.
 - (3) The functional insulin has A and B chains linked together by hydrogen bonds.
 - (4) Genetically engineered insulin is produced in *E-Coli*.

156. Select the **correct** statement.

- (1) Glucocorticoids stimulate gluconeogenesis.
- (2) Glucagon is associated with hypoglycemia.
- (3) Insulin acts on pancreatic cells and adipocytes.
- (4) Insulin is associated with hyperglycemia.

157. Match the following with respect to meiosis :

- | | |
|----------------|---------------------|
| (a) Zygotene | (i) Terminalization |
| (b) Pachytene | (ii) Chiasmata |
| (c) Diplotene | (iii) Crossing over |
| (d) Diakinesis | (iv) Synapsis |

Select the **correct** option from the following :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (ii) | (iv) | (iii) | (i) |

158. If the head of cockroach is removed, it may live for few days because :

- (1) the supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.
- (2) the cockroach does not have nervous system.
- (3) the head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.
- (4) the head holds a 1/3rd of a nervous system while the rest is situated along the dorsal part of its body.

159. Match the following columns and select the **correct** option.

- | Column - I | | Column - II | |
|--------------------------------|-------|----------------|--|
| (a) 6 - 15 pairs of gill slits | (i) | <i>Trygon</i> | |
| (b) Heterocercal caudal fin | (ii) | Cyclostomes | |
| (c) Air Bladder | (iii) | Chondrichthyes | |
| (d) Poison sting | (iv) | Osteichthyes | |
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (iv) | (ii) | (iii) | (i) |
| (4) | (i) | (iv) | (iii) | (ii) |

160. The roots that originate from the base of the stem are :

- (1) Fibrous roots
- (2) Primary roots
- (3) Prop roots
- (4) Lateral roots

161. Identify the **correct** statement with reference to human digestive system.

- (1) Ileum opens into small intestine.
- (2) Serosa is the innermost layer of the alimentary canal.
- (3) Ileum is a highly coiled part.
- (4) Vermiform appendix arises from duodenum.

162. Match the following columns and select the **correct** option.

- | Column - I | | Column - II | |
|-----------------------------------|-------|----------------------------------|--|
| (a) <i>Clostridium butylicum</i> | (i) | Cyclosporin-A | |
| (b) <i>Trichoderma polysporum</i> | (ii) | Butyric Acid | |
| (c) <i>Monascus purpureus</i> | (iii) | Citric Acid | |
| (d) <i>Aspergillus niger</i> | (iv) | Blood cholesterol lowering agent | |
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|-------|
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

163. In relation to Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is **correct** ?

- (1) Gross primary productivity is always less than net primary productivity.
- (2) Gross primary productivity is always more than net primary productivity.
- (3) Gross primary productivity and Net primary productivity are one and same.
- (4) There is no relationship between Gross primary productivity and Net primary productivity.

164. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Pituitary gland	(i)	Grave's disease
(b)	Thyroid gland	(ii)	Diabetes mellitus
(c)	Adrenal gland	(iii)	Diabetes insipidus
(d)	Pancreas	(iv)	Addison's disease
		(a)	(b)
(1)	(iv)	(iii)	(i)
(2)	(iii)	(ii)	(i)
(3)	(iii)	(i)	(iv)
(4)	(ii)	(i)	(iv)

165. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Placenta	(i)	Androgens
(b)	Zona pellucida	(ii)	Human Chorionic Gonadotropin (hCG)
(c)	Bulbo-urethral glands	(iii)	Layer of the ovum
(d)	Leydig cells	(iv)	Lubrication of the Penis
		(a)	(b)
(1)	(iv)	(iii)	(i)
(2)	(i)	(iv)	(ii)
(3)	(iii)	(ii)	(iv)
(4)	(ii)	(iii)	(iv)

166. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Organ of Corti	(i)	Connects middle ear and pharynx
(b)	Cochlea	(ii)	Coiled part of the labyrinth
(c)	Eustachian tube	(iii)	Attached to the oval window
(d)	Stapes	(iv)	Located on the basilar membrane
		(a)	(b)
(1)	(ii)	(iii)	(i)
(2)	(iii)	(i)	(iv)
(3)	(iv)	(ii)	(i)
(4)	(i)	(ii)	(iv)

167. Meiotic division of the secondary oocyte is completed :

- (1) Prior to ovulation
- (2) At the time of copulation
- (3) After zygote formation
- (4) At the time of fusion of a sperm with an ovum

168. Match the trophic levels with their **correct** species examples in grassland ecosystem.

- | | | | |
|-----|----------------------|-------|---------|
| (a) | Fourth trophic level | (i) | Crow |
| (b) | Second trophic level | (ii) | Vulture |
| (c) | First trophic level | (iii) | Rabbit |
| (d) | Third trophic level | (iv) | Grass |

Select the **correct** option :

- | | | | | |
|-----|-------|-------|-------|------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (iv) | (iii) | (ii) | (i) |
| (4) | (i) | (ii) | (iii) | (iv) |

169. Secondary metabolites such as nicotine, strychnine and caffeine are produced by plants for their :

- (1) Nutritive value
- (2) Growth response
- (3) Defence action
- (4) Effect on reproduction

170. Snow-blindness in Antarctic region is due to :

- (1) Freezing of fluids in the eye by low temperature
- (2) Inflammation of cornea due to high dose of UV-B radiation
- (3) High reflection of light from snow
- (4) Damage to retina caused by infra-red rays

171. Which of the following pairs is of unicellular algae ?

- (1) *Laminaria* and *Sargassum*
- (2) *Gelidium* and *Gracilaria*
- (3) *Anabaena* and *Volvox*
- (4) *Chlorella* and *Spirulina*

172. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Eosinophils	(i)	Immune response
(b)	Basophils	(ii)	Phagocytosis
(c)	Neutrophils	(iii)	Release histaminase, destructive enzymes
(d)	Lymphocytes	(iv)	Release granules containing histamine

	(a)	(b)	(c)	(d)
(1)	(iii)	(iv)	(ii)	(i)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(i)	(ii)	(iv)	(iii)
(4)	(ii)	(i)	(iii)	(iv)

173. According to Robert May, the global species diversity is about :

- (1) 1.5 million
- (2) 20 million
- (3) 50 million
- (4) 7 million

174. Flippers of Penguins and Dolphins are examples of :

- (1) Adaptive radiation
- (2) Convergent evolution
- (3) Industrial melanism
- (4) Natural selection

175. The plant parts which consist of two generations - one within the other :

- (a) Pollen grains inside the anther
 - (b) Germinated pollen grain with two male gametes
 - (c) Seed inside the fruit
 - (d) Embryo sac inside the ovule
- (1) (a) only
 - (2) (a), (b) and (c)
 - (3) (c) and (d)
 - (4) (a) and (d)

176. Which of the following is **correct** about viroids ?

- (1) They have RNA with protein coat.
- (2) They have free RNA without protein coat.
- (3) They have DNA with protein coat.
- (4) They have free DNA without protein coat.

177. Match the organism with its use in biotechnology.

(a)	<i>Bacillus thuringiensis</i>	(i)	Cloning vector
(b)	<i>Thermus aquaticus</i>	(ii)	Construction of first rDNA molecule
(c)	<i>Agrobacterium tumefaciens</i>	(iii)	DNA polymerase
(d)	<i>Salmonella typhimurium</i>	(iv)	Cry proteins

Select the **correct** option from the following :

	(a)	(b)	(c)	(d)
(1)	(ii)	(iv)	(iii)	(i)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(iii)	(iv)	(i)	(ii)

178. Which of the following is **not** an attribute of a population ?

- (1) Sex ratio
- (2) Natality
- (3) Mortality
- (4) Species interaction

179. Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G_0). This process occurs at the end of :

- (1) M phase
- (2) G_1 phase
- (3) S phase
- (4) G_2 phase

180. Bilaterally symmetrical and acoelomate animals are exemplified by :

- (1) Ctenophora
- (2) Platyhelminthes
- (3) Aschelminthes
- (4) Annelida

- o o o -

Space For Rough Work

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Test Booklet Code

AKANH

No. :

E5

This Booklet contains 24 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on **side-1** and **side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
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8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. **Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

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Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

1. Flippers of Penguins and Dolphins are examples of :
 - (1) Adaptive radiation
 - (2) Convergent evolution
 - (3) Industrial melanism
 - (4) Natural selection
2. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop.
 - (1) Cytokinin
 - (2) Gibberellin
 - (3) Ethylene
 - (4) Absciscic acid
3. Secondary metabolites such as nicotine, strychnine and caffeine are produced by plants for their :
 - (1) Nutritive value
 - (2) Growth response
 - (3) Defence action
 - (4) Effect on reproduction
4. The body of the ovule is fused within the funicle at :
 - (1) Hilum
 - (2) Micropyle
 - (3) Nucellus
 - (4) Chalaza
5. Match the following columns and select the **correct** option.

Column - I	Column - II
(a) <i>Clostridium butylicum</i>	(i) Cyclosporin-A
(b) <i>Trichoderma polysporum</i>	(ii) Butyric Acid
(c) <i>Monascus purpureus</i>	(iii) Citric Acid
(d) <i>Aspergillus niger</i>	(iv) Blood cholesterol lowering agent

	(a)	(b)	(c)	(d)
(1)	(iii)	(iv)	(ii)	(i)
(2)	(ii)	(i)	(iv)	(iii)
(3)	(i)	(ii)	(iv)	(iii)
(4)	(iv)	(iii)	(ii)	(i)
6. The process responsible for facilitating loss of water in liquid form from the tip of grass blades at night and in early morning is :
 - (1) Transpiration
 - (2) Root pressure
 - (3) Imbibition
 - (4) Plasmolysis
7. Which of the following is **not** an inhibitory substance governing seed dormancy ?
 - (1) Gibberellic acid
 - (2) Absciscic acid
 - (3) Phenolic acid
 - (4) Para-ascorbic acid
8. Identify the **incorrect** statement.
 - (1) Heart wood does not conduct water but gives mechanical support.
 - (2) Sapwood is involved in conduction of water and minerals from root to leaf.
 - (3) Sapwood is the innermost secondary xylem and is lighter in colour.
 - (4) Due to deposition of tannins, resins, oils etc., heart wood is dark in colour.
9. Choose the **correct** pair from the following :

(1) Ligases	-	Join the two DNA molecules
(2) Polymerases	-	Break the DNA into fragments
(3) Nucleases	-	Separate the two strands of DNA
(4) Exonucleases	-	Make cuts at specific positions within DNA
10. By which method was a new breed 'Hisardale' of sheep formed by using Bikaneri ewes and Marino rams ?
 - (1) Out crossing
 - (2) Mutational breeding
 - (3) Cross breeding
 - (4) Inbreeding

11. Dissolution of the synaptonemal complex occurs during :
- (1) Pachytene
 - (2) Zygotene
 - (3) Diplotene
 - (4) Leptotene
12. Match the following diseases with the causative organism and select the **correct** option.

Column - I		Column - II	
(a)	Typhoid	(i)	<i>Wuchereria</i>
(b)	Pneumonia	(ii)	<i>Plasmodium</i>
(c)	Filariasis	(iii)	<i>Salmonella</i>
(d)	Malaria	(iv)	<i>Haemophilus</i>

(a)	(b)	(c)	(d)
(1)	(i)	(iii)	(ii)
(2)	(iii)	(iv)	(i)
(3)	(ii)	(i)	(iii)
(4)	(iv)	(i)	(ii)

13. According to Robert May, the global species diversity is about :
- (1) 1.5 million
 - (2) 20 million
 - (3) 50 million
 - (4) 7 million
14. In light reaction, plastoquinone facilitates the transfer of electrons from :
- (1) PS-II to Cyt_b₆f complex
 - (2) Cyt_b₆f complex to PS-I
 - (3) PS-I to NADP⁺
 - (4) PS-I to ATP synthase

15. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Pituitary gland	(i)	Grave's disease
(b)	Thyroid gland	(ii)	Diabetes mellitus
(c)	Adrenal gland	(iii)	Diabetes insipidus
(d)	Pancreas	(iv)	Addison's disease

(a)	(b)	(c)	(d)
(1)	(iv)	(iii)	(i)
(2)	(iii)	(ii)	(i)
(3)	(iii)	(i)	(iv)
(4)	(ii)	(i)	(iv)

16. Which of the following statements are **true** for the phylum-Chordata ?

- (a) In Urochordata notochord extends from head to tail and it is present throughout their life.
 - (b) In Vertebrata notochord is present during the embryonic period only.
 - (c) Central nervous system is dorsal and hollow.
 - (d) Chordata is divided into 3 subphyla : Hemichordata, Tunicata and Cephalochordata.
- (1) (d) and (c)
 - (2) (c) and (a)
 - (3) (a) and (b)
 - (4) (b) and (c)

17. Select the option including all sexually transmitted diseases.

- (1) Gonorrhoea, Syphilis, Genital herpes
- (2) Gonorrhoea, Malaria, Genital herpes
- (3) AIDS, Malaria, Filaria
- (4) Cancer, AIDS, Syphilis

18. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Organ of Corti	(i)	Connects middle ear and pharynx
(b)	Cochlea	(ii)	Coiled part of the labyrinth
(c)	Eustachian tube	(iii)	Attached to the oval window
(d)	Stapes	(iv)	Located on the basilar membrane

(a)	(b)	(c)	(d)
(1)	(ii)	(iii)	(i)
(2)	(iii)	(i)	(iv)
(3)	(iv)	(ii)	(i)
(4)	(i)	(ii)	(iv)

19. Cuboidal epithelium with brush border of microvilli is found in :

- (1) lining of intestine
- (2) ducts of salivary glands
- (3) proximal convoluted tubule of nephron
- (4) eustachian tube

20. Identify the **wrong** statement with reference to transport of oxygen.

- (1) Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2 .
- (2) Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.
- (3) Higher H^+ conc. in alveoli favours the formation of oxyhaemoglobin.
- (4) Low pCO_2 in alveoli favours the formation of oxyhaemoglobin.

21. Goblet cells of alimentary canal are modified from :

- (1) Squamous epithelial cells
- (2) Columnar epithelial cells
- (3) Chondrocytes
- (4) Compound epithelial cells

22. Identify the **wrong** statement with regard to Restriction Enzymes.

- (1) Each restriction enzyme functions by inspecting the length of a DNA sequence.
- (2) They cut the strand of DNA at palindromic sites.
- (3) They are useful in genetic engineering.
- (4) Sticky ends can be joined by using DNA ligases.

23. Experimental verification of the chromosomal theory of inheritance was done by :

- (1) Mendel
- (2) Sutton
- (3) Boveri
- (4) Morgan

24. Identify the **correct** statement with reference to human digestive system.

- (1) Ileum opens into small intestine.
- (2) Serosa is the innermost layer of the alimentary canal.
- (3) Ileum is a highly coiled part.
- (4) Vermiform appendix arises from duodenum.

25. Identify the **wrong** statement with reference to the gene 'I' that controls ABO blood groups.

- (1) The gene (I) has three alleles.
- (2) A person will have only two of the three alleles.
- (3) When I^A and I^B are present together, they express same type of sugar.
- (4) Allele 'i' does not produce any sugar.

26. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Floating Ribs	(i)	Located between second and seventh ribs
(b)	Acromion	(ii)	Head of the Humerus
(c)	Scapula	(iii)	Clavicle
(d)	Glenoid cavity	(iv)	Do not connect with the sternum

	(a)	(b)	(c)	(d)
(1)	(ii)	(iv)	(i)	(iii)
(2)	(i)	(iii)	(ii)	(iv)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(iv)	(iii)	(i)	(ii)

27. The product(s) of reaction catalyzed by nitrogenase in root nodules of leguminous plants is/are :

- (1) Ammonia alone
- (2) Nitrate alone
- (3) Ammonia and oxygen
- (4) Ammonia and hydrogen

28. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Gregarious, polyphagous pest	(i)	<i>Asterias</i>
(b)	Adult with radial symmetry and larva with bilateral symmetry	(ii)	Scorpion
(c)	Book lungs	(iii)	<i>Ctenoplane</i>
(d)	Bioluminescence	(iv)	<i>Locusta</i>

	(a)	(b)	(c)	(d)
(1)	(i)	(iii)	(ii)	(iv)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(ii)	(i)	(iii)	(iv)

29. Snow-blindness in Antarctic region is due to :

- (1) Freezing of fluids in the eye by low temperature
- (2) Inflammation of cornea due to high dose of UV-B radiation
- (3) High reflection of light from snow
- (4) Damage to retina caused by infra-red rays

30. In relation to Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is **correct** ?

- (1) Gross primary productivity is always less than net primary productivity.
- (2) Gross primary productivity is always more than net primary productivity.
- (3) Gross primary productivity and Net primary productivity are one and same.
- (4) There is no relationship between Gross primary productivity and Net primary productivity.

31. Select the **correct** statement.

- (1) Glucocorticoids stimulate gluconeogenesis.
- (2) Glucagon is associated with hypoglycemia.
- (3) Insulin acts on pancreatic cells and adipocytes.
- (4) Insulin is associated with hyperglycemia.

32. Select the **correct** events that occur during inspiration.

- (a) Contraction of diaphragm
 - (b) Contraction of external inter-costal muscles
 - (c) Pulmonary volume decreases
 - (d) Intra pulmonary pressure increases
- (1) (a) and (b)
 - (2) (c) and (d)
 - (3) (a), (b) and (d)
 - (4) only (d)

33. Match the following concerning essential elements and their functions in plants :

- | | |
|---------------|---|
| (a) Iron | (i) Photolysis of water |
| (b) Zinc | (ii) Pollen germination |
| (c) Boron | (iii) Required for chlorophyll biosynthesis |
| (d) Manganese | (iv) IAA biosynthesis |

Select the **correct** option :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|-------|
| (1) | (ii) | (i) | (iv) | (iii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (iv) | (i) | (ii) | (iii) |

34. In which of the following techniques, the embryos are transferred to assist those females who cannot conceive ?

- (1) ZIFT and IUT
- (2) GIFT and ZIFT
- (3) ICSI and ZIFT
- (4) GIFT and ICSI

35. The infectious stage of *Plasmodium* that enters the human body is :

- (1) Trophozoites
- (2) Sporozoites
- (3) Female gametocytes
- (4) Male gametocytes

36. Which of the following hormone levels will cause release of ovum (ovulation) from the graffian follicle ?

- (1) High concentration of Estrogen
- (2) High concentration of Progesterone
- (3) Low concentration of LH
- (4) Low concentration of FSH

37. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus ?

- (1) Uremia and Ketonuria
- (2) Uremia and Renal Calculi
- (3) Ketonuria and Glycosuria
- (4) Renal calculi and Hyperglycaemia

38. Name the enzyme that facilitates opening of DNA helix during transcription.

- (1) DNA ligase
- (2) DNA helicase
- (3) DNA polymerase
- (4) RNA polymerase

39. Match the trophic levels with their **correct** species examples in grassland ecosystem.

- | | |
|--------------------------|--------------|
| (a) Fourth trophic level | (i) Crow |
| (b) Second trophic level | (ii) Vulture |
| (c) First trophic level | (iii) Rabbit |
| (d) Third trophic level | (iv) Grass |

Select the **correct** option :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (iv) | (iii) | (ii) | (i) |
| (4) | (i) | (ii) | (iii) | (iv) |

40. Match the following :

- | | |
|-------------------------------------|---------------|
| (a) Inhibitor of catalytic activity | (i) Ricin |
| (b) Possess peptide bonds | (ii) Malonate |
| (c) Cell wall material in fungi | (iii) Chitin |
| (d) Secondary metabolite | (iv) Collagen |

Choose the **correct** option from the following :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (ii) | (iv) | (iii) | (i) |
| (2) | (iii) | (i) | (iv) | (ii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (ii) | (iii) | (i) | (iv) |

41. The first phase of translation is :

- (1) Binding of mRNA to ribosome
- (2) Recognition of DNA molecule
- (3) Aminoacylation of tRNA
- (4) Recognition of an anti-codon

42. Identify the substances having glycosidic bond and peptide bond, respectively in their structure :

- (1) Chitin, cholesterol
- (2) Glycerol, trypsin
- (3) Cellulose, lecithin
- (4) Inulin, insulin

43. Which of the following statements about inclusion bodies is **incorrect** ?

- (1) They are not bound by any membrane.
- (2) These are involved in ingestion of food particles.
- (3) They lie free in the cytoplasm.
- (4) These represent reserve material in cytoplasm.

44. Match the following columns and select the **correct** option.

- | | Column - I | | Column - II |
|-----|--------------------------------|-------|-------------------------------|
| (a) | Bt cotton | (i) | Gene therapy |
| (b) | Adenosine deaminase deficiency | (ii) | Cellular defence |
| (c) | RNAi | (iii) | Detection of HIV infection |
| (d) | PCR | (iv) | <i>Bacillus thuringiensis</i> |

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (iv) | (i) | (ii) | (iii) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (i) | (ii) | (iii) | (iv) |

45. Identify the **correct** statement with regard to G₁ phase (Gap 1) of interphase.

- (1) DNA synthesis or replication takes place.
- (2) Reorganisation of all cell components takes place.
- (3) Cell is metabolically active, grows but does not replicate its DNA.
- (4) Nuclear Division takes place.

46. Which of the following is put into Anaerobic sludge digester for further sewage treatment ?

- (1) Primary sludge
- (2) Floating debris
- (3) Effluents of primary treatment
- (4) Activated sludge

47. Which of the following statements is **correct** ?
- (1) Adenine pairs with thymine through two H-bonds.
 - (2) Adenine pairs with thymine through one H-bond.
 - (3) Adenine pairs with thymine through three H-bonds.
 - (4) Adenine does not pair with thymine.
48. The sequence that controls the copy number of the linked DNA in the vector, is termed :
- (1) Selectable marker
 - (2) Ori site
 - (3) Palindromic sequence
 - (4) Recognition site
49. Select the **correct** match.
- | | | |
|-------------------------|---|--|
| (1) Haemophilia | - | Y linked |
| (2) Phenylketonuria | - | Autosomal dominant trait |
| (3) Sickle cell anaemia | - | Autosomal recessive trait, chromosome-11 |
| (4) Thalassemia | - | X linked |
50. Which of the following is **not** an attribute of a population ?
- (1) Sex ratio
 - (2) Natality
 - (3) Mortality
 - (4) Species interaction
51. Strobili or cones are found in :
- (1) *Salvinia*
 - (2) *Pteris*
 - (3) *Marchantia*
 - (4) *Equisetum*
52. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells ?
- (1) Endoplasmic reticulum
 - (2) Peroxisomes
 - (3) Golgi bodies
 - (4) Polysomes
53. Which of the following is **correct** about viroids ?
- (1) They have RNA with protein coat.
 - (2) They have free RNA without protein coat.
 - (3) They have DNA with protein coat.
 - (4) They have free DNA without protein coat.
54. The process of growth is maximum during :
- (1) Log phase
 - (2) Lag phase
 - (3) Senescence
 - (4) Dormancy
55. Which of the following regions of the globe exhibits highest species diversity ?
- (1) Western Ghats of India
 - (2) Madagascar
 - (3) Himalayas
 - (4) Amazon forests
56. The number of substrate level phosphorylations in one turn of citric acid cycle is :
- (1) Zero
 - (2) One
 - (3) Two
 - (4) Three
57. Meiotic division of the secondary oocyte is completed :
- (1) Prior to ovulation
 - (2) At the time of copulation
 - (3) After zygote formation
 - (4) At the time of fusion of a sperm with an ovum
58. Which of the following pairs is of unicellular algae ?
- (1) *Laminaria* and *Sargassum*
 - (2) *Gelidium* and *Gracilaria*
 - (3) *Anabaena* and *Volvox*
 - (4) *Chlorella* and *Spirulina*

59. The QRS complex in a standard ECG represents :

- (1) Repolarisation of auricles
- (2) Depolarisation of auricles
- (3) Depolarisation of ventricles
- (4) Repolarisation of ventricles

60. Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G_0). This process occurs at the end of :

- (1) M phase
- (2) G_1 phase
- (3) S phase
- (4) G_2 phase

61. Match the following with respect to meiosis :

- | | |
|----------------|---------------------|
| (a) Zygotene | (i) Terminalization |
| (b) Pachytene | (ii) Chiasmata |
| (c) Diplotene | (iii) Crossing over |
| (d) Diakinesis | (iv) Synapsis |

Select the **correct** option from the following :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (ii) | (iv) | (iii) | (i) |

62. Which one of the following is the most abundant protein in the animals ?

- (1) Haemoglobin
- (2) Collagen
- (3) Lectin
- (4) Insulin

63. The ovary is half inferior in :

- (1) Brinjal
- (2) Mustard
- (3) Sunflower
- (4) Plum

64. Ray florets have :

- (1) Inferior ovary
- (2) Superior ovary
- (3) Hypogynous ovary
- (4) Half inferior ovary

65. The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of :

- (1) 2 molecules of 3-C compound
- (2) 1 molecule of 3-C compound
- (3) 1 molecule of 6-C compound
- (4) 1 molecule of 4-C compound and 1 molecule of 2-C compound

66. The plant parts which consist of two generations - one within the other :

- (a) Pollen grains inside the anther
 - (b) Germinated pollen grain with two male gametes
 - (c) Seed inside the fruit
 - (d) Embryo sac inside the ovule
- (1) (a) only
 - (2) (a), (b) and (c)
 - (3) (c) and (d)
 - (4) (a) and (d)

67. Match the following columns and select the **correct** option.

- | Column - I | | Column - II | |
|-----------------|-------|--|--|
| (a) Eosinophils | (i) | Immune response | |
| (b) Basophils | (ii) | Phagocytosis | |
| (c) Neutrophils | (iii) | Release histaminase, destructive enzymes | |
| (d) Lymphocytes | (iv) | Release granules containing histamine | |

- | | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|-------|
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (iv) | (i) | (ii) | (iii) |
| (3) | (i) | (ii) | (iv) | (iii) |
| (4) | (ii) | (i) | (iii) | (iv) |

68. Bilaterally symmetrical and acoelomate animals are exemplified by :

- (1) Ctenophora
- (2) Platyhelminthes
- (3) Aschelminthes
- (4) Annelida

69. Identify the basic amino acid from the following.

- (1) Tyrosine
- (2) Glutamic Acid
- (3) Lysine
- (4) Valine

70. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a) Placenta	(i)	Androgens	
(b) Zona pellucida	(ii)	Human Chorionic Gonadotropin (hCG)	
(c) Bulbo-urethral glands	(iii)	Layer of the ovum	
(d) Leydig cells	(iv)	Lubrication of the Penis	
	(a)	(b)	(c)
(1)	(iv)	(iii)	(i)
(2)	(i)	(iv)	(ii)
(3)	(iii)	(ii)	(iv)
(4)	(ii)	(iii)	(iv)

71. Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to :

- (1) Insect pests
- (2) Fungal diseases
- (3) Plant nematodes
- (4) Insect predators

72. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a) 6 - 15 pairs of gill slits	(i)	<i>Trygon</i>	
(b) Heterocercal caudal fin	(ii)	Cyclostomes	
(c) Air Bladder	(iii)	Chondrichthyes	
(d) Poison sting	(iv)	Osteichthyes	
	(a)	(b)	(c)
(1)	(ii)	(iii)	(iv)
(2)	(iii)	(iv)	(i)
(3)	(iv)	(ii)	(iii)
(4)	(i)	(iv)	(iii)

73. Floridean starch has structure similar to :

- (1) Starch and cellulose
- (2) Amylopectin and glycogen
- (3) Mannitol and algin
- (4) Laminarin and cellulose

74. Which of the following statements is **not correct** ?

- (1) In man insulin is synthesised as a proinsulin.
- (2) The proinsulin has an extra peptide called C-peptide.
- (3) The functional insulin has A and B chains linked together by hydrogen bonds.
- (4) Genetically engineered insulin is produced in *E-Coli*.

75. If the head of cockroach is removed, it may live for few days because :

- (1) the supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.
- (2) the cockroach does not have nervous system.
- (3) the head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.
- (4) the head holds a 1/3rd of a nervous system while the rest is situated along the dorsal part of its body.

76. The enzyme enterokinase helps in conversion of :

- (1) protein into polypeptides
- (2) trypsinogen into trypsin
- (3) caseinogen into casein
- (4) pepsinogen into pepsin

77. The transverse section of a plant shows following anatomical features :

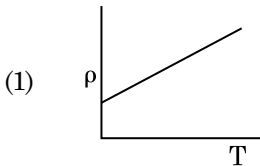
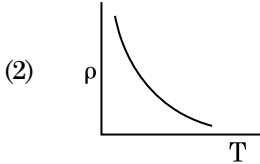
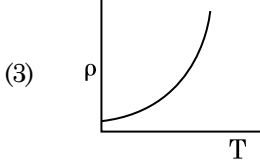
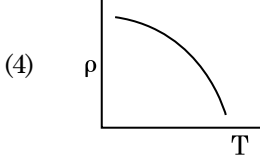
- (a) Large number of scattered vascular bundles surrounded by bundle sheath.
- (b) Large conspicuous parenchymatous ground tissue.
- (c) Vascular bundles conjoint and closed.
- (d) Phloem parenchyma absent.

Identify the category of plant and its part :

- (1) Monocotyledonous stem
- (2) Monocotyledonous root
- (3) Dicotyledonous stem
- (4) Dicotyledonous root

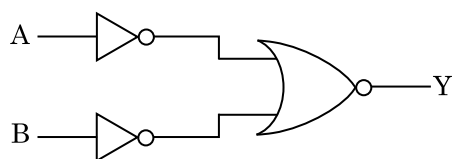
78. In water hyacinth and water lily, pollination takes place by :
- (1) insects or wind
 - (2) water currents only
 - (3) wind and water
 - (4) insects and water
79. In gel electrophoresis, separated DNA fragments can be visualized with the help of :
- (1) Acetocarmine in bright blue light
 - (2) Ethidium bromide in UV radiation
 - (3) Acetocarmine in UV radiation
 - (4) Ethidium bromide in infrared radiation
80. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits ?
- (1) 4
 - (2) 2
 - (3) 14
 - (4) 8
81. Which of the following refer to **correct** example(s) of organisms which have evolved due to changes in environment brought about by anthropogenic action ?
- (a) Darwin's Finches of Galapagos islands.
 - (b) Herbicide resistant weeds.
 - (c) Drug resistant eukaryotes.
 - (d) Man-created breeds of domesticated animals like dogs.
- (1) only (a)
 - (2) (a) and (c)
 - (3) (b), (c) and (d)
 - (4) only (d)
82. Match the organism with its use in biotechnology.
- | | |
|--------------------------------------|--|
| (a) <i>Bacillus thuringiensis</i> | (i) Cloning vector |
| (b) <i>Thermus aquaticus</i> | (ii) Construction of first rDNA molecule |
| (c) <i>Agrobacterium tumefaciens</i> | (iii) DNA polymerase |
| (d) <i>Salmonella typhimurium</i> | (iv) Cry proteins |
- Select the **correct** option from the following :
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (ii) | (iv) | (iii) | (i) |
| (2) | (iv) | (iii) | (i) | (ii) |
| (3) | (iii) | (ii) | (iv) | (i) |
| (4) | (iii) | (iv) | (i) | (ii) |
83. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask :
- (1) CH_4 , H_2 , NH_3 and water vapor at 800°C
 - (2) CH_3 , H_2 , NH_4 and water vapor at 800°C
 - (3) CH_4 , H_2 , NH_3 and water vapor at 600°C
 - (4) CH_3 , H_2 , NH_3 and water vapor at 600°C
84. Embryological support for evolution was disapproved by :
- (1) Karl Ernst von Baer
 - (2) Alfred Wallace
 - (3) Charles Darwin
 - (4) Oparin
85. If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is approximately :
- (1) 2.0 meters
 - (2) 2.5 meters
 - (3) 2.2 meters
 - (4) 2.7 meters

86. Identify the **wrong** statement with reference to immunity.
- (1) When exposed to antigen (living or dead) antibodies are produced in the host's body. It is called "Active immunity".
 - (2) When ready-made antibodies are directly given, it is called "Passive immunity".
 - (3) Active immunity is quick and gives full response.
 - (4) Foetus receives some antibodies from mother, it is an example for passive immunity.
87. The specific palindromic sequence which is recognized by EcoRI is :
- (1) 5' - GAATTC - 3'
3' - CTTAAG - 5'
 - (2) 5' - GGAACC - 3'
3' - CCTTGG - 5'
 - (3) 5' - CTTAAG - 3'
3' - GAATTC - 5'
 - (4) 5' - GGATCC - 3'
3' - CCTAGG - 5'
88. Which of the following would help in prevention of diuresis ?
- (1) More water reabsorption due to undersecretion of ADH
 - (2) Reabsorption of Na^+ and water from renal tubules due to aldosterone
 - (3) Atrial natriuretic factor causes vasoconstriction
 - (4) Decrease in secretion of renin by JG cells
89. Montreal protocol was signed in 1987 for control of :
- (1) Transport of Genetically modified organisms from one country to another
 - (2) Emission of ozone depleting substances
 - (3) Release of Green House gases
 - (4) Disposal of e-wastes
90. The roots that originate from the base of the stem are :
- (1) Fibrous roots
 - (2) Primary roots
 - (3) Prop roots
 - (4) Lateral roots
91. The solids which have the negative temperature coefficient of resistance are :
- (1) metals
 - (2) insulators only
 - (3) semiconductors only
 - (4) insulators and semiconductors
92. A charged particle having drift velocity of $7.5 \times 10^{-4} \text{ m s}^{-1}$ in an electric field of $3 \times 10^{-10} \text{ Vm}^{-1}$, has a mobility in $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$ of :
- (1) 2.25×10^{15}
 - (2) 2.5×10^6
 - (3) 2.5×10^{-6}
 - (4) 2.25×10^{-15}
93. For transistor action, which of the following statements is **correct** ?
- (1) Base, emitter and collector regions should have same doping concentrations.
 - (2) Base, emitter and collector regions should have same size.
 - (3) Both emitter junction as well as the collector junction are forward biased.
 - (4) The base region must be very thin and lightly doped.
94. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be :
- (1) 523 Hz
 - (2) 524 Hz
 - (3) 536 Hz
 - (4) 537 Hz
95. A wire of length L , area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is :
- (1) $\frac{MgL_1}{AL}$
 - (2) $\frac{Mg(L_1 - L)}{AL}$
 - (3) $\frac{MgL}{AL_1}$
 - (4) $\frac{MgL}{A(L_1 - L)}$

96. Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2 . The energy received by the surface during time span of 1 minute is :
- (1) $10 \times 10^3 \text{ J}$
 - (2) $12 \times 10^3 \text{ J}$
 - (3) $24 \times 10^3 \text{ J}$
 - (4) $48 \times 10^3 \text{ J}$
97. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :
- (1) $\pi \text{ rad}$
 - (2) $\frac{3\pi}{2} \text{ rad}$
 - (3) $\frac{\pi}{2} \text{ rad}$
 - (4) zero
98. A capillary tube of radius r is immersed in water and water rises in it to a height h . The mass of the water in the capillary is 5 g . Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is :
- (1) 2.5 g
 - (2) 5.0 g
 - (3) 10.0 g
 - (4) 20.0 g
99. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is :
- (1) zero
 - (2) 0.5
 - (3) 1.0
 - (4) -1.0
100. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes :
- (1) double
 - (2) half
 - (3) four times
 - (4) one-fourth
101. Dimensions of stress are :
- (1) $[\text{MLT}^{-2}]$
 - (2) $[\text{ML}^2\text{T}^{-2}]$
 - (3) $[\text{ML}^0\text{T}^{-2}]$
 - (4) $[\text{ML}^{-1}\text{T}^{-2}]$
102. Find the torque about the origin when a force of $3\hat{j} \text{ N}$ acts on a particle whose position vector is $2\hat{k} \text{ m}$.
- (1) $6\hat{i} \text{ N m}$
 - (2) $6\hat{j} \text{ N m}$
 - (3) $-6\hat{i} \text{ N m}$
 - (4) $6\hat{k} \text{ N m}$
103. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper ?
- (1) 
 - (2) 
 - (3) 
 - (4) 
104. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C . Its density is : ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)
- (1) 0.5 kg/m^3
 - (2) 0.2 kg/m^3
 - (3) 0.1 kg/m^3
 - (4) 0.02 kg/m^3

105. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves)
- (1) $c : 1$
 - (2) $1 : 1$
 - (3) $1 : c$
 - (4) $1 : c^2$
106. For which one of the following, Bohr model is **not** valid ?
- (1) Hydrogen atom
 - (2) Singly ionised helium atom (He^+)
 - (3) Deuteron atom
 - (4) Singly ionised neon atom (Ne^+)
107. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :
- ($\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$)
- (1) $6.28 \times 10^{-4} \text{ T}$
 - (2) $3.14 \times 10^{-4} \text{ T}$
 - (3) $6.28 \times 10^{-5} \text{ T}$
 - (4) $3.14 \times 10^{-5} \text{ T}$
108. The Brewsters angle i_b for an interface should be :
- (1) $0^\circ < i_b < 30^\circ$
 - (2) $30^\circ < i_b < 45^\circ$
 - (3) $45^\circ < i_b < 90^\circ$
 - (4) $i_b = 90^\circ$
109. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth ?
- (1) 48 N
 - (2) 32 N
 - (3) 30 N
 - (4) 24 N
110. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is :
- (1) 0.01 mm
 - (2) 0.25 mm
 - (3) 0.5 mm
 - (4) 1.0 mm
111. The mean free path for a gas, with molecular diameter d and number density n can be expressed as :
- (1) $\frac{1}{\sqrt{2} n \pi d}$
 - (2) $\frac{1}{\sqrt{2} n \pi d^2}$
 - (3) $\frac{1}{\sqrt{2} n^2 \pi d^2}$
 - (4) $\frac{1}{\sqrt{2} n^2 \pi^2 d^2}$
112. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is : ($g = 10 \text{ m/s}^2$)
- (1) 360 m
 - (2) 340 m
 - (3) 320 m
 - (4) 300 m
113. In a certain region of space with volume 0.2 m^3 , the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is :
- (1) zero
 - (2) 0.5 N/C
 - (3) 1 N/C
 - (4) 5 N/C
114. The average thermal energy for a mono-atomic gas is : (k_B is Boltzmann constant and T , absolute temperature)
- (1) $\frac{1}{2} k_B T$
 - (2) $\frac{3}{2} k_B T$
 - (3) $\frac{5}{2} k_B T$
 - (4) $\frac{7}{2} k_B T$

115. For the logic circuit shown, the truth table is :



(1) A B Y

0 0 0

0 1 0

1 0 0

1 1 1

(2) A B Y

0 0 0

0 1 1

1 0 1

1 1 1

(3) A B Y

0 0 1

0 1 1

1 0 1

1 1 0

(4) A B Y

0 0 1

0 1 0

1 0 0

1 1 0

116. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly :

(1) 6

(2) 0.6

(3) 0.06

(4) 0.006

117. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass.

The centre of mass of the system from the 5 kg particle is nearly at a distance of :

(1) 33 cm

(2) 50 cm

(3) 67 cm

(4) 80 cm

118. A spherical conductor of radius 10 cm has a charge of 3.2×10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere ?

$$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$$

(1) 1.28×10^4 N/C

(2) 1.28×10^5 N/C

(3) 1.28×10^6 N/C

(4) 1.28×10^7 N/C

119. Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$?

(1) 9.9801 m

(2) 9.98 m

(3) 9.980 m

(4) 9.9 m

120. A $40 \mu\text{F}$ capacitor is connected to a 200 V, 50 Hz ac supply. The rms value of the current in the circuit is, nearly :

(1) 1.7 A

(2) 2.05 A

(3) 2.5 A

(4) 25.1 A

121. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is :

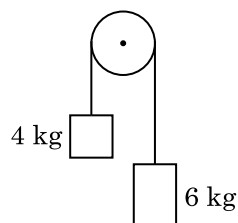
(1) isothermal

(2) adiabatic

(3) isochoric

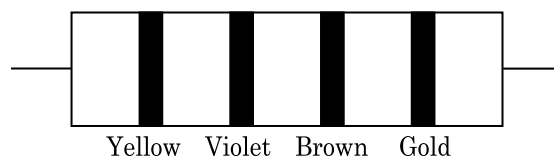
(4) isobaric

122. Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is :



- (1) g
 (2) $g/2$
 (3) $g/5$
 (4) $g/10$
123. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is :
- (1) 10 V
 (2) 10^2 V
 (3) 10^3 V
 (4) 10^4 V
124. When a uranium isotope ${}_{92}^{235}\text{U}$ is bombarded with a neutron, it generates ${}_{36}^{89}\text{Kr}$, three neutrons and :
- (1) ${}_{56}^{144}\text{Ba}$
 (2) ${}_{40}^{91}\text{Zr}$
 (3) ${}_{36}^{101}\text{Kr}$
 (4) ${}_{36}^{103}\text{Kr}$
125. The capacitance of a parallel plate capacitor with air as medium is $6 \mu\text{F}$. With the introduction of a dielectric medium, the capacitance becomes $30 \mu\text{F}$. The permittivity of the medium is :
- ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)
- (1) $0.44 \times 10^{-13} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 (2) $1.77 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 (3) $0.44 \times 10^{-10} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 (4) $5.00 \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$

126. The color code of a resistance is given below :



The values of resistance and tolerance, respectively, are :

- (1) 470 k Ω , 5%
 (2) 47 k Ω , 10%
 (3) 4.7 k Ω , 5%
 (4) 470 Ω , 5%
127. A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of 1Ω of the resistance wire is :
- (1) 1.0×10^{-2} m
 (2) 1.0×10^{-1} m
 (3) 1.5×10^{-1} m
 (4) 1.5×10^{-2} m
128. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled ?
- (1) doubled
 (2) four times
 (3) one-fourth
 (4) zero
129. The energy equivalent of 0.5 g of a substance is :
- (1) 4.5×10^{16} J
 (2) 4.5×10^{13} J
 (3) 1.5×10^{13} J
 (4) 0.5×10^{13} J
130. A short electric dipole has a dipole moment of $16 \times 10^{-9} \text{ C m}$. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is :
- ($\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$)
- (1) 50 V
 (2) 200 V
 (3) 400 V
 (4) zero

131. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , then the angle of incidence is nearly equal to :

- (1) $\frac{A}{2\mu}$
- (2) $\frac{2A}{\mu}$
- (3) μA
- (4) $\frac{\mu A}{2}$

132. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5 r_2$) through 1 K are in the ratio :

- (1) $\frac{27}{8}$
- (2) $\frac{9}{4}$
- (3) $\frac{3}{2}$
- (4) $\frac{5}{3}$

133. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1} . The permeability of the material of the rod is :

- ($\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$)
- (1) $2.4\pi \times 10^{-4} \text{ T m A}^{-1}$
 - (2) $8.0 \times 10^{-5} \text{ T m A}^{-1}$
 - (3) $2.4\pi \times 10^{-5} \text{ T m A}^{-1}$
 - (4) $2.4\pi \times 10^{-7} \text{ T m A}^{-1}$

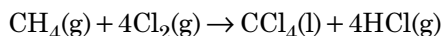
134. Assume that light of wavelength 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is :

- (1) $3.66 \times 10^{-7} \text{ rad}$
- (2) $1.83 \times 10^{-7} \text{ rad}$
- (3) $7.32 \times 10^{-7} \text{ rad}$
- (4) $6.00 \times 10^{-7} \text{ rad}$

135. The increase in the width of the depletion region in a p-n junction diode is due to :

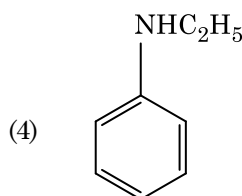
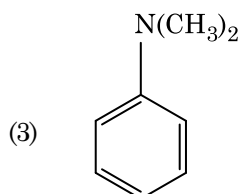
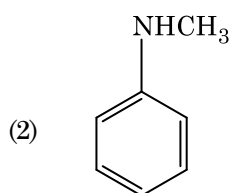
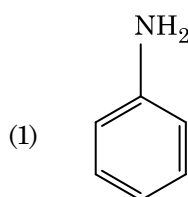
- (1) forward bias only
- (2) reverse bias only
- (3) both forward bias and reverse bias
- (4) increase in forward current

136. What is the change in oxidation number of carbon in the following reaction ?



- (1) + 4 to + 4
- (2) 0 to + 4
- (3) - 4 to + 4
- (4) 0 to - 4

137. Which of the following amine will give the carbylamine test ?

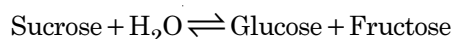


138. The mixture which shows positive deviation from Raoult's law is :

- (1) Ethanol + Acetone
- (2) Benzene + Toluene
- (3) Acetone + Chloroform
- (4) Chloroethane + Bromoethane

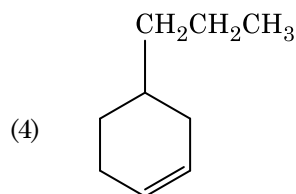
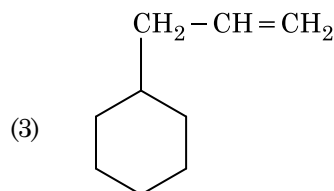
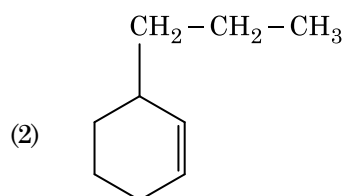
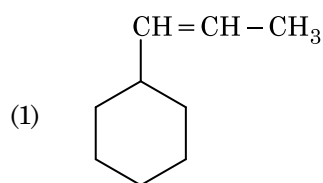
- 139.** An increase in the concentration of the reactants of a reaction leads to change in :
- (1) activation energy
 - (2) heat of reaction
 - (3) threshold energy
 - (4) collision frequency
- 140.** Sucrose on hydrolysis gives :
- (1) β -D-Glucose + α -D-Fructose
 - (2) α -D-Glucose + β -D-Glucose
 - (3) α -D-Glucose + β -D-Fructose
 - (4) α -D-Fructose + β -D-Fructose
- 141.** A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following ?
- (1) -I effect of $-\text{CH}_3$ groups
 - (2) +R effect of $-\text{CH}_3$ groups
 - (3) -R effect of $-\text{CH}_3$ groups
 - (4) Hyperconjugation
- 142.** Identify the **correct** statement from the following :
- (1) Wrought iron is impure iron with 4% carbon.
 - (2) Blister copper has blistered appearance due to evolution of CO_2 .
 - (3) Vapour phase refining is carried out for Nickel by Van Arkel method.
 - (4) Pig iron can be moulded into a variety of shapes.
- 143.** Identify the **incorrect** match.
- | Name | IUPAC Official Name |
|-----------------|---------------------|
| (a) Unnilunium | (i) Mendelevium |
| (b) Unniltrium | (ii) Lawrencium |
| (c) Unnilhexium | (iii) Seaborgium |
| (d) Unununnium | (iv) Darmstadtium |
- (1) (a), (i)
 - (2) (b), (ii)
 - (3) (c), (iii)
 - (4) (d), (iv)
- 144.** The number of Faradays(F) required to produce 20 g of calcium from molten CaCl_2 (Atomic mass of Ca = 40 g mol⁻¹) is :
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4
- 145.** An element has a body centered cubic (bcc) structure with a cell edge of 288 pm. The atomic radius is :
- (1) $\frac{\sqrt{3}}{4} \times 288 \text{ pm}$
 - (2) $\frac{\sqrt{2}}{4} \times 288 \text{ pm}$
 - (3) $\frac{4}{\sqrt{3}} \times 288 \text{ pm}$
 - (4) $\frac{4}{\sqrt{2}} \times 288 \text{ pm}$
- 146.** Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as :
- (1) Aldol condensation
 - (2) Cannizzaro's reaction
 - (3) Cross Cannizzaro's reaction
 - (4) Cross Aldol condensation
- 147.** Find out the solubility of $\text{Ni}(\text{OH})_2$ in 0.1 M NaOH. Given that the ionic product of $\text{Ni}(\text{OH})_2$ is 2×10^{-15} .
- (1) $2 \times 10^{-13} \text{ M}$
 - (2) $2 \times 10^{-8} \text{ M}$
 - (3) $1 \times 10^{-13} \text{ M}$
 - (4) $1 \times 10^8 \text{ M}$
- 148.** For the reaction, $2\text{Cl}(\text{g}) \rightarrow \text{Cl}_2(\text{g})$, the **correct** option is :
- (1) $\Delta_r H > 0$ and $\Delta_r S > 0$
 - (2) $\Delta_r H > 0$ and $\Delta_r S < 0$
 - (3) $\Delta_r H < 0$ and $\Delta_r S > 0$
 - (4) $\Delta_r H < 0$ and $\Delta_r S < 0$
- 149.** Which of the following is a basic amino acid ?
- (1) Serine
 - (2) Alanine
 - (3) Tyrosine
 - (4) Lysine

150. Hydrolysis of sucrose is given by the following reaction.



If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^\ominus$ at the same temperature will be :

- (1) $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(2 \times 10^{13})$
 - (2) $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(2 \times 10^{13})$
 - (3) $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(3 \times 10^{13})$
 - (4) $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(4 \times 10^{13})$
151. An alkene on ozonolysis gives methanal as one of the product. Its structure is :



152. A mixture of N_2 and Ar gases in a cylinder contains 7 g of N_2 and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N_2 is :

- [Use atomic masses (in g mol^{-1}) : N = 14, Ar = 40]
- (1) 9 bar
 - (2) 12 bar
 - (3) 15 bar
 - (4) 18 bar

153. Match the following and identify the **correct** option.

(a)	$\text{CO(g)} + \text{H}_2\text{(g)}$	(i)	$\text{Mg(HCO}_3)_2 + \text{Ca(HCO}_3)_2$
(b)	Temporary hardness of water	(ii)	An electron deficient hydride
(c)	B_2H_6	(iii)	Synthesis gas
(d)	H_2O_2	(iv)	Non-planar structure

	(a)	(b)	(c)	(d)
(1)	(iii)	(i)	(ii)	(iv)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(i)	(iii)	(ii)	(iv)

154. The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

- (1) Iron
- (2) Copper
- (3) Calcium
- (4) Potassium

155. Match the following :

	Oxide		Nature
(a)	CO	(i)	Basic
(b)	BaO	(ii)	Neutral
(c)	Al_2O_3	(iii)	Acidic
(d)	Cl_2O_7	(iv)	Amphoteric

Which of the following is **correct** option ?

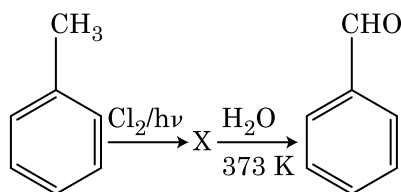
	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(i)	(iv)	(iii)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iv)	(iii)	(ii)	(i)

- 156.** Elimination reaction of 2-Bromo-pentane to form pent-2-ene is :
- β -Elimination reaction
 - Follows Zaitsev rule
 - Dehydrohalogenation reaction
 - Dehydration reaction
- (a), (b), (c)
 - (a), (c), (d)
 - (b), (c), (d)
 - (a), (b), (d)
- 157.** Paper chromatography is an example of :
- Adsorption chromatography
 - Partition chromatography
 - Thin layer chromatography
 - Column chromatography
- 158.** The correct option for free expansion of an ideal gas under adiabatic condition is :
- $q = 0$, $\Delta T = 0$ and $w = 0$
 - $q = 0$, $\Delta T < 0$ and $w > 0$
 - $q < 0$, $\Delta T = 0$ and $w = 0$
 - $q > 0$, $\Delta T > 0$ and $w > 0$
- 159.** Which of the following set of molecules will have zero dipole moment ?
- Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene
 - Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene
 - Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene
 - Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene
- 160.** The number of protons, neutrons and electrons in ${}^{175}_{71}\text{Lu}$, respectively, are :
- 71, 104 and 71
 - 104, 71 and 71
 - 71, 71 and 104
 - 175, 104 and 71
- 161.** On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be :
- Hydrogen gas
 - Oxygen gas
 - H_2S gas
 - SO_2 gas
- 162.** Identify the **correct** statements from the following :
- $\text{CO}_2(\text{g})$ is used as refrigerant for ice-cream and frozen food.
 - The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
 - ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
 - CO is colorless and odourless gas.
- (a), (b) and (c) only
 - (a) and (c) only
 - (b) and (c) only
 - (c) and (d) only

163. Urea reacts with water to form **A** which will decompose to form **B**. **B** when passed through Cu^{2+} (aq), deep blue colour solution **C** is formed. What is the formula of **C** from the following ?

- (1) CuSO_4
- (2) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- (3) $\text{Cu}(\text{OH})_2$
- (4) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$

164. Identify compound X in the following sequence of reactions :



- (1)
- (2)
- (3)
- (4)

165. Anisole on cleavage with HI gives :

- (1) + CH_3I
- (2) + CH_3OH
- (3) + $\text{C}_2\text{H}_5\text{I}$
- (4) + $\text{C}_2\text{H}_5\text{OH}$

166. The freezing point depression constant (K_f) of benzene is $5.12 \text{ K kg mol}^{-1}$. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) :

- (1) 0.20 K
- (2) 0.80 K
- (3) 0.40 K
- (4) 0.60 K

167. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

- (1) Isopropyl alcohol
- (2) Sec. butyl alcohol
- (3) Tert. butyl alcohol
- (4) Isobutyl alcohol

168. The rate constant for a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :

- (1) 100 s
- (2) 200 s
- (3) 500 s
- (4) 1000 s

169. HCl was passed through a solution of CaCl_2 , MgCl_2 and NaCl . Which of the following compound(s) crystallise(s) ?
- (1) Both MgCl_2 and CaCl_2
 - (2) Only NaCl
 - (3) Only MgCl_2
 - (4) NaCl , MgCl_2 and CaCl_2
170. Which of the following oxoacid of sulphur has $-\text{O}-\text{O}-$ linkage ?
- (1) H_2SO_3 , sulphurous acid
 - (2) H_2SO_4 , sulphuric acid
 - (3) $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid
 - (4) $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid
171. Which of the following is a natural polymer ?
- (1) *cis*-1,4-polyisoprene
 - (2) poly (Butadiene-styrene)
 - (3) polybutadiene
 - (4) poly (Butadiene-acrylonitrile)
172. Identify a molecule which does **not** exist.
- (1) He_2
 - (2) Li_2
 - (3) C_2
 - (4) O_2
173. Measuring Zeta potential is useful in determining which property of colloidal solution ?
- (1) Viscosity
 - (2) Solubility
 - (3) Stability of the colloidal particles
 - (4) Size of the colloidal particles
174. The calculated spin only magnetic moment of Cr^{2+} ion is :
- (1) 3.87 BM
 - (2) 4.90 BM
 - (3) 5.92 BM
 - (4) 2.84 BM
175. Which of the following alkane cannot be made in good yield by Wurtz reaction ?
- (1) n-Hexane
 - (2) 2,3-Dimethylbutane
 - (3) n-Heptane
 - (4) n-Butane
176. Which one of the followings has maximum number of atoms ?
- (1) 1 g of Ag(s) [Atomic mass of $\text{Ag} = 108$]
 - (2) 1 g of Mg(s) [Atomic mass of $\text{Mg} = 24$]
 - (3) 1 g of $\text{O}_2(\text{g})$ [Atomic mass of $\text{O} = 16$]
 - (4) 1 g of Li(s) [Atomic mass of $\text{Li} = 7$]
177. Identify the **incorrect** statement.
- (1) $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.
 - (2) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
 - (3) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
 - (4) The oxidation states of chromium in CrO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ are not the same.
178. Which of the following is the **correct** order of increasing field strength of ligands to form coordination compounds ?
- (1) $\text{SCN}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 - (2) $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$
 - (3) $\text{F}^- < \text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 - (4) $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{SCN}^- < \text{F}^-$
179. Which of the following is a cationic detergent ?
- (1) Sodium lauryl sulphate
 - (2) Sodium stearate
 - (3) Cetyltrimethyl ammonium bromide
 - (4) Sodium dodecylbenzene sulphonate
180. Which of the following is **not** correct about carbon monoxide ?
- (1) It forms carboxyhaemoglobin.
 - (2) It reduces oxygen carrying ability of blood.
 - (3) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
 - (4) It is produced due to incomplete combustion.

Space For Rough Work

Space For Rough Work

Space For Rough Work

Test Booklet Code

HAKAN

No. :

F4

This Booklet contains 24 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on **side-1** and **side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The CODE for this Booklet is **F4**. Make sure that the CODE printed on **Side-2** of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. **Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Roll Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

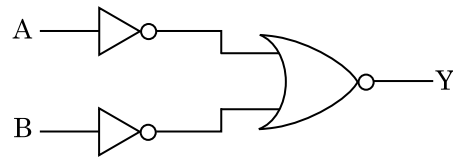
Candidate's Signature : _____ Invigilator's Signature : _____

Facsimile signature stamp of

Centre Superintendent : _____

1. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5 r_2$) through 1 K are in the ratio :
- (1) $\frac{3}{2}$
 - (2) $\frac{5}{3}$
 - (3) $\frac{27}{8}$
 - (4) $\frac{9}{4}$
2. Find the torque about the origin when a force of $3\hat{j}$ N acts on a particle whose position vector is $2\hat{k}$ m .
- (1) $-6\hat{i}$ N m
 - (2) $6\hat{k}$ N m
 - (3) $6\hat{i}$ N m
 - (4) $6\hat{j}$ N m
3. For transistor action, which of the following statements is **correct** ?
- (1) Both emitter junction as well as the collector junction are forward biased.
 - (2) The base region must be very thin and lightly doped.
 - (3) Base, emitter and collector regions should have same doping concentrations.
 - (4) Base, emitter and collector regions should have same size.
4. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is : ($g = 10 \text{ m/s}^2$)
- (1) 320 m
 - (2) 300 m
 - (3) 360 m
 - (4) 340 m
5. The Brewsters angle i_b for an interface should be :
- (1) $45^\circ < i_b < 90^\circ$
 - (2) $i_b = 90^\circ$
 - (3) $0^\circ < i_b < 30^\circ$
 - (4) $30^\circ < i_b < 45^\circ$

6. For the logic circuit shown, the truth table is :



(1)	A	B	Y
	0	0	1
	0	1	1
	1	0	1
	1	1	0
(2)	A	B	Y
	0	0	1
	0	1	0
	1	0	0
	1	1	0
(3)	A	B	Y
	0	0	0
	0	1	0
	1	0	0
	1	1	1
(4)	A	B	Y
	0	0	0
	0	1	1
	1	0	1
	1	1	1

7. The solids which have the negative temperature coefficient of resistance are :
- (1) semiconductors only
 - (2) insulators and semiconductors
 - (3) metals
 - (4) insulators only
8. The increase in the width of the depletion region in a p-n junction diode is due to :
- (1) both forward bias and reverse bias
 - (2) increase in forward current
 - (3) forward bias only
 - (4) reverse bias only
9. Dimensions of stress are :
- (1) $[ML^0T^{-2}]$
 - (2) $[ML^{-1}T^{-2}]$
 - (3) $[MLT^{-2}]$
 - (4) $[ML^2T^{-2}]$

10. Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$?
- 9.980 m
 - 9.9 m
 - 9.9801 m
 - 9.98 m
11. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be :
- 536 Hz
 - 537 Hz
 - 523 Hz
 - 524 Hz
12. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , then the angle of incidence is nearly equal to :
- μA
 - $\frac{\mu A}{2}$
 - $\frac{A}{2\mu}$
 - $\frac{2A}{\mu}$
13. The capacitance of a parallel plate capacitor with air as medium is $6 \mu\text{F}$. With the introduction of a dielectric medium, the capacitance becomes $30 \mu\text{F}$. The permittivity of the medium is :
- ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)
- $0.44 \times 10^{-10} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - $5.00 \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - $0.44 \times 10^{-13} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - $1.77 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
14. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of :
- 67 cm
 - 80 cm
 - 33 cm
 - 50 cm
15. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is :
- 0.5 mm
 - 1.0 mm
 - 0.01 mm
 - 0.25 mm
16. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes :
- four times
 - one-fourth
 - double
 - half
17. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1} . The permeability of the material of the rod is :
- ($\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$)
- $2.4\pi \times 10^{-5} \text{ T m A}^{-1}$
 - $2.4\pi \times 10^{-7} \text{ T m A}^{-1}$
 - $2.4\pi \times 10^{-4} \text{ T m A}^{-1}$
 - $8.0 \times 10^{-5} \text{ T m A}^{-1}$
18. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves)
- 1 : c
 - 1 : c^2
 - c : 1
 - 1 : 1
19. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :
- $\frac{\pi}{2}$ rad
 - zero
 - π rad
 - $\frac{3\pi}{2}$ rad

20. A wire of length L , area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is :

- (1) $\frac{MgL}{AL_1}$
 (2) $\frac{MgL}{A(L_1 - L)}$
 (3) $\frac{MgL_1}{AL}$
 (4) $\frac{Mg(L_1 - L)}{AL}$

21. The mean free path for a gas, with molecular diameter d and number density n can be expressed as :

- (1) $\frac{1}{\sqrt{2} n^2 \pi d^2}$
 (2) $\frac{1}{\sqrt{2} n^2 \pi^2 d^2}$
 (3) $\frac{1}{\sqrt{2} n \pi d}$
 (4) $\frac{1}{\sqrt{2} n \pi d^2}$

22. The energy equivalent of 0.5 g of a substance is :

- (1) $1.5 \times 10^{13} \text{ J}$
 (2) $0.5 \times 10^{13} \text{ J}$
 (3) $4.5 \times 10^{16} \text{ J}$
 (4) $4.5 \times 10^{13} \text{ J}$

23. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is $1.227 \times 10^{-2} \text{ nm}$, the potential difference is :

- (1) 10^3 V
 (2) 10^4 V
 (3) 10 V
 (4) 10^2 V

24. A short electric dipole has a dipole moment of $16 \times 10^{-9} \text{ C m}$. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is :

- $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$
 (1) 400 V
 (2) zero
 (3) 50 V
 (4) 200 V

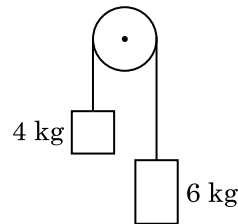
25. A $40 \mu\text{F}$ capacitor is connected to a 200 V, 50 Hz ac supply. The rms value of the current in the circuit is, nearly :

- (1) 2.5 A
 (2) 25.1 A
 (3) 1.7 A
 (4) 2.05 A

26. The average thermal energy for a mono-atomic gas is : (k_B is Boltzmann constant and T , absolute temperature)

- (1) $\frac{5}{2} k_B T$
 (2) $\frac{7}{2} k_B T$
 (3) $\frac{1}{2} k_B T$
 (4) $\frac{3}{2} k_B T$

27. Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is :



- (1) $g/5$
 (2) $g/10$
 (3) g
 (4) $g/2$

28. Assume that light of wavelength 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is :

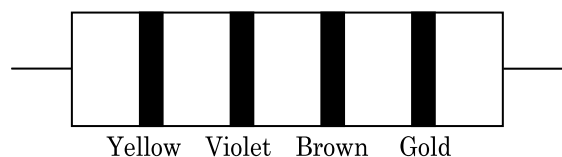
- (1) $7.32 \times 10^{-7} \text{ rad}$
 (2) $6.00 \times 10^{-7} \text{ rad}$
 (3) $3.66 \times 10^{-7} \text{ rad}$
 (4) $1.83 \times 10^{-7} \text{ rad}$

29. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C.

Its density is : ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) 0.1 kg/m^3
- (2) 0.02 kg/m^3
- (3) 0.5 kg/m^3
- (4) 0.2 kg/m^3

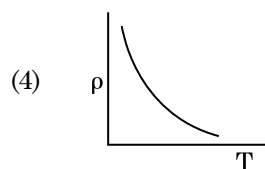
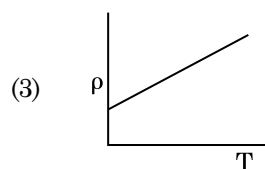
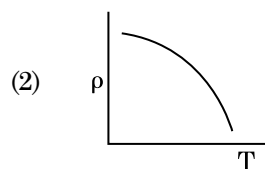
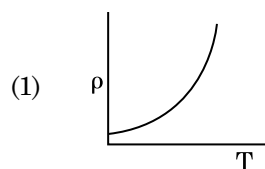
30. The color code of a resistance is given below :



The values of resistance and tolerance, respectively, are :

- (1) 4.7 k Ω , 5%
- (2) 470 Ω , 5%
- (3) 470 k Ω , 5%
- (4) 47 k Ω , 10%

31. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper ?



32. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is :

- (1) isochoric
- (2) isobaric
- (3) isothermal
- (4) adiabatic

33. When a uranium isotope ${}^{235}_{92}\text{U}$ is bombarded with a neutron, it generates ${}^{89}_{36}\text{Kr}$, three neutrons and :

- (1) ${}^{101}_{36}\text{Kr}$
- (2) ${}^{103}_{36}\text{Kr}$
- (3) ${}^{144}_{56}\text{Ba}$
- (4) ${}^{91}_{40}\text{Zr}$

34. A resistance wire connected in the left gap of a metre bridge balances a 10 Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of 1 Ω of the resistance wire is :

- (1) $1.5 \times 10^{-1} \text{ m}$
- (2) $1.5 \times 10^{-2} \text{ m}$
- (3) $1.0 \times 10^{-2} \text{ m}$
- (4) $1.0 \times 10^{-1} \text{ m}$

35. A charged particle having drift velocity of $7.5 \times 10^{-4} \text{ m s}^{-1}$ in an electric field of $3 \times 10^{-10} \text{ Vm}^{-1}$, has a mobility in $\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$ of :

- (1) 2.5×10^{-6}
- (2) 2.25×10^{-15}
- (3) 2.25×10^{15}
- (4) 2.5×10^6

36. Light with an average flux of 20 W/cm² falls on a non-reflecting surface at normal incidence having surface area 20 cm². The energy received by the surface during time span of 1 minute is :

- (1) $24 \times 10^3 \text{ J}$
- (2) $48 \times 10^3 \text{ J}$
- (3) $10 \times 10^3 \text{ J}$
- (4) $12 \times 10^3 \text{ J}$

37. For which one of the following, Bohr model is **not** valid ?
- (1) Deuteron atom
 - (2) Singly ionised neon atom (Ne^+)
 - (3) Hydrogen atom
 - (4) Singly ionised helium atom (He^+)
38. A spherical conductor of radius 10 cm has a charge of 3.2×10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere ?
- $$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$$
- (1) 1.28×10^6 N/C
 - (2) 1.28×10^7 N/C
 - (3) 1.28×10^4 N/C
 - (4) 1.28×10^5 N/C
39. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly :
- (1) 0.06
 - (2) 0.006
 - (3) 6
 - (4) 0.6
40. In a certain region of space with volume 0.2 m^3 , the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is :
- (1) 1 N/C
 - (2) 5 N/C
 - (3) zero
 - (4) 0.5 N/C
41. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :
- $$(\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1})$$
- (1) 6.28×10^{-5} T
 - (2) 3.14×10^{-5} T
 - (3) 6.28×10^{-4} T
 - (4) 3.14×10^{-4} T
42. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled ?
- (1) one-fourth
 - (2) zero
 - (3) doubled
 - (4) four times
43. A capillary tube of radius r is immersed in water and water rises in it to a height h . The mass of the water in the capillary is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is :
- (1) 10.0 g
 - (2) 20.0 g
 - (3) 2.5 g
 - (4) 5.0 g
44. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth ?
- (1) 30 N
 - (2) 24 N
 - (3) 48 N
 - (4) 32 N
45. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is :
- (1) 1.0
 - (2) -1.0
 - (3) zero
 - (4) 0.5
46. The ovary is half inferior in :
- (1) Sunflower
 - (2) Plum
 - (3) Brinjal
 - (4) Mustard

47. Identify the **wrong** statement with regard to Restriction Enzymes.
- (1) They are useful in genetic engineering.
 - (2) Sticky ends can be joined by using DNA ligases.
 - (3) Each restriction enzyme functions by inspecting the length of a DNA sequence.
 - (4) They cut the strand of DNA at palindromic sites.
48. Identify the **wrong** statement with reference to transport of oxygen.
- (1) Higher H^+ conc. in alveoli favours the formation of oxyhaemoglobin.
 - (2) Low pCO_2 in alveoli favours the formation of oxyhaemoglobin.
 - (3) Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2 .
 - (4) Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.
49. In water hyacinth and water lily, pollination takes place by :
- (1) wind and water
 - (2) insects and water
 - (3) insects or wind
 - (4) water currents only
50. If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is approximately :
- (1) 2.2 meters
 - (2) 2.7 meters
 - (3) 2.0 meters
 - (4) 2.5 meters
51. Dissolution of the synaptonemal complex occurs during :
- (1) Diplotene
 - (2) Leptotene
 - (3) Pachytene
 - (4) Zygotene
52. Match the following concerning essential elements and their functions in plants :
- | | |
|---------------|---|
| (a) Iron | (i) Photolysis of water |
| (b) Zinc | (ii) Pollen germination |
| (c) Boron | (iii) Required for chlorophyll biosynthesis |
| (d) Manganese | (iv) IAA biosynthesis |
- Select the **correct** option :
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|-------|
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (iv) | (i) | (ii) | (iii) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |
53. The body of the ovule is fused within the funicle at :
- (1) Nucellus
 - (2) Chalaza
 - (3) Hilum
 - (4) Micropyle
54. In relation to Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is **correct** ?
- (1) Gross primary productivity and Net primary productivity are one and same.
 - (2) There is no relationship between Gross primary productivity and Net primary productivity.
 - (3) Gross primary productivity is always less than net primary productivity.
 - (4) Gross primary productivity is always more than net primary productivity.
55. Which of the following refer to **correct** example(s) of organisms which have evolved due to changes in environment brought about by anthropogenic action ?
- (a) Darwin's Finches of Galapagos islands.
 - (b) Herbicide resistant weeds.
 - (c) Drug resistant eukaryotes.
 - (d) Man-created breeds of domesticated animals like dogs.
- (1) (b), (c) and (d)
 - (2) only (d)
 - (3) only (a)
 - (4) (a) and (c)

- 56.** Identify the **correct** statement with reference to human digestive system.
- (1) Ileum is a highly coiled part.
 - (2) Vermiform appendix arises from duodenum.
 - (3) Ileum opens into small intestine.
 - (4) Serosa is the innermost layer of the alimentary canal.
- 57.** The product(s) of reaction catalyzed by nitrogenase in root nodules of leguminous plants is/are :
- (1) Ammonia and oxygen
 - (2) Ammonia and hydrogen
 - (3) Ammonia alone
 - (4) Nitrate alone
- 58.** The transverse section of a plant shows following anatomical features :
- (a) Large number of scattered vascular bundles surrounded by bundle sheath.
 - (b) Large conspicuous parenchymatous ground tissue.
 - (c) Vascular bundles conjoint and closed.
 - (d) Phloem parenchyma absent.
- Identify the category of plant and its part :
- (1) Dicotyledonous stem
 - (2) Dicotyledonous root
 - (3) Monocotyledonous stem
 - (4) Monocotyledonous root
- 59.** Bilaterally symmetrical and acoelomate animals are exemplified by :
- (1) Aschelminthes
 - (2) Annelida
 - (3) Ctenophora
 - (4) Platyhelminthes
- 60.** Goblet cells of alimentary canal are modified from :
- (1) Chondrocytes
 - (2) Compound epithelial cells
 - (3) Squamous epithelial cells
 - (4) Columnar epithelial cells
- 61.** Which of the following is **not** an attribute of a population ?
- (1) Mortality
 - (2) Species interaction
 - (3) Sex ratio
 - (4) Natality
- 62.** Embryological support for evolution was disapproved by :
- (1) Charles Darwin
 - (2) Oparin
 - (3) Karl Ernst von Baer
 - (4) Alfred Wallace
- 63.** Which one of the following is the most abundant protein in the animals ?
- (1) Lectin
 - (2) Insulin
 - (3) Haemoglobin
 - (4) Collagen
- 64.** Match the following columns and select the **correct** option.
- | Column - I | Column - II |
|-----------------|--|
| (a) Eosinophils | (i) Immune response |
| (b) Basophils | (ii) Phagocytosis |
| (c) Neutrophils | (iii) Release histaminase, destructive enzymes |
| (d) Lymphocytes | (iv) Release granules containing histamine |
- | | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|-------|
| (1) | (i) | (ii) | (iv) | (iii) |
| (2) | (ii) | (i) | (iii) | (iv) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (iv) | (i) | (ii) | (iii) |
- 65.** Which of the following hormone levels will cause release of ovum (ovulation) from the graffian follicle ?
- (1) Low concentration of LH
 - (2) Low concentration of FSH
 - (3) High concentration of Estrogen
 - (4) High concentration of Progesterone

66. The sequence that controls the copy number of the linked DNA in the vector, is termed :
- (1) Palindromic sequence
 - (2) Recognition site
 - (3) Selectable marker
 - (4) Ori site
67. The plant parts which consist of two generations - one within the other :
- (a) Pollen grains inside the anther
 - (b) Germinated pollen grain with two male gametes
 - (c) Seed inside the fruit
 - (d) Embryo sac inside the ovule
- (1) (c) and (d)
 - (2) (a) and (d)
 - (3) (a) only
 - (4) (a), (b) and (c)
68. Match the following columns and select the **correct** option.
- | Column - I | Column - II |
|-----------------------------------|---------------------------------------|
| (a) <i>Clostridium butylicum</i> | (i) Cyclosporin-A |
| (b) <i>Trichoderma polysporum</i> | (ii) Butyric Acid |
| (c) <i>Monascus purpureus</i> | (iii) Citric Acid |
| (d) <i>Aspergillus niger</i> | (iv) Blood cholesterol lowering agent |
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|-------|
| (1) | (i) | (ii) | (iv) | (iii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (ii) | (i) | (iv) | (iii) |
69. The roots that originate from the base of the stem are :
- (1) Prop roots
 - (2) Lateral roots
 - (3) Fibrous roots
 - (4) Primary roots
70. Identify the **wrong** statement with reference to the gene 'I' that controls ABO blood groups.
- (1) When I^A and I^B are present together, they express same type of sugar.
 - (2) Allele 'i' does not produce any sugar.
 - (3) The gene (I) has three alleles.
 - (4) A person will have only two of the three alleles.
71. Which of the following would help in prevention of diuresis ?
- (1) Atrial natriuretic factor causes vasoconstriction
 - (2) Decrease in secretion of renin by JG cells
 - (3) More water reabsorption due to undersecretion of ADH
 - (4) Reabsorption of Na^+ and water from renal tubules due to aldosterone
72. Montreal protocol was signed in 1987 for control of :
- (1) Release of Green House gases
 - (2) Disposal of e-wastes
 - (3) Transport of Genetically modified organisms from one country to another
 - (4) Emission of ozone depleting substances
73. Meiotic division of the secondary oocyte is completed :
- (1) After zygote formation
 - (2) At the time of fusion of a sperm with an ovum
 - (3) Prior to ovulation
 - (4) At the time of copulation
74. By which method was a new breed 'Hisardale' of sheep formed by using Bikaneri ewes and Marino rams ?
- (1) Cross breeding
 - (2) Inbreeding
 - (3) Out crossing
 - (4) Mutational breeding

75. If the head of cockroach is removed, it may live for few days because :

- (1) the head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.
- (2) the head holds a $\frac{1}{3}$ rd of a nervous system while the rest is situated along the dorsal part of its body.
- (3) the supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.
- (4) the cockroach does not have nervous system.

76. Identify the **incorrect** statement.

- (1) Sapwood is the innermost secondary xylem and is lighter in colour.
- (2) Due to deposition of tannins, resins, oils etc., heart wood is dark in colour.
- (3) Heart wood does not conduct water but gives mechanical support.
- (4) Sapwood is involved in conduction of water and minerals from root to leaf.

77. Select the option including all sexually transmitted diseases.

- (1) AIDS, Malaria, Filaria
- (2) Cancer, AIDS, Syphilis
- (3) Gonorrhoea, Syphilis, Genital herpes
- (4) Gonorrhoea, Malaria, Genital herpes

78. Identify the **wrong** statement with reference to immunity.

- (1) Active immunity is quick and gives full response.
- (2) Foetus receives some antibodies from mother, it is an example for passive immunity.
- (3) When exposed to antigen (living or dead) antibodies are produced in the host's body. It is called "Active immunity".
- (4) When ready-made antibodies are directly given, it is called "Passive immunity".

79. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a)	Bt cotton	(i)	Gene therapy
(b)	Adenosine deaminase deficiency	(ii)	Cellular defence
(c)	RNAi	(iii)	Detection of HIV infection
(d)	PCR	(iv)	<i>Bacillus thuringiensis</i>

	(a)	(b)	(c)	(d)
(1)	(ii)	(iii)	(iv)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(iii)	(ii)	(i)	(iv)

80. Which of the following statements is **correct** ?

- (1) Adenine pairs with thymine through three H-bonds.
- (2) Adenine does not pair with thymine.
- (3) Adenine pairs with thymine through two H-bonds.
- (4) Adenine pairs with thymine through one H-bond.

81. According to Robert May, the global species diversity is about :

- (1) 50 million
- (2) 7 million
- (3) 1.5 million
- (4) 20 million

82. Secondary metabolites such as nicotine, strychnine and caffeine are produced by plants for their :

- (1) Defence action
- (2) Effect on reproduction
- (3) Nutritive value
- (4) Growth response

83. Which of the following pairs is of unicellular algae ?

- (1) *Anabaena* and *Volvox*
- (2) *Chlorella* and *Spirulina*
- (3) *Laminaria* and *Sargassum*
- (4) *Gelidium* and *Gracilaria*

84. The enzyme enterokinase helps in conversion of :

- (1) caseinogen into casein
- (2) pepsinogen into pepsin
- (3) protein into polypeptides
- (4) trypsinogen into trypsin

85. Match the following columns and select the correct option.

Column - I				Column - II
(a)	6 - 15 pairs of gill slits			(i) <i>Trygon</i>
(b)	Heterocercal caudal fin			(ii) Cyclostomes
(c)	Air Bladder			(iii) Chondrichthyes
(d)	Poison sting			(iv) Osteichthyes
	(a)	(b)	(c)	(d)
(1)	(iv)	(ii)	(iii)	(i)
(2)	(i)	(iv)	(iii)	(ii)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(iii)	(iv)	(i)	(ii)

86. The process responsible for facilitating loss of water in liquid form from the tip of grass blades at night and in early morning is :

- (1) Imbibition
- (2) Plasmolysis
- (3) Transpiration
- (4) Root pressure

87. Which of the following is put into Anaerobic sludge digester for further sewage treatment ?

- (1) Effluents of primary treatment
- (2) Activated sludge
- (3) Primary sludge
- (4) Floating debris

88. Match the following columns and select the correct option.

Column - I		Column - II	
(a)	Floating Ribs	(i)	Located between second and seventh ribs
(b)	Acromion	(ii)	Head of the Humerus
(c)	Scapula	(iii)	Clavicle
(d)	Glenoid cavity	(iv)	Do not connect with the sternum

(a) (b) (c) (d)

- | | | | | |
|-----|-------|-------|------|-------|
| (1) | (iii) | (ii) | (iv) | (i) |
| (2) | (iv) | (iii) | (i) | (ii) |
| (3) | (ii) | (iv) | (i) | (iii) |
| (4) | (i) | (iii) | (ii) | (iv) |

89. Select the correct statement.

- (1) Insulin acts on pancreatic cells and adipocytes.
- (2) Insulin is associated with hyperglycemia.
- (3) Glucocorticoids stimulate gluconeogenesis.
- (4) Glucagon is associated with hypoglycemia.

90. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus ?

- (1) Ketonuria and Glycosuria
- (2) Renal calculi and Hyperglycaemia
- (3) Uremia and Ketonuria
- (4) Uremia and Renal Calculi

91. Which of the following is correct about viroids ?

- (1) They have DNA with protein coat.
- (2) They have free DNA without protein coat.
- (3) They have RNA with protein coat.
- (4) They have free RNA without protein coat.

92. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits ?

- (1) 14
- (2) 8
- (3) 4
- (4) 2

93. Which of the following regions of the globe exhibits highest species diversity ?

- (1) Himalayas
- (2) Amazon forests
- (3) Western Ghats of India
- (4) Madagascar

94. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a) Placenta	(i)	Androgens	
(b) Zona pellucida	(ii)	Human Chorionic Gonadotropin (hCG)	
(c) Bulbo-urethral glands	(iii)	Layer of the ovum	
(d) Leydig cells	(iv)	Lubrication of the Penis	

	(a)	(b)	(c)	(d)
(1)	(iii)	(ii)	(iv)	(i)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(i)	(iv)	(ii)	(iii)

95. Match the trophic levels with their **correct** species examples in grassland ecosystem.

(a) Fourth trophic level	(i)	Crow
(b) Second trophic level	(ii)	Vulture
(c) First trophic level	(iii)	Rabbit
(d) Third trophic level	(iv)	Grass

Select the **correct** option :

	(a)	(b)	(c)	(d)
(1)	(iv)	(iii)	(ii)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(iii)	(ii)	(i)	(iv)

96. Flippers of Penguins and Dolphins are examples of :

- (1) Industrial melanism
- (2) Natural selection
- (3) Adaptive radiation
- (4) Convergent evolution

97. The number of substrate level phosphorylations in one turn of citric acid cycle is :

- (1) Two
- (2) Three
- (3) Zero
- (4) One

98. Which of the following statements is **not** correct ?

- (1) The functional insulin has A and B chains linked together by hydrogen bonds.
- (2) Genetically engineered insulin is produced in *E-Coli*.
- (3) In man insulin is synthesised as a proinsulin.
- (4) The proinsulin has an extra peptide called C-peptide.

99. Match the following columns and select the **correct** option.

Column - I		Column - II	
(a) Organ of Corti	(i)	Connects middle ear and pharynx	
(b) Cochlea	(ii)	Coiled part of the labyrinth	
(c) Eustachian tube	(iii)	Attached to the oval window	
(d) Stapes	(iv)	Located on the basilar membrane	

	(a)	(b)	(c)	(d)
(1)	(iv)	(ii)	(i)	(iii)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(iii)	(i)	(iv)	(ii)

- 100.** Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G_0). This process occurs at the end of :
- (1) S phase
 - (2) G_2 phase
 - (3) M phase
 - (4) G_1 phase
- 101.** The process of growth is maximum during :
- (1) Senescence
 - (2) Dormancy
 - (3) Log phase
 - (4) Lag phase
- 102.** The QRS complex in a standard ECG represents :
- (1) Depolarisation of ventricles
 - (2) Repolarisation of ventricles
 - (3) Repolarisation of auricles
 - (4) Depolarisation of auricles
- 103.** Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to :
- (1) Plant nematodes
 - (2) Insect predators
 - (3) Insect pests
 - (4) Fungal diseases
- 104.** In which of the following techniques, the embryos are transferred to assist those females who cannot conceive ?
- (1) ICSI and ZIFT
 - (2) GIFT and ICSI
 - (3) ZIFT and IUT
 - (4) GIFT and ZIFT
- 105.** Floridean starch has structure similar to :
- (1) Mannitol and algin
 - (2) Laminarin and cellulose
 - (3) Starch and cellulose
 - (4) Amylopectin and glycogen
- 106.** Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop.
- (1) Ethylene
 - (2) Absciscic acid
 - (3) Cytokinin
 - (4) Gibberellin
- 107.** Match the following columns and select the **correct** option.
- | Column - I | | Column - II | |
|--|-------|--------------------|--|
| (a) Gregarious, polyphagous pest | (i) | <i>Asterias</i> | |
| (b) Adult with radial symmetry and larva with bilateral symmetry | (ii) | Scorpion | |
| (c) Book lungs | (iii) | <i>Ctenoplanea</i> | |
| (d) Bioluminescence | (iv) | <i>Locusta</i> | |
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (iii) | (ii) | (i) | (iv) |
| (2) | (ii) | (i) | (iii) | (iv) |
| (3) | (i) | (iii) | (ii) | (iv) |
| (4) | (iv) | (i) | (ii) | (iii) |
- 108.** Which of the following statements are **true** for the phylum-Chordata ?
- (a) In Urochordata notochord extends from head to tail and it is present throughout their life.
 - (b) In Vertebrata notochord is present during the embryonic period only.
 - (c) Central nervous system is dorsal and hollow.
 - (d) Chordata is divided into 3 subphyla : Hemichordata, Tunicata and Cephalochordata.
- (1) (a) and (b)
 - (2) (b) and (c)
 - (3) (d) and (c)
 - (4) (c) and (a)
- 109.** The first phase of translation is :
- (1) Aminoacylation of tRNA
 - (2) Recognition of an anti-codon
 - (3) Binding of mRNA to ribosome
 - (4) Recognition of DNA molecule

- 110.** Identify the basic amino acid from the following.
- (1) Lysine
 - (2) Valine
 - (3) Tyrosine
 - (4) Glutamic Acid
- 111.** The infectious stage of *Plasmodium* that enters the human body is :
- (1) Female gametocytes
 - (2) Male gametocytes
 - (3) Trophozoites
 - (4) Sporozoites
- 112.** Identify the **correct** statement with regard to G₁ phase (Gap 1) of interphase.
- (1) Cell is metabolically active, grows but does not replicate its DNA.
 - (2) Nuclear Division takes place.
 - (3) DNA synthesis or replication takes place.
 - (4) Reorganisation of all cell components takes place.
- 113.** In light reaction, plastoquinone facilitates the transfer of electrons from :
- (1) PS-I to NADP⁺
 - (2) PS-I to ATP synthase
 - (3) PS-II to Cytb₆f complex
 - (4) Cytb₆f complex to PS-I
- 114.** The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of :
- (1) 1 molecule of 6-C compound
 - (2) 1 molecule of 4-C compound and 1 molecule of 2-C compound
 - (3) 2 molecules of 3-C compound
 - (4) 1 molecule of 3-C compound
- 115.** Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells ?
- (1) Golgi bodies
 - (2) Polysomes
 - (3) Endoplasmic reticulum
 - (4) Peroxisomes
- 116.** Match the following :
- | | |
|-------------------------------------|---------------|
| (a) Inhibitor of catalytic activity | (i) Ricin |
| (b) Possess peptide bonds | (ii) Malonate |
| (c) Cell wall material in fungi | (iii) Chitin |
| (d) Secondary metabolite | (iv) Collagen |
- Choose the **correct** option from the following :
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (ii) | (iii) | (i) | (iv) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (iii) | (i) | (iv) | (ii) |
- 117.** Identify the substances having glycosidic bond and peptide bond, respectively in their structure :
- (1) Cellulose, lecithin
 - (2) Inulin, insulin
 - (3) Chitin, cholesterol
 - (4) Glycerol, trypsin
- 118.** Experimental verification of the chromosomal theory of inheritance was done by :
- (1) Boveri
 - (2) Morgan
 - (3) Mendel
 - (4) Sutton
- 119.** The specific palindromic sequence which is recognized by EcoRI is :
- (1) 5' - CTTAAG - 3'
3' - GAATTC - 5'
 - (2) 5' - GGATCC - 3'
3' - CCTAGG - 5'
 - (3) 5' - GAATTC - 3'
3' - CTTAAG - 5'
 - (4) 5' - GGAACC - 3'
3' - CCTTGG - 5'
- 120.** Name the enzyme that facilitates opening of DNA helix during transcription.
- (1) DNA polymerase
 - (2) RNA polymerase
 - (3) DNA ligase
 - (4) DNA helicase

121. Select the **correct** match.

- | | | | |
|-----|---------------------|---|--|
| (1) | Sickle cell anaemia | - | Autosomal recessive trait, chromosome-11 |
| (2) | Thalassemia | - | X linked |
| (3) | Haemophilia | - | Y linked |
| (4) | Phenylketonuria | - | Autosomal dominant trait |

122. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask :

- (1) CH_4 , H_2 , NH_3 and water vapor at 600°C
- (2) CH_3 , H_2 , NH_3 and water vapor at 600°C
- (3) CH_4 , H_2 , NH_3 and water vapor at 800°C
- (4) CH_3 , H_2 , NH_4 and water vapor at 800°C

123. Match the following columns and select the **correct** option.

Column - I			Column - II		
(a)	Pituitary gland			(i)	Grave's disease
(b)	Thyroid gland			(ii)	Diabetes mellitus
(c)	Adrenal gland			(iii)	Diabetes insipidus
(d)	Pancreas			(iv)	Addison's disease
	(a)	(b)	(c)	(d)	
(1)	(iii)	(i)	(iv)	(ii)	
(2)	(ii)	(i)	(iv)	(iii)	
(3)	(iv)	(iii)	(i)	(ii)	
(4)	(iii)	(ii)	(i)	(iv)	

124. Cuboidal epithelium with brush border of microvilli is found in :

- (1) proximal convoluted tubule of nephron
- (2) eustachian tube
- (3) lining of intestine
- (4) ducts of salivary glands

125. Strobili or cones are found in :

- (1) *Marchantia*
- (2) *Equisetum*
- (3) *Salvinia*
- (4) *Pteris*

126. Snow-blindness in Antarctic region is due to :

- (1) High reflection of light from snow
- (2) Damage to retina caused by infra-red rays
- (3) Freezing of fluids in the eye by low temperature
- (4) Inflammation of cornea due to high dose of UV-B radiation

127. Match the following diseases with the causative organism and select the **correct** option.

Column - I			Column - II	
(a)	Typhoid		(i)	<i>Wuchereria</i>
(b)	Pneumonia		(ii)	<i>Plasmodium</i>
(c)	Filariasis		(iii)	<i>Salmonella</i>
(d)	Malaria		(iv)	<i>Haemophilus</i>
	(a)	(b)	(c)	(d)
(1)	(ii)	(i)	(iii)	(iv)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(i)	(iii)	(ii)	(iv)
(4)	(iii)	(iv)	(i)	(ii)

128. Choose the **correct** pair from the following :

- | | | | |
|-----|--------------|---|--|
| (1) | Nucleases | - | Separate the two strands of DNA |
| (2) | Exonucleases | - | Make cuts at specific positions within DNA |
| (3) | Ligases | - | Join the two DNA molecules |
| (4) | Polymerases | - | Break the DNA into fragments |

129. Which of the following statements about inclusion bodies is **incorrect** ?

- (1) They lie free in the cytoplasm.
- (2) These represent reserve material in cytoplasm.
- (3) They are not bound by any membrane.
- (4) These are involved in ingestion of food particles.

130. Select the **correct** events that occur during inspiration.

- (a) Contraction of diaphragm
- (b) Contraction of external inter-costal muscles
- (c) Pulmonary volume decreases
- (d) Intra pulmonary pressure increases
- (1) (a), (b) and (d)
- (2) only (d)
- (3) (a) and (b)
- (4) (c) and (d)

131. Ray florets have :

- (1) Hypogynous ovary
- (2) Half inferior ovary
- (3) Inferior ovary
- (4) Superior ovary

132. Match the organism with its use in biotechnology.

- | | | |
|--------------------------------------|-------|-------------------------------------|
| (a) <i>Bacillus thuringiensis</i> | (i) | Cloning vector |
| (b) <i>Thermus aquaticus</i> | (ii) | Construction of first rDNA molecule |
| (c) <i>Agrobacterium tumefaciens</i> | (iii) | DNA polymerase |
| (d) <i>Salmonella typhimurium</i> | (iv) | Cry proteins |

Select the **correct** option from the following :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (iii) | (ii) | (iv) | (i) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (iv) | (iii) | (i) | (ii) |

133. Which of the following is **not** an inhibitory substance governing seed dormancy ?

- (1) Phenolic acid
- (2) Para-ascorbic acid
- (3) Gibberellic acid
- (4) Absciscic acid

134. Match the following with respect to meiosis :

- | | | |
|----------------|-------|-----------------|
| (a) Zygotene | (i) | Terminalization |
| (b) Pachytene | (ii) | Chiasmata |
| (c) Diplotene | (iii) | Crossing over |
| (d) Diakinesis | (iv) | Synapsis |

Select the **correct** option from the following :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (i) | (ii) | (iv) | (iii) |
| (2) | (ii) | (iv) | (iii) | (i) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

135. In gel electrophoresis, separated DNA fragments can be visualized with the help of :

- (1) Acetocarmine in UV radiation
- (2) Ethidium bromide in infrared radiation
- (3) Acetocarmine in bright blue light
- (4) Ethidium bromide in UV radiation

136. Which of the following is a natural polymer ?

- (1) polybutadiene
- (2) poly (Butadiene-acrylonitrile)
- (3) *cis*-1,4-polyisoprene
- (4) poly (Butadiene-styrene)

137. On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be :

- (1) H₂S gas
- (2) SO₂ gas
- (3) Hydrogen gas
- (4) Oxygen gas

138. An element has a body centered cubic (bcc) structure with a cell edge of 288 pm. The atomic radius is :

- (1) $\frac{4}{\sqrt{3}} \times 288 \text{ pm}$
- (2) $\frac{4}{\sqrt{2}} \times 288 \text{ pm}$
- (3) $\frac{\sqrt{3}}{4} \times 288 \text{ pm}$
- (4) $\frac{\sqrt{2}}{4} \times 288 \text{ pm}$

139. The correct option for free expansion of an ideal gas under adiabatic condition is :

- (1) $q < 0$, $\Delta T = 0$ and $w = 0$
- (2) $q > 0$, $\Delta T > 0$ and $w > 0$
- (3) $q = 0$, $\Delta T = 0$ and $w = 0$
- (4) $q = 0$, $\Delta T < 0$ and $w > 0$

140. Which of the following set of molecules will have zero dipole moment ?

- (1) Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene
- (2) Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene
- (3) Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene
- (4) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene

141. A mixture of N_2 and Ar gases in a cylinder contains 7 g of N_2 and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N_2 is :

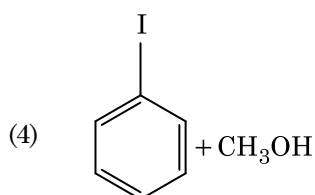
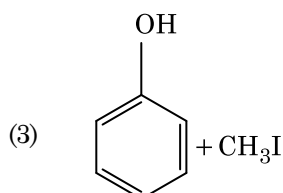
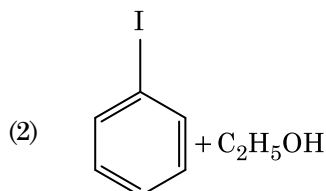
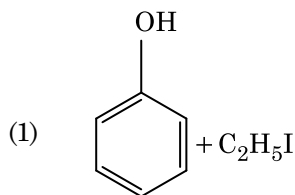
[Use atomic masses (in $g\ mol^{-1}$) : N = 14, Ar = 40]

- (1) 15 bar
- (2) 18 bar
- (3) 9 bar
- (4) 12 bar

142. Identify the **correct** statements from the following :

- (a) $CO_2(g)$ is used as refrigerant for ice-cream and frozen food.
 - (b) The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
 - (c) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
 - (d) CO is colorless and odourless gas.
- (1) (b) and (c) only
 - (2) (c) and (d) only
 - (3) (a), (b) and (c) only
 - (4) (a) and (c) only

143. Anisole on cleavage with HI gives :



144. The mixture which shows positive deviation from Raoult's law is :

- (1) Acetone + Chloroform
- (2) Chloroethane + Bromoethane
- (3) Ethanol + Acetone
- (4) Benzene + Toluene

145. Which of the following is the **correct** order of increasing field strength of ligands to form coordination compounds ?

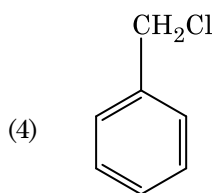
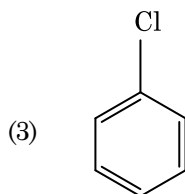
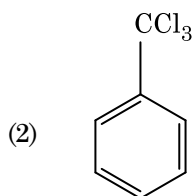
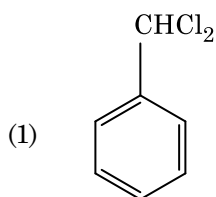
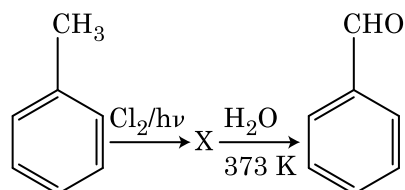
- (1) $F^- < SCN^- < C_2O_4^{2-} < CN^-$
- (2) $CN^- < C_2O_4^{2-} < SCN^- < F^-$
- (3) $SCN^- < F^- < C_2O_4^{2-} < CN^-$
- (4) $SCN^- < F^- < CN^- < C_2O_4^{2-}$

146. Which one of the followings has maximum number of atoms ?

- (1) 1 g of $O_2(g)$ [Atomic mass of O = 16]
- (2) 1 g of Li(s) [Atomic mass of Li = 7]
- (3) 1 g of Ag(s) [Atomic mass of Ag = 108]
- (4) 1 g of Mg(s) [Atomic mass of Mg = 24]

- 147.** Which of the following alkane cannot be made in good yield by Wurtz reaction ?
- (1) n-Heptane
 - (2) n-Butane
 - (3) n-Hexane
 - (4) 2,3-Dimethylbutane
- 148.** Which of the following is **not** correct about carbon monoxide ?
- (1) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
 - (2) It is produced due to incomplete combustion.
 - (3) It forms carboxyhaemoglobin.
 - (4) It reduces oxygen carrying ability of blood.
- 149.** For the reaction, $2\text{Cl(g)} \rightarrow \text{Cl}_2\text{(g)}$, the **correct** option is :
- (1) $\Delta_r H < 0$ and $\Delta_r S > 0$
 - (2) $\Delta_r H < 0$ and $\Delta_r S < 0$
 - (3) $\Delta_r H > 0$ and $\Delta_r S > 0$
 - (4) $\Delta_r H > 0$ and $\Delta_r S < 0$
- 150.** Match the following and identify the **correct** option.
- | | | | |
|-----|---------------------------------------|-------|---|
| (a) | $\text{CO(g)} + \text{H}_2\text{(g)}$ | (i) | $\text{Mg}(\text{HCO}_3)_2 + \text{Ca}(\text{HCO}_3)_2$ |
| (b) | Temporary hardness of water | (ii) | An electron deficient hydride |
| (c) | B_2H_6 | (iii) | Synthesis gas |
| (d) | H_2O_2 | (iv) | Non-planar structure |
- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|------|
| (1) | (iii) | (iv) | (ii) | (i) |
| (2) | (i) | (iii) | (ii) | (iv) |
| (3) | (iii) | (i) | (ii) | (iv) |
| (4) | (iii) | (ii) | (i) | (iv) |
- 151.** Urea reacts with water to form **A** which will decompose to form **B**. **B** when passed through Cu^{2+} (aq), deep blue colour solution **C** is formed. What is the formula of **C** from the following ?
- (1) $\text{Cu}(\text{OH})_2$
 - (2) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
 - (3) CuSO_4
 - (4) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- 152.** Hydrolysis of sucrose is given by the following reaction.
- $$\text{Sucrose} + \text{H}_2\text{O} \rightleftharpoons \text{Glucose} + \text{Fructose}$$
- If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^\ominus$ at the same temperature will be :
- (1) $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(3 \times 10^{13})$
 - (2) $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(4 \times 10^{13})$
 - (3) $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(2 \times 10^{13})$
 - (4) $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(2 \times 10^{13})$
- 153.** The calculated spin only magnetic moment of Cr^{2+} ion is :
- (1) 5.92 BM
 - (2) 2.84 BM
 - (3) 3.87 BM
 - (4) 4.90 BM
- 154.** Measuring Zeta potential is useful in determining which property of colloidal solution ?
- (1) Stability of the colloidal particles
 - (2) Size of the colloidal particles
 - (3) Viscosity
 - (4) Solubility
- 155.** Which of the following is a cationic detergent ?
- (1) Cetyltrimethyl ammonium bromide
 - (2) Sodium dodecylbenzene sulphonate
 - (3) Sodium lauryl sulphate
 - (4) Sodium stearate
- 156.** Find out the solubility of $\text{Ni}(\text{OH})_2$ in 0.1 M NaOH. Given that the ionic product of $\text{Ni}(\text{OH})_2$ is 2×10^{-15} .
- (1) $1 \times 10^{-13} \text{ M}$
 - (2) $1 \times 10^{-8} \text{ M}$
 - (3) $2 \times 10^{-13} \text{ M}$
 - (4) $2 \times 10^{-8} \text{ M}$

157. Identify compound X in the following sequence of reactions :



158. The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

- (1) Calcium
- (2) Potassium
- (3) Iron
- (4) Copper

159. An increase in the concentration of the reactants of a reaction leads to change in :

- (1) threshold energy
- (2) collision frequency
- (3) activation energy
- (4) heat of reaction

160. The rate constant for a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :

- (1) 500 s
- (2) 1000 s
- (3) 100 s
- (4) 200 s

161. The number of Faradays(F) required to produce 20 g of calcium from molten CaCl_2 (Atomic mass of Ca = 40 g mol^{-1}) is :

- (1) 3
- (2) 4
- (3) 1
- (4) 2

162. Identify the **correct** statement from the following :

- (1) Vapour phase refining is carried out for Nickel by Van Arkel method.
- (2) Pig iron can be moulded into a variety of shapes.
- (3) Wrought iron is impure iron with 4% carbon.
- (4) Blister copper has blistered appearance due to evolution of CO_2 .

163. Identify a molecule which does **not** exist.

- (1) C_2
- (2) O_2
- (3) He_2
- (4) Li_2

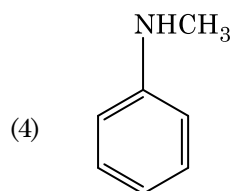
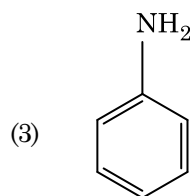
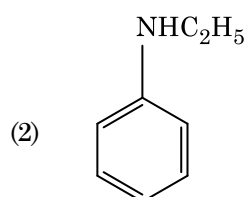
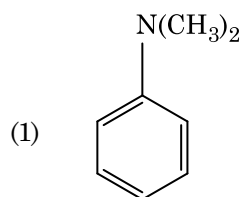
164. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following ?

- (1) – R effect of $-\text{CH}_3$ groups
- (2) Hyperconjugation
- (3) – I effect of $-\text{CH}_3$ groups
- (4) + R effect of $-\text{CH}_3$ groups

165. HCl was passed through a solution of CaCl_2 , MgCl_2 and NaCl. Which of the following compound(s) crystallise(s) ?

- (1) Only MgCl_2
- (2) NaCl, MgCl_2 and CaCl_2
- (3) Both MgCl_2 and CaCl_2
- (4) Only NaCl

166. Which of the following amine will give the carbylamine test ?



167. Identify the **incorrect** match.

	Name		IUPAC Official Name
(a)	Unnilunium	(i)	Mendelevium
(b)	Unniltrium	(ii)	Lawrencium
(c)	Unnilhexium	(iii)	Seaborgium
(d)	Unununnium	(iv)	Darmstadtium
(1)	(c), (iii)		
(2)	(d), (iv)		
(3)	(a), (i)		
(4)	(b), (ii)		

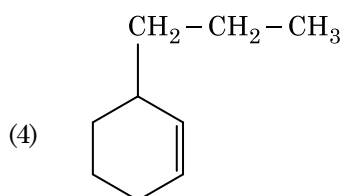
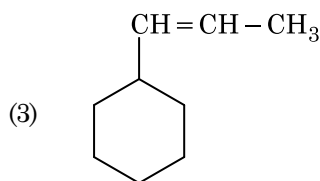
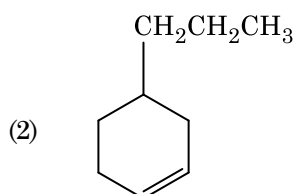
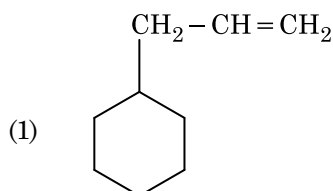
168. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

- (1) Tert. butyl alcohol
- (2) Isobutyl alcohol
- (3) Isopropyl alcohol
- (4) Sec. butyl alcohol

169. Sucrose on hydrolysis gives :

- (1) α -D-Glucose + β -D-Fructose
- (2) α -D-Fructose + β -D-Fructose
- (3) β -D-Glucose + α -D-Fructose
- (4) α -D-Glucose + β -D-Glucose

170. An alkene on ozonolysis gives methanal as one of the product. Its structure is :



171. Identify the **incorrect** statement.

- (1) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
- (2) The oxidation states of chromium in CrO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ are not the same.
- (3) $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.
- (4) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.

172. Which of the following is a basic amino acid ?

- (1) Tyrosine
- (2) Lysine
- (3) Serine
- (4) Alanine

173. Which of the following oxoacid of sulphur has – O – O – linkage ?

- (1) $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid
- (2) $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid
- (3) H_2SO_3 , sulphurous acid
- (4) H_2SO_4 , sulphuric acid

174. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is :

- (a) β -Elimination reaction
- (b) Follows Zaitsev rule
- (c) Dehydrohalogenation reaction
- (d) Dehydration reaction

- (1) (b), (c), (d)
- (2) (a), (b), (d)
- (3) (a), (b), (c)
- (4) (a), (c), (d)

175. Match the following :

Oxide		Nature	
(a)	CO	(i)	Basic
(b)	BaO	(ii)	Neutral
(c)	Al_2O_3	(iii)	Acidic
(d)	Cl_2O_7	(iv)	Amphoteric

Which of the following is **correct** option ?

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (i) | (ii) | (iii) | (iv) |
| (4) | (ii) | (i) | (iv) | (iii) |

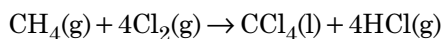
176. The freezing point depression constant (K_f) of benzene is $5.12 \text{ K kg mol}^{-1}$. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) :

- (1) 0.40 K
- (2) 0.60 K
- (3) 0.20 K
- (4) 0.80 K

177. Paper chromatography is an example of :

- (1) Thin layer chromatography
- (2) Column chromatography
- (3) Adsorption chromatography
- (4) Partition chromatography

178. What is the change in oxidation number of carbon in the following reaction ?



- (1) – 4 to + 4
- (2) 0 to – 4
- (3) + 4 to + 4
- (4) 0 to + 4

179. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as :

- (1) Cross Cannizzaro's reaction
- (2) Cross Aldol condensation
- (3) Aldol condensation
- (4) Cannizzaro's reaction

180. The number of protons, neutrons and electrons in ${}^{175}_{71}\text{Lu}$, respectively, are :

- (1) 71, 71 and 104
- (2) 175, 104 and 71
- (3) 71, 104 and 71
- (4) 104, 71 and 71

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Space For Rough Work

Space For Rough Work

Space For Rough Work



This Booklet contains 24 pages.

EE

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The CODE for this Booklet is **EE**. Make sure that the CODE printed on **Side-2** of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
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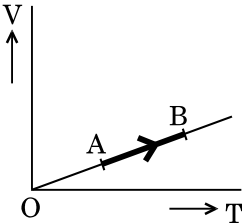
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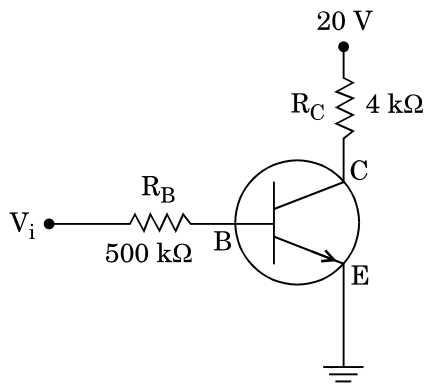
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Centre Superintendent : _____

1. At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere ?
(Given :
Mass of oxygen molecule (m) = 2.76×10^{-26} kg
Boltzmann's constant $k_B = 1.38 \times 10^{-23}$ J K $^{-1}$)
(1) 2.508×10^4 K
(2) 5.016×10^4 K
(3) 8.360×10^4 K
(4) 1.254×10^4 K
2. The volume (V) of a monatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to state B, is

(1) $\frac{2}{5}$
(2) $\frac{1}{3}$
(3) $\frac{2}{3}$
(4) $\frac{2}{7}$
3. The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is
(1) 26.8%
(2) 6.25%
(3) 20%
(4) 12.5%
4. The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is
(1) 13.2 cm
(2) 12.5 cm
(3) 8 cm
(4) 16 cm
5. A metallic rod of mass per unit length 0.5 kg m $^{-1}$ is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is
(1) 7.14 A
(2) 14.76 A
(3) 5.98 A
(4) 11.32 A
6. An inductor 20 mH, a capacitor 100 μ F and a resistor 50 Ω are connected in series across a source of emf, $V = 10 \sin 314 t$. The power loss in the circuit is
(1) 0.79 W
(2) 2.74 W
(3) 0.43 W
(4) 1.13 W
7. Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is
(1) 40 Ω
(2) 250 Ω
(3) 25 Ω
(4) 500 Ω
8. A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from
(1) the current source
(2) the lattice structure of the material of the rod
(3) the magnetic field
(4) the induced electric field due to the changing magnetic field

9. In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0$ and $V_{CE} = 0$. The values of I_B , I_C and β are given by



- (1) $I_B = 40 \mu\text{A}$, $I_C = 10 \text{ mA}$, $\beta = 250$
 (2) $I_B = 20 \mu\text{A}$, $I_C = 5 \text{ mA}$, $\beta = 250$
 (3) $I_B = 25 \mu\text{A}$, $I_C = 5 \text{ mA}$, $\beta = 200$
 (4) $I_B = 40 \mu\text{A}$, $I_C = 5 \text{ mA}$, $\beta = 125$
10. In the combination of the following gates the output Y can be written in terms of inputs A and B as
-
- (1) $\overline{A \cdot B}$
 (2) $\overline{A \cdot B} + A \cdot B$
 (3) $A \cdot \bar{B} + \bar{A} \cdot B$
 (4) $\overline{A + B}$
11. In a p-n junction diode, change in temperature due to heating
- (1) affects only reverse resistance
 (2) does not affect resistance of p-n junction
 (3) affects only forward resistance
 (4) affects the overall V – I characteristics of p-n junction

12. Unpolarised light is incident from air on a plane surface of a material of refractive index ' μ '. At a particular angle of incidence ' i ', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation ?

- (1) Reflected light is polarised with its electric vector parallel to the plane of incidence
 (2) $i = \sin^{-1}\left(\frac{1}{\mu}\right)$
 (3) Reflected light is polarised with its electric vector perpendicular to the plane of incidence
 (4) $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

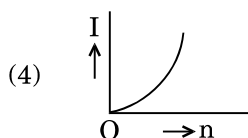
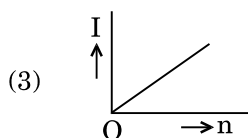
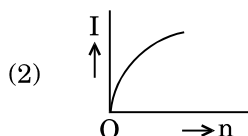
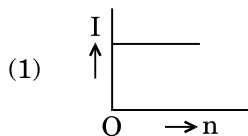
13. In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is $5896 \text{ \AA}$ and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20° . To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to

- (1) 1.8 mm
 (2) 2.1 mm
 (3) 1.9 mm
 (4) 1.7 mm

14. An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of
- (1) small focal length and large diameter
 (2) large focal length and large diameter
 (3) large focal length and small diameter
 (4) small focal length and small diameter

15. An electron of mass m with an initial velocity $\vec{V} = V_0 \hat{i}$ ($V_0 > 0$) enters an electric field $\vec{E} = -E_0 \hat{i}$ ($E_0 = \text{constant} > 0$) at $t = 0$. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is
- $\frac{\lambda_0}{\left(1 + \frac{eE_0}{mV_0} t\right)}$
 - $\lambda_0 t$
 - $\lambda_0 \left(1 + \frac{eE_0}{mV_0} t\right)$
 - λ_0
16. For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the disintegration of 450 nuclei is
- 20
 - 30
 - 10
 - 15
17. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is
- 1 : 1
 - 2 : -1
 - 1 : -1
 - 1 : -2
18. When the light of frequency $2\nu_0$ (where ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1 . When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is v_2 . The ratio of v_1 to v_2 is
- 1 : 2
 - 4 : 1
 - 1 : 4
 - 2 : 1
19. An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be
- 30 cm away from the mirror
 - 30 cm towards the mirror
 - 36 cm away from the mirror
 - 36 cm towards the mirror
20. The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30° . One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is
- 60°
 - 30°
 - 45°
 - zero
21. An em wave is propagating in a medium with a velocity $\vec{V} = V \hat{i}$. The instantaneous oscillating electric field of this em wave is along $+y$ axis. Then the direction of oscillating magnetic field of the em wave will be along
- $-z$ direction
 - $-y$ direction
 - $+z$ direction
 - $-x$ direction
22. The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance
- 0.138 H
 - 1.389 H
 - 138.88 H
 - 13.89 H

23. A battery consists of a variable number 'n' of identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n ?



24. A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10 I. The value of 'n' is

- (1) 10
- (2) 20
- (3) 11
- (4) 9

25. A carbon resistor of $(47 \pm 4.7) \text{ k}\Omega$ is to be marked with rings of different colours for its identification. The colour code sequence will be

- (1) Violet – Yellow – Orange – Silver
- (2) Yellow – Green – Violet – Gold
- (3) Yellow – Violet – Orange – Silver
- (4) Green – Orange – Violet – Gold

26. A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is

- (1) 330 m/s
- (2) 350 m/s
- (3) 339 m/s
- (4) 300 m/s

27. The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

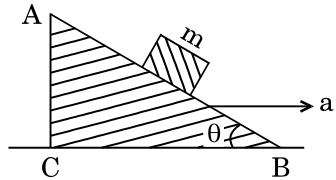
- (1) independent of the distance between the plates.
- (2) proportional to the square root of the distance between the plates.
- (3) linearly proportional to the distance between the plates.
- (4) inversely proportional to the distance between the plates.

28. An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

- (1) smaller
- (2) 10 times greater
- (3) 5 times greater
- (4) equal

29. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is

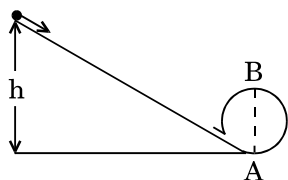
- (1) $2\pi \text{ s}$
- (2) 2 s
- (3) $\pi \text{ s}$
- (4) 1 s

30. The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_0 . If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP . The value of n is
- $\frac{3}{4}$
 - $\frac{256}{81}$
 - $\frac{4}{3}$
 - $\frac{81}{256}$
31. Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area $3A$. If the length of the first wire is increased by Δl on applying a force F , how much force is needed to stretch the second wire by the same amount ?
- $9F$
 - $4F$
 - $6F$
 - F
32. A small sphere of radius ' r ' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to
- r^3
 - r^5
 - r^2
 - r^4
33. A sample of 0.1 g of water at 100°C and normal pressure ($1.013 \times 10^5\text{ Nm}^{-2}$) requires 54 cal of heat energy to convert to steam at 100°C . If the volume of the steam produced is 167.1 cc , the change in internal energy of the sample, is
- 104.3 J
 - 42.2 J
 - 208.7 J
 - 84.5 J
34. The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at $(2, 0, -3)$, about the point $(2, -2, -2)$, is given by
- $-8\hat{i} - 4\hat{j} - 7\hat{k}$
 - $-7\hat{i} - 8\hat{j} - 4\hat{k}$
 - $-4\hat{i} - \hat{j} - 8\hat{k}$
 - $-7\hat{i} - 4\hat{j} - 8\hat{k}$
35. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm . The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of -0.004 cm , the correct diameter of the ball is
- 0.521 cm
 - 0.053 cm
 - 0.525 cm
 - 0.529 cm
36. A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration ' a ' towards the right. The relation between a and θ for the block to remain stationary on the wedge is
- 
- $a = \frac{g}{\operatorname{cosec} \theta}$
 - $a = g \cos \theta$
 - $a = \frac{g}{\sin \theta}$
 - $a = g \tan \theta$
37. A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\vec{E}$. Due to the force $q\vec{E}$, its velocity increases from 0 to 6 m/s in one second duration. At that instant the direction of the field is reversed. The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 seconds are respectively
- $2\text{ m/s}, 4\text{ m/s}$
 - $1\text{ m/s}, 3.5\text{ m/s}$
 - $1\text{ m/s}, 3\text{ m/s}$
 - $1.5\text{ m/s}, 3\text{ m/s}$

38. A moving block having mass m , collides with another stationary block having mass $4m$. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v , then the value of coefficient of restitution (e) will be

(1) 0.5
(2) 0.8
(3) 0.25
(4) 0.4

39. A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter $AB = D$. The height h is equal to



(1) $\frac{3}{2}D$
(2) $\frac{7}{5}D$
(3) D
(4) $\frac{5}{4}D$

40. Three objects, A : (a solid sphere), B : (a thin circular disk) and C : (a circular ring), each have the same mass M and radius R . They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation

(1) $W_C > W_B > W_A$
(2) $W_B > W_A > W_C$
(3) $W_A > W_B > W_C$
(4) $W_A > W_C > W_B$

41. Which one of the following statements is **incorrect** ?

(1) Rolling friction is smaller than sliding friction.
(2) Frictional force opposes the relative motion.
(3) Limiting value of static friction is directly proportional to normal reaction.
(4) Coefficient of sliding friction has dimensions of length.

42. A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is

(1) 7 : 10
(2) 10 : 7
(3) 5 : 7
(4) 2 : 5

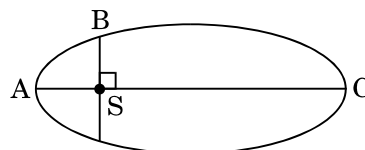
43. A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere ?

(1) Angular velocity
(2) Rotational kinetic energy
(3) Moment of inertia
(4) Angular momentum

44. If the mass of the Sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is **not** correct ?

(1) Raindrops will fall faster.
(2) Time period of a simple pendulum on the Earth would decrease.
(3) Walking on the ground would become more difficult.
(4) 'g' on the Earth will not change.

45. The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A , K_B and K_C , respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then



(1) $K_A < K_B < K_C$
(2) $K_B < K_A < K_C$
(3) $K_A > K_B > K_C$
(4) $K_B > K_A > K_C$

46. Given van der Waals constant for NH_3 , H_2 , O_2 and CO_2 are respectively 4.17, 0.244, 1.36 and 3.59, which one of the following gases is most easily liquefied ?
- NH_3
 - O_2
 - H_2
 - CO_2
47. Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations :
- 60 mL $\frac{\text{M}}{10}$ HCl + 40 mL $\frac{\text{M}}{10}$ NaOH
 - 55 mL $\frac{\text{M}}{10}$ HCl + 45 mL $\frac{\text{M}}{10}$ NaOH
 - 75 mL $\frac{\text{M}}{5}$ HCl + 25 mL $\frac{\text{M}}{5}$ NaOH
 - 100 mL $\frac{\text{M}}{10}$ HCl + 100 mL $\frac{\text{M}}{10}$ NaOH
- pH of which one of them will be equal to 1 ?
- b
 - d
 - a
 - c
48. The solubility of BaSO_4 in water is $2.42 \times 10^{-3} \text{ gL}^{-1}$ at 298 K. The value of its solubility product (K_{sp}) will be
(Given molar mass of $\text{BaSO}_4 = 233 \text{ g mol}^{-1}$)
- $1.08 \times 10^{-10} \text{ mol}^2 \text{ L}^{-2}$
 - $1.08 \times 10^{-14} \text{ mol}^2 \text{ L}^{-2}$
 - $1.08 \times 10^{-12} \text{ mol}^2 \text{ L}^{-2}$
 - $1.08 \times 10^{-8} \text{ mol}^2 \text{ L}^{-2}$
49. On which of the following properties does the coagulating power of an ion depend ?
- The magnitude of the charge on the ion alone
 - Both magnitude and sign of the charge on the ion
 - Size of the ion alone
 - The sign of charge on the ion alone
50. Considering Ellingham diagram, which of the following metals can be used to reduce alumina ?
- Fe
 - Mg
 - Zn
 - Cu
51. The correct order of atomic radii in group 13 elements is
- $\text{B} < \text{Al} < \text{In} < \text{Ga} < \text{Tl}$
 - $\text{B} < \text{Ga} < \text{Al} < \text{Tl} < \text{In}$
 - $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 - $\text{B} < \text{Ga} < \text{Al} < \text{In} < \text{Tl}$
52. In the structure of ClF_3 , the number of lone pairs of electrons on central atom 'Cl' is
- one
 - four
 - two
 - three
53. The correct order of N-compounds in its decreasing order of oxidation states is
- HNO_3 , NO, N_2 , NH_4Cl
 - HNO_3 , NH_4Cl , NO, N_2
 - HNO_3 , NO, NH_4Cl , N_2
 - NH_4Cl , N_2 , NO, HNO_3
54. Which of the following statements is **not** true for halogens ?
- All form monobasic oxyacids.
 - All but fluorine show positive oxidation states.
 - All are oxidizing agents.
 - Chlorine has the highest electron-gain enthalpy.
55. Which one of the following elements is unable to form MF_6^{3-} ion ?
- Ga
 - B
 - Al
 - In

56. Regarding cross-linked or network polymers, which of the following statements is **incorrect** ?

- (1) They contain covalent bonds between various linear polymer chains.
- (2) Examples are bakelite and melamine.
- (3) They are formed from bi- and tri-functional monomers.
- (4) They contain strong covalent bonds in their polymer chains.

57. Nitration of aniline in strong acidic medium also gives m-nitroaniline because

- (1) In spite of substituents nitro group always goes to only m-position.
- (2) In absence of substituents nitro group always goes to m-position.
- (3) In electrophilic substitution reactions amino group is meta directive.
- (4) In acidic (strong) medium aniline is present as anilinium ion.

58. Which of the following oxides is most acidic in nature ?

- (1) MgO
- (2) BaO
- (3) BeO
- (4) CaO

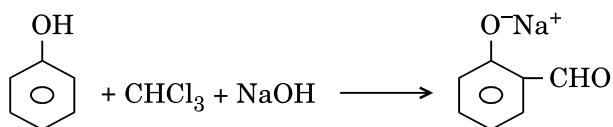
59. The difference between amylose and amylopectin is

- (1) Amylopectin have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ α -linkage
- (2) Amylopectin have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ β -linkage
- (3) Amylose have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ β -linkage
- (4) Amylose is made up of glucose and galactose

60. A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with conc. H_2SO_4 . The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be

- (1) 1.4
- (2) 2.8
- (3) 3.0
- (4) 4.4

61. In the reaction



the electrophile involved is

- (1) dichloromethyl cation (CHCl_2^+)
- (2) dichloromethyl anion (CHCl_2^-)
- (3) formyl cation (CHO^+)
- (4) dichlorocarbene ($:\text{CCl}_2$)

62. Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their

- (1) formation of intramolecular H-bonding
- (2) more extensive association of carboxylic acid via van der Waals force of attraction
- (3) formation of carboxylate ion
- (4) formation of intermolecular H-bonding

63. Compound A, $\text{C}_8\text{H}_{10}\text{O}$, is found to react with NaOI (produced by reacting Y with NaOH) and yields a yellow precipitate with characteristic smell.

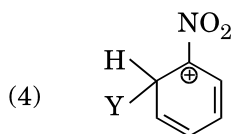
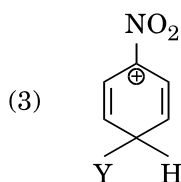
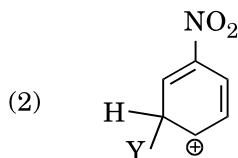
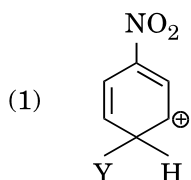
A and Y are respectively

- (1) $\text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}_2-\text{OH}$ and I_2
- (2) $\text{C}_6\text{H}_5-\text{CH}(\text{OH})-\text{CH}_3$ and I_2
- (3) $\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{OH}$ and I_2
- (4) $\text{CH}_3-\text{C}_6\text{H}_3(\text{OH})_2-\text{CH}_3$ and I_2

64. Which of the following molecules represents the order of hybridisation sp^2 , sp^2 , sp , sp from left to right atoms ?

- (1) $HC \equiv C - C \equiv CH$
- (2) $CH_2 = CH - CH = CH_2$
- (3) $CH_2 = CH - C \equiv CH$
- (4) $CH_3 - CH = CH - CH_3$

65. Which of the following carbocations is expected to be most stable ?



66. Which of the following is correct with respect to -I effect of the substituents ? (R = alkyl)

- (1) $-NH_2 < -OR < -F$
- (2) $-NH_2 > -OR > -F$
- (3) $-NR_2 < -OR < -F$
- (4) $-NR_2 > -OR > -F$

67. The type of isomerism shown by the complex $[CoCl_2(en)_2]$ is

- (1) Geometrical isomerism
- (2) Ionization isomerism
- (3) Coordination isomerism
- (4) Linkage isomerism

68. Which one of the following ions exhibits d-d transition and paramagnetism as well ?

- (1) CrO_4^{2-}
- (2) MnO_4^-
- (3) $Cr_2O_7^{2-}$
- (4) MnO_4^{2-}

69. Iron carbonyl, $Fe(CO)_5$ is

- (1) tetranuclear
- (2) trinuclear
- (3) mononuclear
- (4) dinuclear

70. Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the **correct** code :

	Column I	Column II
a.	Co^{3+}	i. $\sqrt{8}$ B.M.
b.	Cr^{3+}	ii. $\sqrt{35}$ B.M.
c.	Fe^{3+}	iii. $\sqrt{3}$ B.M.
d.	Ni^{2+}	iv. $\sqrt{24}$ B.M.
		v. $\sqrt{15}$ B.M.

	a	b	c	d
(1)	iv	v	ii	i
(2)	iv	i	ii	iii
(3)	i	ii	iii	iv
(4)	iii	v	i	ii

71. The geometry and magnetic behaviour of the complex $[Ni(CO)_4]$ are

- (1) square planar geometry and diamagnetic
- (2) square planar geometry and paramagnetic
- (3) tetrahedral geometry and diamagnetic
- (4) tetrahedral geometry and paramagnetic

72. The correct difference between first- and second-order reactions is that
- (1) the rate of a first-order reaction does not depend on reactant concentrations; the rate of a second-order reaction does depend on reactant concentrations
 - (2) a first-order reaction can be catalyzed; a second-order reaction cannot be catalyzed
 - (3) the half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second-order reaction does depend on $[A]_0$
 - (4) the rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations

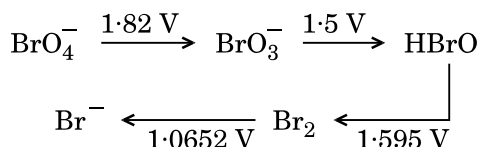
73. Among CaH_2 , BeH_2 , BaH_2 , the order of ionic character is

- (1) $\text{BeH}_2 < \text{CaH}_2 < \text{BaH}_2$
- (2) $\text{BeH}_2 < \text{BaH}_2 < \text{CaH}_2$
- (3) $\text{CaH}_2 < \text{BeH}_2 < \text{BaH}_2$
- (4) $\text{BaH}_2 < \text{BeH}_2 < \text{CaH}_2$

74. In which case is the number of molecules of water maximum?

- (1) 18 mL of water
- (2) 0.00224 L of water vapours at 1 atm and 273 K
- (3) 0.18 g of water
- (4) 10^{-3} mol of water

75. Consider the change in oxidation state of Bromine corresponding to different emf values as shown in the diagram below :



Then the species undergoing disproportionation is

- (1) BrO_3^-
- (2) Br_2
- (3) BrO_4^-
- (4) HBrO

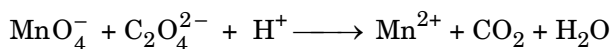
76. The bond dissociation energies of X_2 , Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is -200 kJ mol^{-1} . The bond dissociation energy of X_2 will be

- (1) 200 kJ mol^{-1}
- (2) 800 kJ mol^{-1}
- (3) 100 kJ mol^{-1}
- (4) 400 kJ mol^{-1}

77. When initial concentration of the reactant is doubled, the half-life period of a zero order reaction

- (1) is halved
- (2) is tripled
- (3) is doubled
- (4) remains unchanged

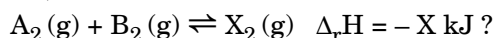
78. For the redox reaction



the correct coefficients of the reactants for the balanced equation are

	MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(1)	16	5	2
(2)	2	16	5
(3)	2	5	16
(4)	5	16	2

79. Which one of the following conditions will favour maximum formation of the product in the reaction,



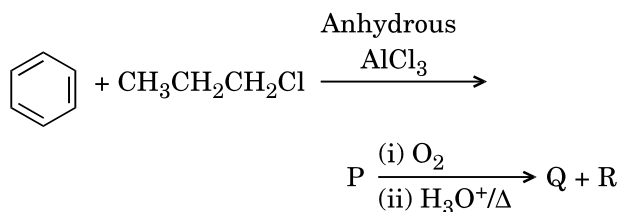
- (1) Low temperature and high pressure
- (2) High temperature and high pressure
- (3) Low temperature and low pressure
- (4) High temperature and low pressure

80. The correction factor 'a' to the ideal gas equation corresponds to

- (1) density of the gas molecules
- (2) electric field present between the gas molecules
- (3) volume of the gas molecules
- (4) forces of attraction between the gas molecules

81. Magnesium reacts with an element (X) to form an ionic compound. If the ground state electronic configuration of (X) is $1s^2 2s^2 2p^3$, the simplest formula for this compound is
- Mg_2X_3
 - Mg_2X
 - MgX_2
 - Mg_3X_2
82. Iron exhibits bcc structure at room temperature. Above 900°C , it transforms to fcc structure. The ratio of density of iron at room temperature to that at 900°C (assuming molar mass and atomic radii of iron remains constant with temperature) is
- $\frac{\sqrt{3}}{\sqrt{2}}$
 - $\frac{3\sqrt{3}}{4\sqrt{2}}$
 - $\frac{4\sqrt{3}}{3\sqrt{2}}$
 - $\frac{1}{2}$
83. Consider the following species :
 CN^+ , CN^- , NO and CN
 Which one of these will have the highest bond order ?
- NO
 - CN^+
 - CN^-
 - CN
84. Which one is a **wrong** statement ?
- Total orbital angular momentum of electron in 's' orbital is equal to zero.
 - The electronic configuration of N atom is
- | | | | | |
|------------------------------|------------------------------|--------------------|--------------------|----------------------|
| $1s^2$ | $2s^2$ | $2p_x^1$ | $2p_y^1$ | $2p_z^1$ |
| $\boxed{\uparrow\downarrow}$ | $\boxed{\uparrow\downarrow}$ | $\boxed{\uparrow}$ | $\boxed{\uparrow}$ | $\boxed{\downarrow}$ |
- An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.
 - The value of m for d_{z^2} is zero.
85. Which oxide of nitrogen is **not** a common pollutant introduced into the atmosphere both due to natural and human activity ?
- N_2O_5
 - N_2O
 - NO_2
 - NO
86. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is
- $\text{CH} \equiv \text{CH}$
 - $\text{CH}_3 - \text{CH}_3$
 - $\text{CH}_2 = \text{CH}_2$
 - CH_4
87. The compound A on treatment with Na gives B, and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order
- $\text{C}_2\text{H}_5\text{OH}$, C_2H_6 , $\text{C}_2\text{H}_5\text{Cl}$
 - $\text{C}_2\text{H}_5\text{Cl}$, C_2H_6 , $\text{C}_2\text{H}_5\text{OH}$
 - $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_2\text{H}_5\text{Cl}$, $\text{C}_2\text{H}_5\text{ONa}$
 - $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_2\text{H}_5\text{ONa}$, $\text{C}_2\text{H}_5\text{Cl}$
88. The compound C_7H_8 undergoes the following reactions :
- $$\text{C}_7\text{H}_8 \xrightarrow{3 \text{ Cl}_2 / \Delta} \text{A} \xrightarrow{\text{Br}_2 / \text{Fe}} \text{B} \xrightarrow{\text{Zn} / \text{HCl}} \text{C}$$
- The product 'C' is
- m*-bromotoluene
 - 3-bromo-2,4,6-trichlorotoluene
 - o*-bromotoluene
 - p*-bromotoluene

89. Identify the major products P, Q and R in the following sequence of reactions :



P

Q

R

- (1)  ,  , $\text{CH}_3\text{CH}_2 - \text{OH}$
- (2)  ,  , $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- (3)  ,  , 
- (4)  ,  , $\text{CH}_3 - \text{CO} - \text{CH}_3$

90. Which of the following compounds can form a zwitterion ?

- (1) Aniline
- (2) Benzoic acid
- (3) Acetanilide
- (4) Glycine

91. Which of the following statements is **correct** ?

- (1) Ovules are not enclosed by ovary wall in gymnosperms.
- (2) Horsetails are gymnosperms.
- (3) *Selaginella* is heterosporous, while *Salvinia* is homosporous.
- (4) Stems are usually unbranched in both *Cycas* and *Cedrus*.

92. Pneumatophores occur in

- (1) Halophytes
- (2) Carnivorous plants
- (3) Free-floating hydrophytes
- (4) Submerged hydrophytes

93. Sweet potato is a modified

- (1) Stem
- (2) Tap root
- (3) Adventitious root
- (4) Rhizome

94. Plants having little or no secondary growth are

- (1) Grasses
- (2) Conifers
- (3) Deciduous angiosperms
- (4) Cycads

95. Casparian strips occur in

- (1) Epidermis
- (2) Cortex
- (3) Pericycle
- (4) Endodermis

96. Secondary xylem and phloem in dicot stem are produced by

- (1) Apical meristems
- (2) Phellogen
- (3) Vascular cambium
- (4) Axillary meristems

97. Select the **wrong** statement :

- (1) Cell wall is present in members of Fungi and Plantae.
- (2) Pseudopodia are locomotory and feeding structures in Sporozoans.
- (3) Mushrooms belong to Basidiomycetes.
- (4) Mitochondria are the powerhouse of the cell in all kingdoms except Monera.

98. The experimental proof for semiconservative replication of DNA was first shown in a
- (1) Fungus
 - (2) Plant
 - (3) Bacterium
 - (4) Virus
99. Select the **correct** match :
- (1) Alec Jeffreys – *Streptococcus pneumoniae*
 - (2) Matthew Meselson and F. Stahl – *Pisum sativum*
 - (3) Alfred Hershey and Martha Chase – TMV
 - (4) Francois Jacob and Jacques Monod – *Lac operon*
100. Select the **correct** statement :
- (1) Franklin Stahl coined the term “linkage”.
 - (2) Spliceosomes take part in translation.
 - (3) Punnett square was developed by a British scientist.
 - (4) Transduction was discovered by S. Altman.
101. Which of the following pairs is **wrongly** matched ?
- (1) Starch synthesis in pea : Multiple alleles
 - (2) XO type sex : Grasshopper determination
 - (3) ABO blood grouping : Co-dominance
 - (4) T.H. Morgan : Linkage
102. Offsets are produced by
- (1) Meiotic divisions
 - (2) Parthenocarpy
 - (3) Mitotic divisions
 - (4) Parthenogenesis
103. Which of the following flowers only once in its life-time ?
- (1) Bamboo species
 - (2) Mango
 - (3) Jackfruit
 - (4) Papaya
104. Which of the following has proved helpful in preserving pollen as fossils ?
- (1) Pollenkitt
 - (2) Oil content
 - (3) Cellulosic intine
 - (4) Sporopollenin
105. Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes ?
- (1) Retrovirus
 - (2) λ phage
 - (3) Ti plasmid
 - (4) pBR 322
106. The correct order of steps in Polymerase Chain Reaction (PCR) is
- (1) Extension, Denaturation, Annealing
 - (2) Denaturation, Extension, Annealing
 - (3) Annealing, Extension, Denaturation
 - (4) Denaturation, Annealing, Extension
107. In India, the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is
- (1) Indian Council of Medical Research (ICMR)
 - (2) Research Committee on Genetic Manipulation (RCGM)
 - (3) Council for Scientific and Industrial Research (CSIR)
 - (4) Genetic Engineering Appraisal Committee (GEAC)
108. Use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called
- (1) Bio-infringement
 - (2) Biodegradation
 - (3) Biopiracy
 - (4) Bioexploitation
109. A ‘new’ variety of rice was patented by a foreign company, though such varieties have been present in India for a long time. This is related to
- (1) Co-667
 - (2) Lerma Rojo
 - (3) Sharbati Sonora
 - (4) Basmati
110. Select the **correct** match :
- (1) Ribozyme – Nucleic acid
 - (2) T.H. Morgan – Transduction
 - (3) $F_2 \times$ Recessive parent – Dihybrid cross
 - (4) G. Mendel – Transformation

- 111.** Niche is
- (1) all the biological factors in the organism's environment
 - (2) the range of temperature that the organism needs to live
 - (3) the physical space where an organism lives
 - (4) the functional role played by the organism where it lives
- 112.** Which of the following is a secondary pollutant ?
- (1) CO
 - (2) SO₂
 - (3) CO₂
 - (4) O₃
- 113.** World Ozone Day is celebrated on
- (1) 5th June
 - (2) 16th September
 - (3) 21st April
 - (4) 22nd April
- 114.** Natalty refers to
- (1) Death rate
 - (2) Number of individuals leaving the habitat
 - (3) Birth rate
 - (4) Number of individuals entering a habitat
- 115.** In stratosphere, which of the following elements acts as a catalyst in degradation of ozone and release of molecular oxygen ?
- (1) Carbon
 - (2) Fe
 - (3) Cl
 - (4) Oxygen
- 116.** What type of ecological pyramid would be obtained with the following data ?
 Secondary consumer : 120 g
 Primary consumer : 60 g
 Primary producer : 10 g
- (1) Inverted pyramid of biomass
 - (2) Upright pyramid of numbers
 - (3) Pyramid of energy
 - (4) Upright pyramid of biomass
- 117.** The Golgi complex participates in
- (1) Fatty acid breakdown
 - (2) Respiration in bacteria
 - (3) Formation of secretory vesicles
 - (4) Activation of amino acid
- 118.** Which of the following is **not** a product of light reaction of photosynthesis ?
- (1) ATP
 - (2) NADPH
 - (3) NADH
 - (4) Oxygen
- 119.** Which among the following is **not** a prokaryote ?
- (1) *Saccharomyces*
 - (2) *Nostoc*
 - (3) *Mycobacterium*
 - (4) *Oscillatoria*
- 120.** Stomatal movement is **not** affected by
- (1) Temperature
 - (2) O₂ concentration
 - (3) Light
 - (4) CO₂ concentration
- 121.** Which of the following is true for nucleolus ?
- (1) Larger nucleoli are present in dividing cells.
 - (2) It takes part in spindle formation.
 - (3) It is a membrane-bound structure.
 - (4) It is a site for active ribosomal RNA synthesis.
- 122.** The stage during which separation of the paired homologous chromosomes begins is
- (1) Pachytene
 - (2) Diakinesis
 - (3) Diplotene
 - (4) Zygotene
- 123.** The two functional groups characteristic of sugars are
- (1) hydroxyl and methyl
 - (2) carbonyl and phosphate
 - (3) carbonyl and methyl
 - (4) carbonyl and hydroxyl
- 124.** Stomata in grass leaf are
- (1) Dumb-bell shaped
 - (2) Rectangular
 - (3) Kidney shaped
 - (4) Barrel shaped

- 125.** Which one of the following plants shows a very close relationship with a species of moth, where none of the two can complete its life cycle without the other ?
- (1) *Hydrilla*
 - (2) Banana
 - (3) *Yucca*
 - (4) *Viola*
- 126.** Pollen grains can be stored for several years in liquid nitrogen having a temperature of
- (1) – 120°C
 - (2) – 196°C
 - (3) – 80°C
 - (4) – 160°C
- 127.** Double fertilization is
- (1) Fusion of two male gametes of a pollen tube with two different eggs
 - (2) Fusion of two male gametes with one egg
 - (3) Fusion of one male gamete with two polar nuclei
 - (4) Syngamy and triple fusion
- 128.** Oxygen is **not** produced during photosynthesis by
- (1) Green sulphur bacteria
 - (2) *Cycas*
 - (3) *Nostoc*
 - (4) *Chara*
- 129.** Which of the following elements is responsible for maintaining turgor in cells ?
- (1) Magnesium
 - (2) Potassium
 - (3) Sodium
 - (4) Calcium
- 130.** What is the role of NAD^+ in cellular respiration ?
- (1) It functions as an enzyme.
 - (2) It is a nucleotide source for ATP synthesis.
 - (3) It functions as an electron carrier.
 - (4) It is the final electron acceptor for anaerobic respiration.
- 131.** In which of the following forms is iron absorbed by plants ?
- (1) Ferric
 - (2) Free element
 - (3) Ferrous
 - (4) Both ferric and ferrous

- 132.** Winged pollen grains are present in
- (1) Mustard
 - (2) Mango
 - (3) *Cycas*
 - (4) *Pinus*
- 133.** After karyogamy followed by meiosis, spores are produced exogenously in
- (1) *Neurospora*
 - (2) *Agaricus*
 - (3) *Alternaria*
 - (4) *Saccharomyces*
- 134.** Which one is **wrongly** matched ?
- (1) Uniflagellate gametes – *Polysiphonia*
 - (2) Gemma cups – *Marchantia*
 - (3) Biflagellate zoospores – Brown algae
 - (4) Unicellular organism – *Chlorella*
- 135.** Match the items given in Column I with those in Column II and select the **correct** option given below :
- | <i>Column I</i> | | <i>Column II</i> | |
|-----------------|------|---|--|
| a. Herbarium | i. | It is a place having a collection of preserved plants and animals. | |
| b. Key | ii. | A list that enumerates methodically all the species found in an area with brief description aiding identification. | |
| c. Museum | iii. | Is a place where dried and pressed plant specimens mounted on sheets are kept. | |
| d. Catalogue | iv. | A booklet containing a list of characters and their alternates which are helpful in identification of various taxa. | |
- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | i | iv | iii | ii |
| (2) | ii | iv | iii | i |
| (3) | iii | ii | i | iv |
| (4) | iii | iv | i | ii |

- 136.** Which of the following is an amino acid derived hormone ?
- (1) Epinephrine
 - (2) Estradiol
 - (3) Ecdysone
 - (4) Estriol
- 137.** Which of the following structures or regions is **incorrectly** paired with its function ?
- (1) Medulla oblongata : controls respiration and cardiovascular reflexes.
 - (2) Hypothalamus : production of releasing hormones and regulation of temperature, hunger and thirst.
 - (3) Limbic system : consists of fibre tracts that interconnect different regions of brain; controls movement.
 - (4) Corpus callosum : band of fibers connecting left and right cerebral hemispheres.
- 138.** The transparent lens in the human eye is held in its place by
- (1) ligaments attached to the ciliary body
 - (2) smooth muscles attached to the iris
 - (3) ligaments attached to the iris
 - (4) smooth muscles attached to the ciliary body
- 139.** Which of the following hormones can play a significant role in osteoporosis ?
- (1) Aldosterone and Prolactin
 - (2) Estrogen and Parathyroid hormone
 - (3) Progesterone and Aldosterone
 - (4) Parathyroid hormone and Prolactin
- 140.** Among the following sets of examples for divergent evolution, select the **incorrect** option :
- (1) Forelimbs of man, bat and cheetah
 - (2) Brain of bat, man and cheetah
 - (3) Heart of bat, man and cheetah
 - (4) Eye of octopus, bat and man
- 141.** Which of the following is **not** an autoimmune disease ?
- (1) Psoriasis
 - (2) Alzheimer's disease
 - (3) Rheumatoid arthritis
 - (4) Vitiligo
- 142.** Which of the following characteristics represent 'Inheritance of blood groups' in humans ?
- a. Dominance
 - b. Co-dominance
 - c. Multiple allele
 - d. Incomplete dominance
 - e. Polygenic inheritance
- (1) b, c and e
 - (2) b, d and e
 - (3) a, b and c
 - (4) a, c and e
- 143.** In which disease does mosquito transmitted pathogen cause chronic inflammation of lymphatic vessels ?
- (1) Elephantiasis
 - (2) Ringworm disease
 - (3) Ascariasis
 - (4) Amoebiasis
- 144.** The similarity of bone structure in the forelimbs of many vertebrates is an example of
- (1) Homology
 - (2) Convergent evolution
 - (3) Analogy
 - (4) Adaptive radiation
- 145.** Conversion of milk to curd improves its nutritional value by increasing the amount of
- (1) Vitamin D
 - (2) Vitamin B₁₂
 - (3) Vitamin A
 - (4) Vitamin E

146. Which one of the following population interactions is widely used in medical science for the production of antibiotics ?

- (1) Commensalism
- (2) Parasitism
- (3) Mutualism
- (4) Amensalism

147. All of the following are included in 'Ex-situ conservation' *except*

- (1) Wildlife safari parks
- (2) Botanical gardens
- (3) Sacred groves
- (4) Seed banks

148. Match the items given in Column I with those in Column II and select the **correct** option given below :

<i>Column I</i>		<i>Column II</i>	
a. Eutrophication		i. UV-B radiation	
b. Sanitary landfill		ii. Deforestation	
c. Snow blindness		iii. Nutrient enrichment	
d. Jhum cultivation		iv. Waste disposal	
a	b	c	d
(1) ii	i	iii	iv
(2) iii	iv	i	ii
(3) i	iii	iv	ii
(4) i	ii	iv	iii

149. In a growing population of a country,

- (1) pre-reproductive individuals are more than the reproductive individuals.
- (2) reproductive and pre-reproductive individuals are equal in number.
- (3) reproductive individuals are less than the post-reproductive individuals.
- (4) pre-reproductive individuals are less than the reproductive individuals.

150. Which part of poppy plant is used to obtain the drug "Smack" ?

- (1) Flowers
- (2) Roots
- (3) Latex
- (4) Leaves

151. Hormones secreted by the placenta to maintain pregnancy are

- (1) hCG, hPL, progestogens, prolactin
- (2) hCG, hPL, progestogens, estrogens
- (3) hCG, hPL, estrogens, relaxin, oxytocin
- (4) hCG, progestogens, estrogens, glucocorticoids

152. The contraceptive 'SAHEL'

- (1) blocks estrogen receptors in the uterus, preventing eggs from getting implanted.
- (2) is an IUD.
- (3) increases the concentration of estrogen and prevents ovulation in females.
- (4) is a post-coital contraceptive.

153. The amnion of mammalian embryo is derived from

- (1) ectoderm and mesoderm
- (2) mesoderm and trophoblast
- (3) endoderm and mesoderm
- (4) ectoderm and endoderm

154. The difference between spermiogenesis and spermiation is

- (1) In spermiogenesis spermatids are formed, while in spermiation spermatozoa are formed.
- (2) In spermiogenesis spermatozoa from sertoli cells are released into the cavity of seminiferous tubules, while in spermiation spermatozoa are formed.
- (3) In spermiogenesis spermatozoa are formed, while in spermiation spermatids are formed.
- (4) In spermiogenesis spermatozoa are formed, while in spermiation spermatozoa are released from sertoli cells into the cavity of seminiferous tubules.

155. Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively ?

- (1) Inflammation of bronchioles; Decreased respiratory surface
- (2) Increased respiratory surface; Inflammation of bronchioles
- (3) Increased number of bronchioles; Increased respiratory surface
- (4) Decreased respiratory surface; Inflammation of bronchioles

156. Match the items given in Column I with those in Column II and select the **correct** option given below :

<i>Column I</i>		<i>Column II</i>	
a. Tricuspid valve	i.	Between left atrium and left ventricle	
b. Bicuspid valve	ii.	Between right ventricle and pulmonary artery	
c. Semilunar valve	iii.	Between right atrium and right ventricle	

- | | a | b | c |
|-----|----------|----------|----------|
| (1) | iii | i | ii |
| (2) | i | ii | iii |
| (3) | i | iii | ii |
| (4) | ii | i | iii |

157. Match the items given in Column I with those in Column II and select the **correct** option given below :

<i>Column I</i>		<i>Column II</i>	
a. Tidal volume	i.	2500 – 3000 mL	
b. Inspiratory Reserve volume	ii.	1100 – 1200 mL	
c. Expiratory Reserve volume	iii.	500 – 550 mL	
d. Residual volume	iv.	1000 – 1100 mL	

- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | iii | ii | i | iv |
| (2) | i | iv | ii | iii |
| (3) | iii | i | iv | ii |
| (4) | iv | iii | ii | i |

158. Match the items given in Column I with those in Column II and select the **correct** option given below :

<i>Column I</i>		<i>Column II</i>	
a. Glycosuria	i.	Accumulation of uric acid in joints	
b. Gout	ii.	Mass of crystallised salts within the kidney	
c. Renal calculi	iii.	Inflammation in glomeruli	
d. Glomerular nephritis	iv.	Presence of glucose in urine	

- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | iii | ii | iv | i |
| (2) | ii | iii | i | iv |
| (3) | i | ii | iii | iv |
| (4) | iv | i | ii | iii |

159. Match the items given in Column I with those in Column II and select the **correct** option given below :

<i>Column I</i> (Function)		<i>Column II</i> (Part of Excretory System)	
a. Ultrafiltration	i.	Henle's loop	
b. Concentration of urine	ii.	Ureter	
c. Transport of urine	iii.	Urinary bladder	
d. Storage of urine	iv.	Malpighian corpuscle	
	v.	Proximal convoluted tubule	

- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | iv | v | ii | iii |
| (2) | v | iv | i | ii |
| (3) | iv | i | ii | iii |
| (4) | v | iv | i | iii |

- 160.** Which of the following events does **not** occur in rough endoplasmic reticulum ?
- (1) Protein folding
 - (2) Cleavage of signal peptide
 - (3) Protein glycosylation
 - (4) Phospholipid synthesis
- 161.** Which of these statements is **incorrect** ?
- (1) Enzymes of TCA cycle are present in mitochondrial matrix.
 - (2) Glycolysis operates as long as it is supplied with NAD that can pick up hydrogen atoms.
 - (3) Glycolysis occurs in cytosol.
 - (4) Oxidative phosphorylation takes place in outer mitochondrial membrane.
- 162.** Nissl bodies are mainly composed of
- (1) Proteins and lipids
 - (2) Nucleic acids and SER
 - (3) DNA and RNA
 - (4) Free ribosomes and RER
- 163.** Which of the following terms describe human dentition ?
- (1) Thecodont, Diphyodont, Homodont
 - (2) Pleurodont, Monophyodont, Homodont
 - (3) Thecodont, Diphyodont, Heterodont
 - (4) Pleurodont, Diphyodont, Heterodont
- 164.** Select the **incorrect** match :
- (1) Lampbrush – Diplotene bivalents chromosomes
 - (2) Submetacentric – L-shaped chromosomes
 - (3) Allosomes – Sex chromosomes
 - (4) Polytene – Oocytes of amphibians Chromosomes
- 165.** Many ribosomes may associate with a single mRNA to form multiple copies of a polypeptide simultaneously. Such strings of ribosomes are termed as
- (1) Polysome
 - (2) Plastidome
 - (3) Polyhedral bodies
 - (4) Nucleosome
- 166.** According to Hugo de Vries, the mechanism of evolution is
- (1) Multiple step mutations
 - (2) Phenotypic variations
 - (3) Saltation
 - (4) Minor mutations
- 167.** Match the items given in Column I with those in Column II and select the **correct** option given below :
- | <i>Column I</i> | <i>Column II</i> |
|------------------------|------------------------------------|
| a. Proliferative Phase | i. Breakdown of endometrial lining |
| b. Secretory Phase | ii. Follicular Phase |
| c. Menstruation | iii. Luteal Phase |
- | | a | b | c |
|-----|----------|----------|----------|
| (1) | iii | ii | i |
| (2) | ii | iii | i |
| (3) | i | iii | ii |
| (4) | iii | i | ii |
- 168.** All of the following are part of an operon *except*
- (1) an operator
 - (2) an enhancer
 - (3) structural genes
 - (4) a promoter
- 169.** AGGTATCGCAT is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed mRNA ?
- (1) AGGUAUCGCAU
 - (2) ACCUAUGCGAU
 - (3) UGGTUTCGCAT
 - (4) UCCAUAGCGUA
- 170.** A woman has an X-linked condition on one of her X chromosomes. This chromosome can be inherited by
- (1) Only daughters
 - (2) Only grandchildren
 - (3) Only sons
 - (4) Both sons and daughters

171. Which of the following gastric cells indirectly help in erythropoiesis ?

- (1) Chief cells
- (2) Goblet cells
- (3) Mucous cells
- (4) Parietal cells

172. Match the items given in Column I with those in Column II and select the **correct** option given below :

Column I

Column II

- | | |
|---------------|------------------------|
| a. Fibrinogen | i. Osmotic balance |
| b. Globulin | ii. Blood clotting |
| c. Albumin | iii. Defence mechanism |

a b c

- | | | |
|---------|-----|-----|
| (1) iii | ii | i |
| (2) i | iii | ii |
| (3) i | ii | iii |
| (4) ii | iii | i |

173. Which of the following is an occupational respiratory disorder ?

- (1) Anthracis
- (2) Botulism
- (3) Silicosis
- (4) Emphysema

174. Calcium is important in skeletal muscle contraction because it

- (1) binds to troponin to remove the masking of active sites on actin for myosin.
- (2) detaches the myosin head from the actin filament.
- (3) activates the myosin ATPase by binding to it.
- (4) prevents the formation of bonds between the myosin cross bridges and the actin filament.

175. Identify the vertebrate group of animals characterized by crop and gizzard in its digestive system.

- (1) Amphibia
- (2) Aves
- (3) Reptilia
- (4) Osteichthyes

176. Ciliates differ from all other protozoans in

- (1) using flagella for locomotion
- (2) using pseudopodia for capturing prey
- (3) having a contractile vacuole for removing excess water
- (4) having two types of nuclei

177. Which of the following features is used to identify a male cockroach from a female cockroach ?

- (1) Presence of a boat shaped sternum on the 9th abdominal segment
- (2) Forewings with darker tegmina
- (3) Presence of caudal styles
- (4) Presence of anal cerci

178. Which one of these animals is **not** a homeotherm ?

- (1) *Macropus*
- (2) *Camelus*
- (3) *Chelone*
- (4) *Psittacula*

179. Which of the following animals does **not** undergo metamorphosis ?

- (1) Earthworm
- (2) Moth
- (3) Tunicate
- (4) Starfish

180. Which of the following organisms are known as chief producers in the oceans ?

- (1) Dinoflagellates
- (2) Cyanobacteria
- (3) Diatoms
- (4) Euglenoids

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

Read carefully the following instructions :

1. Each candidate must show on demand his/her Admit Card to the Invigilator.
2. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. **Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.**
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.